3.137. Is carbon dioxide at 31°C and 70 atm a two-phase or a single-phase system? Repeat for 25°C.

PROBLEMS THAT REQUIRE WRITING COMPUTER PROGRAMS

3.1. You need to fill a reactor with 8.00 lb of NH_3 gas to a pressure of 150 psia at a temperature of 250°F. Apply the van der Waals equation to find the volume of the reactor. Prepare a computer program to solve Eq. (3.13) for V. How do you select the initial estimate of V to start the calculations?

3.2. An experiment was designed to test the validity of Charles' law using the expansion of a fixed amount of air in a balloon in a flask of water. The data taken were as follows at 751 mm Hg constant pressure:

Temp. (°C)	Incremental vol. of balloon (cm^3)	Temp. (°C)	Incremental vol. of balloon (cm^3)
22	0.0	44	4.2
23	0.3	47	4.9
24	0.45	50	5.5
25	0.8	53	6.5
26	0.8	55	7.0
27	1.2	59	9.0
29	2.0	61	10.0
34	3.0	63	11.9
37	3.2	65	13.5
39	3.8	67	14.7
43	3.8		

Ascertain how well the ideal gas law (Charles' law) is obeyed. A least-squares computer code should be used to evaluate the experimental data. What was the initial volume of gas (at 22°C)? Assume that the pressure on the air in the balloon is constant.

3.3. Use the Benedict–Webb–Rubin (BWR) equation of state (see Table 3.2)

$$p = RT\rho + \left(B_0RT - A_0 - \frac{C_0}{T^2}\right)\rho^2 + (bRT - a)\rho^3$$

$$+ a\alpha\rho^6 + \frac{c\rho^3}{T^2}(1 + \gamma\rho^2)\exp(-\gamma\rho^2)$$

where ρ is the density, to predict the density of a mixture of two nonideal gases. Let the pressure be in atmospheres and calculate the density in both gram moles per liter and pound moles per cubic foot for selected pairs of p and T. Use the following rule illustrated for A_0 to obtain the coefficients A_0, B_0, C_0, a, b, c, α, and γ for the mixture:

$$A_0 = \left(\sum_{i=1}^{n} x_i A_{0i}^{1/2}\right)^2$$

where n is the number of components in the mixture and x_i is the mole fraction. Values of the BWR constants for 38 pure components have been tabulated by H. W. Cooper and J. C. Goldfrank, *Hydrocarbon Processing*, v. 46, no. 12, p. 141 (1967). For reduced temperatures above 0.6 and reduced specific volumes less than 0.5, the BWR equation should be accurate to within 1%.

3.4. A cylinder containing 11 lb of CO_2 initially at 200°F and 1500 psia is heated to a final temperature of 400°F. Prepare a computer program to calculate the final pressure in psia using the van der Waals equation.

3.5. Write a complete program to convert mm Hg to psia for the range 100 to 760 mm Hg at each 10-mm increment. Print the results of each conversion.

ENERGY BALANCES 4

In this chapter we take up the second prominent topic in this book, energy balances. It is reported that the average power consumption in the United States per capita is equivalent to three dozen hard-working male slaves or nine horses running at full gallop! Clearly, conversion of inexpensive sources of energy has made the industrial revolution possible. Figure 4.1 indicates what these sources of energy are and how they are used.

To provide publicly acceptable, effective, and yet economical conversion of our resources into energy and to properly utilize the energy so generated, you must understand the basic principles underlying the generation, uses, and transformation of energy in its different forms. The answers to questions such as "Is thermal pollution inherently necessary"; "What is the most economic source of fuel"; "What can be done with waste heat"; "How much steam at what temperature and pressure are needed to heat a process" and related questions can only arise from an understanding of the treatment of energy transfer by natural processes or machines. As an example, examine Fig. 4.2 and try to answer the question: What can be done economically to reduce the loss of energy rejected as heat? Can you offer reasonable suggestions at this stage in your professional life?

In this chapter we discuss energy balances together with the accessory background information needed to understand and apply them correctly. Before taking up the energy balance itself, we discuss the types of energy that are of major interest to engineers and scientists and some of the methods that are used to measure and evaluate these forms of energy. Our main attention will be devoted to heat, work, enthalpy, and internal energy. Next, the energy balance will be described and applied to practical problems. Finally, we shall discuss the energy balance associated with reaction and how energy, evolving from reactions, fits into industrial process calculations. Figure 4.3 shows the relationships among the topics discussed in this chapter and in previous chapters.

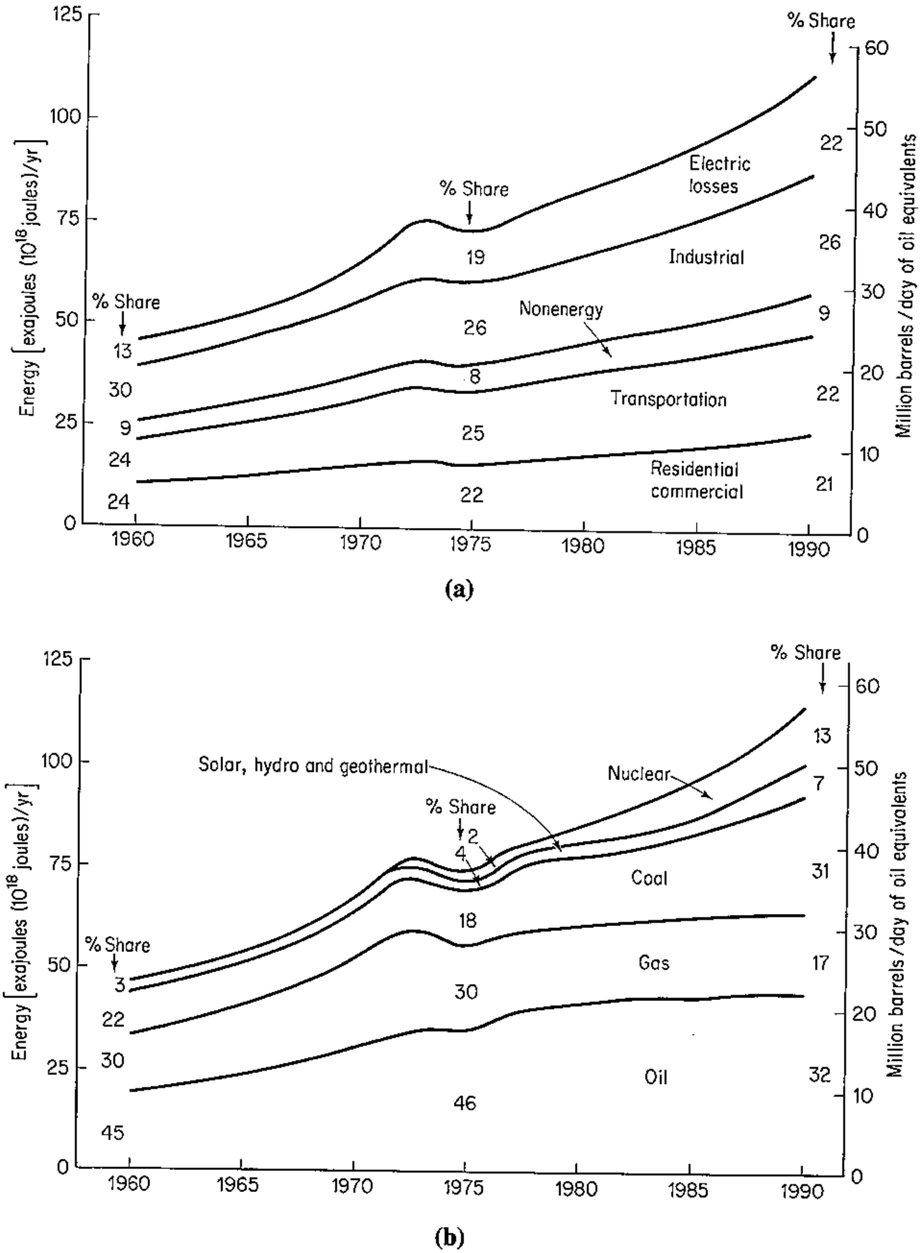


Figure 4.1 (a) United States energy demand by consuming sector; (b) United States energy supply by source.

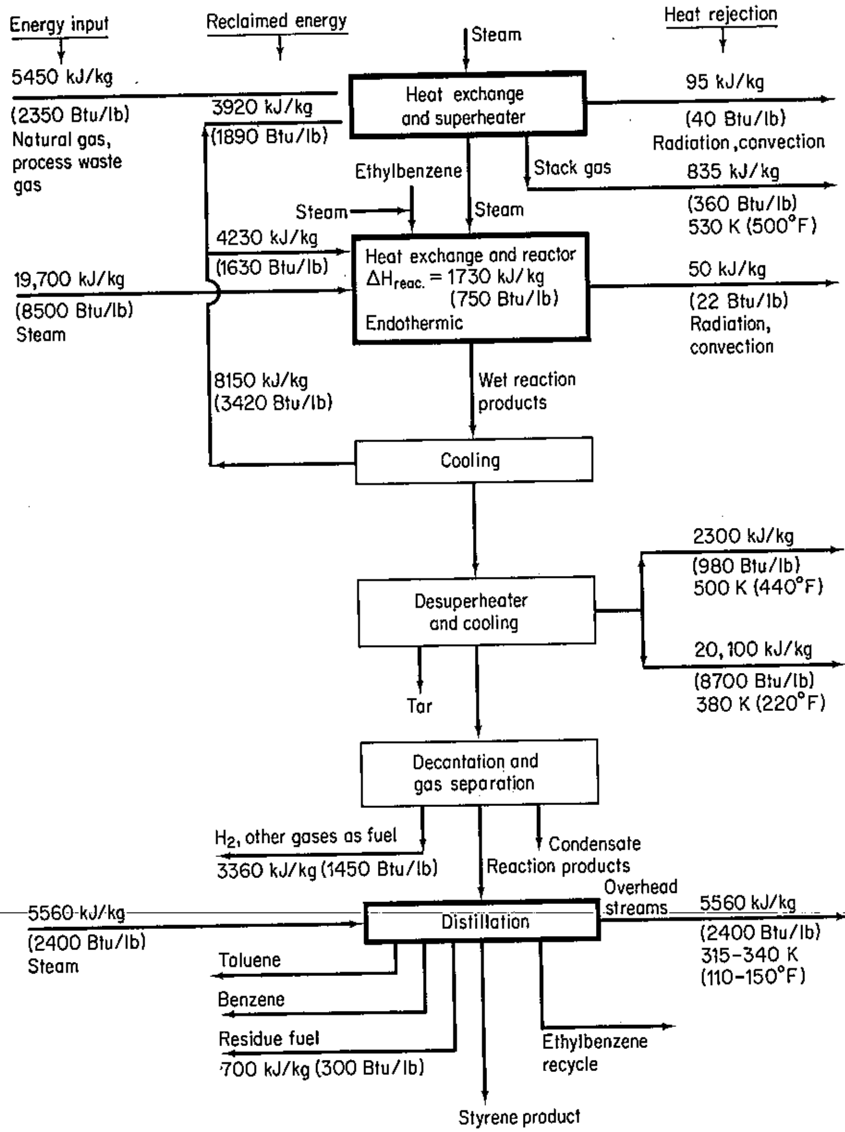


Figure 4.2 Energy balance in styrene production in the United States: 40% conversion of ethylbenzene to products, 90% selectivity to styrene. (From J. T. Reding and B. P. Shepherd, *Energy Consumption: The Chemical Industry*, Report EPA-650/2-75-032a, Environmental Protection Agency, Washington, D.C., April 1975.)

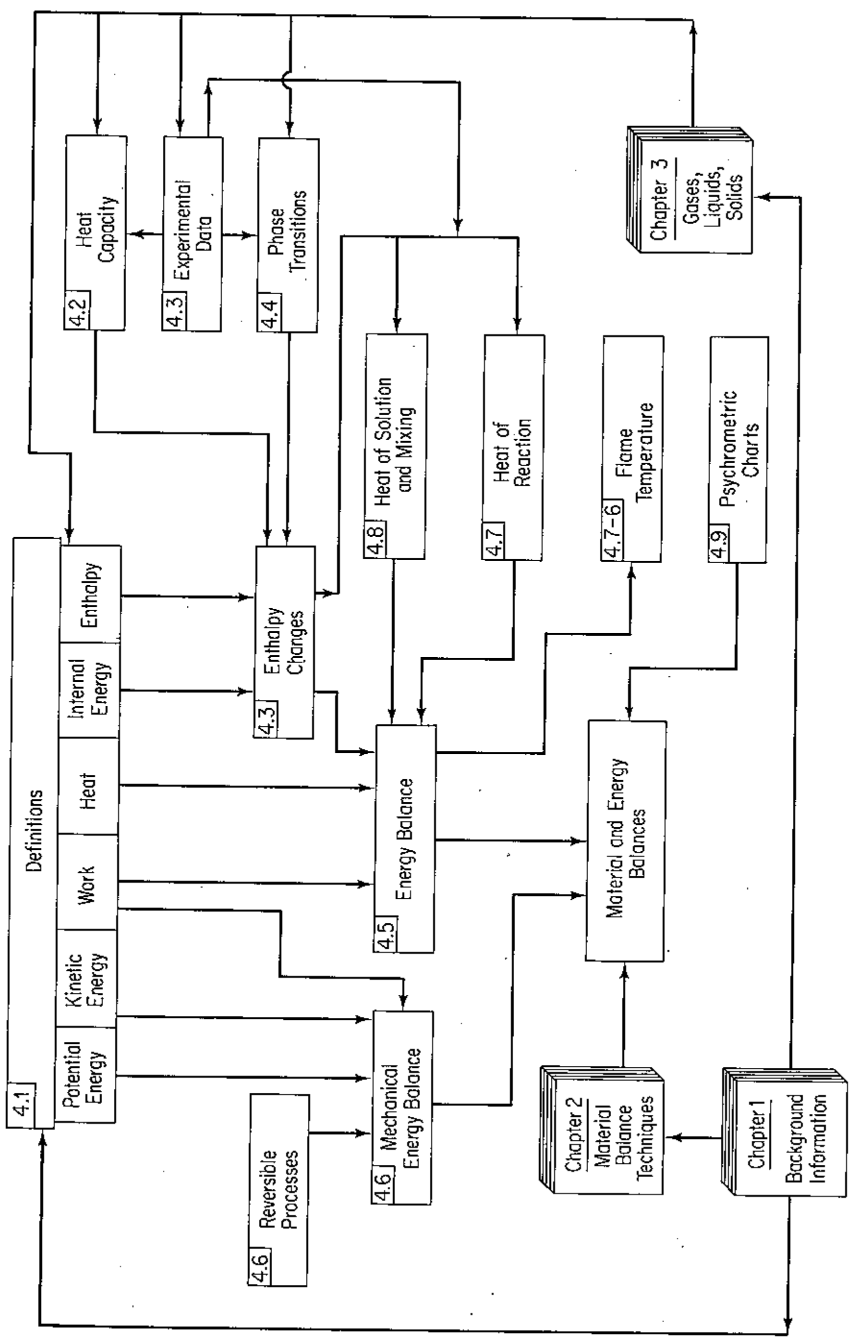


Figure 4.3 Hierarchy of topics to be studied in this chapter (section numbers are in the upper left-hand corner of the boxes).

4.1 CONCEPTS AND UNITS

Your objectives in studying this section are to be able to:

1. Define or explain the following terms: energy, system, closed system, nonflow system, open system, flow system, surroundings, property, extensive property, intensive property, state, heat, work, kinetic energy, potential energy, internal energy, enthalpy, initial state, final state, point (state) function, state variable, cyclical process, and path function.
2. Select and define a system suitable for problem solution, either closed or open, steady or unsteady state, and fix the system boundary.
3. Distinguish among potential, kinetic, and internal energy.
4. Convert energy in one set of units to another set.
5. List and apply the equations used to calculate kinetic energy, potential energy, and work.

As you already know, energy exists in many different forms. In Sec. 4.5 we assemble the various types of energy in the form of an energy balance. Before doing so, we need to clearly distinguish between the common types of energy, describe the units used to express energy, and learn how to calculate the values of various forms of energy.

What units are associated with energy? If you have forgotten, refer back to Table 1.1. No confusion exists with respect to the joule, but it is necessary to be careful in specifying what type of calorie or British thermal unit (Btu) is under consideration (there are four or five common kinds). For example, the type of calorie that is found in older tables of thermochemical properties of substances is the *thermochemical calorie* (equal to 4.184 J), whereas a second type of calorie is the *International Steam Table (I.T.) calorie* (equal to 4.1867 J). A kcal in nutrition is just called a "calorie"; that is, a hamburger containing 500 "calories" really contains 500 kcal.

In this section we characterize and define several common forms of energy. Unfortunately, many of the terms described below are used loosely in our ordinary conversation and writing, and thus have different connotations than those presented below. You may have the impression that you understand the terms from long acquaintance—be sure that you really do. For effective learning you must feel comfortable with them.

Certain terms that have been described in earlier chapters occur repeatedly in this chapter; these terms are summarized below with some elaboration in view of their importance.

System. Any arbitrarily specified mass of material or segment of apparatus to which we would like to devote our attention. A system must be defined by surrounding it with a system boundary. A system enclosed by a boundary that prohibits the transfer of mass across the boundary is termed a ***closed* system, or *nonflow* system,** in distinction to an ***open* system, or *flow* system,** in which the exchange of mass is permitted. All the mass or apparatus external to the defined system is termed the ***surroundings.*** Reexamine some of the example problems in Chap. 2 for illustrations of the location of system boundaries. You should always draw similar boundaries in the solution of your problems, since this step will fix clearly the system and surroundings.

Property. A characteristic of material that can be measured, such as pressure, volume, or temperature—or calculated, if not directly measured, such as certain types of energy. The properties of a system are dependent on its condition at any given time and not on what has happened to the system in the past.

An ***extensive property*** (variable, parameter) is one whose value is the sum of the values of each of the subsystems comprising the whole system. For example, a gaseous system can be divided into two subsystems which have volumes or masses different from the original system. Consequently, mass or volume is an extensive property.

An ***intensive property*** (variable, parameter) is one whose value is not additive and does not vary with the quantity of material in the subsystem. For example, temperature, pressure, density (mass per volume), and so on, do not change if the system is sliced in half or if the halves are put together.

Two properties are *independent* of each other if at least one variation of state for the system can be found in which one property varies while the other remains fixed. The number of independent intensive properties necessary and sufficient to fix the state of the system can be ascertained from the phase rule of Sec. 3.7-1.

State. The given set of properties of material at a given time. The state of a system does not depend on the shape or configuration of the system but only on its intensive properties.

Now that we have reviewed the concepts of system, property, and state, we can discuss the various types of energy with which we will be involved in this chapter. What forms can energy take? We shall consider here six quantities: work, heat, kinetic energy, potential energy, internal energy, and enthalpy. You probably have encountered many of these terms before.

Work. Work (W) is a term that has wide usage in everyday life (such as "I am going to work"), but has a specialized meaning in connection with energy balances. Work is a form of energy that represents a **transfer** between the system and surroundings. Work cannot be stored. For a mechanical force

$$W = \int_{\text{state 1}}^{\text{state 2}} \mathbf{F} \cdot d\mathbf{s} \tag{4.1}$$

where **F** is an external force in the direction of s acting on the system (or a system force acting on the surroundings).

(a) The displacement may not be easy to define.
(b) Integration of $\mathbf{F} \cdot ds$ as shown in Eq. (4.1) does not always result in an equal amount of work being done by the system.
(c) Work can be exchanged without a mechanical force acting on the system boundaries (such as through magnetic or electric effects).

Note that unless the process (or path) under which work is carried out is specified from the initial to the final state of the system, you are not able to calculate the value of the work done. In other words, work done in going between the initial and final states can have *any* value, depending on the path taken. **Work is therefore called a *path function*,** and the value of W depends on the initial state, the path, and the final state of the system, as illustrated in the next example.

EXAMPLE 4.1 Calculation of Work by a Gas on a Piston

Suppose that an ideal gas at 300 K and 200 kPa is enclosed in a cylinder by a frictionless piston, and the gas slowly forces the piston up so that the volume of gas expands from 0.1 to 0.2 m³. Examine Fig. E4.1a. Calculate the work done by the gas on the piston if two different paths are used to go from the initial state to the final state:

Path A: expansion occurs at constant pressure (p = 200 kPa)
Path B: expansion occurs at constant temperature (T = 300 K)

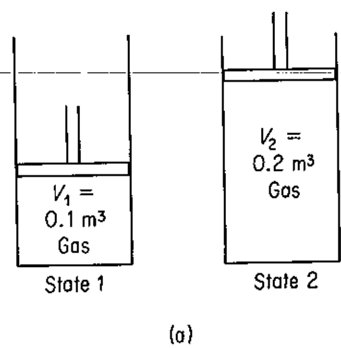


Figure E4.1a

Solution

The work is

$$W = \int_{\text{state 1}}^{\text{state 2}} \frac{\mathbf{F}}{A} \cdot A\, ds = \int_{V_1}^{V_2} p\, dV$$

because p is exerted normally on the piston face.

Path A

$$W = p\int_{V_1}^{V_2} dV = p(V_2 - V_1)$$

$$= \frac{200 \times 10^3\ \text{Pa}}{} \left| \frac{1\ \frac{\text{N}}{\text{m}^2}}{1\ \text{Pa}} \right| \frac{0.1\ \text{m}^3}{} \left| \frac{1\ \frac{\text{J}}{\text{m}}}{1\ \text{N}} \right.$$

$$= 20\ \text{kJ}$$

Path B

$$W = \int_{V_1}^{V_2} \frac{nRT}{V}\, dV = nRT \ln\left(\frac{V_2}{V_1}\right)$$

$$n = \frac{200\ \text{kPa}}{} \left| \frac{0.1\ \text{m}^3}{} \right| \frac{}{300\ \text{K}} \left| \frac{(\text{kg mol})(\text{K})}{8.314(\text{kPa})(\text{m}^3)} \right.$$

$$= 0.00802\ \text{kg mol}$$

$$W = \frac{0.00802\ \text{kg mol}}{} \left| \frac{8.314\ \text{kJ}}{(\text{kg mol})(\text{K})} \right| \frac{300\ \text{K}}{} \ln 2$$

$$= 20 \ln 2 = 13.86\ \text{kJ}$$

Figure E4.1b shows the two different quantities of work on a p–V plot.

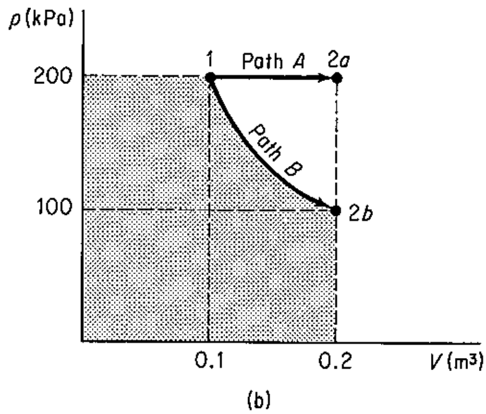


Figure E4.1b

Heat. In a discussion of *heat* we enter an area in which our everyday use of the term may cause confusion, since we are going to use heat in a very restricted sense when we apply the laws governing energy changes. Heat (Q) is commonly defined as that part of the total energy flow across a system boundary that is caused by a temperature difference between the system and the surroundings. Heat may be exchanged by conduction, convection, or radiation. Heat, as is work, is a path function. To evaluate heat transfer *quantitatively*, unless given a priori, you must apply the energy balance that is discussed in Sec. 4.5, evaluate all the terms except Q, and then solve for Q. Heat transfer can be *estimated* for engineering purposes by many empirical relations, which can be found in books treating heat transfer or transport processes. A typical relation to estimate the rate of heat transfer is that $\dot{Q}$ is proportional to the area for heat transfer and the temperature difference between the system at T_2 and its surroundings at T_1:

$$\dot{Q} = UA(T_2 - T_1)$$

where U is an empirical coefficient determined from experimental data.

Since heat and work are by definition mutually exclusive exchanges of energy between the system and the surroundings, we shall qualitatively classify work as energy that can be transferred to or from a mechanical state, or mode, of the system, while heat is the transfer of energy to atomic or molecular states, or modes, which are not macroscopically observable.

Kinetic energy. Kinetic energy (K) is the energy a system possesses because of its velocity relative to the surroundings at rest. Kinetic energy may be calculated from the relation

$$K = \tfrac{1}{2}mv^2 \tag{4.2a}$$

or

$$\hat{K} = \tfrac{1}{2}v^2 \tag{4.2b}$$

where the superscript caret (^) refers to the energy per unit mass (or sometimes per mole) and not the total kinetic energy as in Eq. (4.2a).

EXAMPLE 4.2 Calculation of Kinetic Energy

Water is pumped from a storage tank into a tube of 3.00 cm inner diameter at the rate of 0.001 m^3/s. See Fig. E4.2. What is the specific kinetic energy of the water?

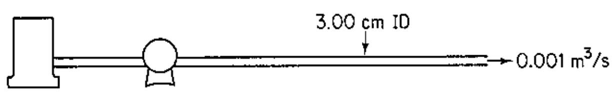


Figure E4.2

Solution

Assume that $\rho = \dfrac{1000 \text{ kg}}{\text{m}^3}$ $\quad r = \tfrac{1}{2}(3.00) = 1.50$ cm

$$v = \frac{0.001 \text{ m}^3}{\text{s}} \left| \frac{1}{\pi(1.50)^2 \text{ cm}^2} \right| \left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^2 = 1.415 \text{ m/s}$$

$$\hat{K} = \frac{1}{2} \left| \left(\frac{1.415 \text{ m}}{\text{s}}\right)^2 \right| \frac{1 \text{ N}}{1 \dfrac{(\text{kg})(\text{m})}{\text{s}^2}} \left| \frac{1 \text{ J}}{1(\text{N})(\text{m})} \right. = 1.00 \text{ J/kg}$$

Potential energy. Potential energy (P) is energy the system possesses because of the body force exerted on its mass by a gravitational or electromagnetic field with respect to a reference surface. Potential energy for a gravitational field can be calculated from

$$P = mgh \tag{4.3a}$$

or

$$\hat{P} = gh \tag{4.3b}$$

where h is the distance from the reference surface and where the symbol ($\hat{\ }$) again means potential energy per unit mass (or sometimes per mole).

EXAMPLE 4.3 Calculation of Potential Energy

Water is pumped from one reservoir to another 300 ft away, as shown in Fig. E4.3. The water level in the second reservoir is 40 ft above the water level of the first reservoir. What is the increase in specific potential energy of the water in Btu/lb$_m$?

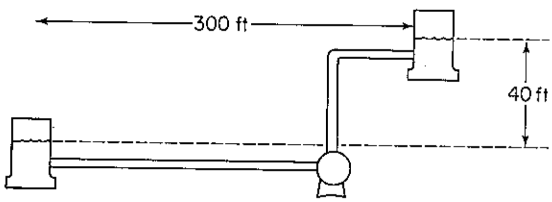


Figure E4.3

Solution

Let the water level in the first reservoir be the reference plane. Then h = 40 ft.

$$\hat{P} = \frac{32.2\ \text{ft}}{\text{s}^2}\left|\frac{40\ \text{ft}}{}\right|\frac{}{\dfrac{32.2(\text{lb}_m)(\text{ft})}{(\text{lb}_f)(\text{s}^2)}}\left|\frac{1\ \text{Btu}}{778.2(\text{ft})(\text{lb}_f)}\right. = 0.0514\ \text{Btu/lb}_m$$

Internal energy. Internal energy (U) is a macroscopic measure of the molecular, atomic, and subatomic energies, all of which follow definite microscopic conservation rules for dynamic systems. Because no instruments exist with which to measure internal energy directly on a macroscopic scale, internal energy must be calculated from certain other variables that can be measured macroscopically, such as pressure, volume, temperature, and composition.

To calculate the internal energy per unit mass ($\hat{U}$) from the variables that can be measured, we make use of a special property of internal energy, namely, that it is an exact differential (because it is a *point* or *state* property, a matter to be described shortly) and, for a pure component, can be expressed in terms of just two intensive variables according to the phase rule for one phase:

$$F = C - \mathcal{P} + 2 = 1 - 1 + 2 = 2$$

Custom dictates the use of temperature and specific volume as the variables. If we say that $\hat{U}$ is a function of T and $\hat{V}$,

$$\hat{U} = \hat{U}(T, \hat{V})$$

by taking the total derivative, we find that

$$d\hat{U} = \left(\frac{\partial \hat{U}}{\partial T}\right)_{\hat{V}} dT + \left(\frac{\partial \hat{U}}{\partial \hat{V}}\right)_T d\hat{V} \tag{4.4}$$

By definition $(\partial \hat{U}/\partial T)_{\hat{V}}$ is the heat capacity[1] at constant volume, given the special symbol C_v. For most practical purposes in this text the term $(\partial \hat{U}/\partial \hat{V})_T$ is so small that the second term on the right-hand side of Eq. (4.4) can be neglected. Consequently, *changes in the internal energy* can be computed by integrating Eq. (4.4) as follows:

$$\hat{U}_2 - \hat{U}_1 = \int_{T_1}^{T_2} C_v \, dT \tag{4.5}$$

Note that you can only calculate *differences* in internal energy, or calculate the internal energy relative to a reference state, *but not absolute values* of internal energy.

EXAMPLE 4.4 Internal Energy

An entrance examination for graduate school asked the following two multiple choice questions:

(a) The internal energy of a solid is equal to the
  (1) absolute temperature of the solid
  (2) total kinetic energy of its molecules
  (3) total potential energy of its molecules
  (4) sum of the kinetic and potential energy of its molecules

(b) The internal energy of an object depends on its
  (1) temperature, only
  (2) mass, only
  (3) phase, only
  (4) temperature, mass, and phase

Which answers would you chose?

Solution

Neither of these questions can be answered with the choices given. Internal energy itself cannot be evaluated—only its change can be. And such changes would include changes in potential energy of atoms, electron energy levels, vibrations inside the molecules, and so on.

In addition to using Eq. (4.4), internal energy changes can be calculated from enthalpy values, the next topic of discussion.

Enthalpy. In applying the energy balance you will encounter a variable which is given the symbol H and the name *enthalpy* (pronounced en´-thal-py). This

[1] The term *heat capacity* has a long history of usage, but is not a scientifically proper connotation. Heat is, by definition, not "stored" in material.

variable is defined as the combination of two variables which will appear very often in the energy balance:

$$H = U + pV \tag{4.6}$$

where p is the pressure and V is the volume. (The term "enthalpy" has replaced the now obsolete terms "heat content" or "total heat," to eliminate any connection whatsoever with heat as defined earlier.)

To calculate the enthalpy per unit mass, we use the property that the enthalpy is also an exact differential. For a pure substance, the enthalpy for a single phase can be expressed in terms of the temperature and pressure (a more convenient variable than specific volume) alone. If we let

$$\hat{H} = \hat{H}(T, p)$$

by taking the total derivative of $\hat{H}$, we can form an expression corresponding to Eq. (4.4):

$$d\hat{H} = \left(\frac{\partial \hat{H}}{\partial T}\right)_p dT + \left(\frac{\partial \hat{H}}{\partial p}\right)_T dp \tag{4.7}$$

By definition $(\partial\hat{H}/\partial T)_p$ is the heat capacity at constant pressure, given the special symbol C_p. For most practical purposes $(\partial\hat{H}/\partial p)_T$ is so small at modest pressures that the second term on the right-hand side of Eq. (4.7) can be neglected. Changes in enthalpy can then be calculated by integration of Eq. (4.7) as follows:

$$\hat{H}_2 - \hat{H}_1 = \int_{T_1}^{T_2} C_p\, dT \tag{4.8}$$

However, in high-pressure processes the second term on the right-hand side of Eq. (4.7) cannot necessarily be neglected, but must be evaluated from experimental data. Consult the references at the end of the chapter for details. One property of ideal gases that should be noted is that their enthalpies and internal energies are functions of temperature only and are not influenced by changes in pressure or specific volume.

As with internal energy, *enthalpy has no absolute value;* **only changes in enthalpy can be calculated.** Often you will use a reference set of conditions (perhaps implicitly) in computing enthalpy changes. For example, the reference conditions used in the steam tables are liquid water at 0°C (32°F) and its vapor pressure. This does not mean that the enthalpy is actually zero under these conditions but merely that the enthalpy has arbitarily been assigned a value of zero at these conditions. In computing enthalpy changes, the reference conditions cancel out as can be seen from the following:

initial state of system	*final state of system*
enthalpy $= \hat{H}_1 - \hat{H}_{\text{ref}}$	enthalpy $= \hat{H}_2 - \hat{H}_{\text{ref}}$

net enthalphy change $= (\hat{H}_2 - \hat{H}_{\text{ref}}) - (\hat{H}_1 - \hat{H}_{\text{ref}}) = \hat{H}_2 - \hat{H}_1$

Point, or state, functions. The variables specific enthalpy and specific internal energy (as well as temperature, pressure, and density) are called *point func-*

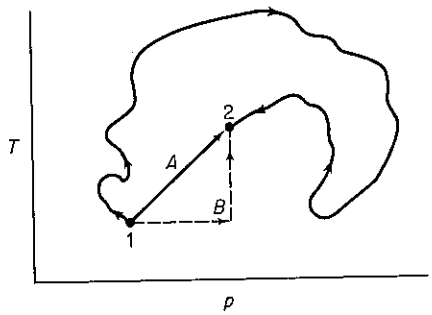


Figure 4.4 Point function.

tions, or *state variables,* meaning that their values depend *only* on the state of the material (temperature, pressure, phase, and composition) and *not* on how the material reached that state. Figure 4.4 illustrates the concept of a state variable. In proceeding from state 1 to state 2, the actual process conditions of temperature and pressure are shown by the wiggly line. However, you may calculate $\Delta\hat{H}$ by route A or B, or any other route, and still obtain the same net enthalpy change as for the route shown by the wiggly line. The change of enthalpy depends only on the initial and final states of the system. A process that proceeds first at constant pressure and then at constant temperature from 1 to 2 will yield exactly the same $\Delta\hat{H}$ as one that takes place first at constant temperature and then at constant pressure as long as the end point is the same. The concept of the point function is the same as that of an airplane passenger who plans to go straight to Chicago from New York but is detoured because of bad weather by way of Cincinnati. His trip costs him the same whatever way he flies, and he eventually arrives at his destination. The fuel consumption of the plane may vary considerably, and in analogous fashion heat (Q) or work (W), the two "path" functions with which we deal, may vary depending on the specific path chosen, while $\Delta\hat{H}$ is the same regardless of path. If the plane were turned back by mechanical problems and landed at New York, the passenger might be irate, but at least could get a refund. Thus $\Delta\hat{H} = 0$ if a cyclical process is involved which goes from state 1 to 2 and back to state 1 again, or

$$\oint d\hat{H} = 0$$

All the intensive properties we shall work with, such as $\hat{P}$, T, $\hat{U}$, p, $\hat{H}$, and so on, are state variables and depend only on the state of the substance of interest, so we can say, for example, that

$$\oint dT = 0$$

$$\oint d\hat{U} = 0$$

To sum up, always keep in mind that the values for a difference in a point function can be calculated by taking the value in the final state and subtracting the value in the initial state, regardless of the actual path.

In the next section we focus attention on the details of how to express the heat capacity as a function of temperature, and in Sec. 4.3 discuss how to use Eq. (4.8) to calculate enthalpy differences.

Self-Assessment Test

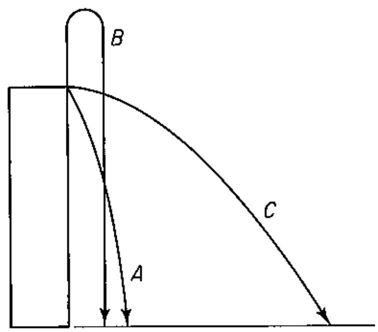
1. Contrast the following property classifications: extensive–intensive, measurable–unmeasurable, state–path.
2. Define heat and work.
3. Consider the hot-water heater in your house. Classify each case below as an open system, closed system, neither, or both.
   (a) The tank is being filled with cold water.
   (b) Hot water is being drawn from the tank.
   (c) The tank leaks.
   (d) The heater is turned on to heat the water.
   (e) The tank is full and the heater is turned off.
4. The units of potential energy or kinetic energy are (select all the correct expressions):
   (a) $(\text{ft})(\text{lb}_f)$
   (b) $(\text{ft})(\text{lb}_m)$
   (c) $(\text{ft})(\text{lb}_f)/(\text{lb}_m)$
   (d) $(\text{ft})(\text{lb}_m)/(\text{lb}_f)$
   (e) $(\text{ft})(\text{lb}_f)/(\text{hr})$
   (f) $(\text{ft})(\text{lb}_m)/(\text{hr})$
5. Review the selection of a system and surroundings by reading two or three examples in Secs. 2.3 to 2.6, covering up the solution, and designating the system. Compare with the system shown in the example.
6. Will the kinetic energy per unit mass of an incompressible fluid flowing in a pipe increase, decrease, or remain the same if the pipe diameter is increased at some place in the line?
7. A 100-kg ball initially on the top of a 5-m ladder is dropped and hits the ground. With reference to the ground:
   (a) What is the initial kinetic and potential enegy of the ball?
   (b) What is the final kinetic and potential energy of the ball?
   (c) What is the change in kinetic and potential energy for the process?
   (d) If all the initial potential energy were somehow converted to heat, how many calories would this amount to? how many Btu? How many joules?
8. In expanding a balloon, two types of work are done by the air in the balloon (the system). One is stretching the balloon $dW = \sigma\, dA$, where σ is the surface tension of the balloon. The other is the work of pushing back the atmosphere. If the balloon is spherical and expands slowly from a diameter of 1 m to 1.5 m, what is the work done in pushing back the atmosphere? What assumptions must you make about T and p?

Thought Problems

1. A proposed goal is to replace 10% of the U.S. domestic gasoline production with ethanol grown from corn. What is your estimate of the fraction of the available U.S. cropland that would be required to support such a proposal? Assume 90.0 bu/acre and 2.6 gal ethanol/bu. Gasoline production is 1.2×10^{10} gal/year.
2. Another proposal is to supply 10% of the U. S. oil usage by coal liquefaction. What is

your estimate of the percentage of the coal now mined in the U.S. that would have to be added in order to fulfill this proposal? Assume 3.26 bbl of liquid per ton of coal.

3. Three baseballs are thrown from the top of a four-story building as shown in the figure. All have the same initial speed. Which of the following answers would you say best reflects the speed of the balls when they hit the ground?
   (1) A is greatest
   (2) B is greatest
   (3) C is greatest
   (4) A and C are the same and greatest
   (5) A and B are the same and greatest
   (6) All have the same speed



4.2 HEAT CAPACITY

Your objectives in studying this section are to be able to:

1. Define heat capacity.
2. Convert an expression for the heat capacity from one set of units to another.
3. Look up from a reference source an equation that expresses the heat capacity as a function of temperature, and compute the heat capacity at a given temperature.
4. Estimate the value of the heat capacity for solids and liquids.
5. Fit empirical heat capacity data with a suitable function of temperature by estimating the values of the coefficients in the function.

Before formulating the general energy balance, we shall discuss in some detail the calculation of enthalpy changes and provide some typical examples of such calculations. Our discussion will be initiated with consideration of the heat capacity C_p.

The two heat capacities have been defined as

(a) $$C_p = \left(\frac{\partial \hat{H}}{\partial T}\right)_p$$

(b)
$$C_v = \left(\frac{\partial \hat{U}}{\partial T}\right)_{\hat{V}}$$

To give these two quantities some physical meaning, you can think of them as representing the amount of energy required to increase the temperature of a substance by 1 degree, energy that might be provided by heat transfer in certain specialized processes, but can be provided by other means as well. To determine from experiments values of C_p (or C_v), the enthalpy (or internal energy) change must first be calculated from an energy balance, and then the heat capacity evaluated. We will discuss C_p as C_v is not used very often.

Suppose that you want to calculate the heat capacity of steam at 10.0 kPa (45.8°C). You can determine the enthalpy change at essentially constant pressure from the steam tables (which list experimental values) as

$$H_{47.7°C} - H_{43.8°C} = (2588.1 - 2581.1)\text{ kJ/kg} = 7.00\text{ kJ/kg}$$

and if you assume C_p is essentially constant over the small temperature range indicated by the subscripts, then

$$C_p \cong \frac{\Delta H}{\Delta T} = \frac{7.00\text{ kJ}}{\text{kg}}\Bigg|\frac{}{3.9\Delta°\text{C}} = 1.79\frac{\text{kJ}}{(\text{kg})(\Delta°\text{C})}$$

What are the units of the heat capacity? From the definitions of heat capacity you can see that the units are (energy)/(temperature difference) (mass or moles). The common units found in engineering practice are (we suppress the Δ symbol)

$$\frac{\text{J}}{(\text{kg mol})(\text{K})} \qquad \frac{\text{cal}}{(\text{g mol})(°\text{C})} \qquad \frac{\text{Btu}}{(\text{lb mol})(°\text{F})}$$

Because of the definition of the calorie or Btu, the heat capacity can be expressed in certain different systems of units and still have the same numerical value; for example, heat capacity may be expressed in the units of

$$\frac{\text{cal}}{(\text{g mol})(°\text{C})} = \frac{\text{kcal}}{(\text{kg mol})(°\text{C})} = \frac{\text{Btu}}{(\text{lb mol})(°\text{F})}$$

and still have the same numerical value.

Alternatively, heat capacity may be in terms of

$$\frac{\text{cal}}{(\text{g})(°\text{C})} = \frac{\text{Btu}}{(\text{lb})(°\text{F})} \qquad \text{or} \qquad \frac{\text{J}}{(\text{kg})(\text{K})}$$

Note that

$$\frac{1\text{ Btu}}{(\text{lb})(°\text{F})} = \frac{4.184\text{ J}}{(\text{g})(\text{K})}$$

and that the heat capacity of water in the SI system is 4184 J/(kg)(K) at 17°C. These relations are worth memorizing.

Figure 4.5 illustrates the behavior of the heat capacity of a pure substance over a wide range of absolute temperatures. Observe that at zero degrees absolute the heat

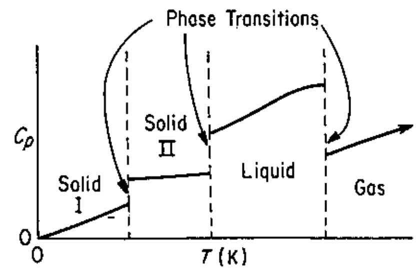


Figure 4.5 Heat capacity as a function of temperature for a pure substance.

capacity is zero. As the temperature rises, the heat capacity also increases until a certain temperature is reached at which a phase transition takes place. The phase transitions are shown also on a related p–T diagram in Fig. 4.6 for water. The phase transition may take place between two solid states, or between a solid and a liquid state, or between a solid and a gaseous state, or between a liquid and a gaseous state. Figure 4.5 shows first a transition between solid state I and solid state II, then the transition between solid state II and the liquid state, and finally the transition between the liquid and the gaseous state. Note that the heat capacity is a continuous function *only* in the region between the phase transitions; consequently, it is not possible to have a heat capacity equation for a substance that will go from absolute zero up to any desired temperature. What an engineer does is to determine experimentally the heat capacity between the temperatures at which the phase transitions occur, fit the data with an equation, and then determine a new heat capacity equation for the next range of temperatures between the succeeding phase transitions.

Experimental evidence indicates that the heat capacity of a substance is not constant with temperature, although at times we may assume that it is constant in order to get approximate results. For the ideal monoatomic gas, of course, the heat capacity at constant pressure is constant even though the temperature varies (see Table 4.1). For typical real gases, see Fig. 4.7; the heat capacities shown are for pure components.

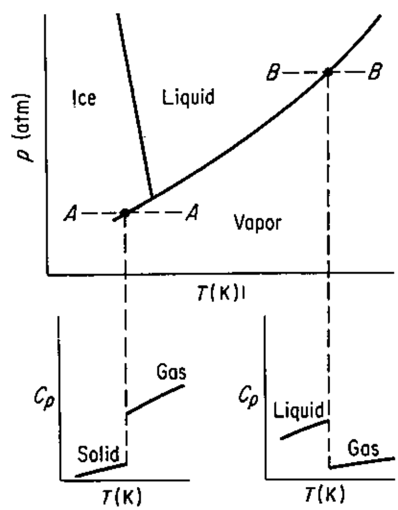


Figure 4.6 Heat capacity and phase transitions.

TABLE 4.1 Heat Capacities of Ideal Gases

	Approximate heat capacity*, C_p	
Type of molecule	High temperature (translational, rotational, and vibrational degrees of freedom)	Room temperature (translational and rotational degrees of freedom only)
Monoatomic	$\frac{5}{2}R$	$\frac{5}{2}R$
Polyatomic, linear	$(3n - \frac{3}{2})R$	$\frac{7}{2}R$
Polyatomic, nonlinear	$(3n - 2)R$	$4R$

*n, number of atoms per molecule; R, gas constant defined in Sec. 3.1.

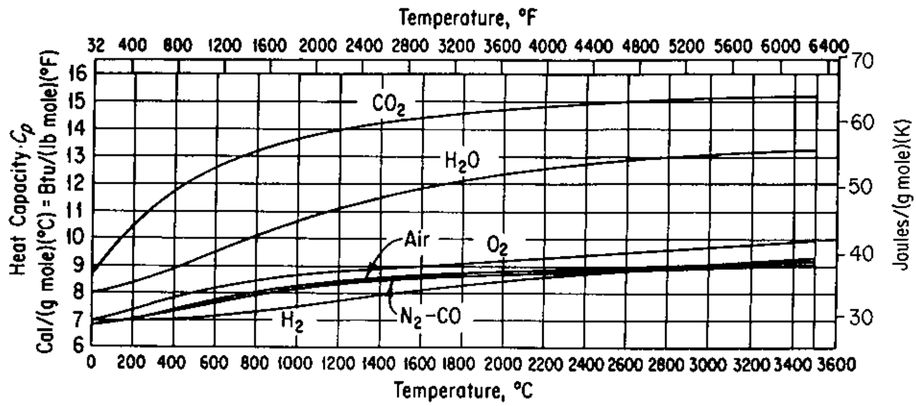


Figure 4.7 Heat capacity curves for the combustion gases.

For ideal gas mixtures, the heat capacity (per mole) of the mixture is the mole weighted average of the heat capacities of the components:

$$C_{p_{avg}} = \sum_{i=1}^{n} x_i C_{p_i} \tag{4.9}$$

For nonideal mixtures, particularly liquids, you should refer to experimental data or some of the estimation techniques listed in the literature (see the supplementary references at the end of chapter). Also, refer to Sec. 4.8 for details regarding mixtures.

Most of the equations for the heat capacities of solids, liquids, and gases are empirical. We usually express the heat capacity at constant pressure C_p as a function of temperature in a power series, with constants a, b, c, and so on; for example,

$$C_p = a + bT$$

or

$$C_p = a + bT + cT^2$$

where the temperature may be expressed in degrees Celsius, degrees Fahrenheit, degrees Rankine, or degrees kelvin. If C_p is expressed in the form of

$$C_p = a + bT + cT^{-1/2}$$

$$C_p = a + bT - cT^{-2}$$

or a form such that you divide by T, then it is necessary to use kelvin or degrees Rankine in the heat capacity equations, because if degrees Celsius or Fahrenheit were to be used, you might divide at some point in the temperature range by zero. Since these heat capacity equations are valid only over moderate temperature ranges, it is possible to have equations of different types represent the experimental heat capacity data with almost equal accuracy. The task of fitting heat capacity equations to heat capacity data is greatly simplified by the use of digital computers, which can determine the constants of best fit by means of a standard prepared program and at the same time determine how precise the predicted heat capacities are. Heat capacity information can be found in Appendix E. The change of C_p with pressure at high pressures is beyond the scope of our work here. Sources of heat capacity data can be found in several of the references listed at the end of the chapter.

Specific heat is a term often considered synonymous with heat capacity, but this connotation has arisen from loose usage. In principle, specific heat is the ratio of the heat capacity of a substance to the heat capacity of a reference substance, such as

$$\frac{C_{p_A}}{C_{p_{H_2O}}} = \frac{\text{Btu}/(\text{lb}_A)(°\text{F})}{\text{Btu}/(\text{lb}_{H_2O})(°\text{F})}$$

The reference substance temperature must be specified. Because water has a heat capacity of 1.00 Btu/(lb)(°F) at about 17°C, numerical values of specific heats and heat capacities in the American engineering and thermochemical systems are about the same, although their units are not.

EXAMPLE 4.5 Heat Capacity Equation

The heat capacity equation for CO_2 gas is

$$C_p = 6.393 + 10.100T \times 10^{-3} - 3.405T^2 \times 10^{-6}$$

with C_p expressed in cal/(g mol)(ΔK) and T in K. Convert this equation into a form so that the heat capacity will be expressed over the entire temperature range in

(a) Cal/(g mol)(Δ°C) with T in °C
(b) Btu/(lb mol)(Δ°F) with T in °F
(c) J/(kg mol)(ΔK) with T in K

Solution

Changing a heat capacity equation from one set of units to another is merely a problem in the conversion of units. Each term in the heat capacity equation must have the same units as the left-hand side of the equation. To avoid confusion in the conversion, you must remember to distinguish between the temperature symbols that represent temperature and the temperature symbols that represent temperature difference even though the same symbol often is used for each concept. In the conversions below we shall distinguish between the temperature and the temperature difference for clarity.

(a) The heat capacity equation with T in °C and ΔT in Δ°C is

$$C_p \frac{\text{cal}}{(\text{g mol})(\Delta°\text{C})} = 6.393\,\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})}\left|\frac{1\ \Delta\text{K}}{1\ \Delta°\text{C}}\right.$$

$$+\ 10.1000 \times 10^{-3}\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})(\text{K})}\left|\frac{1\ \Delta\text{K}}{1\ \Delta°\text{C}}\right|\frac{(T_{°\text{C}} + 273)\ \text{K}}{}$$

$$-3.405 \times 10^{-6}\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})(\text{K})^2}\left|\frac{1\ \Delta\text{K}}{1\ \Delta°\text{C}}\right|\frac{(T_{°\text{C}} + 273)^2\ \text{K}^2}{}$$

$$= 6.393 + 10.100 \times 10^{-3}T_{°\text{C}} + 2.757 - 3.405 \times 10^{-6}T_{°\text{C}}^2$$
$$-1.860 \times 10^{-3}T_{°\text{C}} - 0.254$$

$$= 8.896 + 8.240 \times 10^{-3}T_{°\text{C}} - 3.405 \times 10^{-6}T_{°\text{C}}^2$$

(b)

$$C_p \frac{\text{Btu}}{(\text{lb mol})(\Delta°\text{F})} = 6.393\,\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})}\left|\frac{454\ \text{g mol}}{1\ \text{lb mol}}\right|\frac{1\ \text{Btu}}{252\ \text{cal}}\left|\frac{1\ \Delta\text{K}}{1.8\ \Delta°\text{F}}\right.$$

$$+\ 10.100 \times 10^{-3}\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})(\text{K})}\left|\frac{454\ \text{g mol}}{1\ \text{lb mol}}\right|\frac{1\ \text{Btu}}{252\ \text{cal}}\left|\frac{1\ \Delta\text{K}}{1.8\ \Delta°\text{F}}\right|\frac{\left(273 + \dfrac{T_{°\text{F}} - 32}{1.8}\right)\text{K}}{}$$

$$-3.405 \times 10^{-6}\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})(\text{K})^2}\left|\frac{454\ \text{g mol}}{1\ \text{lb mol}}\right|\frac{1\ \text{Btu}}{252\ \text{cal}}\left|\frac{1\ \Delta\text{K}}{1.8\ \Delta°\text{F}}\right|\frac{\left(273 + \dfrac{T_{°\text{F}} - 32}{1.8}\right)^2\text{K}^2}{}$$

$$= 6.393 + 10.100 \times 10^{-3}[273 + (T_{°\text{F}} - 32)/1.8]$$
$$-3.405 \times 10^{-6}[273 + (T_{°\text{F}} - 32)/1.8]^2$$

$$= 6.393 + 2.575 + 5.61 \times 10^{-3}T_{°\text{F}} - 0.222$$
$$-0.964 \times 10^{-3}T_{°\text{F}} - 1.05 \times 10^{-6}T_{°\text{F}}^2$$

$$= 8.746 + 4.646 \times 10^{-3}T_{°\text{F}} - 1.05 \times 10^{-6}T_{°\text{F}}^2$$

Note: $T_\text{K} = 273 + \dfrac{T_{°\text{F}} - 32}{1.8}$.

(c)

$$C_p \frac{\text{J}}{(\text{kg mol})(\Delta\text{K})} = 6.393\,\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})}\left|\frac{4.184\ \text{J}}{1\ \text{cal}}\right|\frac{1000\ \text{g}}{1\ \text{kg}}$$

$$+\ 10.1000 \times 10^{-3}\frac{\text{cal}}{(\text{g mol})(\Delta\text{K})(\text{K})}\left|\frac{4.184\ \text{J}}{1\ \text{cal}}\right|\frac{1000\ \text{g}}{1\ \text{kg}}\left|\frac{(T_\text{K})\text{K}}{}\right.$$

$$-3.405 \times 10^{-6}\frac{\text{cal}}{(\text{g mol})(\text{K})^2(\Delta\text{K})}\left|\frac{4.184\ \text{J}}{1\ \text{cal}}\right|\frac{1000\ \text{g}}{1\ \text{kg}}\left|\frac{(T_\text{K}^2)\text{K}^2}{}\right.$$

$$= 2.675 \times 10^4 + 42.27T_\text{K} - 1.425 \times 10^{-2}T_\text{K}^2$$

EXAMPLE 4.6 Heat Capacity of the Ideal Gas

Show that $C_p = C_v + \hat{R}$ for the ideal monoatomic gas.

Solution

The heat capacity at constant volume is defined as

$$C_v = \left(\frac{\partial \hat{U}}{\partial T}\right)_{\hat{V}} \tag{a}$$

For any gas,

$$\begin{aligned} C_p &= \left(\frac{\partial \hat{H}}{\partial T}\right)_p = \left[\frac{\partial \hat{U} + \partial(p\hat{V})}{\partial T}\right]_p = \left[\frac{\partial \hat{U} + p\,\partial \hat{V}}{\partial T}\right]_p \\ &= \left(\frac{\partial \hat{U}}{\partial T}\right)_p + p\left(\frac{\partial \hat{V}}{\partial T}\right)_p \end{aligned} \tag{b}$$

For the *ideal gas*, since $\hat{U}$ is a function of temperature only,

$$\left(\frac{\partial \hat{U}}{\partial T}\right)_p = \left(\frac{\partial \hat{U}}{\partial T}\right)_{\hat{V}} = C_v \tag{c}$$

and from $p\hat{V} = \hat{R}T$ we can calculate

$$\left(\frac{\partial \hat{V}}{\partial T}\right)_p = \frac{\hat{R}}{p} \tag{d}$$

so that

$$C_p = C_v + \hat{R}$$

EXAMPLE 4.7 Calculation of C_p from an Equation for Enthalpy

Kelley, U.S. Bureau of Mines, gave an equation for the relative enthalpy of gaseous titanium tetrachloride as

$$H - H_{298} = 25.45T + 0.12 \times 10^{-3}T^2 + 2.36 \times 10^{-5}T^{-1} - 8390$$

where T is in K, H is cal/g mol, and H_{298} is a reference state for enthalpy. What is an equation for C_p in cal/(g mol)(K)?

Solution

$$C_p = \frac{dH}{dT} = 25.45 + 0.24 \times 10^{-3}T - 2.36 \times 10^{-5}T^{-2}$$

EXAMPLE 4.8 Fitting Heat Capacity Data

The heat capacity of carbon dioxide gas as a function of temperature has been found by a series of experiments to be as follows:

	1	2	3	4	5	6
T (K)	300	400	500	600	700	800
C_p [J/(g mol)(K)]	39.87	45.16	50.72	56.85	63.01	69.52
	39.85	45.23	51.03	56.80	63.09	69.68
	39.90	45.17	50.90	57.02	63.14	69.63

Find the values of the coefficients in the equation

$$C_p = a + bT + CT^2$$

that best fit the data.

Solution

Use a least-squares program (or directly minimize using the minimization code in the back of this book) to minimize the sum of the squares of the deviations between the predicted values of C_p and the experimental ones (refer to Appendix M):

$$\text{Minimize} \sum_{i=1}^{6} (C_{p_{\text{predicted},i}} - C_{p_{\text{experimental},i}})^2$$

The variables are a, b, and c.

The solution is

$$C_p = 25.47 + 4.367 \times 10^{-2}T - 1.44 \times 10^{-5}T^2$$

4.2-1 Estimation of Heat Capacities

We now mention a few ways by which to *estimate* heat capacities of solids, liquids, and gases. For the most accurate results, you should employ actual experimental heat capacity data or equations derived from such data in your calculations. However, if experimental data are not available, there are a number of equations and estimation techniques that you may use which give estimates of values for the heat capacities.

Solids. Only very rough approximations of solid heat capacities can be made. *Kopp's rule* (1864) should only be used as a last resort when experimental data cannot be located or new experiments carried out. Kopp's rule states that at room temperature the sum of the heat capacities of the individual elements is approximately equal to the heat capacity of a solid compound. For elements below potassium, numbers have been assigned from experimental data for the heat capacity for each element as shown in Table 4.2. For liquids Kopp's rule can be applied with a modified series of values for the various elements, as shown also in Table 4.2. For example, the heat capacity at room temperature of $Na_2SO_4 \cdot 10H_2O$ would be $2(6.2) + 1(5.4) + 14(4.0) + 20(2.3) = 119.8$ cal/(g mol)(°C). The heat capacity of coal can be estimated from equations in the *Coal Conversion Systems Technical Data Book* cited in the supplementary references. Consult Reid or Perry's *Handbook* for tables of heat capacity data for solids.

TABLE 4.2 Values for Modified Kopp's Rule: Atomic Heat Capacity at 20°C [cal/(g atom)(°C)]

Element	Solids	Liquids
C	1.8	2.8
H	2.3	4.3
B	2.7	4.7
Si	3.8	5.8
O	4.0	6.0
F	5.0	7.0
P or S	5.4	7.4
All others	6.2	8.0

Liquids

Aqueous solutions. For the special but very important case of aqueous solutions, a rough rule in the absence of experimental data is to use the heat capacity of the water only. For example, a 21.6% solution of NaCl is assumed to have a heat capacity of 0.784 cal/(g)(°C); the experimental value at 25°C is 0.806 cal/(g)(°C).

Hydrocarbons. An equation for the heat capacity of liquid hydrocarbons and petroleum products will be found in Appendix K.

Organic liquids. A simple and reasonably accurate relation between C_p in cal/(g)(°C) at 25°C and molecular weight is

$$C_p = kM^a$$

where M is the molecular weight and k and a are constants. Pachaiyappan et al.[2] give a number of values of the set (k, a). For example

Compounds	k	a
Alcohols	0.85	−0.1
Acids	0.91	−0.152
Ketones	0.587	−0.0135
Esters	0.60	−0.0573
Hydrocarbons, aliphatic	0.873	−0.113

Reid and San Jose[3] review a number of estimation methods for liquids and indicate their precision.

Gases and vapors

Petroleum vapors. The heat capacity of petroleum vapors can be estimated from[4]

[2] V. Pachaiyappan, S. H. Ibrahim, and N. R. Kuloor, *Chem. Eng.*, p. 241 (October 9, 1967).

[3] R. C. Reid and J. L. San Jose, *Chem. Eng.*, pp. 67–71 (December 20, 1976).

[4] W. H. Bahlke and W. B. Kay, *Ind. Eng. Chem.*, v. 21, p. 942 (1929).

$$C_p = \frac{(4.0 - s)(T + 670)}{6450}$$

where C_p is in Btu/(lb)(°F), T is in °F, and s is the specific gravity at 60°F/60°F, with air as the reference gas.

Kothari–Doraiswamy equation. Kothari and Doraiswamy[5] recommend plotting

$$C_p = a + b \log_{10} T_r$$

Given two values of C_p at known temperatures, values of C_p can be predicted at other temperatures with good precision. The most accurate methods of estimating heat capacities of vapors are those of Dobratz[6] based on spectroscopic data and generalized correlations based on reduced properties.[7]

Additional methods of estimating solid and liquid heat capacities may be found in Reid et al.[8], who compare various techniques we do not have the space to discuss and make recommendations as to their use.

Self-Assessment Test

1. A problem indicates that the enthalpy of a compound can be predicted by an empirical equation $H\,(\text{J/g}) = -30.2 + 4.25T + 0.001T^2$, where T is in kelvin. What is the heat capacity at constant pressure for the compound?
2. What is the heat capacity at constant pressure at room temperature of O_2 if the O_2 is assumed to be an ideal gas?
3. A heat capacity equation in cal/(g mol)(K) for ammonia gas is

$$C_p = 8.4017 + 0.70601 \times 10^{-2}T + 0.10567 \times 10^{-5}T^2 - 1.5981 \times 10^{-9}T^3$$

   where T is in °C. What are the units of each of the coefficients in the equation?
4. Calculate the heat capacity (C_p) of N_2 gas at 1 atm and 500 K.
5. Estimate the heat capacity (C_p) of acetic acid by (a) Kopp's rule and (b) Pachaiyappan's constants. Compare with the experimental value at room temperature.
6. Convert the following equation for the heat capacity of carbon monoxide gas, where C_p is in Btu/(lb mol)(°F) and T is in °F:

$$C_p = 6.865 + 0.08024 \times 10^{-2}T - 0.007367 \times 10^{-5}T^2$$

   to yield C_p in J/(kg mol)(K) with T in kelvin.

Thought Problem

1. Textbooks often indicate that for solids and liquids the difference $(C_p - C_v)$ is so small that you can say that $C_p = C_v$. Is this generally true?

[5] M. S. Kothari and L. K. Doraiswamy, *Hydrocarbon Process. Pet. Refiner,* v. 43, no. 3, p. 133 (1964).

[6] C. J. Dobratz, *Ind. Eng. Chem.*, v. 33, p. 759 (1941).

[7] R. C. Reid, J. M. Prausnitz, and B. E. Poling, *The Properties of Gases and Liquids,* 4th ed., McGraw-Hill, New York, 1987.

[8] Ibid.

4.3 CALCULATION OF ENTHALPY CHANGES (WITHOUT CHANGE OF PHASE)

Your objectives in studying this section are to be able to:

1. Calculate enthalpy (and internal energy) changes (excluding phase changes) from heat capacity equations, graphs and charts, tables, and computer data bases given the initial and final states of the material.
2. Become familiar with the steam tables and their use both in SI and American engineering units.
3. Ascertain the reference state for enthalpy values from the data source

Now that we have examined sources of heat capacity values, we turn to the details of the calculation of enthalpy and internal energy changes. We omit for the moment consideration of phase changes and examine solely the problem of how to calculate enthalpy (or internal energy) changes that take place in a single phase, that is, how to calculate the so-called **"sensible heat"** changes. Sensible heat is the enthaply difference (normally for a gas) between some reference temperature and the temperature of the material under consideration, excluding any enthalpy differences for phase changes that are termed latent heats and discussed in Sec. 4.4. We examine four procedures:

1. Use of heat capacity equations
2. Use of tables
3. Use of enthalpy charts
4. Use of computer data bases

4.3-1 How to Employ Heat Capacity Equations

Recall that if we use Eq.(4.8), $\Delta\hat{H}$ is the area under the curve in Fig. 4.8:

$$\int_{\hat{H}_1}^{\hat{H}_2} d\hat{H} = \Delta\hat{H} = \int_{T_1}^{T_2} C_p\, dT$$

If the heat capacity is expressed in the form $C_p = a + bT + cT^2$, then

$$\Delta\hat{H} = \int_{T_1}^{T_2} (a + bT + cT^2)\, dT = a(T_2 - T_1) + \frac{b}{2}(T_2^2 - T_1^2) + \frac{c}{3}(T_2^3 - T_1^3) \tag{4.10}$$

If a different functional form of the heat capacity is available, the integration result

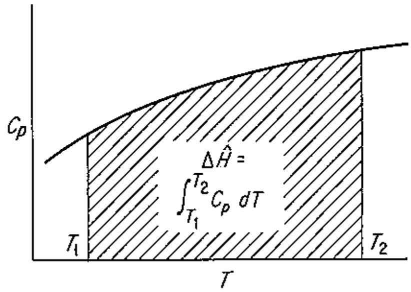


Figure 4.8 Calculation of enthalpy change.

will be different. Equation (4.10) can be stored in a data base and used to calculate enthalpy changes for a single phase as easily as the heat capacity equation can be stored. Can you develop an expression similar to Eq. (4.10) for $\Delta\hat{U}$ based on a polynomial expression for C_v as a function of temperature? Will the right-hand side look exactly the same as Eq. (4.10)?

EXAMPLE 4.9 Calculation of $\Delta\hat{H}$ for a Gas Mixture Using Heat Capacity Equations

The conversion of solid wastes to innocuous gases can be accomplished in incinerators in an environmentally acceptable fashion. However, the hot exhaust gases must be cooled or diluted with air. An economic feasibility study indicates that solid municipal waste can be burned to a gas of the following composition (on a dry basis):

CO_2	9.2%
CO	1.5%
O_2	7.3%
N_2	82.0%
	100.0%

What is the enthalpy difference for this gas per lb mol between the bottom and the top of the stack if the temperature at the bottom of the stack is 550°F and the temperature at the top is 200°F? Ignore the water vapor in the gas. Because these are ideal gases, you can neglect any energy effects resulting from the mixing of the gaseous components.

Solution

Heat capacity equations from Table E.2 [T in °F; C_p = Btu/(lb mol)(°F)] are

N_2: $C_p = 6.895 + 0.7624 \times 10^{-3}T - 0.7009 \times 10^{-7}T^2$

O_2: $C_p = 7.104 + 0.7851 \times 10^{-3}T - 0.5528 \times 10^{-7}T^2$

CO_2: $C_p = 8.448 + 5.757 \times 10^{-3}T - 21.59 \times 10^{-7}T^2 + 3.059 \times 10^{-10}T^3$

CO: $C_p = 6.865 + 0.8024 \times 10^{-3}T - 0.7367 \times 10^{-7}T^2$

Basis: 1.00 lb mol of gas

By multiplying these equations by the respective mole fraction of each component, and then adding them together, you can save time in the integration.

N_2: $0.82(6.895 + 0.7624 \times 10^{-3}T - 0.7009 \times 10^{-7}T^2)$

O_2: $0.073(7.104 + 0.7851 \times 10^{-3}T - 0.5528 \times 10^{-7}T^2)$

CO_2: $0.092(8.448 + 5.757 \times 10^{-3}T - 21.59 \times 10^{-7}T^2 + 3.059 \times 10^{-10}T^3)$

CO: $0.015(6.865 + 0.8024 \times 10^{-3}T - 0.7367 \times 10^{-7}T^2)$

$$C_{p_{\text{net}}} = 7.053 + 1.2242 \times 10^{-3}T - 2.6124 \times 10^{-7}T^2 + 0.2814 \times 10^{-10}T^3$$

$$\Delta\hat{H} = \int_{550}^{200} C_p\, dT = \int_{550}^{200} (7.053 + 1.2242 \times 10^{-3}T - 2.6124 \times 10^{-7}T^2 + 0.2814 \times 10^{-10}T^3)\, dT$$

$$= 7.053[(200) - (550)] + \frac{1.2242 \times 10^{-3}}{2}[(200)^2 - (550)^2] - \frac{2.6124 \times 10^{-7}}{3}[(200)^3 - (550)^3] + \frac{0.2814 \times 10^{-10}}{4}[(200)^4 - (550)^4]$$

$$= -2468.6 - 160.7 + 13.8 - 0.633$$

$$= -2616 \text{ Btu/lb mol gas}$$

4.3-2 Tabular Data

When the values of the physical properties used in your calculations must be accurate, you might well turn to tables. Tables can cover ranges of physical properties well beyond the range applicable for a single equation. Because the most commonly measured properties are temperature and pressure, tables for pure compounds usually are organized in columns and rows, with T and p being the independent variables. If the intervals between table entries are close enough, linear interpolation between entries is reasonably accurate.

Steam tables of all varieties are cited in several of the references in Table 4.5. Tables of the enthalpies for some important compounds at 1 atmosphere will be found in Appendix D.

If you remember that enthalpy values are all relative to some reference state, you can make enthalpy difference calculations merely by subtracting the initial enthalpy from the final enthalpy for any two sets of conditions as shown in the following examples.

EXAMPLE 4.10 Calculation of Enthalpy Change Using Tabulated Enthalpy Values

Calculate the enthalpy change for 1 kg mol of N_2 gas which is heated at a constant pressure of 100 kPa from 18°C to 1100°C.

Solution

We will use the data in Table 4.4 (the pressure is about 1 atm).

$$\text{at } 1100°\text{C} \Leftrightarrow 1373\text{K:} \quad \Delta\hat{H} = 34{,}715 \text{ kJ/kg mol (by interpolation)}$$

$$\text{at } 18°\text{C} \Leftrightarrow 291 \text{ K:} \quad \Delta\hat{H} = 524 \text{ kJ/kg mol}$$

Basis: 1 kg mol of N_2

$$\Delta\hat{H} = 34{,}715 - 524 = 34{,}191 \text{ kJ/kg mol}$$

Tables 4.3 and 4.4 list typical enthalpy data for the combustion gases. Some sources of enthalpy data are listed in Table 4.5. The most common source of enthalpy data for water is the steam tables which are reproduced in Appendix C1 and on a sheet that can be found inside the back cover.

To make use of the steam tables, you must first locate the region of the phase diagram in which the state lies. The tables are organized so that the saturation properties are given separately from the properties of superheated steam and subcooled liquid. Examine the large set of tables in the back of the book. In addition, the saturation properties are presented in two ways: (1) the saturation pressure is given at even intervals for easy interpolation, and (2) the saturation temperature given at even intervals for the same reason.

You can find the region in which a particular state lies by referring to one of the two tables for saturation properties. If, at the given T or P, the given specific intensive property lies outside the range of properties that can exist for saturated liquid, saturated vapor, or their mixtures, the state must be in either the superheated or the subcooled region. For example, look at a brief extract from the steam tables in SI units:

		Specific volume (m³/kg)			Enthalpy (kJ/kg)		
P_{sat}(kPa)	T_{sat}(°C)	v_l	v_{lg}	v_g	h_l	h_{lg}	h_g
101.325	100.0	0.001043	1.672	1.673	419.5	2256.0	2675.6
200.0	120.2	0.001061	0.8846	0.8857	504.7	2201.5	2706.2

For each saturation pressure, the corresponding saturation temperature (boiling point) is given along with the values of specific volume and enthalpy for both saturated liquid and saturated vapor. The volume and enthalpy value in the middle column designated by the subscript lg is the difference between the saturated vapor and saturated liquid states, and will be discussed in the next section.

TABLE 4.3 Enthalpies of Combustion Gases* (Btu/lb mol)

°R	N_2	O_2	Air	H_2	CO	CO_2	H_2O
492	0.0	0.0	0.0	0.0	0.0	0.0	0.0
500	55.67	55.93	55.57	57.74	55.68	68.95	64.02
520	194.9	195.9	194.6	191.9	194.9	243.1	224.2
537	313.2	315.1	312.7	308.9	313.3	392.2	360.5
600	751.9	758.8	751.2	744.4	752.4	963	867.5
700	1,450	1,471	1,450	1,433	1,451	1,914	1,679
800	2,150	2,194	2,153	2,122	2,154	2,915	2,501
900	2,852	2,931	2,861	2,825	2,863	3,961	3,336
1,000	3,565	3,680	3,579	3,511	3,580	5,046	4,184
1,100	4,285	4,443	4,306	4,210	4,304	6,167	5,047
1,200	5,005	5,219	5,035	4,917	5,038	7,320	5,925
1,300	5,741	6,007	5,780	5,630	5,783	8,502	6,819
1,400	6,495	6,804	6,540	6,369	6,536	9,710	7,730
1,500	7,231	7,612	7,289	7,069	7,299	10,942	8,657
1,600	8,004	8,427	8,068	7,789	8,072	12,200	9,602
1,700	8,774	9,251	8,847	8,499	8,853	13,470	10,562
1,800	9,539	10,081	9,623	9,219	9,643	14,760	11,540
1,900	10,335	10,918	10,425	9,942	10,440	16,070	12,530
2,000	11,127	11,760	11,224	10,689	11,243	17,390	13,550
2,100	11,927	12,610	12,030	11,615	12,050	18,730	14,570
2,200	12,730	13,460	12,840	12,160	12,870	20,070	15,610
2,300	13,540	14,320	13,660	12,890	13,690	21,430	16,660
2,400	14,350	15,180	14,480	13,650	14,520	22,800	17,730
2,500	15,170	16,040	15,300	14,400	15,350	24,180	18,810

*Pressure = 1 atm.

SOURCE: Page 30 Kobe, K.A., et al., *Thermochemistry of Petrochemicals,* Reprint No. 44 from the *Petroleum Refiner*, Gulf Publ. Co., Houston, TX (1958).

TABLE 4.4 Enthalpies of Combustion Gases* (J/g Mol)†

K	N_2	O_2	Air	H_2	CO	CO_2	H_2O
273	0	0	0	0	0	0	0
291	524	527	523	516	525	655	603
298	728	732	726	718	728	912	837
300	786	790	784	763	786	986	905
400	3,695	3,752	3,696	3,655	3,699	4,903	4,284
500	6,644	6,811	6,660	6,589	6,652	9,204	7,752
600	9,627	9,970	9,673	9,518	9,665	13,807	11,326
700	12,652	13,225	12,736	12,459	12,748	18,656	15,016
800	15,756	16,564	15,878	15,413	15,899	23,710	18,823
900	18,961	19,970	19,116	18,384	19,125	28,936	22,760
1,000	22,171	23,434	22,367	21,388	22,413	34,308	26,823
1,100	25,472	26,940	25,698	24,426	25,760	39,802	31,011
1,200	28,819	30,492	29,078	27,509	29,154	45,404	35,312
1,300	32,216	34,078	32,501	30,626	32,593	51,090	39,722
1,400	35,639	37,693	35,953	33,789	36,070	56,860	44,237
1,500	39,145	41,337	39,463	36,994	39,576	62,676	48,848
1,750	47,940	50,555	48,325	45,275	48,459	77,445	60,751
2,000	56,902	59,914	57,320	53,680	57,488	92,466	73,136
2,250	65,981	69,454	66,441	62,341	66,567	107,738	85,855
2,500	75,060	79,119	75,646	71,211	75,772	123,176	98,867
2,750	84,265	88,910	84,935	80,290	85,018	138,699	112,089
3,000	93,512	98,826	94,265	89,453	94,265	154,347	125,520
3,500	112,131	119,034	113,135	108,030	112,968	185,895	152,799
4,000	130,875	141,410	132,172	127,528	131,796	217,777	180,414

†To convert to cal/g mol multiply by 0.2390.
*Pressure = 1 atm.
SOURCE: Page 30 of Reference in Table 4.3.

TABLE 4.5 Sources of Enthalpy Data

1. American Petroleum Institute, Division of Refining, *Technical Data Book—Petroleum Refining,* 2nd ed., API, Washington, D.C., 1970.
2. American Petroleum Institute, Research Project 44, *Selected Values of Physical and Thermodynamic Properties of Hydrocarbons and Related Compounds,* API, Washington, D.C., 1953. See also Reference 30.
3. American Society of Mechanical Engineers, *Thermodynamic Data for Waste Incineration,* Book No. H00141, New York, 1979.
4. Barner, H.E., and R.V. Scheuerman, *Handbook of Thermochemical Data for Compounds and Aqueous Species,* Wiley-Interscience, New York, 1978.
5. *Bulletin of Chemical Thermodynamics,* published by Department of Chemistry, Oklahoma State University, Stillwater, Okla., 74074. (Index, bibliography, reports.)
6. Chase, M.W., "JANAF Thermochemical Tables," *J. Phys. Chem. Ref. Data,* v. 11, p. 695 (1987).
7. Danner, R.P., and T.E. Daubert, eds., *Manual for Predicting Chemical Process Design Data,* American Institute of Chemical Engineers, New York (1984).
8. Engineering Sciences Data Unit Ltd., London. (Extensive series of property data.)
9. Fratzscher, W., et al., "The Acquisition, Collection, and Tabulation of Substance Data on Fluid Systems for Calculations in Chemical Engineering," *Int. Chem. Eng.,* v. 20, no. 1, pp. 19–28 (1980). (A list of references for data.)
10. Freeman, R.D., *Chemical Thermodynamics,* Bulletin No. 54 in CODATA Directory of Data Sources for Science and Technology, CODATA Secretariat, Paris, 1983.
11. Grigull, U., T. Straug, and P. Schiebener, *Steam Tables in SI Units,* Springer-Verlag, Berlin, 1987.
12. Haar, L., J.S.Gallagher, and G.S. Kell, *NBS/NRC Steam Tables,* Hemisphere, New York, 1984.
13. Hamblin, F.D., *Abridged Thermodynamic and Thermochemical Tables (S.I. Units),* Pergamon Press, Elmsford, N.Y. (paperback edition available), 1972.
14. Haywood, R.W., *Thermodynamic Tables in SI (Metric) Units,* 2nd ed., Cambridge University Press, Cambridge, 1972.
15. Hisham, M.W.M., and S.W. Benson, "Thermochemistry of Inorganic Solids, *J. Chem. Eng. Data,* v. 32, p. 243 (1987).
16. Irvine, T.F., and P.E. Liley, eds., *Microcomputer Program for the Thermodynamic Properties of Air/Steam,* Rumford Publishing Co., W. Lafayette, IN, 1980. (For personal computers.)
17. *Journal of Physical and Chemical Reference Data,* American Chemical Society, 1972 and following.
18. Keenan, J.H., et al., *Steam Tables,* Wiley, New York, 1978. (SI units.) Also J.H. Keenan and F.G. Keyes, *Thermodynamic Properties of Steam,* Wiley, New York, 1936.
19. Keenan, J.H., J. Chao, and J. Kaye, *Gas Tables (English Units),* 2nd ed., Wiley, New York, 1980.
20. Lin, C.T., et al., "Data Bank for Synthetic Fuels," *Hydrocarbon Process,* Parts 1, 2, 3 (August and November 1980).
21. Naumov, G.B., *Handbook of Thermodynamic Data* (translation into English of

TABLE 4.5 *Continued*

Handbook of Thermodynamic Data, Atomizdat, Moscow, 1971). (Available as NTIS PB-226722/7.)

22. Perry, J.H., *Chemical Engineers' Handbook,* 6th ed., McGraw-Hill, New York, 1986.
23. Raznjevic, K., *Handbook of Thermodynamic Tables and Charts,* Hemisphere, New York, 1976.
24. Rossini, F.K., et al., "Tables of Selected Values of Chemical Thermodynamic Properties," *National Bureau of Standards, Circular* 500, 1952. (Revisions are being issued periodically under the Technical Note 270 series by other authors.)
25. Selover, T.B., National *Standard Reference Data Service of the USSR,* Hemisphere, New York (continuing series).
26. Selover, T.B., and E. Buck, "DIPPR in Its Seventh Year," *Chem. Eng. Prog.,* p. 18 (July 1987).
27. Stephenson, F.M. and S. Malanowski, *Handbook of the Thermodynamics of Organic Compounds,* Elsevier, New York (1987).
28. Stewart, R.B., and Victor J. Johnson, *A Compendium of the Properties of Materials at Low Temperatures (Phase II),* National Bureau of Standards Cryogenic Engineering Lab. Rep. WADD GO-56, Part IV, U.S. Air Force Systems Command, Wright-Patterson Air Force Base, Ohio (December 1961).
29. Stull, D.R., and H. Prophet, *JANAF Thermochemical Tables,* 2nd ed., U.S. Government Printing Office, Washington, D.C., No. 0303–0872, C13, 48:37 (AD-732 043), 1971. (Data for over 1000 compounds.)
30. Thermodynamics Research Center, Texas A & M University. Continuation of API Project 44 plus extensive additional data. Write Data Distribution Office, TRC, College Station, Tex. 77843.
31. Touloukian, Y.S., and E.H. Buyco, *Thermophysical Properties of Matter—The TPRC Data Series,* Plenum, New York, 1983.
32. Wagman, D.D., *NBS Tables of Chemical Thermodynamic Properties,* American Chemical Society, Washington, D.C., 1983.
33. Yaws, C.L., "Physical and Thermodynamic Properties" (Graphs), *Chem. Eng.,* Initiated with Part I in the June 10, 1974, issue. Subsequent issues: July 3, Aug. 19, Sept. 30, Oct. 28, Nov. 25, Dec. 23, 1974; Jan. 20, Feb. 17, Mar. 31, May 12, July 21, Sept. 1, Sept. 29, 1975. See "Physical Properties: A Guide. . .," published 1977, McGraw-Hill, New York; "Enthalpics of Formation for 700 Major Organic Compounds," *Chem. Eng.,* p. 81 (Sept. 26, 1988).

EXAMPLE 4.11 Use of the Steam Tables

Steam is cooled from 640°F and 92 psia to 480°F and 52 psia. What is $\Delta\hat{H}$ in Btu/lb?

Solution

Use the tables for the American engineering units in the pocket in the back of the book. You must employ double interpolation to get the specific enthalpies, $\Delta\hat{H}$, relative to the reference for the table:

	T (°F)					
p (psia)	600	700	p	600	640	700
90	1328.7	1378.1	92	1328.6		1378.0
95	1328.4	1377.8			1348.4	
	450	500	p	450	480	500
50	1258.7	1282.6	52	1258.4		1282.4
55	1258.2	1282.2			1272.8	

Note that the steam table values include the effect of pressure on $\Delta\hat{H}$ as well as temperature. An example of the interpolation needed at 600°F is

$$\tfrac{2}{5}(1328.7 - 1328.4) = 0.4(0.3) = 0.12$$

At $p = 92$ psia and $T = 600$, $\Delta\hat{H} = 1328.7 - 0.12 = 1328.6$.

The enthalpy change is

$$\Delta\hat{H} = 1272.8 - 1348.4 = -75.6 \text{ Btu/lb}$$

EXAMPLE 4.12 Calculation of the Change in Enthalpy for Water from the Steam Tables

Four kilograms of water at 27°C and 200 kPa are heated at constant pressure until the volume of the water becomes 1000 times the original value. What is the final temperature of the water?

Solution

The effect of pressure on the volume of liquid water can be neglected (refer to the table of the properties of liquid water), hence the initial specific volume is that of saturated liquid water at 300 K, or 0.001004 m³/kg. The final specific volume is

$$0.001004(1000) = 1.004 \text{ m}^3/\text{kg}$$

At 200 kPa, using interpolation, between 400 and 450 K we find T by solving

$$0.9024\frac{\text{m}^3}{\text{kg}} + \frac{(1.025 - 0.9024)\text{ m}^3/\text{kg}}{(450 - 400)\text{ K}}(T - 400)\text{ K} = 1.004\frac{\text{m}^3}{\text{kg}}$$

$$T = 400 + 41 = 441 \text{ K}$$

EXAMPLE 4.13 Enthalpy Change for Liquid Water

Water is heated from 400 K and 2000 kPa to 475 K and 5000 kPa. What are $\Delta\hat{H}$ and $\Delta\hat{U}$ in kJ/kg?

Solution

From the steam tables in SI units in the back of the book, the data are:

	$\Delta\hat{H}$	$\Delta\hat{U}$
400 K and 2000 kPa (kJ/kg)	533.76	531.63
475 K and 5000 kPa (kJ/kg)	861.96	856.18

$$\Delta\hat{H} = 861.96 - 533.76 = 328.20 \text{ kJ/kg}$$

$$\Delta\hat{U} = 856.18 - 531.63 = 324.55 \text{ kJ/kg}$$

4.3-3 Graphical Presentation of Enthalpy Data

You have often heard of saying: a picture is worth a 1000 words. Something similar might be said of two-dimensional charts, namely that you can get an excellent idea of the characteristics of the enthalpy of a substance in all regions of interest via a chart. Although the accuracy of the readings of values from a chart may be limited (depending on the scale of the chart), tracing out various processes on a chart enables you to rapidly visualize and analyze what is taking place. Primarily, charts are a simple and quick method of getting data to compute enthalpy changes. Figure 4.9 is an example chart. A number of the sources listed in the references in Table 4.6 and in the Appendices, as well as Perry's *Handbook*, include property charts. Reynolds[9] has an excellent collection of charts of 40 substances in his volume. Other sources of such charts are listed in Table 4.6. Appendix J contains charts for toluene and carbon dioxide.

Charts are drawn with various coordinates, such as

$$p \text{ versus } \hat{H}$$

$$p \text{ versus } \hat{V}$$

$$p \text{ versus } T$$

$$\hat{H} \text{ versus } \hat{S}^{10}$$

Since a chart has only two dimensions, the other parameters of interest have to be plotted as lines of constant value across the face of the chart. Recall, for example, that the p–$\hat{V}$ diagram for CO_2, Fig. 3.3, lines of constant temperature were shown as parameters. Similarly, on a chart with pressure and enthalpy as the axes, lines of constant volume and/or temperature might be drawn.

How many properties have to be specified for a single component gas to definitely fix the state of the gas? Because we know from our discussion in Sec. 3.7

[9] W. C. Reynolds, *Thermodynamic Properties in SI*, Department of Mechanical Engineering, Stanford University, Stanford, Calif., 1979.

[10] S is *entropy*—this diagram is called a *Mollier* diagram.

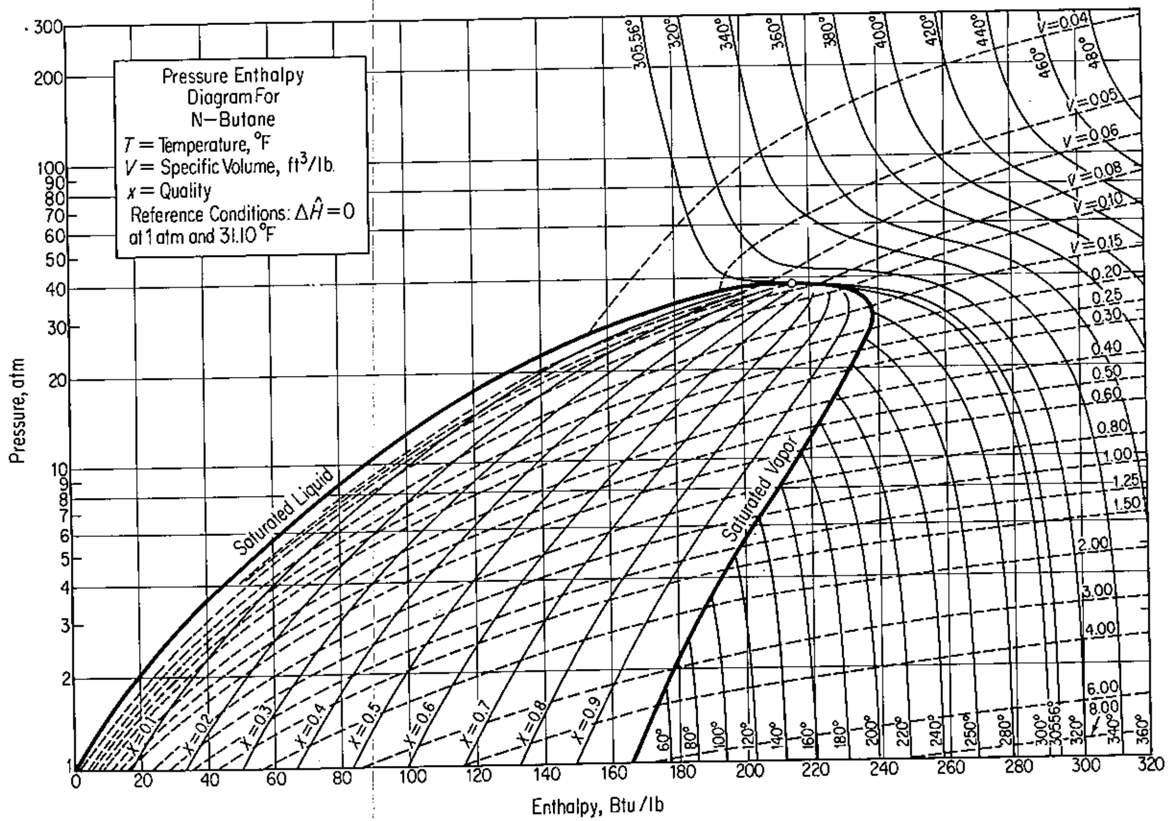


Figure 4.9 Pressure enthalpy diagram for *n*-butane.

TABLE 4.6 Thermodynamic Charts Showing Enthalpy Data for Pure Compounds*

Compound	Reference†
Acetone	2
Acetylene	1
Air	V.C. Williams, *AIChE Trans.*, v. 39, p. 93 (1943); *AIChE J.*, v. 1, p. 302 (1955).
Benzene	1
1,3-Butadiene	C.H. Meyers, *J. Res. Natl. Bur. Stand.*, v. A39, p. 507 (1947).
i-Butane	1, 3
n-Butane	1, 3, 4
n-Butanol	L.W. Shemilt, in *Proceedings of the Conference on Thermodynamic Transport Properties of Fluids*, London, 1957, Institute of Mechanical Engineers, London, 1958.
t-Butanol	F. Maslan, *AIChE J.*, v. 7, p. 172 (1961).
n-Butene	1
Chlorine	R.E. Hulme and A.B. Tilman, *Chem. Eng.* (January 1949).
Ethane	1, 3, 4
Ethanol	R.C. Reid and J.M. Smith, *Chem. Eng. Prog.*, v. 47, p. 415 (1951).
Ethyl ether	2
Ethylene	1, 3
Ethylene oxide	J.E. Mock and J.M. Smith, *Ind. Eng. Chem.*, v. 42, p. 2125 (1950).
Fatty acids	J.D. Chase, *Chem. Eng.*, p. 107 (March 24, 1980).
n-Heptane	E.B. Stuart et al., *Chem Eng. Prog.*, v. 46, p. 311 (1950).
n-Hexane	1
Hydrogen sulfide	J.R. West, *Chem. Eng. Prog.*, v. 44, p. 287 (1948).
Isopropyl ether	2
Mercury	General Electric Company Report GET-1879A, 1949.
Methane	1, 3, 4
Methanol	J.M. Smith, *Chem. Eng. Prog.*, v. 44, p. 52 (1948).
Methyl ethyl ketone	2
Monomethyl hydrazine	F. Bizjak and D.F. Stai, *AIAA J.*, v. 2, p. 954 (1964).
Neon	Cryogenic Data Center, National Bureau of Standards, Boulder, Colo.
Nitrogen	G.S. Lin, *Chem. Eng. Prog.*, v. 59, no. 11, p. 69 (1963).

*For mixtures, see V.F. Lesavage et al., *Ind. Eng. Chem.*, v. 59, no. 11, p. 35 (1967).

†1. L.N. Cajar et al., *Thermodynamic Properties and Reduced Correlations for Gases*, Gulf Publishing Company, Houston, 1967. (Series of articles which appeared in the magazine *Hydrocarbon Processing* from 1962 to 1965.)

2. P.T. Eubank and J.M. Smith, *J. Chem. Eng. Data*, v. 7, p. 75 (1962).

3. W.C. Edmister, *Applied Hydrocarbon Thermodynamics*, Gulf Publishing Company, Houston, 1987.

4. K.E. Starling et al., *Hydrocarbon Processing*, 1971 and following. *Note:* Charts available separately from Gulf Publishing Company, Houston.

TABLE 4.6 ***Continued***

Compound	Reference[†]
n-Pentane	1, 3, 4
Propane	1, 3, 4
n-Propanol	Shemilt (*see n*-Butanol).
Propylene	1, 3
Refrigerant 245	R.L. Shank, *J. Chem. Eng. Data,* v. 12, p. 474 (1967).
Sulfur dioxide	J.R. West and G.P. Giusti, *J. Phys.Colloid Chem.*, v. 54, p. 601 (1950).
Combustion gases	H.C. Hottel, G.C. Williams, and C.N. Satterfield, *Thermodynamic Charts for Combustion Processes* (I) Text, (II) Charts, Wiley, New York, 1949.
Hydrocarbons	3

that specifying two intensive properties for a pure gas will ensure that all the other intensive properties will have definite values, any two properties can be chosen at will. Since a particular state for a gas can be defined by any two independent properties, a two-dimensional thermodynamic chart can be seen to be a handy way to present many combinations of physical properties.

EXAMPLE 4.14 Use of Pressure–Enthalpy Chart for Butane

Calculate ΔH, ΔV, and ΔT changes for 1 lb of saturated vapor of *n*-butane going from 2 atm to 20 atm (saturated).

Solution

Obtain the necessary data from Fig. 4.9.

	$\hat{H}$ (Btu/lb)	$\hat{V}$ (ft³/lb)	T (°F)
Saturated vapor at 2 atm:	179	3.00	72
Saturated vapor at 20 atm:	233	0.30	239

$$\Delta \hat{H} = 233 - 179 = 54 \text{ Btu/lb}$$

$$\Delta \hat{V} = 3.00 - 0.30 = 2.70 \text{ ft}^3/\text{lb}$$

$$\Delta T = 239 - 72 = 167°\text{F}$$

Enthalpies and other thermodynamic properties can be *estimated* by generalized methods based on the theory of corresponding states or additive bond contributions.[11]

[11] R. R. Tarakad and R. P. Palmer, "A Comparison of Enthalpy Prediction methods," *AIChE J.*, v. 22, p. 409 (1976).

4.3-4 Retrieval of Data from Computer Data Bases

Values of the properties of hundreds of substances are available in the form of computer programs that can provide the values at any given state. Thus you avoid the need for interpolation and/or auxiliary computation, such as when only ΔH is given in a table and ΔU must be calculated from $\Delta U = \Delta H - \Delta pV$. One such program, that for the properties of water, is included on the disk in the back of this book. The program is abridged so that it does not have the accuracy nor cover the range of commercially available programs, but it is quite suitable for solving problems in this text.

Computer tapes can be purchased providing information on the physical properties of large numbers of compounds, and immediate access to computer-based information systems via the telephone can be obtained from computer service bureaus. Table 4.7 lists a number of such programs and data bases.

TABLE 4.7 Sources of Computer Based Property Data

American Institute of Chemical Engineers, *Data Compilation, Tables of Properties of Pure Compounds,* STN International (on-line access), Ohio.

Aslam, S., and M. Charmchi, *STEAM,* Mechanical Engineering Department, University of Lowell, Lowell, Mass., 1986.

Benedek, P., and F. Ogli, *Computer Aided Chemical Thermodynamics of Gases and Liquids,* Wiley-Interscience, New York, 1985.

CTS Engineering, *Steam Tables,* CTS Engineering, Monroe, La., 1986.

Ganapathy, U., *BASIC Programs for Steam Plant Engineers,* Marcel Dekker, New York, 1986.

GASPROPS, *A Computer Program for the Calculation of the Thermodynamic Properties of Ideal Gases and Combustion Products,* Wiley, New York, 1984.

Garvin, D., V.B. Parkin, and D.D. Wagman, "Thermodynamic Data Banks," *Chemtech,* p. 691 (November 1982).

Impulse Engineering, *IMP-STM 85/PC,* Impulse Engineering, San Francisco, 1985. (Steam and water-for PC.)

Irvine, T.F., and P.E. Liley, *Steam and Gas Tables with Computer Equations,* Academic Press, Orlando, Fla., 1984.

Moore, K.V., *4STEM: Thermodynamic Properties of Water and Steam,* DE86048580/GAR, tape, National Technical Information Service, Springfield, Va., 1986.

STEAMCALC, *A Computer Program for the Calculation of the Thermodynamic Properties of Steam,* Wiley, New York, 1984.

Thermodynamics Research Center, Texas A & M University, *Chemical Information System,* Computer Sciences Corp., Falls Church, Va. (on line).

Self-Assessment Test

1. Calculate the enthalpy change in 24 g of N_2 if heated from 300 K to 1500 K at constant pressure.

2. What is the enthalpy change when 2 lb of n-butane gas is cooled from 320°F and 2 atm to saturated vapor at 6 atm?
3. You are told that 4.3 kg of water at 200 kPa occupies (a) 4.3, (b) 43, (c) 430, (d) 4300, and (e) 43,000 liters. State for each case whether the water is in the solid, liquid, liquid-vapor, or vapor regions.
4. Two hundred pounds of a 35° API distillate is heated from 130°F to 275°F. Estimate the enthalpy change (in Btu). See Appendix K.
5. Water at 400 kPa and 500 K is cooled to 200 kPa and 400 K. What is the enthalpy change? Use the steam tables.

4.4 ENTHALPY CHANGES FOR PHASE TRANSITIONS

Your objectives in studying this section are to be able to:

1. Estimate the heat of fusion or heat of vaporization from empirical formulas, or look up the value in a reference table.
2. Estimate the heat of vaporization from the Clausius–Clapeyron equation or the Othmer plot.
3. Calculate an enthalpy change of a substance including the phase transitions.

In making enthalpy calculations, we noted in Fig. 4.5 that the heat capacity data are discontinuous at the points of a phase transition. The name usually given to the enthalpy changes for these phase transitions is *latent heat* changes, latent meaning "hidden" in the sense that the substance (e.g., water) can absorb a large amount of heat without any noticeable increase in temperature. Unfortunately, the word *heat* is still associated with these enthalpy changes for historical reasons, although they have nothing directly to do with heat as defined in Sec. 4.1. Ice at 0°C can absorb energy amounting to 334 J/g without undergoing a temperature rise or a pressure change, and similarly, liquid water at 1 atmosphere can absorb 2256.1 J/g before the temperature and pressure will change. Figure 4.10 shows, by the vertical lines at constant temperature, the enthalpies for the phase changes for water at 1 atmosphere pressure. The various enthalpy changes are termed:

Enthalpy change	Phase change
Heat of fusion	Solid to liquid
Heat of vaporization	Liquid to vapor
Heat of condensation	Vapor to liquid
Heat of sublimation	Solid to vapor

You can see from the chart for n-butane (Fig. 4.9) that the heats of vaporization (and

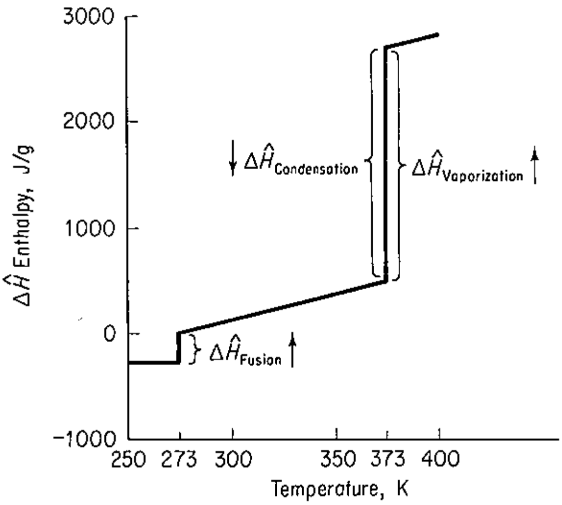


Figure 4.10 Enthalpy change for water at 1 atmosphere showing the phase transitions.

accordingly condensation) changes with temperature all the way up to the critical point, where it vanishes. Calculation of the *quality* of a liquid–vapor mixture was illustrated by Example 3.18.

You can find experimental values of latent heats in the references in Table 4.5, and a brief tabulation is listed in Appendix D. The symbols used for latent heat changes vary, but you usually find one or more of the following employed: $\Delta\hat{H}$, L, λ, Λ. Keep in mind that the enthalpy changes for vaporization given in the steam tables are for water under its vapor pressure at the indicated temperature.

In the absence of experimental values for the latent heats of transition, the following approximate methods will provide a rough *estimate of the molar latent heats*. Reid, Prausnitz and Poling, *The Properties of Gases and Liquids,* give many more methods.[12]

4.4-1 Heat of Fusion

No accurate, simple way to estimate ΔH_f exists. The heat of fusion for many elements and compounds can be roughly approximated by

$$\frac{\Delta\hat{H}_f}{T_f} = \text{constant} = \begin{cases} 2\text{–}3 & \text{for elements} \\ 5\text{–}7 & \text{for inorganic compounds} \\ 9\text{–}11 & \text{for organic compounds} \end{cases} \tag{4.11}$$

where $\Delta\hat{H}_f$ = molar heat of fusion, cal/g mol
T_f = melting point, K

4.4-2 Heat of Vaporization

Because the heat of vaporization is so large, it is important to estimate ΔH_v accurately. Three techniques are discussed below.

[12] See the supplementary references in Chap. 3.

Clausius–Clapeyron equation. The Clapeyron equation itself is an exact thermodynamic relationship between the slope of the vapor-pressure curve and the molar heat of vaporization and the other variables listed below:

$$\frac{dp^*}{dT} = \frac{\Delta\hat{H}_v}{T(\hat{V}_g - \hat{V}_l)} \tag{4.12}$$

where p^* = vapor pressure
T = absolute temperature
$\Delta\hat{H}_v$ = molar heat of vaporization at T
$\hat{V}_i$ = molar volume of gas or liquid as indicated by the subscript g or l

Any consistent set of units may be used.

If experimental vapor-pressure data are available for a span of temperatures, or a correlation is available, dp^*/dT can be evaluated in the vicinity of T. Furthermore, $(\hat{V}_g - \hat{V}_l)$ can be estimated solely from $\hat{V}_g$ if we neglect V_l; hence for a nonideal gas

$$\frac{dp^*}{dT} = -\frac{\Delta\hat{H}_v}{z(RT^2/p^*)} \tag{4.13}$$

Eq. (4.13) can be solved for $\Delta\hat{H}_v$.

Another variation of Eq. (4.12) is as follows. Assume that:

(a) $\hat{V}_l$ is negligible in comparison with $\hat{V}_g$.
(b) The ideal gas law is applicable for the vapor:

$$\hat{V}_g = RT/p^*$$

Then

$$\frac{dp^*}{p^*} = \frac{\Delta\hat{H}_v\, dT}{RT^2} \tag{4.14}$$

Rearrange to

$$\frac{d\ln p^*}{d(1/T)} = 2.303\frac{d\log_{10} p^*}{d(1/T)} = -\frac{\Delta\hat{H}_v}{R} \tag{4.15}$$

You can plot the $\log_{10} p^*$ vs. $1/T$ and obtain the slope $-(\Delta\hat{H}_v/2.303R)$.

If we further assume that $\Delta\hat{H}_v$ is constant over the temperature range of interest, integration of Eq. (4.15) yields an indefinite integral

$$\log_{10} p^* = -\frac{\Delta\hat{H}_v}{2.303RT} + B \tag{4.16}$$

The indefinite integral, Eq. (4.16), is known as the Clausius–Clapeyron equation. Unfortunately, a plot of $\ln p^*$ versus $1/T$ over a significant range of $1/T$ does not give a straight line. Consequently, Eq. (4.16) often is modified; one result is the Antoine equation discussed in Sec. 3.3. A definite integral of Eq. (4.15) is

$$\log_{10}\frac{p_1^*}{p_2^*} = \frac{\Delta\hat{H}_v}{2.303R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \tag{4.17}$$

Either of these equations can be used graphically or analytically to obtain $\Delta\hat{H}_v$ for a short temperature interval.

EXAMPLE 4.15 Heat of Vaporization from the Clausius–Clapeyron Equation

Estimate the heat of vaporization of isobutyric acid at 200°C.

Solution

The vapor-pressure data for isobutyric acid (from Perry's *Handbook*) are

Pressure (mm Hg)	Temp. (°C)	Pressure (atm)	Temp. (°C)
100	98.0	1	154.5
200	115.8	2	179.8
400	134.5	5	217.0
760	154.5	10	250.0

Basis: 1 g mol of isobutyric acid

Since $\Delta\hat{H}_v$ remains essentially constant for short temperature intervals, the heat of vaporization can be estimated from Eq. (4.17) and the vapor-pressure data at 179.8°C and 217.0°C.

$$179.8°\text{C} \Leftrightarrow 453.0 \text{ K} \qquad 217°\text{C} \Leftrightarrow 490.2\text{K}$$

$$\log_{10}\frac{2}{5} = \frac{\Delta\hat{H}_v}{(2.303)(8.314)}\left(\frac{1}{490.2} - \frac{1}{453.0}\right)$$

$$\Delta\hat{H}_v = 45{,}483 \text{ J/g mol at } 200°\text{C}$$

The experimental value of $\Delta\hat{H}_v$ is not known at 200°C. At the normal boiling point (154.5°C), $\Delta\hat{H}_v = 41{,}300$ J/g mol, hence the value calculated is high. It should be lower than 41,300.

Reduced form of the Clapeyron equation. This is an equation in terms of the reduced pressure and temperature which gives good results:

$$d \ln p^* = -\frac{\Delta\hat{H}_v}{zRT_c} d\left(\frac{1}{T_r}\right)$$

or

$$\frac{\Delta\hat{H}_v}{zRT_c} = -\frac{d \ln p^*}{d(1/T_r)}$$

If we compute the right-hand side, say, from the Antoine equation (see Appendix G), we get

$$\frac{\Delta\hat{H}_v}{zRT_c} = \frac{B}{T_c}\left[\frac{T_r}{T_r + (C/T_c)}\right]^2 \tag{4.18}$$

Chen's equation. An equation that yields values of $\Delta\hat{H}_v$ (in kJ/g mol) to within 2% is *Chen's equation:*

$$\Delta\hat{H}_v = \frac{T_b[0.0331(T_b/T_c) + 0.0297 \log_{10} p_c - 0.0327]}{1.07 - T_b/T_c}$$

where T_b is the normal boiling point of the liquid in K, and p_c is the critical pressure in atmospheres.

Prediction using enthalpy of vaporization at the normal boiling point. As an example, Watson[13] found empirically that

$$\frac{\Delta\hat{H}_{v_2}}{\Delta\hat{H}_{v_1}} = \left(\frac{1 - T_{r_2}}{1 - T_{r_1}}\right)^{0.38}$$

where $\Delta\hat{H}_{v_2}$ = heat of vaporization of a pure liquid at T_2
$\Delta\hat{H}_{v_1}$ = heat of vaporization of the same liquid at T_1

Yaws[14] lists various other values of the exponent for various substances.

4.4-3 Reference Substance Plots

A number of graphical techniques have been proposed to estimate the molal heat of vaporization of a liquid at any temperature by comparing the $\Delta\hat{H}_v$ for the unknown liquid with that of a known liquid such as water. Two of these methods are described below.

Duhring plot. The temperature of the wanted compound A is plotted against the temperature of the known (reference) liquid at *equal vapor pressure*. For example, if the temperature of A (isobutyric acid) and the reference substance (water) are determined at 760, 400, and 200 mm Hg pressure, then a plot of the temperatures of A vs. the temperatures of the reference substance will be approximately a straight line over a wide temperature range, and have a slope of $(\Delta\hat{H}_{v_A}/\Delta\hat{H}_{v_{H_2O}})(T_{H_2O}/T_A)^2$.

Othmer plot. The Othmer plot[15] is based on the same concepts as the Duhring plot except that the *logarithms* of vapor pressures are plotted against each other at *equal temperatures*. As illustrated in Fig. 4.11, a plot of the $\log_{10}(p_A^*)$ against $\log_{10}(p_{\text{ref}}^*)$ chosen at the same temperature yields a straight line over a very wide temperature range.

To indicate how to apply the Othmer plot to estimate the heat of vaporization of compound A, we apply the Clapeyron equation to each substance and take the

[13] K. M. Watson, *Ind. Eng. Chem.*, v. 23, p. 360 (1931); v. 35, p. 398 (1943).

[14] C. L. Yaws, *Physical Properties*, McGraw-Hill, New York, 1977.

[15] D. F. Othmer, *Ind. Eng. Chem.*, v. 32, p. 841 (1940). For a complete review of the technique and a comparative statistical analysis among various predictive methods, refer to D. F. Othmer and H. N. Huang, *Ind. Eng. Chem.*, v. 57, p. 40 (1965); D. F. Othmer and E. S. Yu, *Ind. Eng. Chem.*, v. 60, no. 1, p. 22 (1968); and D. F. Othmer and H. T. Chem, *Ind. Eng. Chem.*, v. 60, no. 4, p. 39 (1968).

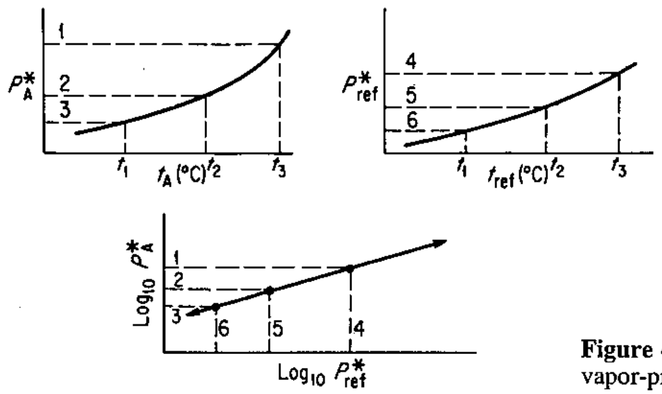


Figure 4.11 Othmer plot and vapor-pressure curves.

ratio:

$$\frac{d(\ln p_A^*)}{d(\ln p_{\text{ref}}^*)} = \frac{\dfrac{-\Delta \hat{H}_{v_A}}{R\,d(1/T_A)}}{\dfrac{-\Delta \hat{H}_{v_{\text{ref}}}}{R\,d(1/T_{\text{ref}})}} \tag{4.19}$$

By choosing values of vapor pressure at equal temperature ($T_A = T_{\text{ref}}$) in Eq. (4.19), we obtain

$$\frac{d \ln p_A^*}{d \ln p_{\text{ref}}^*} = \frac{\Delta \hat{H}_{v_A}}{\Delta \hat{H}_{v_{\text{ref}}}} = m = \text{slope of Othmer plot as in Fig. 4.11} \tag{4.20}$$

The Othmer plot works well because the errors inherent in the assumptions made in deriving Eq. (4.20) cancel out to a considerable extent. At very high pressures the Othmer plot is not too effective. Incidentally, this type of plot has been applied to estimate a wide variety of thermodynamic and transport relations, such as equilibrium constants, diffusion coefficients, solubility relations, ionization and dissociation constants, and so on, with considerable success.

Othmer recommended another type of relationship that can be used to estimate the heat of vaporization:

$$\ln p_{r_A}^* = \frac{\Delta \hat{H}_{v_A}}{\Delta H_{v_{\text{ref}}}} \left(\frac{T_{c_{\text{ref}}}}{T_{c_A}}\right) \ln p_{r_{\text{ref}}}^* \tag{4.21}$$

where p_r^* is the reduced vapor pressure. Equation (4.21) is effective and gives a straight line through the point $p_{c_A}^* = 1$ and $T_{c_A} = 1$. The Gordon method plots the log of the vapor pressure of A versus that of the reference substance at equal *reduced temperature*.

EXAMPLE 4.16 Use of an Othmer Plot

Repeat the calculation for the heat of vaporization of isobutyric acid at 200°C using an Othmer plot.

Solution

Data are as follows:

Temp. (°C)	p^*_{iso} (atm)	$\log p^*_{iso}$	$p^*_{H_2O}$ (atm)	$\log p^*_{H_2O}$
154.5	1	0	5.28	0.723
179.8	2	0.3010	9.87	0.994
217.0	5	0.698	21.6	1.334
250.0	10	1.00	39.1	1.592

Figure E4.16 is the Othmer plot.

$$\text{slope} = \frac{\Delta\hat{H}_{v_{iso}}}{\Delta\hat{H}_{v_{H_2O}}} = \frac{0.80 - 0.20}{1.42 - 0.90} = 1.15$$

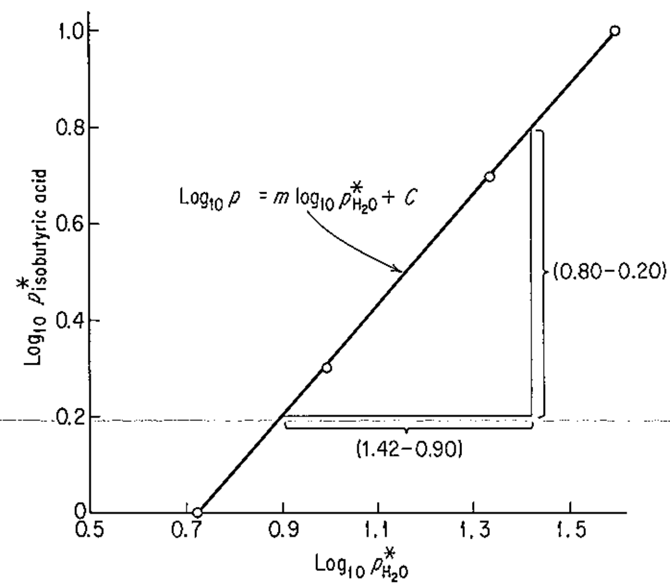


Figure E4.16

At 200°C(392°F), from the steam tables,

$$\Delta\hat{H}_{v_{H_2O}} = 34{,}895 \text{ kJ/kg mol} \qquad (1938.6 \text{ kJ/kg})$$

Then

$$\Delta\hat{H}_{v_{iso}} = (34{,}895)(1.15) = 40{,}130 \text{ kJ/kg mol}$$

The answer in this case is lower than that in Example 4.15. Without the experimental value, it is difficult to say what the proper answer is, but 40,130 is more reasonable than the value of 45,483 calculated in Example 4.15.

We are now equipped to include in the calculation of an enthalpy change the occurrence of a phase change. Because enthalpy is a state variable, any arbitary path from the initial to the final state will suffice for the actual computation. Simply choose the simplest possible path.

EXAMPLE 4.17 Calculation of Enthalpy Change Including Phase Transition

What is the enthalpy change in British thermal units when 1 gal of water is heated from 60°F to 1150°F and 240 psig?

Solution

We will use the steam tables for this problem.

Basis: 1 lb of H_2O at 60°F

From steam tables (ref. temp. = 32°F):

$$\hat{H} = 28.07 \text{ Btu/lb} \qquad \text{at } 60°\text{F}$$

$$\hat{H} = 1604.5 \text{ Btu/lb} \qquad \text{at } 1150°\text{F and 240 psig (254.7 psia)}$$

$$\Delta\hat{H} = (1604.5 - 28.07) = 1576.4 \text{ Btu/lb}$$

$$\Delta H = 1576(8.345) = 13{,}150 \text{ Btu/gal}$$

Note: The enthalpy value which has been used for liquid was taken from the steam tables for the saturated liquid under its own vapor pressure. Since the enthalpy of liquid water changes negligibly with pressure, no loss of accuracy is encountered for engineering purposes if the initial pressure on the water is not stated.

EXAMPLE 4.18 Calculation of Enthalpy Change Including a Phase Change

What is the enthalpy change of 1 kg of water from ice at 0°C to vapor at 120°C and 100 kPa?

Solution

In this problem we will use ΔH_f and ΔH_v in the calculations to illustrate how to calculate the overall enthalpy change when a table or chart is not available. Recall that enthalpy is a state function. Consequently, we can choose any convenient path between the initial and final state for the calculations. Figure E4.18 shows two different possible paths to be used to calculate ΔH. Which should be selected? The answer is the one with the best data. Usually, phase transitions are evaluated at the melting point (for a solid) and the normal boiling point of 1 atm (for a liquid) because you can easily find tabulated data at those temperatures, but any temperature would suffice. We will choose the path $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ for which the data are

$$\Delta\hat{H}_f = 335 \text{ J/g at } 0°\text{C and } p^* \text{ or at } 101.3 \text{ kPa}$$

$$\Delta\hat{H}_v = 2256 \text{ J/g at } 100°\text{C (101.3 kPa)}$$

The heat capacity equations can be located in Table E.1 in the Appendix.

Liquid [J/(g mol)(K)]: $18.296 + 47.212 \times 10^{-2}T_K - 133.88 \times 10^{-5}T_K^2 + 1314.2 \times 10^{-9}T_K^3$

Vapor [J/(g mol)(K)]: $33.46 + 0.6880 \times 10^{-2}T_{°C} + 0.7604 \times 10^{-5}T_{°C}^2 - 3.593 \times 10^{-9}T_{°C}^3$

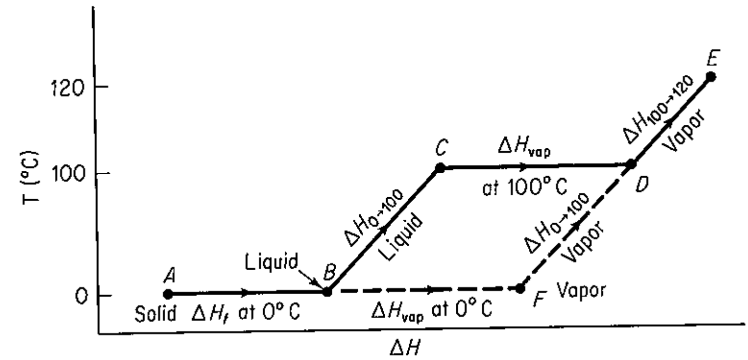


Figure E4.18 (not to scale)

$$\Delta\hat{H}_{\text{overall}} = \Delta\hat{H}_{\text{fusion at 0°C}} + \Delta\hat{H}_{\text{liquid at 0°C}\to100°C} + \Delta\hat{H}_{\text{vaporization at 100°C}} + \Delta\hat{H}_{\text{vapor 100}\to120°C}$$

$$= 335 + \tfrac{1}{18}\int_{273}^{373} (18.296 + 47.212 \times 10^{-2}T - 133.88 \times 10^{-5}T^2$$

$$+ 1{,}314.2 \times 10^{-9}T^3)\,dT + 2256 + \tfrac{1}{18}\int_{100}^{120} (33.46 + 0.6880 \times 10^{-2}T$$

$$+ 0.7604 \times 10^{-5}T^2 - 3.593 \times 10^{-9}T^3)\,dT$$

$$= 335 + 418.6 + 2256 + 38.1 = 3048 \text{ J/g}$$

From the steam tables, the value of $\Delta\hat{H}$ is

$$\Delta\hat{H} = \overset{\Delta\hat{H}_f}{335} + \overset{\Delta\hat{H}_{0\to120°C}}{2716} = 3051 \text{ J/g}$$

Can you offer a reason for the small difference?

Now that we have finished describing ways to calculate the quantities that will appear in the energy balance, it is time to turn to consideration of the balance itself.

Self-Assessment Test

1. Calculate the enthalpy change in Btu/lb of benzene as it goes from a vapor at 300°F and 1 atm to a solid at 0°F and 1 atm.
2. What is the enthalpy change that occurs when 5 lb of water is heated from ice at 32°F to vapor at 250°F and 1 atm? Use the steam tables.
3. Estimate the heat of vaporization of water at 450 K from Watson's relation given that $\Delta H_v = 2256.1$ kJ/kg at 373.14 K, and compare with experimental data.
4. Prepare an Othmer chart to predict the heat of vaporization of methyl alcohol at 90°C. Use water as the reference substance. Repeat using the Clausius–Clapeyron equation.
5. Air conditioners use Freon-12 as the working fluid. A charging when unit full holds 10 kg of saturated liquid at 20°C. If the unit contains 3 kg at 20°C, what is the quality? Use the following extract from the Freon-12 table in solving the problem.

Satn. pres., p^*(kPa)	Satn. temp., t_s(°C)	Volume (cm³/g)		Enthalpy (J/g)	
		v_l	v_g	h_l	h_g
500	15.6	0.7438	34.82	50.66	194.0
600	22	0.7566	29.13	56.80	196.6

Thought Problems

1. Fire walkers with bare feet walk across beds of glowing coals without apparent harm. The rite is found in many parts of the world today and was practiced in classical Greece and ancient India and China, according to the *Encyclopaedia Britannica.*

   The temperature of glowing coals is about 1000°C. What are some of the possible reasons that fire walkers are not seriously burned?

2. A fire-induced BLEVE (boiling liquid expanding vapor explosion) in a storage tank can result in a catastrophy. The scenario is somewhat as follows: A pressure vessel (e.g., a pressurized storage tank), partially filled with liquid, is subjected to high heat flux from a fire. The temperature of the liquid starts to increase, causing an increase in pressure within the tank. When the vapor pressure reaches the safety relief valve pressure setting, the relief valve opens and starts to vent vapor (or liquid) to the outside. Concurrent with the previous step, the temperature of the portion of the tank shell not in contact with the liquid (i.e., the ullage space) increases dramatically.

   The heat weakens the tank shell around the ullage space. Thermally induced stresses are created in the tank shell near the vapor/liquid interface, and the heat-weakened tank plus the high internal pressure combine to cause a sudden, violent tank rupture. Fragments of the tank are propelled away from the tank at great force. Most of the remaining superheated liquid vaporizes rapidly due to the pressure release. The rest is mechanically atomized to small drops due to the force of the explosion. A fireball is created by the burning vapor and liquid.

   What steps would you recommend to prevent a BLEVE in the case of fire near a storage tank? [*Hint:* Two possible routes are: (1) prevent the fire from heating the tank; and (2) prevent the buildup of presure in the tank.]

4.5 THE GENERAL ENERGY BALANCE

Scientists did not begin to write energy balances for physical systems prior to the latter half of the nineteenth century. Before 1850 they were not sure what energy was or even if it was important. But in the 1850s the concepts of energy and the energy balance became clearly formulated. We do not have the space here to outline the historical development of the energy balance and of special cases of it, but it truly makes a most interesting story and can be found elsewhere.[16–19] Today we consider

[16] E. J. Hoffman, *The Concept of Energy: An Inquiry into Origins and Applications,* Ann Arbor Science Publishers, Ann Arbor, Mich., 1977.

[17] V. V. Raman, "Where Credit Is Due—The Energy Conservation Principle," *Phys. Teacher,* p. 80(February 1965).

[18] T. M. Brown, *Am. J. Physics,* v. 33, no. 10, p. 1 (1965).

[19] L. K. Nash, *J. Chem. Educ.,* v. 42, p. 64 (1965) (a resource paper).

the energy balance to be so fundamental a physical principle that we invent new classes of energy to make sure that the equation indeed does balance. Equation (4.22) as written below is a generalization of the results of numerous experiments on relatively simple special cases. We universally believe the equation is valid because we cannot find exceptions to it in practice, taking into account the precision of the measurements.

It is necessary to keep in mind two important points as you read what follows. First, we examine only systems that are homogeneous, not charged, and without surface effects, in order to make the energy balance as simple as possible. Second, the energy balance is developed and applied from the macroscopic viewpoint (overall about the system) rather than from a microscopic viewpoint (i.e., an elemental volume within the system).

The concept of the macroscopic energy balance is similar to the concept of the macroscopic material balance, namely,

$$\left\{\begin{array}{l}\textbf{accumulation of}\\ \textbf{energy within the}\\ \textbf{system}\end{array}\right\} = \left\{\begin{array}{l}\textbf{transfer of energy}\\ \textbf{into system through}\\ \textbf{system boundary}\end{array}\right\} - \left\{\begin{array}{l}\textbf{transfer of energy out}\\ \textbf{of system through}\\ \textbf{system boundary}\end{array}\right\}$$
$$+ \left\{\begin{array}{l}\textbf{energy genera-}\\ \textbf{tion within}\\ \textbf{system}\end{array}\right\} - \left\{\begin{array}{l}\textbf{energy con-}\\ \textbf{sumption}\\ \textbf{within system}\end{array}\right\} \qquad (4.22)$$

Equation (4.22) can be applied to a single piece of equipment or to a complex plant such as that shown in Fig. 4.2.

While the formulation of the energy balance in words as outlined in Eq. (4.22) is easily understood and rigorous, you will discover in later courses that to express each term of Eq. (4.22) in mathematical notation may require certain simplifications to be introduced, a discussion of which is beyond our scope here, but they have a quite minor influence on our final balance.

In what follows we first treat systems in which no material flows in and out. Do you remember the name for such a system? Then we examine systems in which material does flow in and out. What is the name for such a system? Finally, in Sec. 4.7 we discuss energy balances for processes in which chemical reactions take place. Thus the energy generation and consumption terms will not play a role in this chapter unless sources such as radioactive decay or the slowing down of neutrons enter into the process.

4.5-1 Energy Balances for Closed Systems (without Chemical Reaction)

Your objectives in studying this section are to be able to:

1. Write down the general energy balance in words [Eq. (4.22)].
2. Write down the energy balance for a closed system in symbols [Eq. (4.23)], and apply it to solve energy balance problems.

3. Cite the signs for work and heat entering and leaving the system.
4. Calculate the total energy or any of its components (internal energy, kinetic energy, potential energy) associated with the mass of the system.

Figure 4.12 shows the various types of energy to be accounted for in Eq. (4.22). As to the notation, the subscripts t_1 and t_2 refer to the initial and final times for the period over which the accumulation is to be evaluated, with $t_2 > t_1$. The superscript caret (ˆ) means that the symbol stands for energy per unit mass; without the caret, the symbol means energy of the total mass present. Other notation is evident from Table 4.8 and is repeated in the notation list at the end of the book.

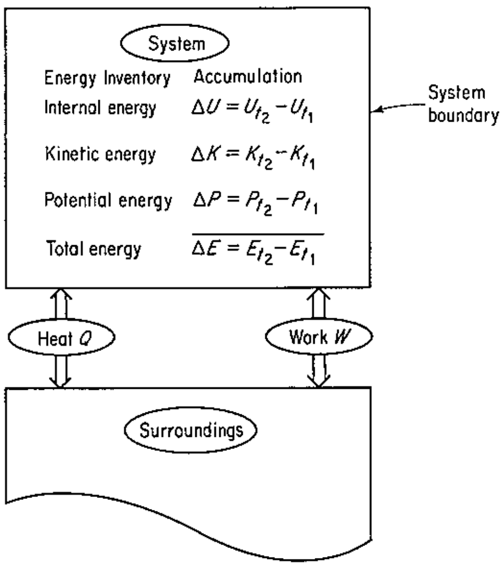


Figure 4.12 Terms in the energy balance for a closed system.

As is commonly done, we have split the total energy (E) associated with the mass in the system into three categories: internal energy (U), kinetic energy (K), and potential energy (P). Energy transported across the system boundary can be transferred by two modes: heat (Q) and work (W). (We discuss energy transferred with mass flow in Sec. 4.5-2.) Note that Q and W here are defined as the *net* transfer of heat and work, respectively, between the system and the surroundings, and respectively equal the integral of the net rate of flow of heat or work over the time interval t_1 to t_2:

$$Q = \int_{t_1}^{t_2} \dot{Q}\, dt \qquad W = \int_{t_1}^{t_2} \dot{W}\, dt$$

Other forms of energy can be split from Q, W, or ΔE, and included as separate terms in Eq. (4.23) if they are important enough to be distinguished. Keep in mind that the internal energies U_{t_1} and U_{t_2} in the system cannot be evaluated as absolute quantities—only $(U_{t_2} - U_{t_1}) = \Delta U$ can be calculated. The reference state cancels.

Equation (4.22) when translated into mathematical symbols becomes

$$\Delta E = E_{t_2} - E_{t_1} = Q - W \tag{4.23}$$

where Δ = **difference operator signifying *final* minus *initial* in time**
Q = **heat absorbed *by* the system *from* the surroundings** (by definition Q is ***positive for heat entering the system***)
W = **mechanical work done *by* the system *on* the surroundings** (by definition W is ***positive for work going from the system to the surroundings***)

[Equation (4.23) is known as the ***first law of thermodynamics*** for a closed system.]

Keep in mind that a system may do work, or have work done on it, without some obvious mechanical device such as a pump, shaft, and so on, being present. Often the nature of the work is implied rather than explicity stated. For example, a cylinder filled with gas enclosed by a movable piston implies that the surrounding atmosphere can do work on the piston or the reverse; a batch fuel cell does no mechanical work, unless it produces bubbles, but does deliver a current at a potential difference; electromagnetic radiation can impinge on or leave a system; and so forth.

Now for a word of warning: Be certain you use *consistent units* for all terms; in the American engineering system the use of foot-pound, for example, and Btu in different places in Eq. (4.23) is a common error for the beginner.

Let us look now at some examples of applications of the energy balance for closed systems. Remember to follow the checklist presented in Chap. 2 in analyzing the problem.

EXAMPLE 4.19 Application of the Energy Balance

Ten pounds of CO_2 at room temperature (80°F) are stored in a fire extinguisher having a volume of 4.0 ft^3. How much heat must be removed from the extinguisher so that 40% of the CO_2 becomes liquid?

Solution

This problem involves a closed system (Fig. E4.19) so that Eq. (4.23) applies. We can use the CO_2 chart in Appendix J to get the necessary property values.

Steps 1, 2, and 3 The specific volume of the CO_2 is $4.0/10 = 0.40$ ft^3/lb, hence CO_2 is a gas at the start of the process. The pressure is 300 psia and $\Delta\hat{H} = 160$ Btu/lb.

Step 4

Basis: 10 lb CO_2

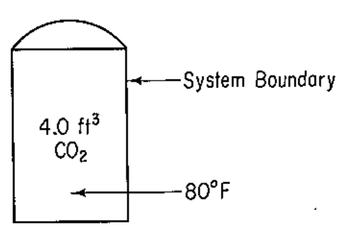


Figure E4.19

Steps 5 and 6 In the energy balance

$$\Delta E = Q - W$$

W is zero because the volume of the system is fixed, hence with $\Delta K = \Delta P = 0$,

$$Q = \Delta U = \Delta H - \Delta(pV)$$

We do not have values of $\Delta\hat{U}$, just values of $\Delta\hat{H}$, on the CO_2 chart. We can find $\Delta\hat{H}_{\text{final}}$ from the CO_2 chart by following the constant-volume line of 0.40 ft^3/lb to the spot where the quality is 0.6. Hence the final state is fixed, and all the final properties can be identified, namely

$$\Delta\hat{H}_{\text{final}} = 81 \text{ Btu/lb}$$

$$p_{\text{final}} = 140 \text{ psia}$$

Steps 7, 8, and 9

$$Q = (81 - 160) - \left[\frac{(140)(144)(0.40)}{778.2} - \frac{(300)(144)(0.40)}{778.2}\right]$$

$$= -67.2 \text{ Btu} \quad \text{(heat is removed)}$$

EXAMPLE 4.20 Application of the Energy Balance

Argon gas in an insulated plasma deposition chamber with a volume of 2 L is to be heated by an electric resistance heater. Initially the gas, which can be treated as an ideal gas, is at 1.5 Pa and 300 K. The 1000-ohm heater draws current at 40 V for 5 minutes (i.e., 480 J of work is done by the surroundings). What is the final gas temperature and pressure at equilibrium? The mass of the heater is 12 g and its heat capacity is 0.35 J/(g)(K). Assume that the heat transfer to the chamber from the gas at this low pressure and in the short time period is negligible.

Solution

The system does not exchange mass with the surroundings, so Eq. (4.23) applies with $\Delta P = \Delta K = 0$:

$$\Delta E = Q - W = \Delta U$$

Steps 1, 2, and 3 The system is the gas plus the heater as shown in Fig. E4.20. Because of the assumption about the heat transfer, $Q = 0$. W is given as -480 J (work done on the system) in 5 minutes.

Step 4

$$\text{Basis} = 5 \text{ minutes}$$

Steps 5 and 6 For an ideal gas

$$pV = nRT$$

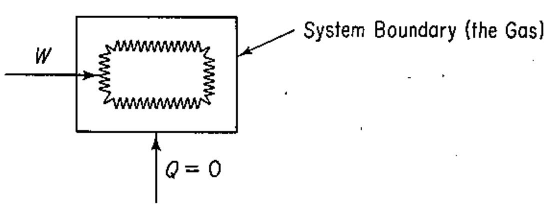


Figure E4.20

and initially we know p, V, and T, and thus can calculate the mass of the gas:

$$n = \frac{pV}{RT} = \frac{1.5 \text{ Pa}}{} \left| \frac{2 \text{ L}}{} \right| \frac{10^{-3} \text{ m}^3}{1 \text{ L}} \left| \frac{10^3(\text{g mol})(\text{K})}{8.314 \times 10^3(\text{Pa})(\text{m}^3)} \right| \frac{}{300 \text{ K}}$$

$$= 1.203 \times 10^{-6} \text{ g mol}$$

The heater mass and heat capacity are given, and the C_V of the gas is (see Sec. 4.2) $C_V = C_p - R$ since $C_p = \frac{5}{2}R$,

$$C_V = \tfrac{5}{2}R - R = \tfrac{3}{2}R$$

Assume that the heat capacity of the heater is C_V also. We know that

$$\Delta U = n \int_{300}^{T_p} C_V \, dT = nC_V(T - 300)$$

hence we can find T once we calculate ΔU from $-W = \Delta U$.

Steps 7, 8, and 9

$$\Delta U = -(-480 \text{ J}) = 480 \text{ J}$$

$$= \overbrace{(12)(0.35)(T - 300)}^{\textbf{heater}} + \overbrace{(1.203 \times 10^{-6})\,(\tfrac{3}{2})(8.314)(T - 300)}^{\textbf{gas}}$$

$$T = 414.3 \text{ K}$$

The final pressure is

$$\frac{p_2 V_2}{p_1 V_1} = \frac{n_2 RT_2}{n_1 RT_1}$$

or

$$p_2 = p_1 \left(\frac{T_2}{T_1}\right) = 1.5 \left(\frac{414.3}{300}\right) = 2.07 \text{ Pa}$$

EXAMPLE 4.21 Energy Balance

Ten pounds of water at 35°F, 4.00 lb of ice at 32°F, and 6.00 lb of steam at 250°F and 20 psia are mixed together in a container of fixed volume. What is the final temperature of the mixture? How much steam condenses? Assume that the volume of the vessel is constant with a value equal to the volume of the steam and that the vessel is insulated.

Solution

Steps 1, 2, and 3 We can assume that the overall batch process takes place with $Q = 0$ and $W = 0$ if we define the system as in Fig. E4.21. Let T_2 be the final temperature. The system consists of 20 lb of H_2O in one or two phases. ΔK and ΔP equal 0. The energy balance reduces to $\Delta U = 0$.

The initial properties can be obtained from the steam tables. Unfortunately, we can only retrieve $\Delta \hat{H}$ values, not the $\Delta \hat{U}$ values that we want (and could get from the SI steam tables) for the energy balance, hence we need also to collect values of p and $\hat{V}$.

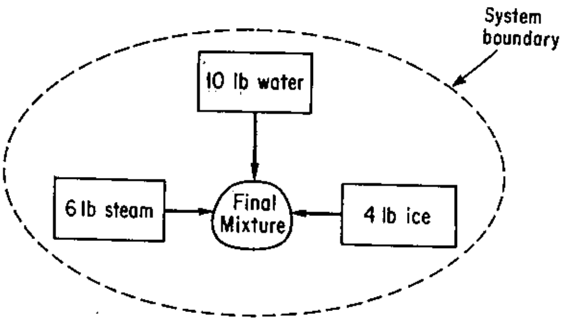


Figure E4.21

	$\Delta \hat{H}$ (Btu/lb)	$\hat{V}$ (ft^3/lb)	p (psia)	T (°F)
Ice	−143.6*	—	—	32
Water	3.02	0.0162		35
Steam	1168	20.81	20	250

*Heat of fusion (pressure and volume have no significant effect).

We know that $\Delta\hat{U} = \Delta\hat{H} - \Delta p\ \hat{V}$, so that the energy balance becomes

$$\Delta H - \Delta p\ V = 0 \qquad \text{(a)}$$

The volume of the container is fixed at the initial volume of the steam, namely (6) (20.81) = 124.86 ft^3; we ignore the volumes of the ice and water in the calculations, as they are so small.

Step 4 The basis is 20 lb of water at the given conditions.

$$\text{Basis: } \begin{cases} \text{4 lb of ice at 32°F} \\ \text{10 lb of } H_2O \text{ at 35°F} \\ \text{6 lb of steam at 250°F and 20 psia} \end{cases}$$

Steps 5 and 6 If we assume that the final state of the water is liquid water in equilibrium with water vapor, an assumption to be checked out, we can calculate the final $\Delta\hat{H}$ of the 20 lb of water. We also know that the final water is saturated. The second condition to fix the state (quality, temperature, and pressure) of the system can evolve from the energy balance, hence the problem has a unique solution. However, the final state must be calculated indirectly if tables are to be used.

Steps 7, 8, and 9

$$\overbrace{20\ \Delta\hat{H}^f_{T_f}}^{\textbf{final } \Delta H} - \overbrace{(6\Delta\hat{H}^s_{T_s} + 10\Delta\hat{H}^w_{T_w} + 4\Delta\hat{H}^i_{T_i})}^{\textbf{initial } \Delta H}$$
$$= 20(p\hat{V})^f_{T_f} - 6(p\hat{V})^s_{T_s} - 10(p\hat{V})^w_{T_w} - 4(p\hat{V})^i_{T_i} \qquad \text{(b)}$$

The last two terms on the right-hand side of Eq. (b) cannot be more than 1 Btu at the very most and can safely be neglected. Be sure to check this assumption! Thus the equation to be used is

$$20\Delta\hat{H}^f_{T_f} - 20(p\hat{V})^f_{T_f} = (6\Delta\hat{H}^s_{T_s} + 10\Delta\hat{H}^w_{T_w} + 4\Delta\hat{H}^i_{T_i}) - 6(p\hat{V})^s_{T_s} \qquad \text{(c)}$$

Because of the phase changes that take place as well as the nonlinearity of the heat capacities as a function of temperature, it is not possible to replace the enthalpies in Eq. (c) with functions of temperature and get a linear algebraic equation that is easy to solve. Consequently, the strategy we will use is first to assume a final temperature and pressure, and next we will check the calculation via Eq. (c). We want to bracket the temperature if possible, and then can interpolate for the desired answer.

The pressure in the vessel drops as more steam condenses. Data for the specific volume of the steam as a function of temperature and pressure can be taken from the steam tables, at saturated conditions, and the mass of steam left at any assumed temperature (or pressure) can be calculated by dividing 124.86 ft³ by the specific volume of the steam.

The right-hand side of Eq. (c) is equal to

$$\frac{6\text{ lb}}{}\left|\frac{1168.0\text{ Btu}}{\text{lb}}\right. + \frac{10\text{ lb}}{}\left|\frac{3.02\text{ Btu}}{\text{lb}}\right. + \frac{4\text{ lb}}{}\left|\frac{-143.6\text{ Btu}}{\text{lb}}\right.$$

$$-\frac{6\text{lb}_m}{}\left|\frac{20\text{ lb}_f}{\text{in}^2.}\right|\left(\frac{12\text{ in.}}{1\text{ ft}}\right)^2\left|\frac{20.81\text{ ft}^3}{\text{lb}_m}\right|\frac{1\text{ Btu}}{778(\text{ft})(\text{lb}_f)} = 6001.6\text{ Btu}$$

As an initial guess, suppose that one-half of the steam does not condense. Then the specific volume of the final steam is (124.86 ft³/3 lb) = 41.62 ft³/lb. The closest integer line in the steam tables in the saturated steam column for pressure is 10 psia (T = 193.21°F) with a specific volume of 38.462 ft³/lb and with $\Delta\hat{H}_{\text{steam}}$ = 1143.3 Btu/lb and $\Delta\hat{H}_{\text{liquid}}$ = 161.17 Btu/lb. Let us use these latter data as the initial assumption. We calculate

$$S = \frac{124.86\text{ ft}^3}{}\left|\frac{1\text{ lb steam}}{38.462\text{ ft}^3}\right. = 3.246\text{ lb steam not condensed}$$

We check this assumption by seeing if Eq. (c′) balances:

$$S[\Delta\hat{H}_{T_f} - p\hat{V}]^f_{\text{vapor}} + (20 - S)[\Delta\hat{H}_{T_f} - p\hat{V}]^f_{\text{liquid}} \stackrel{?}{=} 6001.6 \qquad (c')$$

$$(p\hat{V})^f_{\text{vapor}} = \frac{(10)(12)^2(38.462)}{778} = 71.2\text{ Btu/lb}$$

$$(p\hat{V})^f_{\text{liquid}} \cong 0.0$$

$$3.246(1143.3 - 71.2) + (20 - 3.246)(161.17) \stackrel{?}{=} 6001.6$$

$$6180.28 \neq 6001.6$$

The initial guess for T_f and p was too low (we need a bigger specific volume for the steam, hence S is less). Next we assume that T^* = 186°F (p^* = 8.566 psia, $\hat{V}_{\text{vapor}}$ = 44.55 ft³/lb, $\Delta\hat{H}_{\text{liquid}}$ = 153.93 Btu/lb, and $\Delta\hat{H}_{\text{vapor}}$ = 1140.5 Btu/lb).

$$S = \frac{124.86}{44.45} = 2.899\text{ lb not condensed}$$

Again we check Eq. (c′):

$$(p\hat{V})^f_{\text{vapor}} = \frac{(8.566)(12)^2(44.45)}{778} = 70.5\text{ Btu/lb}$$

$$(pV)^f_{\text{liquid}} \cong 0$$

$$2.899(1140.5 - 70.5) + (20 - 2.899)(153.93) \stackrel{?}{=} 6001.6$$

$$5734.29 \neq 6001.6$$

However, we have bracketed the solution. Linear interpolation for S gives

$$S = 2.899 + \left(\frac{6001.6 - 5734.29}{6180.28 - 5734.29}\right)(3.246 - 2.899)$$
$$= 3.11$$

Consequently, the steam condensed is $6 - 3.11 = 2.89$ lb, and the assumption in steps 5 and 6 proved correct.

Self-Assessment Test

1. Liquid oxygen is stored in a 14,000-L storage tank. When charged, the tank contains 13,000 L of liquid in equilibrium with its vapor at 1 atm pressure. What is the **(a)** temperature, **(b)** mass, and **(c)** quality of the oxygen in the vessel? The pressure relief valve of the storage tank is set at 2.5 atm. If heat leaks into the oxygen tank at the rate of 5.0×10^6 J/hr, **(d)** when will the pressure relief valve operate, and **(e)** what will be the temperature in the storage tank at that time?

   Data: at 1 atm, saturated, $\hat{V}_l = 0.0281$ L/g mol, $\hat{V}_g = 7.15$ L/g mol, $\Delta\hat{H} = -133.5$ J/g; at 2.5 atm, saturated, $\Delta\hat{H} = -116.6$ J/g.
2. Suppose that you fill an insulated Thermos to 95% of the volume with ice and water at equilibrium and securely seal the opening.
   (a) Will the pressure in the Thermos go up, down, or remain the same after 2 hr?
   (b) After 2 weeks?
   (c) For the case in which after filling and sealing, the Thermos is shaken vigorously, what will happen to the pressure?
3. An 0.25-liter container initially filled with 0.225 kg of water at a pressure of 20 atm is cooled until the pressure inside the container is 100 kPa.
   (a) What are the initial and final temperatures of the water?
   (b) How much heat was transferred from the water to reach the final state?
4. Compute ΔE in Btu for 2 lb mol of an ideal monoatomic gas heated from 60°F to 160°F.
5. In a shock tube experiment, the gas (air) is held at room temperature at 15 atm in a volume of 0.350 ft^3 by a metal seal. When the seal is broken, the air rushes down the evacuated tube, which has a volume of 20 ft^3. The tube is insulated. In the experiment:
   (a) What is the work done by the air?
   (b) What is the heat transferred to the air?
   (c) What is the internal energy change of the air?
   (d) What is the final temperature of the air after 3 min?
   (e) What is the final pressure of the air?

4.5-2 Energy Balances for Open Systems (without Chemical Reaction)

Your objectives in studying this section are to be able to:

1. Write down the energy balance for an open system [Eq. (4.24)] in words and symbols, explain each term, and apply the equation to

solve energy balance problems using the 10-step strategy plus two new steps.
2. Explain the "pV work" (flow work) concept.
3. Define isothermal, adiabatic, isobaric, and isometric processes.
4. Make the necessary assumptions and approximations to simplify and solve the energy balance for open systems.

We now apply Eq. (4.22) to systems in which mass as well as energy can pass through the system boundaries. Figure 4.13 illustrates the general case. All the notation is listed in Table 4.8. Keep in mind that the symbols m_1 or m_2 represent a *steady* flow of mass. We give an alternative formulation of the energy balance in terms of differentials in Chap. 6 [see Eqs. (6.8) and (6.9)], where emphasis is placed on the instantaneous rate of change of energy of a system rather than just the initial and final states of the system. The formulation in this chapter can be considered to be the result of integrating the differential balance [see Eq. (6.9)]. The **energy balance for the open system is**

$$\underbrace{m_{t_2}(\hat{U} + \hat{K} + \hat{P})_{t_2} - m_{t_1}(\hat{U} + \hat{K} + \hat{P})_{t_1}}_{\text{accumulation}} = \underbrace{(\hat{U}_1 + \hat{K}_1 + \hat{P}_1)m_1}_{\text{transfer in by mass flow}}$$
$$\underbrace{-(\hat{U}_2 + \hat{K}_2 + \hat{P}_2)m_2}_{\text{transfer out by mass flow}} + \underbrace{Q}_{\text{net transfer by heat}} \underbrace{- W + p_1\hat{V}_1 m_1 - p_2\hat{V}_2 m_2}_{\text{net transfer by work}} \tag{4.24}$$

The terms $p_1\hat{V}_1$ and $p_2\hat{V}_2$ in Eq. (4.24) and Table 4.8 represent the so-called "pV work," or "pressure energy" or "flow work," or "flow energy," that is, the work

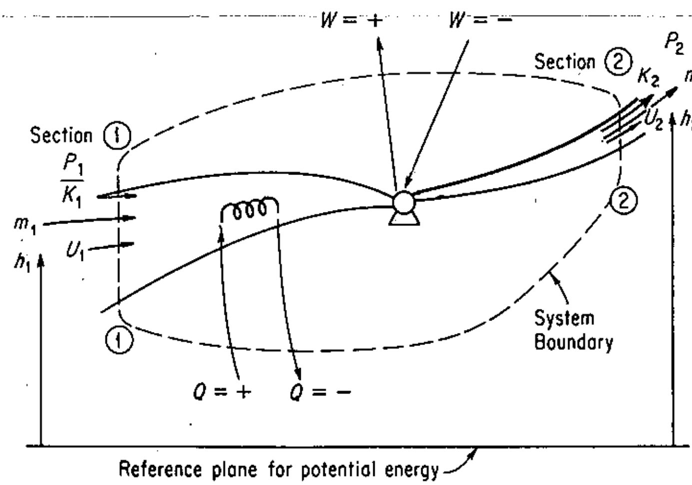


Figure 4.13 General process showing the system boundary and energy transport across the boundary.

TABLE 4.8 Summary of the Symbols to be Used in the General Energy Balance

Accumulation term		
Type of energy	*At time t_1*	*At time t_2*
Internal	U_{t_1}	U_{t_2}
Kinetic	K_{t_1} $\Big\}$ E_{t_1}	K_{t_2} $\Big\}$ E_{t_2}
Potential	P_{t_1}	P_{t_2}
Mass	m_{t_1}	m_{t_2}
Energy accompanying mass transport		
Type of energy	*Transport in*	*Transport out*
Internal	U_1	U_2
Kinetic	K_1	K_2
Potential	P_1	P_2
Mass	m_1	m_2
Net heat input to system	Q	
Net work done by system on surroundings		
Mechanical work or work by moving parts	W	
Work to introduce material into system, less work recovered on removing material from system	$(p_2\hat{V}_2)m_2 - (p_1\hat{V}_1)m_1$	

done by the surroundings to put a unit mass of matter into the system at ① in Fig. 4.13 and the work done by the system on the surroundings as a unit mass leaves the system at ②. Because the pressures at the entrance and exit to the system are constant for differential displacements of mass, the work done by the surroundings on the system adds energy to the system at ①:

$$W_1 = \int_0^{\hat{V}_1} p_1\, d\hat{V} = p_1(\hat{V}_1 - 0) = p_1\hat{V}_1$$

where $\hat{V}$ is the volume per unit mass. Similarly, the work done by the fluid on the surroundings as the fluid leaves the system is $W_2 = p_2\hat{V}_2$, a term that has to be subtracted (why?) from the right-hand side of Eq. (4.24).

In practice we introduce the expression $\Delta\hat{U} + \Delta p\hat{V} = \Delta\hat{H}$ in Eq. (4.24) and thus retain the variable H. Then the energy balance reduces to a form easier to memorize:

$$\Delta E = E_{t_2} - E_{t_1} = -\Delta[(\hat{H} + \hat{K} + \hat{P})m] + Q - W \qquad (4.24a)$$

where $\Delta E = (\hat{U} + \hat{K} + \hat{P})_{t_2}m_{t_2} - (\hat{U} + \hat{K} + \hat{P})_{t_1}m_{t_1}$.

In Eq. (4.24a) the delta symbol (Δ) has two different meanings:

(a) In ΔE, Δ means final minus initial in time.

(b) In ΔH, and so on, Δ means out of the system minus into the system.

Such usage is perhaps initially confusing, but is very common; hence you might as well become accustomed to it. A more rigorous derivation of Eq. (4.24) from a microscopic energy balance may be found in Slattery.[20]

If there is more than one input and output stream to the system, Eq. (4.24a) would become

$$E_{t_2} - E_{t_1} = \sum_{\text{in}} m_i(\hat{H}_i + \hat{K}_i + \hat{P}_i) - \sum_{\text{out}} m_o(\hat{H}_o + \hat{K}_o + \hat{P}_o) + Q - W \qquad (4.25)$$

Here the subscript (o) designates an output stream and the subscript (i) designates an input stream.

In most problems you do not have to use all the terms of the general energy balance equation because certain terms may be zero or may be so small that they can be neglected in comparison with the other terms. Several special cases can be deduced from the general energy balance of considerable industrial importance by introducing certain simplifying assumptions:

(a) No mass transfer **(closed or batch system)** ($m_1 = m_2 = 0$):

$$\Delta E = Q - W \qquad (4.23)$$

(b) No accumulation ($\Delta E = 0$), no mass transfer ($m_1 = m_2 = 0$):

$$Q = W \qquad (4.26)$$

(c) No accumulation ($\Delta E = 0$), but with mass flow:

$$Q - W = \Delta[(\hat{H} + \hat{K} + \hat{P})m] \qquad (4.27)$$

(d) No accumulation, $Q = 0$, $W = 0$, $\hat{K} = 0$, $\hat{P} = 0$:

$$\Delta\hat{H} = 0 \qquad (4.28)$$

[Equation (4.28) is called the **"enthalpy balance."**]

Take, for example, the flow system shown in Fig. 4.14. Overall, between locations 1 and 5, we would find that $\Delta P = 0$. In fact, the only portion of the system where ΔP would be of concern would be between location 4 and some other location. Between 3 and 4, ΔP may be consequential, but between 2 and 4 it may be

[20] J. C. Slattery, *Momentum, Energy, and Mass Transfer in Continua*, McGraw-Hill, New York, 1972.

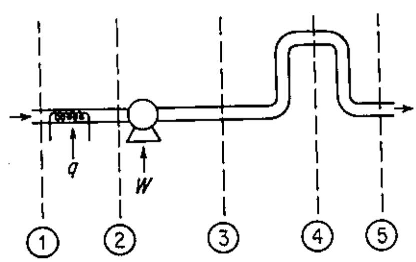


Figure 4.14 Flow system. The dashed lines delineate sub-systems.

negligible in comparison with the work introduced by the pump. Between location 3 and any further downstream point, both Q and W are zero. After reading the problem statements in the examples below but before continuing to read the solution, you should try to apply Eq. (4.24a) yourself to test your ability to simplify the energy balance for particular cases.

Some special process names associated with energy balance problems are worth remembering:

(a) *Isothermal* ($dT = 0$): constant-temperature process
(b) *Isobaric* ($dp = 0$): constant-pressure process
(c) *Isometric* or *isochoric* ($dV = 0$): constant-volume process
(d) *Adiabatic* ($Q = 0$): no heat interchange (i.e., an insulated system). If we inquire as to the circumstances under which a process can be called adiabatic, one of the following is most likely:
  (1) The system is insulated.
  (2) Q is very small in relation to the other terms in the energy equation and may be neglected.
  (3) The process takes place so fast that there is no time for heat to be transferred.

One further remark that we should make is that the energy balance we have presented has included only the most commonly used energy terms. If a change in surface energy, rotational energy, or some other form of energy is important, these more obscure energy terms can be incorporated in an appropriate energy expression by separating the energy of special interest from the term in which it presently is incorporated in Eq. (4.24a). As an example, kinetic energy might be split into linear kinetic energy (translation) and angular kinetic energy (rotation).

To assist you in solving problems involving energy balances, two additional steps should be added to your mental checklist for analyzing problems (see Table 2.4):

(a) *Step 3a:* Always write down the general energy balance, Eq. (4.24) or (4.24a), below your sketch. By this step you will be certain not to neglect any of the terms in your analysis.
(b) *Step 3b:* Examine each term in the general energy balance and eliminate all terms that are zero or can be neglected. Write down why you do so.

We now examine some applications of the general energy balance.

EXAMPLE 4.22 Application of the Energy Balance

Air is being compressed from 100 kPa and 255 K (where it has an enthalpy of 489 kJ/kg) to 1000 kPa and 278 K (where it has an enthalpy of 509 kJ/kg). The exit velocity of the air from the compressor is 60 m/s. What is the power required (in kW) for the compressor if the load is 100 kg/hr of air?

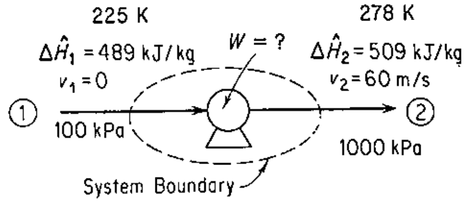


Figure E4.22

Solution

Steps 1, 2, and 3 Figure E4.22 shows the known quantities. The process is clearly a flow process or open system. Let us assume that the entering velocity of the air is zero.

Step 4

Basis: 100 kg of air = 1 hr

Steps 3a and 3b

$$\Delta E = -\Delta[(\hat{H} + \hat{K} + \hat{P})m] + Q - W$$

Let us simplify the energy balance:

(1) The process is in the steady state, hence $\Delta E = 0$.
(2) $m_1 = m_2$.
(3) $\Delta(\hat{P}m) = 0$.
(4) $Q = 0$ by assumption (Q would be small even if the system were not insulated).
(5) $v_1 = 0$ (value is not known but would be small).

The result is

$$W = -\Delta[(\hat{H} + \hat{K})m] = -\Delta H - \Delta K$$

Steps 5 and 6 We have one equation and one unknown, W (ΔK, in effect, and ΔH can be calculated), hence the problem has a unique solution.

Steps 7, 8, and 9

$$\Delta H = \frac{(509 - 489)\text{ kJ}}{\text{kg}} \Bigg| \frac{100\text{ kg}}{} = 2000\text{ kJ}$$

$$\Delta K = \tfrac{1}{2}m(v_2^2 - v_1^2)$$

$$= \left(\frac{1}{2}\right)\frac{100\text{ kg}}{} \Bigg| \frac{(60\text{ m})^2}{\text{s}^2} \Bigg| \frac{1\text{ kJ}}{\frac{1000(\text{kg})(\text{m}^2)}{(\text{s})^2}} = 180\text{ kJ}$$

$$W = -(2000 + 180) = -2180\text{ kJ}$$

(*Note:* The minus sign indicates work is done on the air.)

To convert to power (work/time),

$$\text{kW} = \frac{2180\text{ kJ}}{1\text{ hr}} \Bigg| \frac{1\text{ kW}}{\frac{1\text{ kJ}}{\text{s}}} \Bigg| \frac{1\text{ hr}}{3600\text{ s}} = 0.61\text{ kW}$$

EXAMPLE 4.23 Application of the Energy Balance

Water is being pumped from the bottom of a well 15 ft deep at the rate of 200 gal/hr into a vented storage tank to maintain a level of water in a tank 165 ft above the ground. To prevent freezing in the winter a small heater puts 30,000 Btu/hr into the water during its transfer from the well to the storage tank. Heat is lost from the whole system at the constant rate of 25,000 Btu/hr. What is the temperature of the water as it enters the storage tank, assuming that the well water is at 35°F? A 2-hp pump is being used to pump the water. About 55% of the rated horsepower goes into the work of pumping and the rest is dissipated as heat to the atmosphere.

Solution

Steps 1, 2, and 3 Let the system consist of the well inlet, the piping, and the outlet at the storage tank. The process is a flow process since material continually enters and leaves the system. See Fig. E4.23.

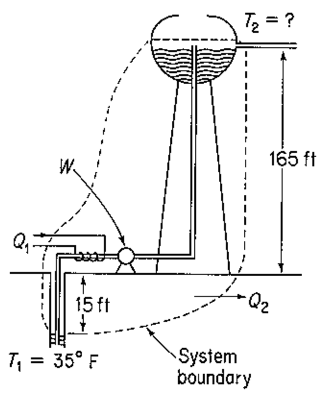


Figure E4.23

Steps 3a, 3b, and 7 The general energy balance is

$$\Delta E = -\Delta[(\hat{H} + \hat{K} + \hat{P})m] + Q - W$$

Step 4

Basis: 1 hr of operation

We will first simplify the energy balance:

(1) The process is in the steady state, so that $\Delta E = 0$.
(2) $m_1 = m_2$.
(3) $\Delta K \cong 0$ because we will assume that $v_1 = v_2 \cong 0$.

Then

$$0 = -\Delta[(\hat{H} + \hat{P})m] + Q - W$$

Steps 5 and 6 Only ΔH at the top of the tank is unknown, and can be calculated from the energy balance. The temperature can be retrieved from

$$\Delta H = m\,\Delta\hat{H} = m\int_{T_1=35°\text{F}}^{T_2} C_p\,dT = mC_p(T_2 - 35)$$

if C_p is an essential constant. Hence the problem has a unique solution.

Steps 7, 8, and 9 The total amount of water pumped is

$$\frac{200\text{ gal}}{\text{hr}}\left|\frac{8.33\text{ lb}}{1\text{ gal}}\right. = 1666\text{ lb/hr}$$

The potential energy change is

$$\Delta P = m\Delta\hat{P} = mg\,\Delta h = \frac{1666\text{ lb}_m}{}\left|\frac{32.2\text{ ft}}{\text{s}^2}\right|\frac{180\text{ ft}}{}\left|\frac{}{\dfrac{32.2\text{ (ft)(lb}_m)}{(\text{s}^2)(\text{lb}_f)}}\right.$$

$$= 300{,}000\text{ (ft)(lb}_f)$$

or

$$\frac{300{,}000\text{ (ft)(lb}_f)}{778\text{ (ft)(lb}_f)/\text{Btu}} = 385.6\text{ Btu}$$

The heat lost by the system is 25,000 Btu while the heater puts 30,000 Btu into the system; hence the net heat exchange is

$$Q = 30{,}000 - 25{,}000 = 5000\text{ Btu}$$

The rate of work being done on the water by the pump is

$$W = -\frac{2\text{ hp}}{}\left|\frac{0.55}{}\right|\frac{33{,}000\text{ (ft)(lb}_f)}{(\text{min})(\text{hp})}\left|\frac{60\text{ min}}{\text{hr}}\right|\frac{\text{Btu}}{778\text{ (ft)(lb}_f)}$$

$$= -2800\text{ Btu/hr}$$

ΔH can be calculated from: $Q - W = \Delta H + \Delta P$:

$$5000 - (-2800) = \Delta H + 386$$

$$\Delta H = 7414\text{ Btu}$$

Because the temperature range considered is small, the heat capacity of liquid water may be assumed to be constant and equal to 1.0 Btu/(lb)(°F) for the problem. Thus

$$7414 = \Delta H = mC_p\,\Delta T = 1666(1.0)(\Delta T)$$

$\Delta T \simeq 4.5$°F temperature rise. Hence, $T = 39.5$°F

EXAMPLE 4.24 Application of the Energy Balance

Steam (that is used to heat a biomass) enters the steam chest, which is segregated from the biomass, at 250°C saturated, and is completely condensed in the steam chest. The rate of the heat loss from the steam chest to the surroundings is 1.5 kJ/s. The reactants are placed in the vessel at 20°C and at the end of the heating the material is at 100°C. If the charge consists of 150 kg of material with an average heat capacity of C_p = 3.26 J/(g)(K), how many kilograms of steam are needed per kilogram of charge? The charge remains in the reaction vessel for 1 hr.

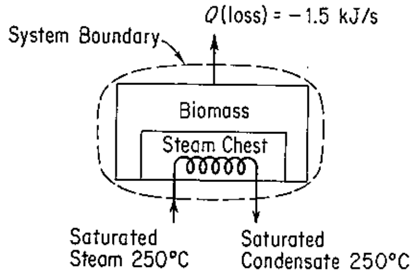


Figure E4.24a

Solution

Steps 1, 2, and 3 Figure E4.24a defines the system and lists the known conditions.
Steps 3a and 3b The energy balance is

$$\Delta E = -\Delta[(\hat{H} + \hat{K} + \hat{P})m] + Q - W \tag{a}$$

Let us simplify the energy balance:

(1) The process is not in the steady state, so $\Delta E \neq 0$.
(2) We can assume that $\Delta K = 0$, and $\Delta P = 0$ in the system, and $W = 0$.
(3) The steam is the only material entering and leaving the system, and $m_1 = m_2$, and ΔK and ΔP of the entering and exit material are zero.

Consequently, Eq. (a) becomes

$$\Delta E = \Delta U = -\Delta[(\hat{H})m] + Q \tag{b}$$

where ΔU refers only to the biomass and not the steam, and $\Delta H = \Delta(\hat{H}m)$ refers only to the steam and not the biomass.

Step 4

Basis: 1 hr of operation

Steps 5 and 6 We can calculate $\Delta H = m_{\text{steam}}\, \Delta\hat{H}_{\text{vaporization}}$ of the steam because we know Q, and can calculate for the biomass $\Delta U = \Delta H - \Delta(pV) = m_{\text{biomass}} C_{p,\text{ biomass}}\, \Delta T$ [because we know that $\Delta(pV)$ for the liquid or solid charge will be negligible]. Thus we can calculate m_{steam} uniquely.

Steps 7, 8, and 9

(a) The heat loss is given as $Q = -1.50$ kJ/s or

$$\frac{-1.50\text{ kJ}}{\text{s}}\left|\frac{3600\text{ s}}{1\text{ hr}}\right|\frac{1\text{ hr}}{} = -5400\text{ kJ}$$

(b) The specific enthalpy change for the steam can be determined from the steam tables. The $\Delta\hat{H}_{\text{vap}}$ of saturated steam at 250°C is 1701 kJ/kg, so that

$$\Delta\hat{H}_{\text{steam}} = -1701\text{ kJ/kg}$$

(c) The enthalpy change of the biomass is

$$\Delta U = \Delta H_{\text{biomass}} = \frac{150\text{ kg}}{}\left|\frac{3.26\text{ kJ}}{(\text{kg})(\text{K})}\right|\frac{(373 - 293)\text{ K}}{} = 39{,}120\text{ kJ}$$

Introduction of all these values into Eq. (b) gives

$$39{,}120 \text{ kJ} = -\left(-1701\frac{\text{kJ}}{\text{kg steam}}\right)(m_{\text{steam}} \text{ kg}) - 5400 \text{ kJ} \tag{c}$$

from which the kilograms of steam per hour, m_{steam}, can be calculated as

$$m_{\text{steam}} = \frac{44{,}520 \text{ kJ}}{} \left| \frac{1 \text{ kg steam}}{1701 \text{ kJ}} \right. = 26.17 \text{ kg steam}$$

or

$$\frac{44{,}520 \text{ kJ}}{} \left| \frac{1 \text{ kg steam}}{1701 \text{ kJ}} \right| \frac{1}{150 \text{ kg charge}} = 0.17\frac{\text{kg steam}}{\text{kg charge}}$$

If the system had been chosen to be everything but the steam chest and lines, we would have a situation as shown in Fig. E4.24b. Under these circumstances heat would be transferred from the steam chest. From a balance on the steam chest (no accumulation),

$$Q_{\text{system II}} = \Delta H_{\text{steam}} \tag{d}$$

As indicated in Fig. E4.24b, the value of $Q_{\text{system II}}$ is negative.

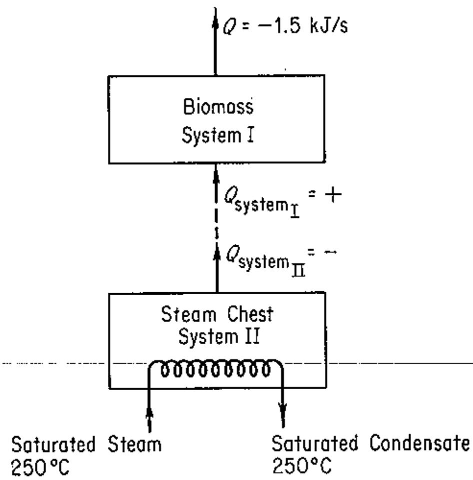


Figure E4.24b

The energy balance for system I with no mass flow in or out of system I is

$$\Delta E_{\text{system I}} = \Delta U_{\text{biomass}} = \Delta H_{\text{biomass}} = Q_{\text{I}} - 5400 \tag{e}$$

Q_{II} and Q_{I} have opposite values because heat is removed from system II and added to system I. Because

$$Q_{\text{II}} = \Delta H_{\text{steam}} = (-1701)m$$

we know that $Q_{\text{I}} = +(1701)m$, and Eq. (e) becomes the same as Eq. (c).

Self-Assessment Test

1. In a refinery a condenser is to cool 1000 lb/hr of benzene that enters at 1 atm, 200°F, and leaves at 171°F. Assume negligible heat loss to the surroundings. How many pounds of cooling water are required per hour if the cooling water enters at 80°F and leaves at 100°F?

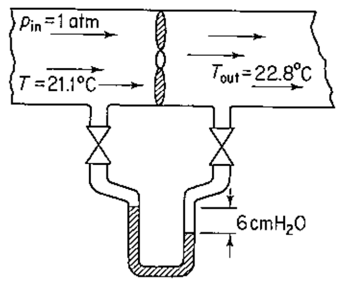
2. In a steady-state process, 10 g mol/s of O_2 at 100°C and 10 g mol/s of nitrogen at 150°C are mixed in a vessel which has a heat loss to the surroundings equal to $209(T - 25)$ J/s, where T is the temperature of the gas mixture in °C. Calculate the gas temperature of the exit stream in °C. Use the following heat capacity equations:

$$O_2: \quad C_p = 6.15 + 3.1 \times 10^{-3}T$$

$$N_2: \quad C_p = 6.5 + 1.25 \times 10^{-3}T$$

where T is in K and C_p is in cal/(g mol)(K).

3. An exhaust fan in a constant-area well-insulated duct delivers air at an exit velocity of 1.5 m/s at a pressure differential of 6 cm H_2O. Thermometers show the inlet and exit temperatures of the air to be 21.1°C and 22.8°C, respectively. The duct area is 0.60 m^2. Determine the actual power requirement for the fan.



4. A water system is fed from a very large tank, large enough so that the water level in the tank is essentially constant. A pump delivers 3000 gal/min in a 12-in.-ID pipe to users 40 ft below the tank level. The rate of work delivered to the water is 1.52 hp. If the exit velocity of the water is 1.5 ft/s and the water temperature in the reservoir is the same as in the exit water, estimate the heat loss per second from the pipeline by the water in transit.

Thought Problems

1. Do you save energy if you
   (a) Let ice build up inside your freezer?
   (b) Use extra laundry detergent?
   (c) Light a fire in your conventional fireplace?
   (d) Turn off your window air conditioner if you will be gone for a couple of hours?
   (e) Turn off your furnace's pilot light during the spring and summer?
   (f) Take baths rather than showers?
   (g) Use long-life incandescent light bulbs?
   (h) Use fluorescent rather than incandescent lights?
   (i) Install your refrigerator beside your range?
   (j) Drive 55 instead of 70 miles per hour?
   (k) Choose a car with air conditioning, power steering, and an automatic transmission over one without these features?
2. Liquid was transferred by gravity from one tank to another tank of about the same height several hundred meters away. The second tank overflowed.
   What might cause such overflow?

4.6 REVERSIBLE PROCESSES AND THE MECHANICAL ENERGY BALANCE

Your objectives in studying this section are to be able to:

1. Define a quasi-static and a reversible process.
2. Identify a process as reversible or irreversible given a description of the process.
3. Define efficiency and apply the concept to calculate the work for an irreversible process.
4. Write down the steady-state mechanical energy balance for an open system and apply it to a problem.

In this section we examine some additional features of various types of energy changes with which you should become familiar.

4.6-1 Quasi-static and Reversible Processes

As explained in Sec. 3.3, a process at equilibrium will not undergo change—no work will be done and the system properties will not change. Suppose that a system at equilibrium is subjected to a differential external force (such as a higher temperature) so that a differential change occurs. The system will pass through nonequilibrium states, but with only very slight deviations from equilibrium if the driving force is infinitesimal. Such a process is said to be a **quasi-static process.** A quasi-static process is an idealization that can rarely be rigorously executed but can be approached with proper physical arrangements.

If there are no dissipative effects, that is, friction, viscosity, inelasticity, electrical resistance, and so on, during a quasi-static process, the process is termed **reversible.** Only an infinitesimal change is required to reverse the process, a concept that leads to the name "reversible." Most industrial processes exhibit heat transfer over finite temperature differences, mixing of dissimilar substances, sudden changes in phase, mass transport under finite concentration differences, free expansion, pipe friction, and other mechanical, chemical, and thermal nonidealities, and consequently are deemed **irreversible.** An irreversible process always involves a degradation of the potential of the process to do work, that is, will not produce the maximum amount of work that would be possible via a reversible process (if such a process could occur).

To illustrate the concept of reversibility, consider a gas confined in a cylinder by the piston shown in Fig. 4.15. During an expansion process the piston moves the distance x and the volume of gas confined in the piston increases from V_1 to V_2. Two forces act on the piston; one is the force exerted by the gas, equal to the pressure times the area of the piston, and the other is the force on the piston shaft and head

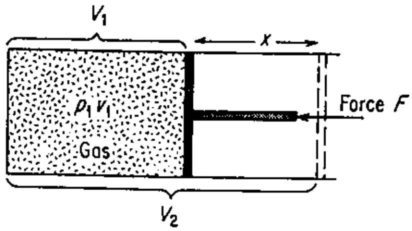


Figure 4.15 Gas expansion.

from outside. If the force exerted by the gas equals the force F, nothing happens. If F is greater than the force of the gas, the gas will be compressed, whereas if F is less than the force of the gas, the gas will expand.

In the latter case, the work done by the expanding gas and the piston will be $W = \int F\,dx$. The work of the gas would be $W = \int_{V_1}^{V_2} p\,dV$ if the process were reversible, that is, if the force F divided by the piston area were differentially less at all times than the pressure of the gas, and if the piston were frictionless; but in a real process some of the work done by the gas is dissipated by viscous effects, and the piston will not be frictionless, so that the work as measured by $\int F\,dx$ will be less than $\int p\,dV$. For these two integrals to be equal, none of the available energy of the system could be degraded to heat or internal energy. To achieve such a situation we have to ensure that the movement of the piston is frictionless and that the motion of the piston proceeds under only a differential imbalance of forces so that no shock or turbulence is present. Naturally, such a process would take a long time to complete.

EXAMPLE 4.25 Calculation of Work for a Batch Process

One kilogram mole of N_2 is in a horizontal cylinder at 1000 kPa and 20°C. A 6-cm^2 piston of 2-kg mass seals the cylinder and is fixed by a pin. The pin is released and the N_2 volume is doubled, at which time the piston is stopped again. What is the work done by the gas in this process?

Solution

Steps 1, 2, and 3 Draw a picture. See Fig. E4.25. From the ideal gas law we can compute the specific volume of the gas at the initial state

$$\hat{V}_1 = \frac{RT}{p} = \frac{8.31\ (\text{kPa})(\text{m}^3)}{(\text{K})(\text{kg mol})} \left| \frac{293\ \text{K}}{1000\ \text{kPa}} \right. = 2.43\ \text{m}^3/\text{kg mol}$$

Choose the gas to be the system.

Step 4 We can use as a basis 2.43 m^3 at 1000 kPa and 20°C.

Steps 5 and 6 We want to calculate the work done by the gas on the piston and the cylinder.

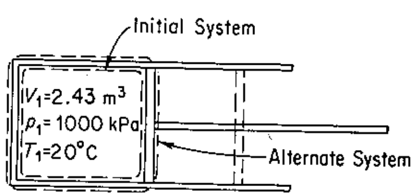


Figure E4.25

(a) Is the process a flow or a nonflow process? Since no material leaves or enters the cylinder, let us analyze the process as a nonflow system.

(b) This process is definitely irreversible because of friction between the piston and the walls of the cylinder, turbulence in the gas, the large pressure drops, and so on.

Consequently, we cannot calculate the work done on the piston by $\int p\, dV$.

Let us change our system to include the gas, the piston, and the cylinder. With this choice, the system expansion is still irreversible, but the pressure in the surroundings can be assumed to be atmospheric pressure and is constant. The work done in pushing back the atmosphere probably is done almost reversibly from the viewpoint of the surroundings and can be closely estimated by calculating the work done on the surroundings:

$$W = \int_{l_1}^{l_2} F\, dl = \int_{V_1}^{V_2} p\, dV = p\, \Delta V$$

ΔV of the gas is $2V_1 - V_1 = V_1 = 2.43\ \text{m}^3$, so that the volume change of surroundings is $-2.43\ \text{m}^3$. Then the work done on the surroundings is

Basis: 2.43 m³ at 101.3 kPa and 20°C (assumed)

$$W = \frac{101.3\ \text{kPa}}{} \left| \frac{-2.44\ \text{m}^3}{} \right| \frac{1(\text{N})(\text{m}^{-2})}{1\ \text{Pa}} \left| \frac{1\ \text{J}}{1\ (\text{N})(\text{m})} \right. = -247\ \text{kJ}$$

The minus sign indicates that work is done on the surroundings. From this we know that the work done by the alternative system is

$$W_{\text{system}} = -W_{\text{surroundings}} = -(-247) = 247\ \text{kJ}$$

This answer still does not answer the question of what work was done by the gas, because our system now includes both the piston and the cylinder as well as the gas. You may wonder what happens to the energy transferred from the gas to the piston itself. Some of it goes into work against the atmosphere, which we have just calculated; where does the rest of it go? Energy can go into raising the temperature of the piston or the cylinder or the gas. With time, part of it can be transferred to the surroundings to raise the temperature of the surroundings. From our macroscopic viewpoint we cannot say specifically what happens to this "lost" energy.

You should also note that if the surroundings were a vacuum instead of air, then *no work* would be done by the system, consisting of the piston plus the cylinder plus the gas, although the gas itself probably still would do some work on the piston.

EXAMPLE 4.26 Evaporation

How much work is done by 1 liter (1 dm³) of liquid water when it evaporates from an open vessel at 25°C?

Solution

Steps 1, 2, 3 and 4 Does the water do work in evaporating? Certainly! It does work against the atmosphere. Furthermore, the process, diagrammed in Fig. E4.26, is a reversible one because the evaporation takes place at constant temperature and pressure, and presumably the conditions in the atmosphere immediately above the open portion of the vessel are in equilibrium with the water surface. Assume that the atmospheric pressure is 100 kPa. The basis will be 1 L liquid water at 25°C.

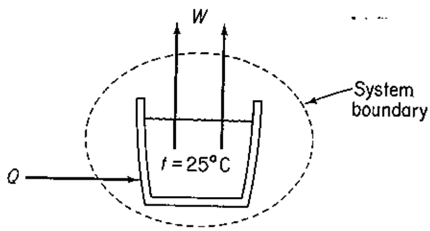


Figure E4.26

Steps 3a, 3b, 5, and 6 The general energy balance

$$\Delta E = -\Delta[(\hat{H} + \hat{K} + \hat{P})m] + Q - W$$

will not be useful in solving this problem because Q is unknown. However, imagine that an expansible bag is placed over the open face of the vessel so that the system now becomes a closed system. Because of the special conditions established for this problem, the work is

$$W = \int_{V_1}^{V_2} p\, dV = p\, \Delta V$$

Furthermore, the work is the reversible work done by the water in pushing back the atmosphere.

Steps 7, 8, and 9

$$W = \frac{100 \times 10^3 \text{ Pa}}{} \left| \frac{1 \text{ L}}{} \right| \frac{10^{-3} \text{ m}^3}{1 \text{ L}} \left| \frac{1 (\text{N})(\text{m}^{-2})}{1 \text{ Pa}} \right| \frac{1 \text{ J}}{1 (\text{N})(\text{m})}$$

$$= 100 \text{ J}$$

Since practically all real processes are irreversible you may wonder why we bother paying attention to the theoretical reversible process. The reason is that it represents the best that can be done, the ideal case, and gives us a measure of our maximum accomplishment. In a real process we cannot do as well and so are less effective. A goal is provided by which to measure our effectiveness. And then, many processes are not too irreversible, so that there is not a big discrepancy between the practical and the ideal process.

Given the concept of an ideal (reversible process) and knowing the work in an actual process, two ways in which we can define mechanical efficiency are

$$\text{efficiency} = \eta_1 = \frac{\text{actual work output for the process}}{\text{work output for a reversible process}} \qquad (4.29a)$$

and

$$\text{efficiency} = \eta_2 = \frac{\text{work input for a reversible process}}{\text{actual work input for the process}} \qquad (4.29b)$$

depending on whether work is done by the system [Eq. (4.29a)] or work is being done on the system [Eq. (4.29b)].

Another type of efficiency is concerned just with the useful energy output divided by the total energy input:

$$\text{efficiency} = \eta_3 = \frac{\text{useful energy out}}{\text{energy in}} \qquad (4.29c)$$

For example, assume that the conversion of fuel in a power plant yields 88 kJ in the steam product per 100 kJ of available energy from the coal being burned, the conversion of the energy in the steam to mechanical energy is 43% efficient, and the conversion of the mechanical to electrical energy is 97% efficient, all based on Eq. (4.29c). The overall efficiency is 1.00(0.88)(0.43)(0.97) = 0.37, meaning that two-thirds of the initial energy is dissipated as heat to the environment. These definitions provide a good way to compare process performance for energy conservation (but not the only way).

EXAMPLE 4.27 Use of Efficiency

Calculate the reversible work required to compress 5 ft^3 of an ideal gas initially at 100°F from 1 to 10 atm in an adiabatic cylinder. Such a gas has an equation of state $pV^{1.40}$ = constant. Then calculate the actual work required if the efficiency of the process is 80%.

Solution

Steps 1, 2, and 3 Figure 4.15 indicates the type of apparatus for the compression. Also,

$$V_2 = V_1\left(\frac{p_1}{p_2}\right)^{1/1.40} = 5\left(\frac{1}{10}\right)^{1/1.40} = 0.965 \text{ ft}^3$$

Step 4 The basis is 5 ft^3 at 100°F and 1 atm.

Steps 5–9

$$W_{rev} = \int_{V_1=5}^{V_2=0.965} p\,dV = \int_{V_1}^{V_2} p_1\left(\frac{V_1}{V}\right)^{1.40} dV = p_1 V_1^{1.4}\int_{V_1}^{V_2} V^{-1.40}\,dV$$

$$= \frac{p_1 V_1^{1.40}}{1-1.40}\left(V_2^{-0.40} - V_1^{-0.40}\right) = \frac{p_2V_2 - p_1V_1}{1-1.40}$$

$$= \frac{[(10)(0.965) - (1)(5)](\text{ft}^3)(\text{atm})}{-0.40}\left|\frac{1.987 \text{ Btu}}{0.7302\ (\text{ft}^3)(\text{atm})}\right.$$

$$= -31.63 \text{ Btu}$$

The negative sign means that work is done on the system.

The actual work required is

$$\frac{31.63}{0.8} = 39.5 \text{ Btu}$$

So far we have looked at reversible work for closed systems. Next we examine reversible work in open systems.

4.6-2 Steady-State Mechanical Energy Balance

In some processes, such as distillation columns or reactors, heat transfer and enthalpy changes are *the* important energy components in the energy balance. Work, potential energy, and kinetic energy are zero or quite minor. However, in other processes, such as the compression of gases and pumping of liquids, work and the mechanical forms of energy are the important factors. For these processes, an energy balance treating solely the mechanical forms of energy becomes a useful tool.

The energy balance of Sec. 4.5 is concerned with various classes of energy without inquiring into how "useful" each form of energy is to human beings. Our experience with machines and thermal processes indicates that some types of energy cannot be transformed completely into other types, and that energy in one state cannot be transformed to another state without the addition of extra work or heat. For example, internal energy cannot be completely converted into mechanical work. To account for these limitations on energy utilization, the second law of thermodynamics was eventually developed as a general principle.

One of the consequences of the second law of thermodynamics is that two categories of energy of different "quality" can be envisioned:

(a) The so-called *mechanical* forms of energy, such as kinetic energy, potential energy, and work, which are *completely* convertible by an *ideal* (reversible) engine from one form to another within the class

(b) Other forms of energy, such as internal energy and heat, which are not so freely convertible

Of course, in any real process with friction, viscous effects, mixing of components, and other dissipative phenomena taking place which prevent the complete conversion of one form of mechanical energy to another, allowances will have to be made in making a balance on mechanical energy for these "losses" in quality.

A balance on mechanical energy can be written on a microscopic basis for an elemental volume by taking the scalar product of the local velocity and the equation of motion.[21] After integration over the entire volume of the system the ***steady-state mechanical energy balance*** **for a system with mass interchange with the surroundings** becomes, on a per unit mass basis,

$$\Delta(\hat{K} + \hat{P}) + \int_{p_1}^{p_2} \hat{V}\,dp + \hat{W} + \hat{E}_v = 0 \tag{4.30}$$

where $\hat{K}$ and $\hat{P}$ are associated with the mass in and out of the system, and E_v represents the loss of mechanical energy, that is, the irreversible conversion *by the flowing fluid* of mechanical energy to internal energy, a term which must in each individual process be evaluated by experiment (or, as occurs in practice, by use of already existing experimental results for a similar process). Equation (4.30) is sometimes called the **Bernoulli equation,** especially for the reversible process for which $\hat{E}_v = 0$. The mechanical energy balance is best applied in fluid-flow calculations when the kinetic and potential energy terms and the work are of importance, and the friction losses can be evaluated from handbooks with the aid of *friction factors* or *orifice coefficients*.

[21] Slattery, *Momentum, Energy, and Mass Transfer* (see ref. 20)

EXAMPLE 4.28 Calculation of Reversible Work for a Flow Process

We will repeat the solution of the preceding example except that in this problem the process will be an open system.

Solution

Steps 1, 2, and 3 Figure E4.28 designates the system and data. The moles of gas are

$$\eta_1 = \frac{p_1 V_1}{RT_1} = \frac{1 \text{ atm} \mid 5 \text{ ft}^3 \mid \quad \mid 1 \text{ (lb mol)(°R)}}{\quad \mid \quad \mid 560\text{°R} \mid 0.7302 \text{ (ft}^3\text{)(atm)}} = 0.0122 \text{ lb mol}$$

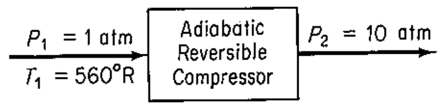


Figure E4.28

Step 4

Basis = 0.0122 lb mol

Steps 3a, 3b, and 7 The mechanical energy balance (per unit mass, or mole here)

$$\Delta(\hat{K} + \hat{P}) + \int_{p_1}^{p_2} \hat{V}\, dp + \hat{W} + \hat{E}_v = 0$$

can be simplified

$$\Delta\hat{K} = 0$$
$$\Delta\hat{P} = 0$$
$$\hat{E}_v = 0 \quad \text{(reversible)}$$

so that

$$\hat{W}_{rev} = -\int_{p_1}^{p_2} \hat{V}\, dp = -\int_{p_1}^{p_2} \hat{V}_1 \left(\frac{p_1}{p}\right)^{1/1.40} dp$$

$$= -\hat{V}_1 p_1^{0.714}[3.50(p_2^{0.286} - p_1^{0.286})]$$

$$W_{rev} = n_1 \hat{W}_{rev} = -n_1\left(\frac{V_1}{n_1}\right) p_1^{0.714}[3.50(p_2^{0.286} - p_1^{0.286})]$$

$$= -(5)(1)^{0.714}(3.50)[(10)^{0.286} - 1^{0.286}](\text{ft}^3)(\text{atm})\left(\frac{1.987 \text{ Btu}}{0.7302 \text{ (ft}^3\text{)(atm)}}\right)$$

$$= -44.3 \text{ Btu}$$

The actual work required to be done on the system is

$$\frac{44.3}{0.8} = 55.4 \text{ Btu}$$

EXAMPLE 4.29 Application of the Mechanical Energy Balance

Calculate the work per minute required to pump 1 lb of water per minute from 100 psia and 80°F to 1000 psia and 100°F. The exit stream is 10 ft above the entrance stream.

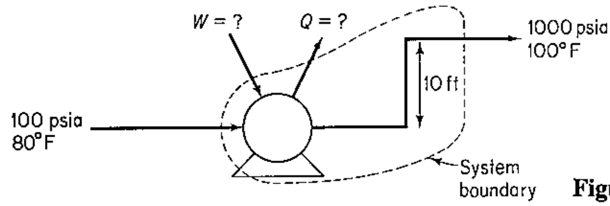


Figure E4.29

Solution

Steps 1, 2, and 3 The system is shown in Fig. E4.29.
Steps 3a, 3b, and 7 The general mechanical energy balance is

$$\Delta(\hat{K} + \hat{P}) + \int_{p_1}^{p_2} \hat{V}\,dp + \hat{W} + \hat{E}_v = 0 \tag{a}$$

We will assume that $\Delta\hat{K}$ is insignificant, and also preliminarily assume that the process is reversible so that $E_v = 0$, and the pump is 100% efficient. (Subsequently, we will consider what to do if the process is not reversible.) Equation (a) reduces to

$$\hat{W} = -\int_{p_1}^{p_2} \hat{V}\,dp - \Delta\hat{P} \tag{b}$$

Step 4

Basis: 1 min of operation = 1 lb H_2O

Steps 5 and 8 From the steam tables, the specific volume of liquid water is 0.01607 ft^3/lb_m at 80°F and 0.01613 ft^3/lb_m at 100°F. For all practical purposes the water is incompressible, and the specific volume can be taken to be 0.0161 ft^3/lb_m. We have only one unknown in Eq. (b): $\hat{W}$.

Step 9

$$\Delta\hat{P} = \frac{1\ \text{lb}_m \;\Big|\; 10\ \text{ft} \;\Big|\; 32.2\ \text{ft} \;\Big|\; \;\Big|\; 1\ \text{Btu}}{\;\Big|\; \;\Big|\; \text{sec}^2 \;\Big|\; \dfrac{32.2(\text{ft})(\text{lb}_m)}{(\text{lb}_f)(\text{sec}^2)} \;\Big|\; 778(\text{ft})(\text{lb}_f)} = 0.0129\ \text{Btu}$$

$$1\ \text{lb}_m \int_{100}^{1000} 0.0161\,dp = \frac{1\ \text{lb}_m \;\Big|\; 0.0161\ \text{ft}^3 \;\Big|\; (1000 - 100)\ \text{lb}_f \;\Big|\; \left(\dfrac{12\ \text{in.}}{1\ \text{ft}}\right)^2 \;\Big|\; 1\ \text{Btu}}{\;\Big|\; \text{lb}_m \;\Big|\; \text{in.}^2 \;\Big|\; \;\Big|\; 778(\text{ft})(\text{lb}_f)}$$

$$= 2.68\ \text{Btu}$$

$$\hat{W} = -2.68 - 0.0129 = -2.69\frac{\text{Btu}}{\text{lb}_m}$$

About the same value can be calculated using Eq. (4.24) if $\hat{Q} = \hat{K} = 0$, because the enthalpy change for a reversible process for 1 lb of water going from 100 psia and 100°F to 1000 psia is 2.70 Btu. Make the computation yourself. However, usually the enthalpy data for liquids other than water are missing, or not of sufficient accuracy to be valid, which forces an engineer to turn to the mechanical energy balance.

We might now well inquire for the purpose of purchasing a pump-motor, say, as to what the work would be for a real process, instead of the fictitious reversible process assumed above. First, you would need to know the efficiency of the combined pump and motor so that the actual input from the surroundings (the electric connection) to the system would be known. Second, the friction losses in the pipe, valves, and fittings must be estimated so that the term $\hat{E}_v$ could be reintroduced into Eq. (4.30). Suppose, for the purposes of illustration, that $\hat{E}_v$ was estimated to be, from an appropriate handbook, 320 (ft)(lb_f)/lb_m and the motor-pump efficiency was 60% (based on 100% efficiency for a reversible pump-motor). Then,

$$\hat{E}_v = \frac{320(\text{ft})(\text{lb}_f)}{1\ \text{lb}_m}\left|\frac{1\ \text{Btu}}{778\ (\text{ft})(\text{lb}_f)}\right. = 0.41\ \text{Btu/lb}_m$$

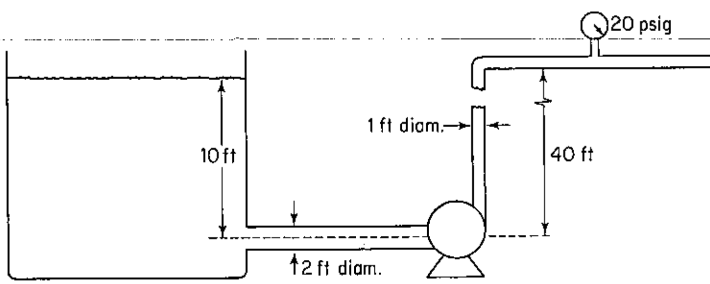
$$\hat{W} = -(2.68 + 0.013 + 0.41) = -3.10\ \text{Btu/lb}_m$$

Remember that the minus sign indicates that work is done on the system. The pump motor must have the capacity

$$\frac{3.10\ \text{Btu}}{1\ \text{min}}\left|\frac{1}{0.60}\right|\frac{1\ \text{min}}{60\ \text{sec}}\left|\frac{1.415\ \text{hp}}{1\ \text{Btu/sec}}\right. = 0.122\ \text{hp}$$

Self-Assessment Test

1. Which process will yield more work: (1) expansion of a gas confined by piston against constant pressure or (2) reversible expansion of a gas confined by a piston?
2. Differentiate between thermal and mechanical energy.
3. Define a reversible process.
4. Find Q, W, ΔE, and ΔH for the reversible compression of 3 moles of an ideal gas from a volume of 100 dm^3 to 2.4 dm^3 at a constant T of 300 K.
5. Water is pumped from a very large storage reservoir as shown in the figure at the rate of 2000 gal/min. Determine the minimum power (i.e., for a reversible process) required by the pump in horsepower.



Thought Problems

1. A news announcement in a professional journal described a 20-hp Stirling engine that was going to be connected to and drive a 68-kW generator. Would you buy one of these at $4500?
2. An author compared the effectiveness of gasoline-powered automobiles with electric-powered automobiles as in the following table, and concluded that electric-powered vehicles overall were more efficient. Is the comparison valid?

Efficiencies	Gasoline (bbl of oil)	Electric (bbl of oil)
KWh equivalent	1700	1700
Refinery efficiency	74% to gasoline	89% to heavy oil
Distribution efficiency	95%	
Power generation efficiency		40%
Power distribution efficiency		91%
Battery efficiency		70%
Motor, drive train efficiency	14.7%	80%
KWh available	175.7	308.8
Vehicle weight	2400 lb	2800 lb
Road load: 50 mph, level	8.4 kW	10 kW
Distance traveled	1045 miles	1545 miles

4.7 ENERGY BALANCES WITH CHEMICAL REACTION

So far we have not discussed how to treat energy balances in which chemical reactions occur. We do so in this section. For any given process, when a reaction takes place, the energy exchange with the surroundings will be different from the same process without a reaction. As you know, the observed heat transfer that occurs in a closed system (with zero work) as a result of a reaction represents the energy associated with the rearrangement of the bonds holding together the atoms of the reacting molecules. For an **exothermic reaction,** the energy required to hold the products of the reaction together is less than that required to hold the reactants together and the surplus energy is released. The reverse is true of an **endothermic reaction.** The energy change that appears directly as a result of a reaction is termed the *heat of reaction*, ΔH_{rxn}, a term that is a legacy of the days of the caloric theory. Note that "heat of reaction" is actually an *enthalpy change* and not heat transfer.

4.7-1 Standard Heat of Formation and Heat of Reaction

Your objectives in studying this section are to be able to:

1. Compute heats of formation from experimental data for the enthalpy change (including phase changes) of a process with a reaction taking place.
2. Look up heats of formation in reference tables for a given compound.
3. List the standard conventions and reference states used for reactions associated with the standard heat of formation.

4. Calculate the standard heat of reaction from tabulated standard heats of formation for a given reaction.

To take account of possible energy changes caused by a reaction, in the energy balance we incorporate in the enthalpy of each individual constituent an additional quantity termed the **standard heat** (really enthalpy) **of formation,**[22] $\Delta\hat{H}_f^\circ$, a quantity that is discussed in detail below. (The superscript ° denotes "standard state" and the subscript f denotes "formation.") Thus for the case of a single species A without any pressure effect on the enthalpy and omitting phase changes, the specific enthalpy change from the reference state is given by

$$\Delta\hat{H}_A = \Delta\hat{H}_{fA}^\circ + \int_{T_{\text{ref}}}^{T} C_{pA}\, dT \tag{4.31}$$

For several species, we would have in a stream

$$\Delta H_{\text{mixture}} = \sum_{i=1}^{s} n_i\, \Delta\hat{H}_{fi}^\circ + \sum_{i=1}^{s} \int_{T_{\text{ref}}}^{T} n_i C_{p_i}\, dT \tag{4.32}$$

where i designates each species, n_i is the number of moles of species i, and s is the total number of species. If phase changes take place, additional terms for the enthalpy of the phase changes have to be added to the right-hand side of Eq. (4.32) to give the **total enthalpy** of substance A:

$$\Delta\hat{H}_A = \Delta\hat{H}_{fA}^\circ + (\hat{H}_{T,p} - \hat{H}_{\text{ref}}^\circ)$$

where $(\hat{H}_{T,p} - \hat{H}_{\text{ref}}^\circ)$ includes both sensible heat and phase changes.

If a mixture enters and leaves a system without a reaction taking place, we would find that the same species entered and left so that the enthalpy change in Eq. (4.24a) would not be any different with the modification described above than what we have used before, because the terms that account for the heat of formation in the energy balance would cancel. For example, for the case of two species in a flow system, the output enthalpy would be

$$\Delta H_{\text{output}} = \underbrace{n_1\, \Delta\hat{H}_{f1}^\circ + n_2\, \Delta\hat{H}_{f2}^\circ}_{\text{"heat of formation"}} + \underbrace{\int_{T_{\text{ref}}}^{T_{\text{out}}} (n_1 C_{p1} + n_2 C_{p2})\, dT}_{\text{"sensible heat"}}$$

and the input enthalpy would be

$$\Delta H_{\text{input}} = \underbrace{n_1\, \Delta\hat{H}_{f1}^\circ + n_2\, \Delta\hat{H}_{f2}^\circ}_{\text{"heat of formation"}} + \underbrace{\int_{T_{\text{ref}}}^{T_{\text{in}}} (n_1 C_{p1} + n_2 C_{p2})\, dT}_{\text{"sensible heat"}}$$

Without reaction, observe that $\Delta H_{\text{output}} - \Delta H_{\text{input}}$ would only involve the sensible

[22] Historically, the name arose because the changes in enthalpy associated with chemical reactions were generally determined in a device called a calorimeter, to which heat is added or removed from the reacting system so as to keep the temperature constant.

heat terms that we have described before; the heat-of-formation terms would be exactly the same in each enthalpy change and would cancel out.

However, if a reaction takes place, the quantities that enter and leave will differ, and the terms involving the heat of formation will not cancel. For example, suppose that species 1 and 2 enter the system, react, and species 3 and 4 leave. Then

$$\Delta H_{\text{out}} - \Delta H_{\text{in}} = (n_3\,\Delta\hat{H}^\circ_{f3} + n_4\,\Delta\hat{H}^\circ_{f4}) - (n_1\,\Delta\hat{H}^\circ_{f1} + n_2\,\Delta\hat{H}^\circ_{f2})$$
$$+ \int_{T_{\text{ref}}}^{T_{\text{out}}} (n_3 C_{p3} + n_4 C_{p4})\,dT - \int_{T_{\text{ref}}}^{T_{\text{in}}} (n_1 C_{p1} + n_2 C_{p2})\,dT \qquad (4.33)$$

You will be retrieving information on heats of formation from reference tables and data bases. The values in the tables have been reconciled from innumerable experiments. To determine the values of the standard heats (enthalpies) of formation, the experimenter usually selects either a simple flow process without kinetic energy, potential energy, or work effects (a flow calorimeter), or a simple batch process (a bomb calorimeter), in which to conduct the reaction. Consider an experiment in a flow process under standard state conditions in which the experimental arrangement is such that the summation of sensible heat terms on the right-hand side of Eq. (4.33) is zero and no work is done. The steady-state (no accumulation term) version of Eq. (4.24a) **for stoichiometric quantities of reactants and products** reduces to

$$Q = \Delta H$$
$$= \left(\sum_{\text{products}} n_i\,\Delta\hat{H}^\circ_{fi} - \sum_{\text{reactants}} n_i\,\Delta\hat{H}^\circ_{fi}\right) \equiv \Delta H^\circ_{\text{rxn}} \qquad (4.34)$$

where $\Delta H^\circ_{\text{rxn}}$ is the symbol used for the **heat of reaction at the standard state.** Observe that the enthalpy change caused by the reaction in the prescribed type of experiment appears as heat transferred to or from the system, and the value of Q that is measured is the heat transferred from the reaction that occurs in the system and is equivalent to the standard heat of reaction at the standard state.

In general, a heat of reaction depends not only on the reaction stoichiometry but also on the number of moles that actually react, the temperature of the reaction, the phases of the products and reactants, and the pressure. The calculations that we shall make here will all be for low pressures, and, although the effect of pressure upon heats of reaction is relatively negligible under most conditions, if exceedingly high pressures are encountered, you should make the necessary corrections as explained in most texts on thermochemistry.

There are certain conventions and symbols which you should always keep in mind concerning thermochemical calculations if you are to avoid difficulty later on. These conventions can be summarized as follows:

(a) The reactants are shown on the left-hand side of the chemical equation, and the products are shown on the right: for example,

$$CH_4(g) + H_2O(l) \longrightarrow CO(g) + 3H_2(g)$$

(b) The conditions of phase, temperature, and pressure must be specified unless the last two are the standard conditions, as presumed in the equation above, when only the phase is required. This is particularly important for compounds such as H_2O, which can exist as more than one phase under common conditions. If the reaction takes place at other than standard conditions, you might write

$$CH_4(g, 1.5\ \text{atm}) + H_2O(l) \xrightarrow{50^\circ C} CO(g, 3\ \text{atm}) + H_2(g, 3\ \text{atm})$$

(c) Unless otherwise specified, all the constituents are at the standard state of 25°C (77°F) and 101.3 kPa (1 atm) total pressure. The heat of reaction under these conditions is called the ***standard heat of reaction,*** and is distinguished by the superscript ° symbol.

(d) Unless the amounts of material reacting are stated, it is assumed that the quantities reacting are the stoichiometric amounts shown in the chemical equation.

Data to compute the standard heat of reaction are reported and tabulated in two different but essentially equivalent forms:

(a) Standard heats of formation

(b) Standard heats of combustion

We first describe the standard heat of formation ($\Delta\hat{H}_f^\circ$) and then the standard heat of combustion ($\Delta\hat{H}_c^\circ$). The units of both quantities are usually tabulated as energy per mole, such as J/g mol, kJ/g mol, kcal/g mol, or Btu/lb mol. The "per mole" refers to the specified reference substance in the related stoichiometric equation.

The ***standard heat of formation*** is the special heat of reaction for the formation of 1 mole of a compound from its constituent elements, for example

$$C(s) + \tfrac{1}{2}O_2(g) \longrightarrow CO(g)$$

in the standard state. The initial reactants and final products must be stable and at 25°C and 1 atm. The reaction does not necessarily represent a real reaction that would proceed at constant temperature but can be a fictitious process for the formation of a compound from the elements. By defining **the heat of formation as zero in the standard state for each *element*,** it is possible to design a system to express the heats of formation for all *compounds* at 25°C and 1 atm. If you use the conventions discussed above, then the thermochemical calculations will all be consistent, and you should not experience any confusion as to signs. Consequently, with the use of well-defined standard heats of formation, it is not necessary to record the experimental heats of reaction for every reaction that can take place. That would be an impossible task.

To sum up, a formation reaction is understood by convention to be a reaction that forms 1 mole of compound from the elements that make it up. Standard heats of reaction at 25°C and 1 atm for any reaction can be calculated from the tabulated (or experimental) standard heats of formation values by using Eq. (4.34), because the standard heat of formation is a state (point) function, as illustrated in the examples below.

EXAMPLE 4.30 Retrieval of Heats of Formation from Reference Data

What is the standard heat of formation of HCL(g)?

Solution

Look in Appendix F. The column heading is $-\Delta\hat{H}_f^\circ$.

$-\Delta\hat{H}_f^\circ$
92.311 kJ/g mol

which means that $\Delta\hat{H}_f^\circ = -92.312$ kJ/g mol.

In the reaction at 25°C and 1 atm,

$$\tfrac{1}{2}H_2(g) + \tfrac{1}{2}Cl_2(g) \longrightarrow HCl(g)$$

both $H_2(g)$ and $Cl_2(g)$ would be assigned $\Delta\hat{H}_f^\circ$ values of 0, and the value shown in Appendix F for HCl(g) of −92,312 J/g mol is the standard heat of formation for this compound as well as the standard heat of reaction for the reaction as written above:

$$\Delta H^\circ_{rxn} = \sum_{\text{products}} n_i\,\Delta\hat{H}^\circ_{fi} - \sum_{\text{reactants}} n_i\,\Delta\hat{H}^\circ_{fi}$$

$$= 1(-92.312) - [\tfrac{1}{2}(0) + \tfrac{1}{2}(0)] = -92.312 \text{ kJ/g mol HCl(g)}$$

The value tabulated in Appendix F might actually be determined by carrying out the reaction shown for HCl(g) and measuring the energy liberated in a calorimeter, or by some other more convenient method.

EXAMPLE 4.31 Determination of Heats of Formation from Experimental Data

Suppose that you want to find the standard heat of formation of CO from experimental data. Can you prepare pure CO from C and O_2 and measure the heat transfer? This would be far too difficult. It would be easier experimentally to find the heat of reaction for the two reactions shown below and subtract them as follows:

Basis: 1 g mol of CO

	Reaction	$\Delta\hat{H}^\circ_{rxn}$*(experimental)*
A:	$C(\beta) + O_2(g) \longrightarrow CO_2(g)$	−393.51 kJ/g mol
B:	$CO(g) + \tfrac{1}{2}O_2(g) \longrightarrow CO_2(g)$	−282.99 kJ/g mol
$A - B$:	$C(\beta) + \tfrac{1}{2}O_2(g) \longrightarrow CO(g)$	

$$\Delta\hat{H}^\circ_{rxn\,A-B} = (-393.51) - (-282.99) = \Delta\hat{H}_f^\circ = -110.52 \text{ kJ/g mol}$$

The energy change for the overall reaction scheme is the desired standard heat of formation per mole of CO(g). See Fig. E4.31.

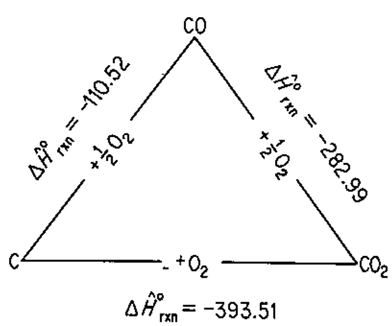


Figure E4.31

EXAMPLE 4.32 Calculation of the Heat of Reaction from Standard Heats of Formation

Calculate $\Delta\hat{H}^\circ_{rxn}$ for the following reaction of 4 g mol of NH_3:

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$$

Solution

Basis: 4 g mol of NH_3

Tabulated data	$NH_3(g)$	$O_2(g)$	$NO(g)$	$H_2O(g)$
$\Delta\hat{H}^\circ_f$ per mole at 25°C and 1 atm (kcal/g mol)	−11.04	0	+21.60	−57.80

We shall use Eq. (4.34) to calculate ΔH°_{rxn} for 4 g mol of NH_3:

$$\Delta H^\circ_{rxn} = [4(21.60) + 6(-57.80)] - [5(0) + 4(-11.04)]$$
$$= -216.24 \text{ kcal/4 g mol } NH_3.$$

EXAMPLE 4.33 Heat of Formation Including a Phase Change

If the standard heat of formation for $H_2O(l)$ is −285.838 kJ/g mol and the heat of evaporation is +44.012 kJ/g mol at 25°C and 1 atm, what is the standard heat of formation of $H_2O(g)$?

Solution

Basis: 1 g mol of H_2O

We shall proceed as in Example 4.31 to add the known chemical equations and the phase transitions to yield the desired chemical equation and carry out the same operations on the enthalpy changes. For reaction A, $\Delta\hat{H}^\circ_{rxn} = \Sigma\, \Delta\hat{H}^\circ_{f\,\text{products}} - \Sigma\, \hat{H}^\circ_{f\,\text{reactants}}$:

$$\begin{array}{llcll}
A: & H_2(g) + \tfrac{1}{2}O_2(g) & \longrightarrow & H_2O(l) & \Delta\hat{H}^\circ_{rxn} = -285.838 \text{ kJ/g mol} \\
B: & H_2O(l) & \longrightarrow & H_2O(g) & \Delta\hat{H}^\circ_{vap} = +44.012 \text{ kJ/g mol} \\
\hline
A + B: & H_2(g) + \tfrac{1}{2}O_2(g) & \longrightarrow & H_2O(g) &
\end{array}$$

$$\Delta\hat{H}^\circ_{rxn\,A} + \Delta\hat{H}^\circ_{vap} = \Delta\hat{H}^\circ_{rxn\,H_2O(g)} = \Delta\hat{H}^\circ_{f\,H_2O(g)} = -241.826 \text{ kJ/g mol}$$

You can see that any number of chemical equations can be treated by algebraic methods, and the corresponding standard heats of reaction can be added or subtracted in the same fashion as are the equations. By carefully following these rules of procedure, you will avoid most of the common errors in thermochemical calculations.

To simplify matters, the value cited for $\Delta\hat{H}_{vap}$ of water in Example 4.33 was at 25°C and 1 atm. To calculate this value, if the final state for water is specified as $H_2O(g)$ at 25°C and 1 atm, the following enthalpy changes should be taken into account if you start with $H_2O(l)$ at 25°C and 1 atm:

$$\Delta H_{vap} \text{ at } 25°\text{C and 1 atm} \begin{cases} H_2O(l)\ 25°\text{C, 1 atm} \\ \quad \downarrow \Delta H_1 \\ H_2O(l)\ 25°\text{C, vapor pressure at } 25°\text{C} \\ \quad \downarrow \Delta H_2 = \Delta H_{vap} \text{ at the vapor pressure of water} \\ H_2O(g)\ 25°\text{C, vapor pressure at } 25°\text{C} \\ \quad \downarrow \Delta H_3 \\ H_2O(g)\ 25°\text{C, 1 atm} \end{cases}$$

For practical purposes the value of ΔH_{vap} at 25°C and the vapor pressure of water, namely 43,911 J/g mol (10,495 cal/g mol), will be adequate for engineering calculations at one atmosphere.

There are many sources of tabulated values for the standard heats of formation listed in Table 4.5. A good source of extensive data is the *National Bureau of Standards Bulletin* 500, and its supplements by F. K. Rossini, or the serial publications of the Thermodynamics Research Center at Texas A & M University (API Research Project No. 44 and the TRC Data Project). A condensed set of values for the heats of formation may be found in Appendix F. If you cannot find a standard heat of formation for a particular compound in reference books or in the chemical literature, $\Delta\hat{H}_f^\circ$ may be estimated by the methods described in Verma and Doraiswamy[23] or by some of the authors listed as references in Table 4.5. Remember that the values for the standard heats of formation are negative for exothermic reactions.

Self-Assessment Test

1. Calculate the standard heat of formation of CH_4 given the following experimental results at 25°C and 1 atm:

$$H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O(l) \qquad \Delta H = -285.84 \text{ kJ/g mol } H_2$$
$$C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -393.51 \text{ kJ/g mol C}$$
$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H = -890.36 \text{ kJ/g mol } CH_4$$

   Compare your answer with that found in the table of heats of formation listed in Appendix F.

2. Calculate the standard heat of reaction for the following reaction from the heats of formation:

$$C_6H_6(g) \longrightarrow 3C_2H_2(g)$$

Thought Problems

1. Clipping from the *Wall Street Journal:*

Technical Disputes

Furnace efficiency is a comparison of the energy that goes into a furnace with the usable heat that comes out. Because oil furnaces use blowers to burn the fuel more efficiently and

[23] K. K. Verma and L. K. Doraiswamy, *Ind. Eng. Chem. Fundam.*, v. 4, p. 389 (1965).

because oil produces less water vapor than gas, less heat is vented up the chimney, the Oil Jobbers Council says. The American Gas Association complains that such calculations ignore variations in oil quality that hurt efficiency. And it contends that oil furnaces are apt to "lose efficiency over time," while it says gas furnaces don't.

Which organization is correct, or is neither correct?

2. A review of additives to gasoline to give blends which improve its octane rating shows that oxygenated compounds necessarily contain lower energy (Btu per gallon). Methanol was the lowest, having a heat of reaction approximately one-half that of gasoline. Methanol costs 40 cents/gal, whereas unleaded premium gas costs 80 cents/gal, so that the result may seem like a standoff (i.e., half the energy at half the cost).

   Are the two fuels really equivalent in practical use?

4.7-2 Standard Heat of Combustion

Your objectives in studying this section are to be able to:

1. Look up heats of combustion in reference tables for a given compound.
2. List the standard conventions and reference states used for reactions associated with the standard heat of combustion.
3. Calculate the standard heat of reaction from tabulated standard heats of combustion for a given reaction.
4. Calculate the standard higher heating value from the lower heating value or the reverse.

Standard heats of combustion are the second method of recording thermochemical data. The standard heats of combustion do not have the same standard states as the standard heats of formation. The conventions used with the standard heats of combustion are

(a) The compound is oxidized with oxygen or some other substance to the products $CO_2(g)$, $H_2O(l)$, $HCl(aq)$,[24] and so on.

(b) The reference conditions are still 25°C and 1 atm.

(c) Zero values of $\Delta\hat{H}_c^\circ$ are assigned to certain of the oxidation products as, for example, $CO_2(g)$, $H_2O(l)$, and $HCl(aq)$, and to $O_2(g)$ itself.

(d) If other oxidizing substances are present, such as S or N_2, or if Cl_2 is present, it is necessary *to make sure that states of the products are carefully specified and are identical to (or can be transformed into) the final conditions which determine the standard state as shown in Appendix F.*

The standard heat of combustion of an unoxidized compound can never be

[24] HCl(aq) represents an infinitely dilute solution of HCl (see Sec. 4.8).

positive but is always negative. A positive value would mean that the substance would not burn or oxidize. Standard heats of reaction can be calculated from standard heats of combustion by an equation analogous to Eq. (4.34):

$$\begin{aligned}\Delta H^\circ_{rxn} &= -\left(\sum \Delta H^\circ_{c\,\text{products}} - \sum H^\circ_{c\,\text{reactants}}\right)\\ &= -\left(\sum_i n_{\text{prod}\,i}\,\Delta \hat{H}^\circ_{c\,\text{prod}\,i} - \sum_i n_{\text{react}\,i}\,\Delta \hat{H}^\circ_{c\,\text{react}\,i}\right)\end{aligned} \tag{4.35}$$

Note: The minus sign in front of the summation expression occurs because the choice of reference states is zero for the right-hand products of the standard reactions. Refer to Appendix F for values of $\Delta \hat{H}^\circ_c$.

For a fuel such as coal or oil, the negative of the standard heat of combustion is known as the **heating value** of the fuel. To determine the heating value, a weighed sample is burned in oxygen in a calorimeter bomb, and the energy given off is detected by measuring the temperature increase of the bomb and surrounding apparatus.

Because the water produced in the calorimeter is in the vicinity of room temperature, although the gas within the bomb is saturated with water vapor at this temperature, essentially all the water formed in the combustion process is condensed into liquid water. This is not like a combustion process in a furnace, where the water remains as a vapor and passes up the stack. Consequently, we have two heating values for fuels containing hydrogen: (1) the *gross*, or *higher, heating value,* in which all the water formed is condensed into the liquid state, and (2) the *net*, or *lower, heating value,* in which all the water formed remains in the vapor state. The value you determine in the calorimeter is the gross heating value; this is the one reported along with the fuel analysis and should be presumed unless the data state specifically that the net heating value is being reported.

You can estimate the heating value of a coal within about 3% from the Dulong formula[25]:

Higher heating value (HHV) in British thermal units per pound

$$= 14{,}544\text{C} + 62{,}028\left(\text{H} - \frac{\text{O}}{8}\right) + 4050\text{S}$$

where C = weight fraction carbon

$\text{H} - \dfrac{\text{O}}{8}$ = weight fraction net hydrogen

= total weight fraction hydrogen $-\frac{1}{8}$ weight fraction oxygen

S = weight fraction net sulfur

The values of C, H, S, and O can be taken from the fuel or flue-gas analysis. If the heating value of a fuel is known and the C and S are known, the approximate value of the net H can be determined from the Dulong formula. A general relation between the gross heating and net heating values is

$$\text{net Btu/lb coal} = \text{gross Btu/lb coal} - 91\ (\%\ \text{total H by weight})$$

[25] H. H. Lowry, ed., *Chemistry of Coal Utilization,* Wiley, New York, 1945, Chap. 4.

Urban refuse or municipal solid waste is produced at a rate of about 11 kg (5 lb) per person per day, and contains both organic as well as nonorganic materials. Combustion of organic waste represents a supplemental source of supply of energy. Table 4.9 lists the HHV for typical wastes. If you "watch calories," Table 4.10 shows the calories in various foods. Table 4.11 indicates some heating values of typical coals.

TABLE 4.9 Higher Heating Value of Municipal Refuse (kJ/kg)

Refuse component	As delivered	Dry basis	Refuse component	As delivered	Dry basis
Lawn grass	4,780	19,320	Average	—	20,050
Meat scraps, cooked	32,260	32,260	Mail	14,150	14,820
Newspaper	18,530	19,710	Cardboard	16,370	17,260
Shrub cuttings	6,290	20,300	Ripe tree leaves	18,550	20,610
Vegetable food waste	4,170	19,220	Magazine	12,210	12,730

TABLE 4.10 Calorie Counts in Various Foods

Food	Portion	Kilo-calories	Food	Portion	Kilo-calories
Beer	12 oz	165	Orange juice	$\frac{1}{2}$ glass	56
Chicken, broiled	$\frac{1}{2}$	308	Soft drink	1 can	73
Coffee	1 cup	0	Toast	2 pieces	120
Martini	1 oz	168	Trout	1 lb	224

TABLE 4.11 Higher Heating Values of Typical Coals

Class	kJ/kg	Btu/lb
Meta-anthracite	26,680	11,480
Anthracite	33,110	14,250
Low-volatile bituminous	35,370	15,220
High-volatile bituminous	27,790	11,960
Subituminous	23,330	10,040
Lignite	17,290	7,440

EXAMPLE 4.34 Heating Value of Coal

Coal gasification consists of the chemical transformation of solid coal into gas. The heating values of coal differ, but the higher the heating value, the higher the value of the gas produced (which is essentially methane, carbon monoxide, hydrogen, etc.). The following coal has a reported heating value of 29,770 kJ/kg as received. Assuming that this is the gross heating value, calculate the net heating value.

Component	Percent
C	71.0
H_2	5.6
N_2	1.6
Net S	2.7
Ash	6.1
O_2	13.0
Total	100.0

Solution

The corrected ultimate analysis shows 5.6% hydrogen on the as-received basis.

Basis: 100 kg of coal as received

The water formed on combustion is

$$\frac{5.6 \text{ kg } H_2}{} \left| \frac{1 \text{ kg mol } H_2}{2.02 \text{ kg } H_2} \right| \frac{1 \text{ kg mol } H_2O}{1 \text{ kg mol } H_2} \left| \frac{18 \text{ kg } H_2O}{1 \text{ kg mol } H_2O} \right. = 49.9 \text{ kg } H_2O$$

The energy required to evaporate the water is

$$\frac{49.9 \text{ kg } H_2O}{100 \text{ kg coal}} \left| \frac{2370 \text{ kJ}}{\text{kg } H_2O} \right. = \frac{1183 \text{ kJ}}{\text{kg coal}}$$

The net heating value is

$$29{,}770 - 1183 = 28{,}587 \text{ kJ/kg}$$

The value 2370 kJ/kg is not the latent heat of vaporization of water at 25°C (2440 kJ/kg) but includes the effect of a change from a heating value at constant volume to one at constant pressure (−70 kJ/kg) as described in a later section of this chapter.

Use the Dulong formulate to calculate the HHV of this coal. Do you get 30,000 kJ/kg?

EXAMPLE 4.35 Calculation of Heat of Reaction from Heat of Combustion Data

Compute the heat of reaction of the following reaction from standard heat of combustion data:

$$\underset{\text{ethyl alcohol}}{C_2H_5OH(l)} + \underset{\text{acetic acid}}{CH_3COOH(l)} \longrightarrow \underset{\text{ethyl acetate}}{C_2H_5OOCCH_3(l)} + H_2O(l)$$

Solution

Basis: 1 g mol of C_2H_5OH

tabulated data	$C_2H_5OH(l)$	$CH_3COOH(l)$	$C_2H_5OOCCH_3(l)$	$H_2O(l)$
$\Delta\hat{H}_c^\circ$ per g mol at 25°C and 1 atm (kJ/g mol)	−1366.91	−871.69	−2274.48	0

We use Eq. (4.35):

$$\Delta H^{\circ}_{rxn} = -[(-2274.48) - (-1366.91 - 871.69)] = +35.9 \text{ kJ/g mol}$$

One of the common errors made in these thermochemical calculations is to forget that the standard state for heat of combustion calculations is liquid water, and that if gaseous water is present, a phase change (the heat of vaporization or heat of condensation) must be included in the calculations. Don't you forget!

With adequate data available you will find it simpler to use only the heats of combustion or only the heats of formation in the same algebraic calculation for a heat of reaction. Mixing these two sources of enthalpy change data, unless you take care, will only lead to error and confusion. Since the heat of combustion values are so large, calculations made by subtracting these large values from each other can be a source of error. To avoid other major errors, carefully check all signs and make sure all equations are not only balanced but are written according to the proper conventions.

Self-Assessment Test

1. Calculate the standard heat of reaction for the Sachse process (in which acetylene is made by partial combustion of LPG) from heat of combustion data:

$$C_3H_8(l) + 2O_2(g) \longrightarrow C_2H_2(g) + CO(g) + 3H_2O(l)$$

2. Can a standard heat of combustion ever be positive?
3. A synthetic gas analyzes 6.1% CO_2, 0.8% C_2H_4, 0.1% O_2, 26.4% CO, 30.2% H_2, 3.8% CH_4, and 32.6% N_2. What is the heating value of the gas measured at 60°F, saturated, when the barometer reads 30.0 in. Hg?

Thought Problem

1. The boiler efficiency and some data for a boiler are outlined in Problem 4.140. Degradation of performance and economic loss result from poor boiler performance. What are some of the steps you might take to improve boiler performance?

4.7-3 Heat of Reaction at Constant Pressure versus Constant Volume

Your objectives in studying this section are to be able to:

1. Calculate the heat transfer, Q, from a bomb calorimeter (constant volume) or a steady flow calorimeter (constant pressure), Q_p, from theory or experimental data.
2. Calculate the heat of reaction for either process.

Consider the heat of reaction of a substance obtained in a bomb calorimeter, such as in a bomb in which the volume is constant but not the pressure. For such a process (the system is the material in the bomb), the general energy balance, Eq. (4.2-4a), reduces to (with no work, mass flow, nor kinetic or potential energy effects)

$$\Delta U = U_{t_2} - U_{t_1} = Q_v \tag{4.36}$$

Here Q_v designates the heat transferred from the bomb. Next, let us assume that the heat of reaction is determined in steady-state flow calorimeter with 1 = entering fluid and 2 = exiting fluid, and with $W = 0$, $\Delta P = 0$, and $\Delta K = 0$. Then if the process takes place at constant pressure the general energy balance reduces to

$$\Delta H = Q_p$$

Here Q_p designates the heat transferred from the flow calorimeter.

If we subtract Q_v from Q_p and use $H = U + pV$, we find that

$$\begin{aligned} Q_p - Q_v &= \Delta H - \Delta U = \Delta H - [\Delta(H - pV)] \\ &= (H_2 - H_1) - [(H - pV)_{t_{12}} - (H - pV)_{t_1}] \end{aligned} \tag{4.37}$$

Suppose, furthermore, that the enthalpy change for the batch, constant-volume process is made identical to the enthalpy difference between the outlet and inlet in the flow process by a suitable adjustment of the temperature of the water baths surrounding the calorimeters. Then

$$H_{t_2} - H_{t_1} = H_2 - H_1$$

and

$$Q_p - Q_v = (pV)_{t_2} - (pV)_{t_1} \tag{4.38}$$

To evaluate the terms on the right-hand side of Eq. (4.38), we can assume for solids and liquids that the $\Delta(pV)$ change is negligible and can be ignored. Therefore, the only change which must be taken into account is for gases present as products and/or reactants. If, for simplicity, the gases are assumed to be ideal, then at constant temperature

$$pV = nRT$$

$$\Delta(pV) = \Delta n\, RT$$

and

$$Q_p - Q_v = \Delta n\, (RT) \tag{4.39}$$

Equation (4.39) gives the difference between the heat of reaction for the constant-pressure experiment and the constant-volume experiment.

EXAMPLE 4.36 Difference between Heat of Reaction at Constant Pressure and at Constant Volume

Find the difference between the heat of reaction at constant pressure and at constant volume for the following reaction at 25°C (assuming that it could take place):

$$C(s) + \tfrac{1}{2}O_2(g) \longrightarrow CO(g)$$

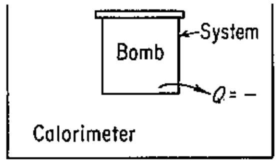


Figure E4.36

Solution

Basis: 1 mole of C(s)

Examine Fig. E4.36. The system is the bomb; $Q = +$ when heat is absorbed by the bomb. From Eq. (4.39), $Q_p - Q_v = \Delta n(RT)$.

$$\Delta n = 1 - \tfrac{1}{2} = +\tfrac{1}{2}$$

since C is a solid.

$$Q_p - Q_v = \tfrac{1}{2}RT = 0.5[8.314 \text{ J/(g mol)(K)}](298 \text{ K}) = 1239 \text{ J/g mol}$$

If the measured heat evolved from the bomb was 111,759 J,

$$Q_p = Q_v + 1239 = -111{,}759 + 1239 = -110{,}520 \text{ J}$$

The size of this correction is relatively insignificant compared to either the quantity Q_p or Q_v. In any case, the heat of reaction calculated from the bomb experiment is

$$\Delta\hat{H}_{\text{rxn}_v} = \frac{Q_v}{n} = \frac{-111{,}759 \text{ J}}{\text{g mol C(s)}}$$

Since reported heats of reaction are normally for constant-pressure processes, the value of $-111{,}759$ would not be reported, but instead you would report that

$$\Delta\hat{H}_{\text{rxn}_p} = \frac{Q_p}{n} = \frac{-110{,}521 \text{ J}}{\text{g mol C(s)}}$$

Self-Assessment Test

1. Is Q_p always bigger than Q_v? Explain.
2. The Claude ammonia synthesis gas reaction

$$CO(g) + 3H_2(g) \longrightarrow CH_4(g) + H_2O(l)$$

is carried out in a constant-volume bomb reactor. The molar ratio of H_2 to CO in the bomb at the start is 3 : 1. The reaction is complete. The initial and final temperature is 25°C. The initial pressure is 4 atm. The volume of the bomb is 1.50 L.
(a) Calculate the final pressure in the bomb reactor.
(b) Calculate the heat transfer to or from the reactor—state which.

Thought Problem

1. Does the vaporization of a given amount of liquid water at a given temperature require the same amount of heat transfer if the process is carried out in a closed system versus an open system?

4.7-4 Incomplete Reactions

Your objectives in studying this section are to be able to:

1. Calculate the heat of reaction at standard conditions for processes in which the reactants are not fed in stoichiometric quantities and the reaction may not go to completion.
2. Solve simple problems combining both material and energy balances.

If an **incomplete reaction occurs,** you should **calculate the standard heat of reaction only for the products which are actually formed from the reactants that actually react.** In other words, only the portion of the reactants that actually undergo some change and liberate or absorb some energy are to be considered in calculating the overall standard heat of the reaction. If some material passes through the reactor unchanged, it can contribute nothing to the *standard heat of reaction* calculations (however, when the reactants or products are at conditions *other* than 25°C and 1 atm, whether they react or not, you must include them in the enthalpy calculations as explained in Sec. 4.7-5). If several reactions occur simultaneously, your material balance must reflect what enters the reactor and is produced via the *independent* reactions.

EXAMPLE 4.37 Incomplete Reactions

An iron pyrite ore containing 85.0% FeS_2 and 15.0% gangue (inert dirt, rock, etc.) is roasted with an amount of air equal to 200% excess air according to the reaction

$$4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$$

in order to produce SO_2. All the gangue plus the Fe_2O_3 end up in the solid waste product (cinder), which analyzes 4.0% FeS_2. Determine the standard heat of reaction per kilogram of ore.

Solution

The material balance for the problem must be worked out prior to determining the standard heat of reaction.

Steps 1, 2, and 3 The process is a steady-state open system (see Fig. E4.37). You calculate the excess air based on the stated reaction as if all the FeS_2 reacted to Fe_2O_3 even though some FeS_2 does not react. Six unknowns exist (including the N_2 in P) and five elemental balances can be written; the mass fraction of FeS_2 is given, hence the material balance problem has a unique solution. The molecular weights are Fe(55.85), Fe_2O_3(159.70), and FeS_2(120.0).

Step 4

Basis: 100 kg of pyrite ore

Steps 5 and 6 Rather than use the stoichiometry to determine generation and consumption of compounds, we will use five elemental balances, O_2, N_2, S, gangue, and Fe, plus

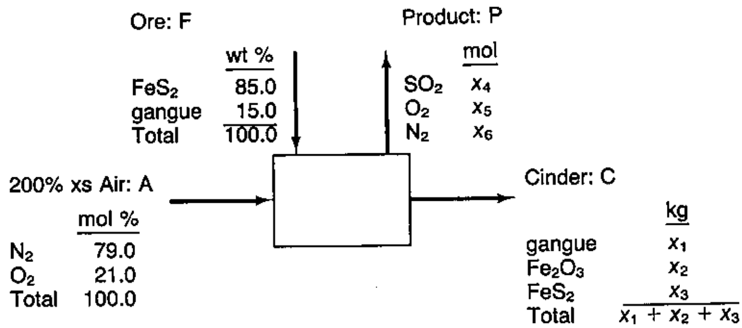


Figure E4.37

the information about the FeS_2 to calculate the moles of SO_2, O_2, and N_2 in the gaseous product, and the kg of gangue, Fe_2O_3, and FeS_2 in the cinder.

Step 3 (Continued) The excess air is

$$\text{Mol FeS}_2 = \frac{85.0}{120.0} = 0.7083 \text{ kg mol}$$

$$\text{Required O}_2 = 0.7083(11/4) = 1.9479 \text{ kg mol}$$

$$\text{Excess O}_2 = 1.9479(2.00) = \underline{3.8958}$$

$$\text{Total O}_2 \text{ in} = 5.8437 \text{ kg mol}$$

$$\text{Total N}_2 \text{ in} = 5.8437(79/21) = 21.983 \text{ kg mol}$$

The element mass balances are

	In	*Out*
Gangue (kg):	15.0	$= x_1$
N_2 (kg mol):	21.983	$= x_6$
S (kg mol):	2(85.0/120.0)	$= x_4 + (x_3/120.0)(2)$
Fe (kg mol):	1(85.0/120.0)	$= (x_2/159.70)2 + (x_3/120.0)(1)$
O_2 (kg mol):	5.8437	$= x_4 + x_5 + (x_2/159.70)(1.5)$

Also,

$$\frac{x_3}{x_1 + x_2 + x_3} = 0.04$$

The solution of these equations is

In P	*In C*
$SO_2 = 1.368$ kg mol	Gangue = 15.0 kg
$O_2 = 3.938$	$Fe_2O_3 = 54.63 \Leftrightarrow 0.342$ kg mol
$N_2 = 21.983$	$FeS_2 = 2.90 \Leftrightarrow 0.0242$ kg mol

Next we calculate the standard heat of reaction. The moles of FeS_2 that react are $0.7083 - 0.0242 = 0.684$ kg mol. Oxygen and nitrogen have zero values for $\Delta\hat{H}_f^\circ$ and the gangue does not react. We will use the following tabulated heats of formation (in kJ/g mol) to calculate ΔH°_{rxn}:

Component:	FeS_2(c)	O_2(g)	Fe_2O_3(c)	SO_2(g)
$\Delta\hat{H}_f^\circ$	−177.9	0	−822.156	−296.90

Two equivalent ways exist to calculate the heat of reaction:

1. Use the material balance data directly.
2. Use the stoichiometric equation multiplied by (0.684/4).

$$\Delta H^\circ_{rxn} = \overbrace{342(-822.156)}^{Fe_2O_3} + \overbrace{24.2(-177.9)}^{FeS_2} + \overbrace{1{,}368(-296.90)}^{SO_2}$$

$$- \overbrace{708.3(-177.90)}^{FeS_2} = -5.656 \times 10^5 \text{ kJ}$$

or

$$\Delta H^\circ_{rxn} = \left[\left(\frac{2}{4}\right)(684)(-822.156) + \left(\frac{8}{4}\right)(684)(-296.90)\right]$$

$$- \left[(684)(-177.9) + \left(\frac{11}{4}\right)(684)(0)\right] = -5.657 \times 10^5 \text{ kJ}$$

Per kg of ore,

$$\Delta\hat{H}_{rxn} = -5.657 \times 10^5 \text{ kJ}/100 \text{ kg} = -5.657 \times 10^3 \text{ kJ/kg}$$

Note that only the moles reacting and produced are involved in computing the heat of reaction. The unoxidized FeS_2 is not included, nor is the N_2.

Self-Assessment Test

1. A dry low-Btu gas with the analysis of CO: 20%, H_2: 20% and N_2: 60% is burned with 200% excess dry air which enters at 25°C. If the exit gases leave at 25°C, calculate the standard heat of reaction per unit volume of entering gas measured at standard conditions (25°C and 1 atm).

Thought Problem

1. Thermal destruction systems have become recognized over the past decade as an increasingly desirable alternative to the more traditional methods of disposing of hazardous wastes in landfills and injection wells. What are some of the problems in the combustion of substances such as methylene chloride, chloroform, trichloroethylene, waste oil, phenol, aniline, and hexachloroethane?

4.7-5 Energy Balance When the Products and Reactants Are Not at 25°C

Your objectives in studying this section are to be able to:

1. Calculate heats of reaction at other than the standard temperature.
2. Calculate how much material must be introduced into a system to provide a prespecified quantity of heat transfer for the system.
3. Apply the general energy balance to processes involving reactions.
4. Determine the temperature of an incoming stream of material given the exit stream temperature (when a reaction occurs).

You no doubt realize that the standard state of 25°C for the heats of formation is only by accident the temperature at which the products enter and the reactants leave a process. In most instances the temperatures of the materials entering and leaving will be higher or lower than 25°C. However, since enthalpies (and hence heats of reaction) are point functions, you can use Eq. (4.24a) together with Eq. (4.33), (4.34), or (4.35), to answer questions about the process being analyzed. Typical questions might be:

(a) What is the heat of reaction at a temperature other than 25°C, but still at 1 atm?
(b) What is the temperature of an incoming or exit stream?
(c) What is the temperature of the reaction?
(d) How much material must be introduced to provide a specified amount of heat transfer for the system?

Consider the process illustrated in Fig. 4.16, for which the reaction is

$$aA + bB \longrightarrow cC + dD$$

Nonstoichiometric amounts of reactants and products, respectively, enter and leave the system.

In employing Eq. (4.24a) you should always **first choose a reference state** at which the heats of formation are known. This usually turns out to be 25°C and 1 atm. (If no reaction takes place, the reference state can be an inlet or outlet stream temperature.) Then the next step is to calculate the enthalpy changes for each stream

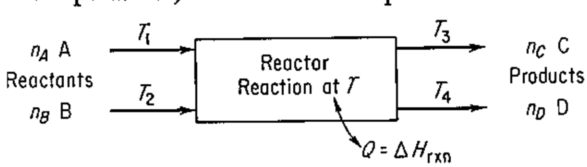


Figure 4.16 Process with reaction.

entering and leaving *relative to this reference state*. Usually, it proves convenient in calculating the enthalpy change to compute the sensible heat changes, the phase changes, and the heat of reaction separately, but this arrangement is not essential. **Any phase changes taking place among the reactants or products not accounted for in the heats of formation must be taken into account in applying the general energy balance.**

Because enthalpy is a state function, you can choose any path you want to execute the calculations for the overall enthalpy change in a process as long as you start and finish at the specified initial and final states, respectively. Figure 4.17 illustrates the idea. For example, suppose that the reference state is chosen to be 25°C and 1 atm, the state for which the $\Delta\hat{H}_f^\circ$'s are known. Figure 4.17 indicates that the path for the calculation for the reactants is from T_A (for A) and T_B (for B) to 25°C; next, the heats of formation are used to get ΔH_{rxn} at 25°C and 1 atm employing A, B, C, and D; and finally, the path for the products goes from 25°C to T_C and T_D. In Fig. 4.17, $T_C = T_D$. Pressure effects can be included along with temperature effects on the enthalpy, but we will omit consideration of the effect of pressure except for problems in which the enthalpy data are retrieved from tables (such as the steam tables) or data bases.

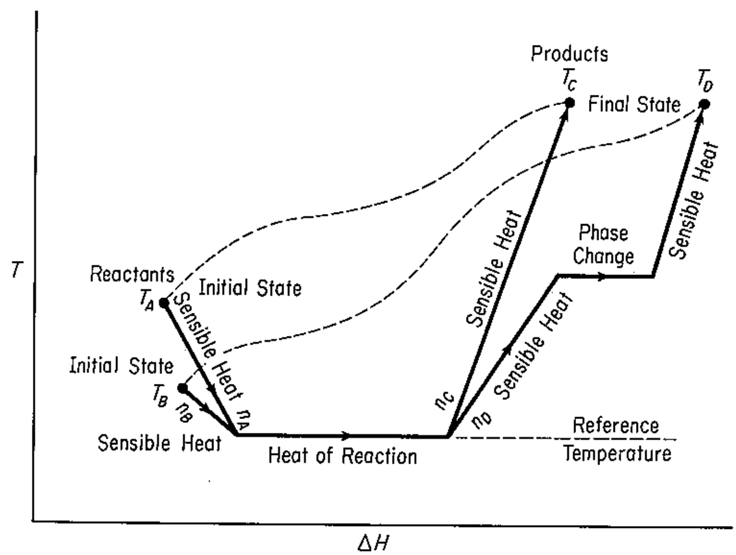


Figure 4.17 Calculation of enthalpy changes for a process in which a reaction takes place. The solid lines indicate an arbitrary selection of path whereas the dashed lines figuratively indicate the actual path taken by the materials—the actual path may not be known in practice.

We have discussed in earlier sections of this chapter methods that you can use to determine the sensible heat ΔH values for the individual streams:

(a) Obtain the enthalpy values from a set of published tables (e.g., the steam tables or tables such as in Appendix D).

(b) Analytically, graphically, or numerically, find

$$\Delta H = \int_{T_1}^{T_2} C_p \, dT$$

for each component individually, using the respective heat capacity equations.

(c) Retrieve values from a data base.

Let us first demonstrate how to calculate the **heat of reaction at a temperature other than 25°C.** By this we mean that stoichiometric quantities of the reactants enter and the products leave at the same temperature—a temperature different from the standard state of 25°C in stoichiometric quantities. Figure 4.18 illustrates the information flow for the calculations corresponding to those associated with Eq. (4.33) assuming a steady-state process ($\Delta E = 0$), no kinetic or potential energy changes, and $W = 0$. The general energy balance [Eq. (4.24)] reduces to

$$Q = \Delta H = [\Delta H_f^\circ + \Delta(H - H^\circ)]_{\text{products}} - [\Delta H_f^\circ + \Delta(H - H^\circ)]_{\text{reactant}}$$

or

$$Q = \overbrace{(\Delta H_{\text{P}} - \Delta H_{\text{R}})}^{\text{sensible}} + (\Delta H_{\text{rxn}_{T_{\text{ref}}}}) \tag{4.40}$$

in terms of the notation in Fig. 4.18. By definition, Q, as calculated by Eq. (4.40), is equal to the "heat of reaction at the temperature T," so that

$$\Delta H_{\text{rxn}_T} = \Delta H_{\text{rxn}_{T_{\text{ref}}}} + \Delta H_{\text{P}} - \Delta H_{\text{R}} \tag{4.41}$$

To indicate a simplified way to calculate $\Delta H_{\text{P}} - \Delta H_{\text{R}}$, particularly using a computer code, suppose that the heat capacity equations are expressed as

$$C_p = \alpha + \beta T + \gamma T^2 \tag{4.42}$$

Then, to obtain $\Delta H_{\text{P}} - \Delta H_{\text{R}}$, we add up the sensible heat enthalpy changes for the

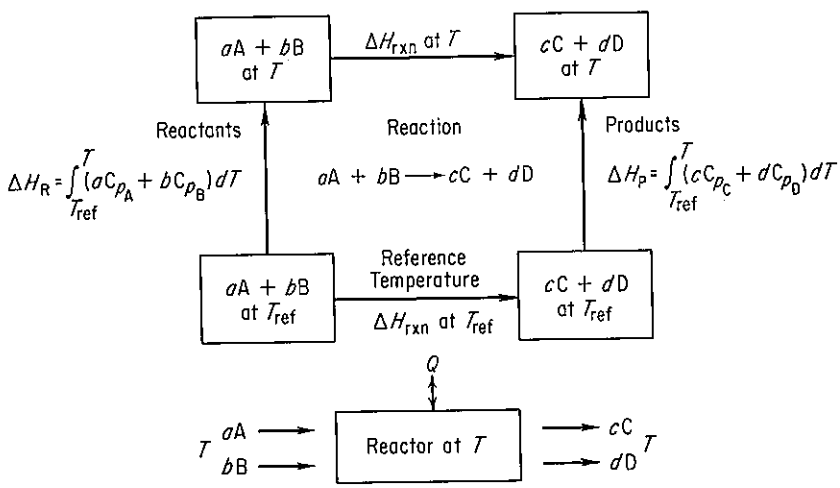


Figure 4.18 Heat of reaction at a temperature other than standard conditions.

products and subtract those for the reactants. Rather than integrate separately, let us consolidate like terms as follows. Each heat capacity equation is multiplied by the proper number of moles:

$$aC_{p_A} = a(\alpha_A + \beta_A T + \gamma_A T^2) \tag{4.43a}$$

$$bC_{p_B} = b(\alpha_B + \beta_B T + \gamma_B T^2) \tag{4.43b}$$

$$cC_{p_C} = c(\alpha_C + \beta_c T + \gamma_c T^2) \tag{4.43c}$$

$$dC_{p_D} = d(\alpha_D + \beta_D T + \gamma_D T^2) \tag{4.43d}$$

Then we define a new term ΔC_p which is equal to

original expression — equivalent new term

$$cC_{p_C} + dC_{p_D} - (aC_{p_A} + bC_{p_B}) = \Delta C_p$$

and

$$\begin{aligned} [(c\alpha_C + d\alpha_D) - (a\alpha_A + b\alpha_B)] &= \Delta\alpha \\ T[(c\beta_C + d\beta_D) - (a\beta_A + b\beta_B)] &= T\,\Delta\beta \\ T^2[(c\gamma_C + d\gamma_D) - (a\gamma_A + b\gamma_B)] &= T^2\,\Delta\gamma \end{aligned} \tag{4.44}$$

Simplified, ΔC_P can be expressed as

$$\Delta C_p = \Delta\alpha + \Delta\beta T + \Delta\gamma T^2 \tag{4.45}$$

Furthermore,

$$\begin{aligned} \Delta H_P - \Delta H_0 &= \int_{T_0}^{T} \Delta C_p\, dT = \int_{T_0}^{T} (\Delta\alpha + \Delta\beta T + \Delta\gamma T^2)\, dT \\ &= \Delta\alpha(T - T_0) + \frac{\Delta\beta}{2}(T^2 - T_0^2) + \frac{\Delta\gamma}{3}(T^3 - T_0^3) \end{aligned} \tag{4.46}$$

where we let $T_0 = T_{\text{ref}}$ for simplicity.

If the integration is carried out without definite limits,

$$\Delta H_P - \Delta H_R = \int (\Delta C_P)\, dT = \Delta\alpha T + \frac{\Delta\beta}{2}T^2 + \frac{\Delta\gamma}{3}T^3 + C \tag{4.47}$$

where C is the integration constant.

Finally, ΔH_{rxn} at the new temperature T is

$$\Delta H_{\text{rxn}_T} = \Delta H_{\text{rxn}_{T_0}} + \Delta\alpha(T - T_0) + \frac{\Delta\beta}{2}(T^2 - T_0^2) + \frac{\Delta\gamma}{3}(T^3 - T_0^3) \tag{4.48a}$$

or

$$\Delta H_{\text{rxn}_T} = \Delta H_{\text{rxn}_{T_0}} + \Delta\alpha T + \frac{\Delta\beta}{2}T^2 + \frac{\Delta\gamma}{3}T^3 + C \tag{4.48b}$$

as the case may be. Using Eq. (4.48a) and knowing ΔH_{rxn} at the reference temperature T_0, you can easily calculate ΔH_{rxn} at any other temperature.

Equation (4.48b) can be consolidated into

$$\Delta H_{rxn_T} = \Delta H_0 + \Delta\alpha T + \frac{\Delta\beta}{2}T^2 + \frac{\Delta\gamma}{3}T^3 \qquad (4.49)$$

where

$$\Delta H_0 = \Delta H_{rxn_{T_0}} + C$$

Now, if ΔH_{rxn} is known at any temperature T, you can calculate ΔH_0 as follows:

$$\Delta H_0 = \Delta H_{rxn_T} - \Delta\alpha(T) - \frac{\Delta\beta}{2}(T^2) - \frac{\Delta\gamma}{3}T^3 \qquad (4.50)$$

Once you calculate the value of ΔH_0, you can use it to calculate ΔH_{rxn} at any other temperature.

Probably the easiest way to compute the necessary enthalpy changes is to use enthalpy data obtained directly from published tables or published formulas. Do not forget to take into account phase changes, if they take place, in the enthalpy calculations. If there is a phase change in one or more of the streams entering or leaving the process, you can conceptually think of the enthalpy changes that take place as shown in Fig. 4.17 if the phase difference is not included in $\Delta\hat{H}_f^\circ$. For example, if the product of combustion is water vapor, use $\Delta\hat{H}_f^\circ$ for water vapor, not for liquid water. Then the phase change from liquid water at 25°C to water vapor at 25°C automatically is taken into account. Otherwise, ΔH of the phase change must be accounted for separately by incorporating it into Eq. (4.46).

EXAMPLE 4.38 Calculation of Heat of Reaction at a Temperature Different from Standard Conditions

An inventor thinks he has developed a new catalyst that can make the gas-phase reaction

$$CO_2 + 4H_2 \longrightarrow 2H_2O + CH_4$$

proceed with 100% conversion. Estimate the heat that must be provided or removed if the gases enter and leave at 500°C.

Solution

Figure 4.18 applies. In effect we need to calculate heat of reaction at 500°C from Eq. (4.41) or Q in Eq. (4.40). For illustrative purposes we use the technique based on Eqs. (4.49) and (4.50). At standard conditions

Basis: 1 g mol of $CO_2(g)$

Tabulated data	$CO_2(g)$	$H_2(g)$	$H_2O(g)$	$CH_4(g)$
$-\Delta\hat{H}_f^\circ$(J/g mol)	393,513	0	241,826	74,848

$$\Delta\hat{H}_{rxn_{298\,K}} = [-74{,}848 - 2(241{,}826)] - [4(0) - 393{,}513]$$
$$= -164{,}987 \text{ J/g mol } CO_2$$

First we shall calculate ΔC_p; units are $\dfrac{\text{J}}{(\text{g mol})(\text{K})}$

$$C_{p\,CO_2} = 26.75 + 42.26 \times 10^{-3}T - 14.25 \times 10^{-6}T^2 \qquad T \text{ in K}$$
$$C_{p\,H_2} = 26.88 + 4.35 \times 10^{-3}T - 0.33 \times 10^{-6}T^2 \qquad T \text{ in K}$$
$$C_{p\,H_2O} = 29.16 + 14.49 \times 10^{-3}T - 2.02 \times 10^{-6}T^2 \qquad T \text{ in K}$$
$$C_{p\,CH_4} = 13.41 + 77.03 \times 10^{-3}T - 18.74 \times 10^{-6}T^2 \qquad T \text{ in K}$$

$$\Delta\alpha = [1(13.41) + 2(29.16)] - [1(26.75) + 4(26.88)]$$
$$= -62.54$$
$$\Delta\beta = [1(77.03) + 2(14.49)](10^{-3}) - [1(42.26) + 4(4.35)](10^{-3})$$
$$= 46.35 \times 10^{-3}$$
$$\Delta\gamma = [1(-18.74) + 2(-2.02)](10^{-6}) - [1(-14.25) + 4(-0.33)](10^{-6})$$
$$= -7.21 \times 10^{-6}$$
$$\Delta C_p = -62.54 + 46.35 \times 10^{-3}T - 7.21 \times 10^{-6}T^2$$

Next we find ΔH_0, using as a reference temperature 298 K.

$$\Delta H_0 = \Delta H_{\text{rxn}_{298}} - \Delta\alpha\, T - \frac{\Delta\beta}{2}T^2 - \frac{\Delta\gamma}{3}T^3$$
$$= -164{,}987 - \left[(-62.54)(298) + \frac{46.35 \times 10^{-3}}{2}(298)^2 + \frac{-7.21 \times 10^{-6}}{3}(298)^3\right]$$
$$= -164{,}987 + 18{,}637 - 2058 + 64 = -148{,}345 \text{ J/g mol } CO_2(g)$$

Then, with ΔH_0 known, the ΔH_{rxn} at 773 K can be determined.

$$\Delta H_{\text{rxn}_{773\,\text{K}}} = \Delta H_0 + \Delta\alpha\, T + \frac{\Delta\beta}{2}T^2 + \frac{\Delta\gamma}{3}T^3$$
$$= -148{,}345 + \left[-62.54(773) + \frac{46.35 \times 10^{-3}}{2}(773)^2 + \frac{-7.21 \times 10^{-6}}{3}(773)^3\right]$$
$$= -183{,}950 \text{ J} = Q$$

Hence 183,950 J/g mol CO_2 must be removed.

EXAMPLE 4.39 Calculation of Heat of Reaction at a Temperature Different from Standard Conditions

Repeat the calculation of the preceding example using enthalpy values from Table 4.4 and Appendix D. Use linear interpolation in the tables.

Solution

The heat of reaction at 25°C and 1 atm from the preceding example is

$$\Delta\hat{H}_{rxn_{298\,K}} = -164{,}987\frac{\text{J}}{\text{g mol}}$$

From Tables D.2 and 4.4b, $\Delta\hat{H}$ (J/g mol) values are [reference is 0°C (273 K)]:

Temp. (°C)	CO_2	H_2	H_2O	CH_4
25	912	718	837	879
500	22,345	14,615	17,795	24,014

From Eq. (4.41),

$$\begin{aligned}\Delta H_{rxn_{500°C}} &= \Delta H_{rxn_{25°C}} + \Delta H_{\text{products}} - \Delta H_{\text{reactants}}\\ &= -164{,}987 + [(1)(24{,}014 - 879) + (2)(17{,}795 - 837)]\\ &\quad -[(1)(22{,}345 - 912) + (4)(14{,}615) - 718)]\\ &= -184{,}957\frac{\text{kJ}}{\text{kg mol } CO_2}\end{aligned}$$

Note that the enthalpies of the products and of the reactants are both based on the reference temperature of 25°C. The answer is not quite the same as in Example 4.38 because the heat capacity data used there were not quite the same as those used in calculating the ΔH values in the tables, and because of the use of linear interpolation in the tables.

In most processes with reaction, the temperature of the entering materials and exit materials is not the same. Such cases can be represented by an information diagram such as Fig. 4.19. Equation (4.40) still applies. Phase changes must still be included, if applicable, as explained previously.

EXAMPLE 4.40 Application of the Energy Balance to a Process in which the Temperatures of the Entering and Exit Streams Differ

Carbon monoxide at 50°F is completely burned at 2 atm pressure with 50% excess air that is at 1000°F. The products of combustion leave the combustion chamber at 800°F. Calculate the heat evolved from the combustion chamber expressed as British thermal units per pound of CO entering.

Solution

Steps 1, 2, and 3 Refer to Fig. E4.40a. A material balance is needed before an energy balance can be made.

Step 4

Basis: 1 lb mol of CO (easier to use than 1 lb of CO)

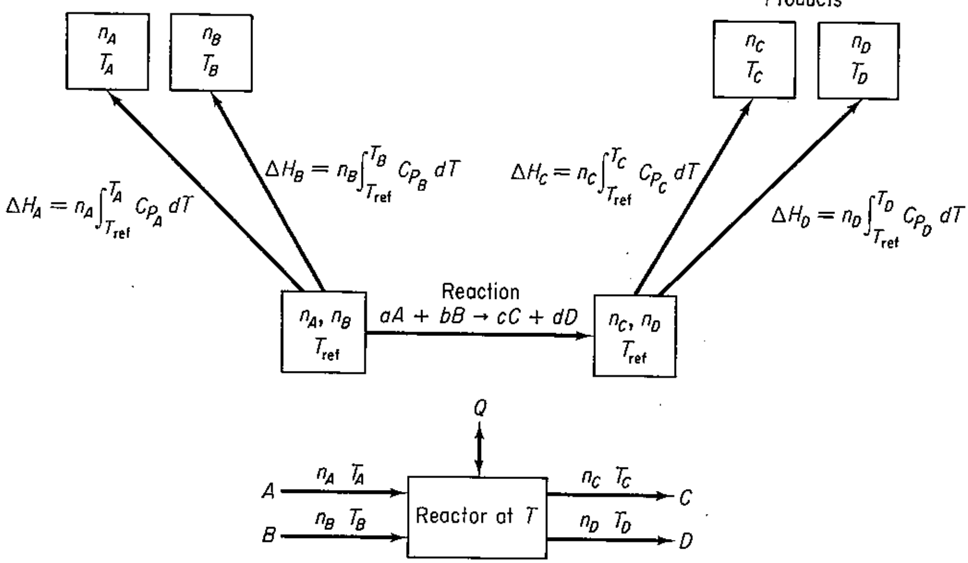


Figure 4.19 Information diagram for the case in which the reactants and products are not at the same temperature.

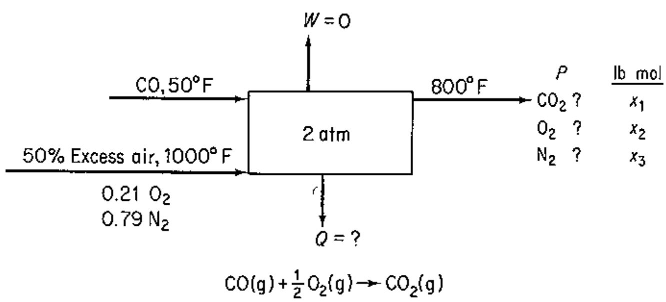


Figure E4.40a

Steps 5, 6, and 8 We have three elements, hence can make three independent balances. Because we have three unknown compositions, a unique solution exists, a solution that can be obtained by direct addition or subtraction.

Steps 2 and 3 (Continued) Amount of air entering:

$$\frac{1 \text{ lb mol CO}}{} \left| \frac{0.5 \text{ lb mol } O_2}{1 \text{ lb mol CO}} \right| \frac{1.5 \text{ lb mol } O_2 \text{ used}}{1.0 \text{ lb } O_2 \text{ mol needed}} \left| \frac{1 \text{ lb mol air}}{0.21 \text{ lb mol } O_2} \right. = 3.57 \text{ lb mol air}$$

composed of

$$3.57(0.79) = 2.82 \text{ lb mol } N_2$$

$$3.57(0.21) = 0.750 \text{ lb mol } O_2$$

Steps 7, 8, and 9 The element balances are

	In		*Out*	*In P* (lb mol)	
C	1	=	x_1	CO_2	1
N_2	2.82	=	x_3	N_2	2.82
O_2	$1(0.5) + 0.750$	=	$x_1 + x_2 = 1 + x_2$	O_2	0.250

Steps 3a and 3b The general energy balance

$$\Delta E = -\Delta[(\hat{H} + \hat{P} + \hat{K})m] + Q - W$$

reduces to $Q = \Delta H$. Why? Only Q is unknown.

Steps 8 and 9 (Continued) Enthalpy data have been taken from Table 4.3. The heat of reaction at 25°C (77°F) and 1 atm from Example 4.31 after conversion of units is −121,672 Btu/lb mol of CO. We can assume that the slightly higher pressure than SC of 2 atm has no effect on the heat of reaction or the enthalpy values. Figure E4.40b shows the sensible heat (enthalpy) values for the entering and exiting materials.

$\Delta\hat{H}$ (Btu/lb mol; reference is 32°F)

Temp. (°F)	*CO*	*Air*	O_2	N_2	CO_2
50	125.2	—	—	—	—
77	313.3	312.7	315.1	313.2	392.2
800	—	—	5690	5443	8026
1000	—	6984	—	—	—

$$Q = \Delta H_{rxn_{77°F}} + \overbrace{\Delta H_{products} - \Delta H_{reactants}}^{\text{sensible}}$$

(a) $$\Delta\hat{H}_{rxn_{77°F}} = -121{,}672 \text{ Btu/lb mol}$$

(b) $$\Delta H_{products} = \Delta H_{800°F} - \Delta H_{77°F}$$

$$= \overbrace{(1)(8026 - 392.2)}^{CO_2} + \overbrace{(2.82)(5443 - 313.2)}^{N_2}$$

$$+ \overbrace{(0.25)(5690 - 315.1)}^{O_2}$$

$$= 7634 + 14{,}466 + 1344$$

$$= 23{,}444 \text{ Btu/lb mol}$$

(c) $$\Delta H_{reactants} = \overbrace{\Delta H_{1000°F} - \Delta H_{77°F}}^{\text{air}} + \overbrace{\Delta H_{50°F} - \Delta H_{77°F}}^{CO}$$

$$= (3.57)(6984 - 312.7) + (1)(125.2 - 313.3)$$

$$= 23{,}817 - 188.1 = 23{,}628 \text{ Btu/lb mol}$$

(d) $$Q = -121{,}672 + 23{,}444 - 23{,}628$$

$$= -121{,}856 \text{ Btu/lb mol}$$

$$\frac{-121{,}856 \text{ Btu}}{1 \text{ lb mol CO}}\left|\frac{1 \text{ lb mol CO}}{28 \text{ lb CO}}\right. = -4352 \text{ Btu/lb CO}$$

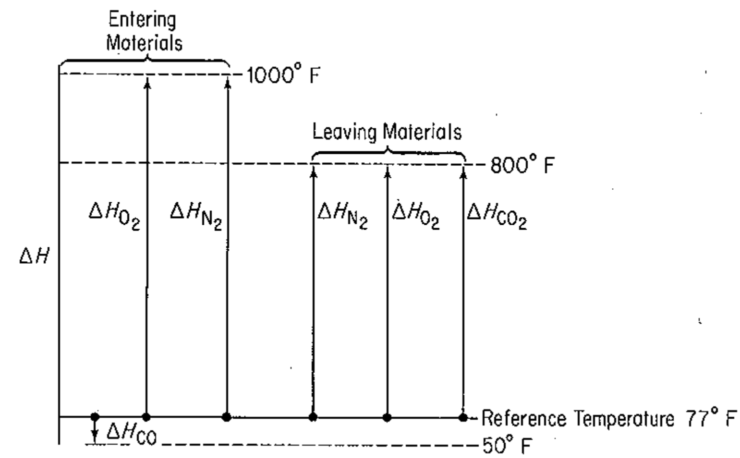


Figure E4.40b

Self-Assessment Test

1. Calculate the heat of reaction at 1000 K and 1 atm for the reaction

$$H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O(g)$$

2. Methane is burned in a furnace with 100% excess dry air to generate steam in a boiler. Both the air and the methane enter the combustion chamber at 500°F and 1 atm, and the products leave the furnace at 2000°F and 1 atm. If the effluent gases contain only CO_2, H_2O, O_2, and N_2, calculate the amount of heat absorbed by the water to make steam per pound of methane burned.
3. A mixture of metallic aluminum and Fe_2O_3 can be used for high-temperature welding. Two pieces of steel are placed end to end and the powdered mixture is applied and ignited. If the temperature desired is 3000°F and the heat loss is 20% of ($\Delta H_{rxn_{25°C}} + \Delta H_{prod} - \Delta H_{react}$) by radiation, what weight in pounds of the mixture (used in the molecular proportions of $2Al + 1Fe_2O_3$) must be used to produce this temperature in 1 lb of steel to be welded? Assume that the starting temperature is 65°F.

$$2Al + Fe_2O_3 \longrightarrow Al_2O_3 + 2Fe$$

Data	C_p, solid [Btu/(lb)(°F)]	Heat of fusion (Btu/lb)	Melting point (°F)	C_p, Liquid [Btu/(lb)(°F)]
Fe and steel	0.12	86.5	2800	0.25
Al_2O_3	0.20	—	—	—

Thought Problem

1. Many different opinions have been expressed as to whether gasohol is a feasible fuel for motor vehicles. An important economic question is: Does 10% grain-based-alcohol-in-gasoline gasohol produce positive net energy? Examine the details of the energy inputs

and outputs, including agriculture (transport, fertilizer, etc.), ethanol processing (fermentation, distilling, drying, etc.), petroleum processing and distribution, and the use of byproducts (corncobs, stalks, mash, etc.). Ignore taxes and tax credits, and assume that economical processing takes place.

4.7-6 Temperature of a Reaction

Your objectives in studying this section are to be able to:

1. Calculate the temperature of an entering or exiting steam of a process given all the other necessary information to obtain a unique solution.
2. Calculate the adiabatic reaction temperature.

We are now equipped to determine what is called the ***adiabatic reaction temperature.*** This is the temperature obtained inside the process when (1) the reaction is carried out under adiabatic conditions, that is, there is no heat interchange between the container in which the reaction is taking place and the surroundings; and (2) when there are no other effects present, such as electrical effects, work, ionization, free radical formation, and so on. In calculations of flame temperatures for combustion reactions, the adiabatic reaction temperature *assumes complete combustion.* Equilibrium considerations may dictate less than complete combustion for an actual case. For example, the adiabatic flame temperature for the combustion of CH_4 with theoretical air has been calculated to be 2010°C; allowing for incomplete combustion, it would be 1920°C. The actual temperature when measured is 1885°C.

The adiabatic reaction temperature tells us the temperature ceiling of a process. We can do no better, but of course the actual temperature may be less. The adiabatic reaction temperature helps us select the types of materials that must be specified for the container in which the reaction is taking place. Chemical combustion with air produces gases at a maximum temperature of 2500 K which can be increased to 3000 K with the use of oxygen and more exotic oxidants, and even this value can be exceeded although handling and safety problems are severe. Applications of such hot gases lie in the preparation of new materials, micromachining, welding using laser beams, and the direct generation of electricity using ionized gases as the driving fluid.

To calculate the adiabatic reaction temperature, you assume that all the energy liberated from the reaction at the reference temperature plus that brought in by the entering stream (relative to the same base temperature) is available to raise the temperature of the products. We assume that the products leave at the temperature of the reaction, and thus if you know the temperature of the products, you automatically know the temperature of the reaction. In effect, for this adiabatic process we can apply Eq. (4.40).

Since no heat escapes and the energy available is allowed only to increase the enthalpy of the products of the reaction, we have

$$\text{sensible } \Delta H_{\text{products}} = \underbrace{\boxed{\text{sensible } \Delta H_{\text{reactants}} - \Delta H_{\text{rxn}}}}_{\text{"energy pool"}} \tag{4.51}$$

Any phase changes that take place and are not accounted for in the heats of formation must be incorporated into the "energy pool." Because of the character of the information available, the determination of the adiabatic reaction temperature or flame temperature may involve a trial-and-error solution; hence the iterative solution of adiabatic flame temperature problems is often carried out on a computer.

EXAMPLE 4.41 Adiabatic Flame Temperature

Calculate the theoretical flame temperature for CO gas burned at constant pressure with 100% excess air, when the reactants enter at 100°C.

Solution

The solution presentation will be abbreviated to save space. The system is shown in Fig. E4.41. We will use Table 4.4 for the data.

$$CO(g) + \tfrac{1}{2}O_2(g) \longrightarrow CO_2(g)$$

Basis: 1 g mol of CO(g); ref. temp. 25°C

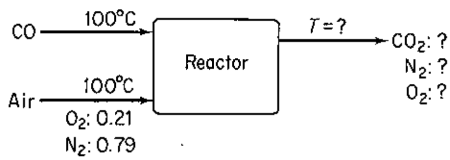


Figure E4.41

Material balance:

Entering reactants		Exit products	
Component	g mol	Component	g mol
CO(g)	1.00	CO_2(g)	1.00
O_2(req.)	0.50	O_2(g)	0.50
O_2(xs)	0.50	N_2(g)	3.76
O_2(total)	1.00		
N_2	3.76		
Air	4.76		

From Example 4.31, ΔH_{rxn} at 25°C = −282,990 J/g mol CO. Then, to fill in Eq. (4.51), $\Delta H_{\text{reactants}}$:

Component	Moles	Reactants (100°C, 373 K) $\Delta\hat{H}$ (J/g mol)	ΔH (J)
CO(g)	1.00	2913 − 728	2,185
Air	4.76	2910 − 726	10,396
		$\Sigma\Delta H_R$ =	12,581

$$\Delta H_{\text{products}} = 12{,}581 + 282{,}990 = 295{,}571 \text{ J}$$

To find the temperature which yields a $\Delta H_{\text{products}}$ of 295,571 J, the simplest procedure is to assume various values of the exit temperature of the products until the $\Sigma\, \Delta H_P = 295{,}571$.

Assume that TFT (theoretical flame temperature) = 2000 K.

Component	Moles	$\Delta\hat{H}$ (J/g mol)	ΔH (J)
CO_2	1.00	(92,466 − 912)	91,554
O_2	0.50	(59,914 − 732)	29,591
N_2	3.76	(56,902 − 728)	211,214
		$\Sigma\Delta H_p$ =	332,359

Assume that TFT = 1750 K.

Component	Moles	$\Delta\hat{H}$ (J/g mol)	ΔH (J)
CO_2	1.00	(77,455 − 912)	76,543
O_2	0.50	(50,555 − 732)	24,912
N_2	3.76	(47,940 − 728)	177,517
		$\Sigma\Delta H_P$ =	278,972

Make a linear interpolation:

$$\text{TFT} = 1750 + \frac{295{,}571 - 278{,}972}{332{,}359 - 278{,}972}(250) = 1750 + 78$$

$$= 1828 \approx 1555°\text{C}$$

If we use heat capacity equations to calculate the "sensible heats," we can obtain an analytic relation to be solved for the flame temperature as follows [C_p values are in J/(g mol)(K) and are good to ±1.5% up to 1500 K]; T is in K:

$$C_{pO_2} = 25.59 + 13.25 \times 10^{-3}T - 4.20 \times 10^{-6}T^2$$

$$C_{pN_2} = 27.02 + 5.81 \times 10^{-3}T - 0.29 \times 10^{-6}T^2$$

$$C_{pCO_2} = 26.75 + 42.26 \times 10^{-3}T - 14.25 \times 10^{-6}T^2$$

For the products

$$\Delta H_{\text{products}} = 1 \int_{298}^{T} (26.75 + 42.26 \times 10^{-3}T - 14.25 \times 10^{-6}T^2)\, dT$$

$$+ 0.5 \int_{298}^{T} (25.59 + 13.25 \times 10^{-3}T - 4.20 \times 10^{-6}T^2)\, dT$$

$$+ 3.76 \int_{298}^{T} (27.02 + 5.81 \times 10^{-3}T - 0.29 \times 10^{-6}T^2)\, dT$$

$$295{,}571 = 141.14T + 35.37 \times 10^{-3}T^2 - 5.81 \times 10^{-6}T^3 - 45{,}047$$

the solution to which is (by Newton's method starting at $T = 2000$ K)

$$T = 1828 \text{ K}$$

Self-Assessment Test

1. Calculate the theoretical flame temperature when hydrogen burns with 400% excess dry air at 1 atm. The reactants enter at 100°C.

Thought Problems

1. Would burning a fuel with oxygen or with air yield a higher adiabatic flame temperature?
2. A recent news article said:

> Two workers were killed and 45 others hurt when a blast at the ____________ refinery shook a neighborhood and shot flames 500 feet into the air. Hydrogen from a "cracker unit" that separates crude oil into such products as gasoline and diesel fuel burned at temperatures from 4,000 to 5,000 degrees in the 9:50 A.M. accident at the refinery. The fire was put out about an hour later.

Is the temperature cited reasonable?

4.8 HEATS OF SOLUTION AND MIXING

4.8-1 Enthalpy Changes for Mixtures

Your objectives in studying this section are to be able to:

1. Distinguish between ideal solutions and real solutions.
2. Calculate the heat of mixing, or the heat of dissolution, at standard conditions given the moles of the materials forming the mixture.
3. Calculate the standard integral heat of solution.
4. Define the standard integral heat of solution at infinite dilution.
5. Apply the energy balance to problems in which the heat of mixing is significant.

So far in our energy calculations, we have been considering each substance to be a completely pure and separate material. The physical properties of an ideal solution or mixture may be computed from the sum of the properties in question for the individual components. For gases, mole fractions can be used as weighting values, or, alternatively, each component can be considered to be independent of the others. In most instances so far in this book we have used the latter procedure. Using the former technique, as we did in a few instances, we could write down, for the heat capacity of an ideal mixture,

$$C_{p\,\text{mixture}} = x_A C_{p_A} + x_B C_{p_B} + x_C C_{p_C} + \cdots \tag{4.52}$$

or, for the enthalpy,

$$\Delta\hat{H}_{\text{mixture}} = x_A \Delta\hat{H}_A + x_B \Delta\hat{H}_B + x_C \Delta\hat{H}_C + \cdots \tag{4.53}$$

These equations are applicable to ideal mixtures only.

When two or more pure substances are mixed to form a gas or liquid solution, we frequently find heat is absorbed or evolved from the system upon mixing. Such a solution would be called a "real" solution. Per mole of solute

$$\Delta\hat{H}^\circ_{\text{final solution}} - \Delta\hat{H}^\circ_{\text{initial components}} = \Delta\hat{H}^\circ_{\text{mixing}} \tag{4.54}$$

The **heat of mixing** ($\Delta\hat{H}^\circ_{\text{mixing}}$) (i.e, the enthaply change on mixing) has to be determined experimentally, but can be retrieved from tabulated experimental (smoothed) results, once such data are available. This type of energy change has been given the formal name ***heat of solution*** when one substance dissolves in another; and there is also the negative of the heat of solution, the ***heat of dissolution,*** for a substance that separates from a solution.

Tabulated data for heats of solution appear in Table 4.12 in terms of energy *per mole of solute* for consecutively added quantities of solvent to the solute; the gram mole refers to the gram mole of solute. Heats of solution are somewhat similar to heats of reaction in that an energy change takes place because of differences in the forces of attraction of the solvent and solute molecules. Of course, these energy changes are much smaller than those we find accompanying the breaking and combining of chemical bonds. Heats of solution are conveniently treated in exactly the same way as are the heats of reaction in the energy balance.

The solution process can be represented by an equation such as the following:

$$HCl(g) + 5H_2O \longrightarrow HCl \cdot 5H_2O$$

or

$$HCl(g) + 5H_2O \longrightarrow HCl(5H_2O)$$

$$\Delta H^\circ_{\text{soln}} = -64{,}047 \text{ J/g mol HCl(g)}$$

The expression $HCl(5H_2O)$ means that 1 mole of HCl has been dissolved in 5 moles of water, and the enthalpy change for the process is −64,047 J/g mol of HCl. Table 4.12 shows the heat of solution for various cumulative numbers of moles of water added to 1 mole of HCl.

The ***standard integral heat of solution*** is the cumulative $\Delta H^\circ_{\text{soln}}$ as shown in the next to last column for the indicated number of molecules of water. As succes-

TABLE 4.12 Heat of Solution of HCl* (at 25°C and 1 atm)

Composition	Total moles H_2O added to 1 mole HCl	$-\Delta\hat{H}°$ for each incremental step (J/g mol HCl) $= -\Delta\hat{H}°_{\text{dilution}}$	Integral heat of solution: cumulative $-\Delta\hat{H}°$ (J/g mol HCl)	Heat of formation $-\Delta\hat{H}_f°$ (J/g mol HCl)
HCl(g)	0			92,311
$HCl \cdot 1H_2O(aq)$	1	26,225	26,225	118,536
$HCl \cdot 2H_2O(aq)$	2	22,593	48,818	141,129
$HCl \cdot 3H_2O(aq)$	3	8,033	56,851	149,161
$HCl \cdot 4H_2O(aq)$	4	4,351	61,202	153,513
$HCl \cdot 5H_2O(aq)$	5	2,845	64,047	156,358
$HCl \cdot 8H_2O(aq)$	8	4,184	68,231	160,542
$HCl \cdot 10H_2O(aq)$	10	1,255	69,486	161,797
$HCl \cdot 15H_2O(aq)$	15	1,503	70,989	163,300
$HCl \cdot 25H_2O(aq)$	25	1,276	72,265	164,576
$HCl \cdot 50H_2O(aq)$	50	1,013	73,278	165,589
$HCl \cdot 100H_2O(aq)$	100	569	73,847	166,158
$HCl \cdot 200H_2O(aq)$	200	356	74,203	166,514
$HCl \cdot 500H_2O(aq)$	500	318	74,521	166,832
$HCl \cdot 1000H_2O(aq)$	1,000	163	74,684	166,995
$HCl \cdot 50{,}000H_2O(aq)$	50,000	146	75,077	167,388
$HCl \cdot \infty H_2O$		67	75,144	167,455

*To convert to cal/g mol multiply by 0.2390.
SOURCE: *National Bureau of Standards Circular 500,* U.S. Government Printing Office, Washington, D.C., 1952.

sive increments of water are added to the mole of HCl, the cumulative heat of solution (the integral heat of solution) increases, but the incremental enthalpy change decreases as shown in Table 4.12. Note that both the reactants and products have to be at standard conditions. The heat of dissolution would be just the negative of these values. The integral heat of the solution is plotted in Fig. 4.20, and you can see that an asymptotic value is approached as the solution becomes more and more dilute. At infinite dilution this value is called the ***standard integral heat of solution at infinite dilution*** and is $-75{,}144$ J/g mol of HCl. What can you conclude about the reference state for the heat of solution of pure HCl from Fig. 4.20? In Appendix H are other tables presenting standard integral heat of solution data and the heats of formation of

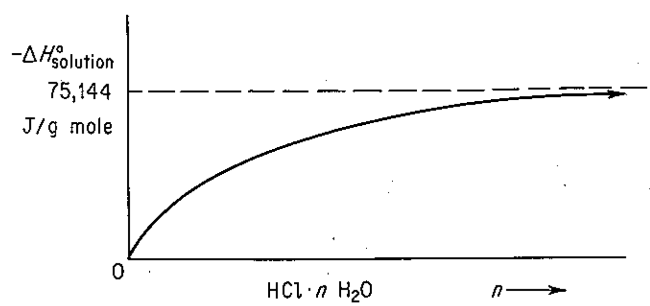


Figure 4.20 Integral heat of solution of HCl in water.

solutions. Since the energy changes for heats of solution are point functions, you can easily look up any two concentrations of HCl and find the energy change caused by adding or subtracting water. For example, if you mix 1 mole of $HCl{\cdot}15\,H_2O$ and 1 mole of $HCl{\cdot}5H_2O$, you obtain 2 moles of $HCl{\cdot}10H_2O$, and the total enthalpy change at 25°C is

$$\Delta H^\circ = [2(-69{,}486)] - [1(-70{,}989) + 1(-64{,}047)]$$
$$= -3936 \text{ J}$$

You would have to remove 3936 J to keep the temperature of the final mixture at 25°C.

To calculate the standard heat of formation of a solute in solution, you proceed as follows. What is the standard heat of formation of 1 g mol of HCl in 5 g mol of H_2O? We treat the solution process in an identical fashion to a chemical reaction:

			J/g mol
$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g)$	$= HCl(g)$	A:	$\Delta\hat{H}^\circ_f = -92{,}311$
$HCl(g) + 5H_2O$	$= HCl(5H_2O)$	B:	$\Delta\hat{H}^\circ_{soln} = -64{,}047$
$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) + 5H_2O$	$= HCl(5H_2O)$	$A + B$:	$\Delta\hat{H}^\circ_f = -156{,}358$

It is important to remember that the heat of formation of H_2O itself does not enter into the calculation. The heat of formation of HCl in an infinitely dilute solution is

$$\Delta\hat{H}^\circ_f = -92{,}311 - 75{,}144 = -167{,}455 \text{ kJ/g mol}$$

Another type of heat of solution which is occasionally encountered is the partial molal heat of solution. Information about this thermodynamic property can be found in most standard thermodynamic texts or in books on thermochemistry, but we do not have the space to discuss it here.

One point of special importance concerns the formation of water in a chemical reaction. When water participates in a chemical reaction in solution as a reactant or product of the reaction, you must include the heat of formation of the water as well as the heat of solution in the energy balance. Thus, if 1 mole of HCl reacts with 1 mole of sodium hydroxide to form water and the reaction is carried out in only 2 moles of water to start with, it is apparent that you will have 3 moles of water at the end of the process. Not only do you have to take into account the heat of reaction when the water is formed, but there is also a heat of solution contribution. If gaseous HCl reacts with crystalline sodium hydroxide and the product is 1 mole of gaseous water vapor, you could employ the energy balance without worrying about the heat of solution effect.

EXAMPLE 4.42 Application of Heat of Solution Data

Hydrochloric acid is an important industrial chemical. To make aqueous solutions of it of commercial grade (known as *muriatic acid*), purified HCl(g) is absorbed in water in a tantalum absorber in a continuous process. How much heat must be removed from the absorber per 100 kg of product if hot HCl(g) at 120°C is fed into water in the absorber as shown in

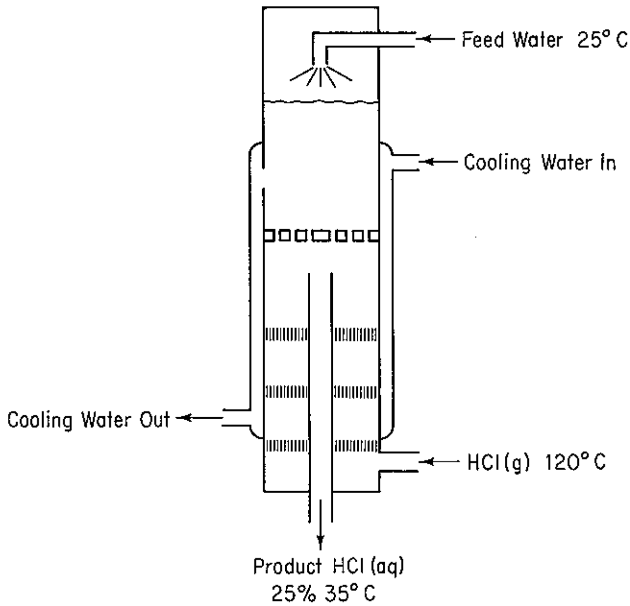


Figure E4.42

Fig. E4.42? The feed water can be assumed to be at 25°C, and the exit product HCl(aq) is 25% HCl (by weight) at 35°C.

Solution

Steps 1, 2, and 3 We need to add to the figure enthalpy data which are easiest to get per mole of HCl. Consequently, we will first convert the product into moles of HCl and moles of H_2O.

Component	kg	Mol. wt.	kg mol	Mole fraction
HCl	25	36.37	0.685	0.141
H_2O	75	18.02	4.163	0.859
Total	100		4.848	1.000

The mole ratio of H_2O to HCl is 4.163/0.685 = 6.077.

Step 4 The system will be the HCl and water (not including the cooling water).

Basis: 100 kg of product

Ref. temperature: 25°C

Steps 5 and 6 Equation (4.40) can be used to calculate $Q = \Delta H$, and both the initial and final enthalpies of all the streams are known or can be calculated directly, hence the problem has a unique solution. The kg and moles of HCl in and out, and the water in and out, have been calculated above.

Step 3 (Continued) The enthalpy values for the streams are [C_p for the HCl(g) is from Table E.1; C_p for the product is approximately 2.7 J/(g)(°C)]:

Stream	ΔH
Feed H_2O	0
HCl(g)	$685 \int_{25}^{120} C_p\, dT = 685(2747) = 1.883 \times 10^6$ J
HCl(aq)	$10^5 \int_{25}^{35} C_p\, dT = 2.7 \times 10^6$ J

By interpolation, the heat of solution data from Table 4.12 give

$$\Delta \hat{H}^\circ_{\text{solution}} = -65{,}442 \text{ J/g mol HCl}$$

$$\Delta H^\circ_{\text{solution}} = (-65{,}442)(685) = -4.4828 \times 10^7 \text{ J}$$

Steps 7, 8, and 9

$$Q = \Delta H = \Delta H_{\text{final products}} - \Delta H_{\text{entering feeds}} + \Delta H^\circ_{\text{solution}}$$

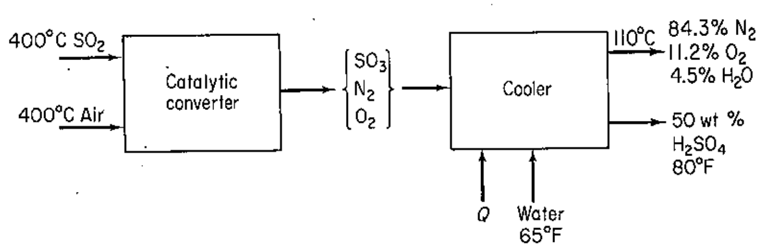
$$= 2.7 \times 10^6 - 1.883 \times 10^6 + (-4.4828 \times 10^7)$$

$$= -4.4 \times 10^7 \text{ J}$$

The negative value of Q means heat is removed from the system.

Self-Assessment Test

1. Is a gas mixture an ideal solution?
2. Give (a) two examples of exothermic mixing of two liquids and (b) two examples of endothermic mixing based on your experience.
3. (a) What is the reference state for H_2O in the table for the heat of solution of HCl?
   (b) What is the value of the enthalpy of H_2O in the reference state?
4. Use the heat of solution data in Appendix H to determine the heat transferred per mole of entering solution into or out of (state which) a process in which 2 g mol of a 50 *mole %* solution of sulfuric acid at 25°C is mixed with water at 25°C to produce a solution at 25°C containing a mole ratio of $10H_2O$ to $1H_2SO_4$.
5. Calculate the heat that must be added or removed per ton of 50 wt % H_2SO_4 produced by the process shown.



Thought Problems

1. A tanker truck of hydrochloric acid was inadvertently unloaded into a large storage tank used for sulfuric acid. After about one-half of the 3000-gal load had been discharged, a violent explosion occurred, breaking the inlet and outlet lines and buckling the tank.
   What might be the cause of the explosion?

2. A concentrated solution (73%) of sodium hydroxide was stored in a vessel. Under normal operations, solution was forced out by air pressure as needed. When application of air pressure did not work, apparently due to solidification of the caustic solution, water was poured through a manhole to dilute the caustic and free up the pressure line. An explosion took place and splashed caustic out of the manhole 15 ft into the air.
   What caused the incident?

4.8-2 Enthalpy–Concentration Charts

Your objectives in studying this section are to be able to:

1. Prepare an enthalpy–concentration chart for a binary mixture given the enthalpy or heat capacity data for the pure components, the heat capacity data for mixtures at various concentrations, and the necessary heats of mixing at various concentrations.
2. Use the enthalpy–concentration chart in solving material and energy balances.

A convenient way to represent enthalpy data for binary solutions is via an enthalpy–concentration diagram. Enthalpy–concentration diagrams (H–x) are plots of specific enthalpy versus concentration (usually weight or mole fraction) with temperature as a parameter. Figure 4.21 illustrates one such plot. If available,[26] such charts are useful in making combined material and energy balances calculations in distillation, crystallization, and all sorts of mixing and separation problems. You will find a few examples of enthalpy–concentration charts in Appendix I.

According to the phase rule $F = C - \mathcal{P} + 2$, hence in the single-phase region for a two-component system (with no reaction), $C = 2$, $\mathcal{P} = 1$, and $F = 2 - 1 + 2 = 3$. The enthalpy–concentration chart is for a fixed pressure; hence the state of the system is determined by specifying two of the following intensive variables: T, $\Delta\hat{H}$, or ω (or x or y).

At some time in your career you may find you have to make numerous repetitive material and energy balance calculations on a given binary system and would like to use an enthalpy-concentration chart for the system, but you cannot find one in the literature or in your files. How do you go about constructing such a chart?

[26] For a literature survey as of 1957, see Robert Lemlich, Chad Gottschlich, and Ronald Hoke, *Chem. Eng. Data Ser.*, v. 2, p. 32 (1957). Additional references: for CCl_4, see M. M. Krishnaiah et al., *J. Chem. Eng. Data*, v. 10, p. 117 (1965); for EtOH–EtAc, see Robert Lemlich, Chad Gottschlich, and Ronald Hoke, *Br. Chem. Eng.*, v. 10, p. 703 (1965); for methanol–toluene, see C. A. Plank and D. E. Burke, *Hydrocarbon Process.*, v. 45, no. 8, p. 167 (1966); for acetone–isopropanol, see S. N. Balasubramanian, *Br. Chem. Eng.*, v. 11, p. 1540 (1966); for alcohol–aromatic systems, see C. C. Reddy and P. S. Murti, *Br. Chem. Eng.*, v. 12, p. 1231 (1967); for acetonitrile–water–ethanol, see Reddy and Murti, ibid., v. 13, p. 1443 (1968); for alcohol–aliphatics, see Reddy and Murti, ibid., v. 16, p. 1036 (1971); and for H_2SO_4, see D. D. Huxtable and D. R. Poole, *Proc. Int. Solar Energy Soc.*, Winnipeg, August 15, 1976, v. 8, p. 178 (1977).

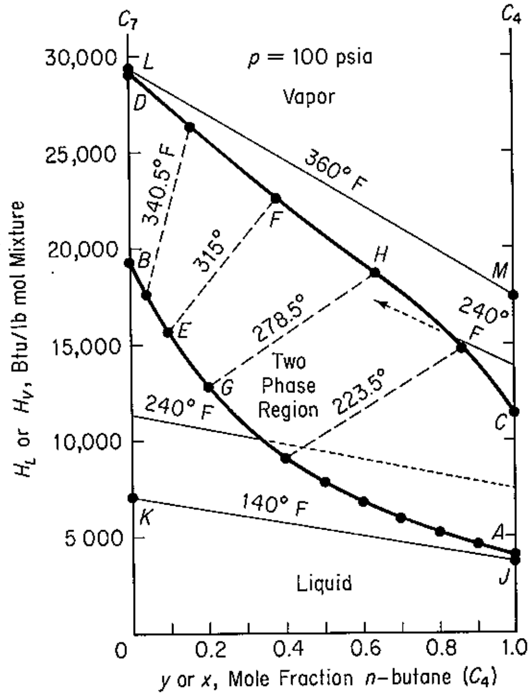


Figure 4.21 Enthalpy concentration chart for n-butane-n-heptane at 100 psia. Curve DFHC is the saturated vapor; curve BEGA is the saturated liquid. The dashed lines are equilibrium tie lines connecting y and x at the same temperature.

As usual, the first thing to do is choose a basis—some given amount of the mixture, usually 1 lb (or kg) or 1 lb mole (or kg mol). Then choose a reference temperature ($\Delta H_0 = 0$ at T_0) for the enthalpy calculations. Assuming that 1 lb is the basis, you then write an energy balance for solutions of various compositions at various temperatures:

$$\Delta\hat{H}_{\text{mixture}} = \omega_A\,\Delta\hat{H}_A + \omega_B\,\Delta\hat{H}_B + \Delta\hat{H}_{\text{mixing}} \tag{4.55}$$

where $\Delta\hat{H}_{\text{mixture}}$ = enthalpy of 1 lb of the mixture

$\Delta\hat{H}_A$, $\Delta\hat{H}_B$ = enthalpies of the pure components per pound relative to the reference temperature (the reference temperature does not have to be the same for A as for B)

$\Delta\hat{H}_{\text{mixing}}$ = heat of mixing (solution) per pound at the temperature of the given calculation

ω = mass fraction

In the common case where the $\Delta\hat{H}_{\text{mixing}}$ is known only at one temperature (usually 77°F), a modified procedure as discussed below would have to be followed to calculate the enthalpy of the mixture.

The choice of the reference temperatures for A and B locates the zero enthalpy datum on each side of the diagram (see Fig. 4.22). From enthalpy tables such as the steam tables, or by finding

$$(\Delta\hat{H}_A - \Delta\hat{H}_{A_0}) = \int_{T_0}^{T=77^\circ\text{F}} C_{p_A}\,dT$$

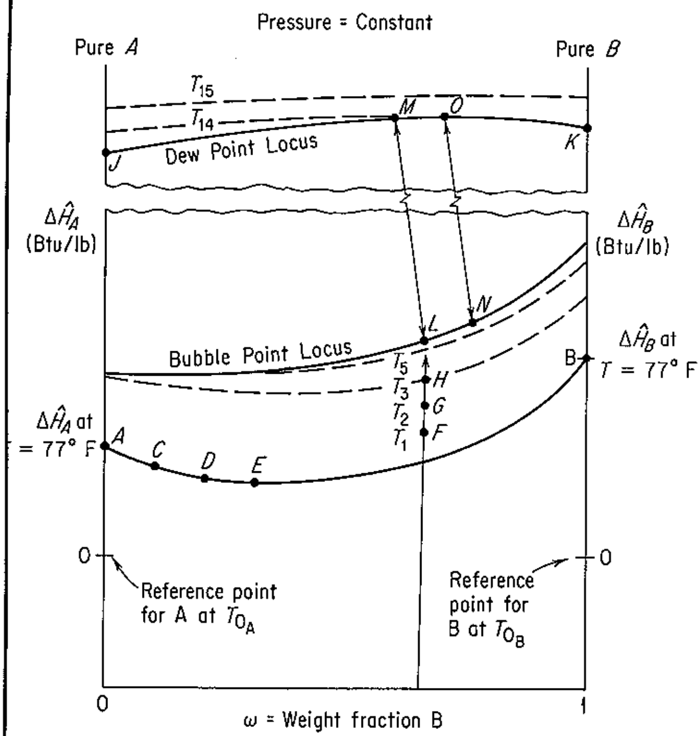


Figure 4.22 Enthalpy–concentration diagram.

you can find $\Delta\hat{H}_A$ at 77°F and similarly obtain $\Delta\hat{H}_B$ at 77°F for the pure substances. Now you have located points A and B. If one of the components is water, it is advisable to choose 32°F as T_0 because then you can use the steam tables as a source of data.

Now that you have $\Delta\hat{H}_A$ and $\Delta\hat{H}_B$ at 77°F (or any other temperature at which $\Delta\hat{H}_{\text{mixing}}$ is known), you can calculate $\Delta\hat{H}_{\text{mixture}}$ at 77°F by Eq. (4.55). For various mixtures such as 10% B_a and 90% A_b, 20% B_a and 80% A_b, and so on, plot the calculated $\Delta\hat{H}_{\text{mixture}}$ as shown by the points C, D, E, and so on, in Fig. 4.22, and connect these points with a continuous line.

You can calculate $\Delta\hat{H}_{\text{mixture}}$ at any other temperature T, once this 77°F isotherm has been constructed, by again calculating an enthalpy change as follows:

$$\Delta\hat{H}_{\text{mixture at any } T} = \Delta\hat{H}_{\text{mixture at 77°F}} + \int_{77°\text{F}}^{T} C_{p_\omega}\, dt \qquad (4.56)$$

where C_{p_ω} is the heat capacity of the solution at concentration ω. These heat capacities must be determined experimentally, although in a pinch they might be estimated. Points F, G, H, and so on, can be determined in this way for a given composition, and then additional like calculations at other fixed concentrations will give you enough points so that all isotherms can be drawn up to the bubble-point line for the mixture at the pressure for which the chart is being constructed.

To include the vapor region on the enthalpy concentration chart, you need to know:

(a) The heat of vaporization of the pure components (to get points J and K)
(b) The composition of the vapor in equilibrium with a given composition of liquid (to get the tie lines L–M, N–O, etc.)
(c) The dew point temperatures of the vapor (to get the isotherms in the vapor region)
(d) The heats of mixing in the vapor region, usually negligible [to fix the points M, O, etc., by means of Eq. (4.55)] (Alternatively, the heat of vaporization of a given composition could be used to fix points M, O, etc.)

The use of enthalpy–concentration diagrams for combination material–energy balance problems will now be illustrated.

EXAMPLE 4.43 Application of the Enthalpy–Concentration Chart

One hundred pounds of a 73% NaOH solution at 350°F is to be diluted to give a 10% solution at 80°F. How many pounds of water at 80°F and ice at 32°F are required if there is no external source of cooling available? See Fig. E4.43. Use the steam tables and the NaOH–H_2O enthalpy–concentration chart in Appendix I as your source of data. (The reference conditions for the latter chart are $\Delta H = 0$ at 32°F for liquid water and $\Delta H = 0$ for an infinitely dilute solution of NaOH, with pure caustic having an enthalpy at 68°F of 455 Btu/lb above this datum.)

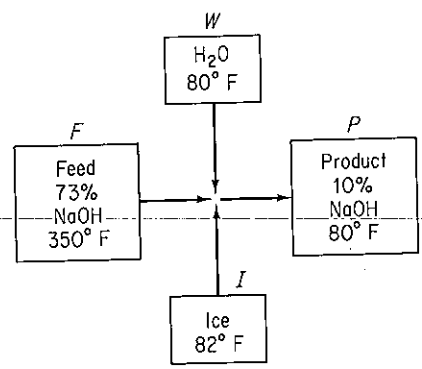


Figure E4.43

Solution

The data required to make a material and an energy balance are as follows:

From Appendix I			
NaOH conc.	Temp. (°F)	$\Delta\hat{H}$ (Btu/lb)	From the steam tables
73	350	468	$\Delta\hat{H}$ of liquid H_2O = 48 Btu/lb
10	80	42	$\Delta\hat{H}$ of ice = −143 Btu/lb (i.e., minus the heat of fusion)

We can make a material balance first. Assume a steady state adiabatic flow process occurs.

Basis: 100 lb of 73% NaOH at 350°F

NaOH balance: $100(0.73) = P(0.10)$ hence $P = 730$ lb

Total balance: $100 + W + I = P = 730$

Hence water plus ice added = 730 − 100 = 630 lb.

Next we make an energy balance; the process is steady state, so that

$$\Delta H_{\text{overall}} = 0 \qquad \text{or} \qquad \Delta H_{\text{in}} = \Delta H_{\text{out}}$$

To differentiate between the ice and the liquid water added, let us designate the ice as I lb, and then the H_2O becomes $(630 - I)$ lb:

$$\overbrace{\underbrace{\frac{100\text{ lb}}{}\Big|\frac{468\text{ Btu}}{\text{lb}}}_{\text{73\% NaOH solution}} + \underbrace{\frac{(630 - I)\text{lb}}{}\Big|\frac{48\text{ Btu}}{\text{lb}}}_{H_2O} + \underbrace{\frac{I\text{ lb}}{}\Big|\frac{-143\text{ Btu}}{\text{lb}}}_{\text{ice}}}^{In} = \overbrace{\underbrace{\frac{730\text{ lb}}{}\Big|\frac{42\text{ Btu}}{\text{lb}}}_{\text{10\% NaOH solution}}}^{Out}$$

$$46{,}800 + 30{,}240 - 191I = 30{,}660$$

$$I = 243 \text{ lb ice at } 32°\text{F}$$

$$\text{liquid } H_2O \text{ at } 80°\text{F} = 387 \text{ lb}$$

EXAMPLE 4.44 Application of the Enthalpy–Concentration Chart

Six hundred pounds of 10% NaOH per hour at 200°F are added to 400 lb/hr of 50% NaOH at the boiling point. Calculate the following:

(a) The final temperature of the exit solution
(b) The final concentration of the exit solution
(c) The pounds of water evaporated per hour during the process

Solution

Basis: 1000 lb of final solution ≡ 1 hr

Use the same NaOH–H_2O enthalpy–concentration chart as in Example 4.43 to obtain the enthalpy data. We can write the following material balance:

Component	*10% solution*	+ *50% solution*	= *Final solution*	*wt %*
NaOH	60	200	260	26
H_2O	540	200	740	74
Total	600	400	1000	100

Next, the energy (enthalpy) balance (in Btu) is

Data:

	10% solution	*50% solution*
$\Delta\hat{H}$ (Btu/lb):	152	290

10% solution		*50% solution*		*Final solution*
600(152)	+	400(290)	=	ΔH
91,200	+	116,000	=	207,200

Note that the enthalpy of 50% solution at its boiling point is taken from the bubble-point curve at $\omega = 0.50$. The enthalpy per pound is

$$\frac{207{,}200 \text{ Btu}}{1000 \text{ lb}} = 207 \text{ Btu/lb}$$

On the enthalpy–concentration chart for NaOH–H_2O, for a 26% NaOH solution with an enthalpy of 207 Btu/lb, you would find that only a two-phase mixture of (1) saturated H_2O vapor and (2) NaOH–H_2O solution at the boiling point could exist. To get the fraction H_2O vapor, we have to make an additional energy (enthalpy) balance. By interpolation, draw the tie line through the point $x = 0.26$, $H = 207$ (make it parallel to the 220° and 250°F tie lines). The final temperature appears from Fig. E4.44 to be 232°F; the enthalpy of the liquid at the bubble point is about 175 Btu/lb. The enthalpy of the saturated water vapor (no NaOH is in the vapor phase) from the steam tables at 232°F is 1158 Btu/lb. Let x = lb of H_2O evaporated.

Basis: 1000 lb of final solution

$$x(1158) + (1000 - x)175 = 1000(207.2)$$

$$x = 32.6 \text{ lb of } H_2O \text{ evaporated/hr}$$

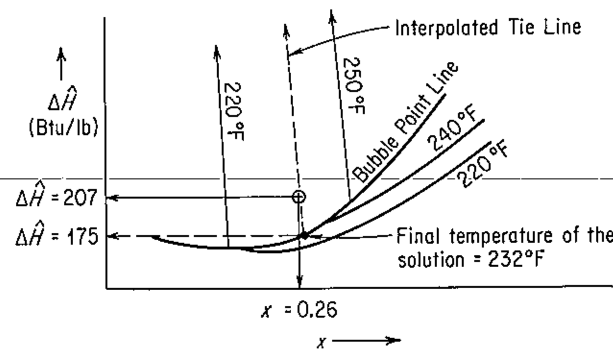


Figure E4.44

EXAMPLE 4.45 Nonadiabatic Equilibrium Vaporization

Enthalpy–concentration charts help make vaporization problems easy to solve. According to the phase rule $F = C - \mathcal{P} + 2$, and for a binary mixture of two phases without reaction, $C = 2$, $\mathcal{P} = 2$, and $F = 2$. If the pressure is fixed at some value, only one more intensive needs to be specified to fix completely the state and other properties of the system. In vaporization, suppose that you continuously feed a vapor–liquid mixture of known composition and temperature to a flash drum, and produce as product saturated liquid and vapor, the latter being of some preset composition. If the vapor is saturated and the mass fraction is given, the temperature of the vapor and liquid and all the other properties of the two products are fixed.

For example, suppose that 1000 lb/hr of a saturated liquid–vapor mixture of NH_3–H_2O at 200°F with an overall mass fraction for the combined phases of 30% NH_3 *by weight* is fed to a flash drum operating at 50 psia. If the exit saturated vapor is to be produced with a composition of 90% *by weight*, what is the composition of the saturated liquid product, the temperature of both phases, and the Btu/hr of heat added to or removed from the process if the products are at equilibrium? Use the ammonia–water chart in Appendix I for the data. See Fig. E4.45.

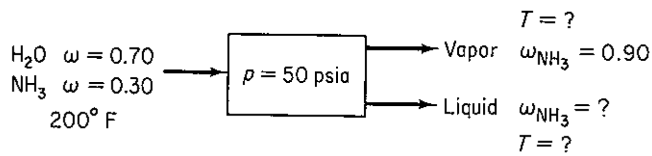


Figure E4.45

Solution

We will just sketch the details of the solution here, skipping the 10-step procedure to save space. At 50 psia, the following data can be retrieved from the ammonia–water chart (the values are not precise because the chart is so small):

	$\Delta\hat{H}$ (Btu/lb mixture)	ω_{NH_3}	T (°F)
Entering mixture	305	0.30	200
Exit vapor	800	0.90	160
Exit liquid	80	0.24	160

To get the data, you need to draw a 200°F tie line and a tie line line from $\omega = 0.90$ for the saturated vapor, both at 50 psia. We also must calculate the lb of liquid and vapor in the product in order to make an energy balance. Let the basis be 1 hr = 1000 lb material.

The material balances are

$$\text{Total:} \quad 1000 = V + L$$

$$NH_3\text{:} \quad 1000(0.30) = V(0.90) + L(0.24)$$

$$V = 91 \text{ lb} \qquad L = 909 \text{ lb}$$

The energy balance reduces to

$$Q = \Delta H = 909(80) + 91(800) - 1000(305) = -159{,}000 \text{ Btu/hr}$$

Heat must be removed.

Enthalpy changes can also be calculated using graphical techniques that are described in texts treating the unit operations of chemical engineering.

Self-Assessment Test

1. Use the NH_3–H_2O enthalpy–concentration chart in Appendix I to determine for 1 atm pressure what the mass fraction ammonia is at equilibrium in the gas and liquid phases for a liquid temperature of 130°F.
2. Estimate the heat of vaporization of an ethanol–water mixture at 1 atm and an ethanol mass fraction of 0.50 from the enthalpy–concentration chart in Appendix I.

3. Prepare an enthalpy–concentration chart for ethanol and water using the data in Table I.1 similar to the one in Appendix I for the acetic acid–water system.
4. For the sulfuric acid–water system, what are the phase(s), composition(s), and enthalpy(ies) existing at $\hat{H} = 120$ Btu/lb and $T = 260°F$?

4.9 HUMIDITY CHARTS AND THEIR USE

Your objectives in studying this section are to be able to:

1. Define humidity, humid heat, humid volume, dry-bulb temperature, wet-bulb temperature, humidity chart, moist volume, and adiabatic cooling line.
2. Explain and show by use of equations why the slope of the wet-bulb lines are the same as the adiabatic cooling lines for water–air mixtures.
3. Prepare a humidity chart given relations for the heat capacities of air and water.
4. Use the humidity chart to determine the properties of moist air, and to calculate enthalpy changes and solve heating and cooling problems involving moist air.

In Chap. 3 we discussed humidity, condensation, and vaporization. In this section we apply simultaneous material and energy balances to solve problems involving humidification, air conditioning, water cooling, and the like. Before proceeding, you should review briefly the sections in Chap. 3 dealing with vapor pressure and partial saturation.

Recall that the *humidity* $\mathcal{H}$ is the mass (lb or kg) of water vapor per mass (lb or kg) of bone-dry air (some texts use moles of water vapor per mole of dry air as the humidity)

$$\mathcal{H} = \frac{18 p_{H_2O}}{29(p_T - p_{H_2O})} = \frac{18 n_{H_2O}}{29(n_T - n_{H_2O})} \tag{4.57}$$

Table 4.13 lists some of the pertinent notation and data needed in preparing humidity charts.

You will also find it indispensable to learn the following definitions and relations.

(a) The ***humid heat*** is the heat capacity of an air–water vapor mixture expressed on the *basis of 1 lb or kg of bone-dry air*. Thus the humid heat C_S is

$$C_S = C_{p\,\text{air}} + (C_{p_{H_2O\,\text{vapor}}})(\mathcal{H}) \tag{4.58}$$

TABLE 4.13 Pertinent Data for Humidity Charts

Symbol	Meaning	SI value	American engineering value
$C_{p_{air}}$	Heat capacity of air	1.00 kJ/(kg)(K)	0.24 Btu/(lb)(°F)
$C_{p_{H_2O\ vapor}}$	Heat capacity of water vapor	1.88 kJ/(kg)(K)	0.45 Btu/(lb)(°F)
$\Delta\hat{H}_{vap}$	Specific heat of vaporization of water at 0°C (32°F)	4502 kJ/kg	1076 Btu/lb
$\Delta\hat{H}_{air}$	Specific enthalpy of air		
$\Delta\hat{H}_{H_2O\ vapor}$	Specific enthalpy of water vapor		

where the heat capacities are all per mass and not per mole. Assuming that the heat capacities of air and water vapor are constant under the narrow range of conditions experienced for air-conditioning and humidification calculations, we can write in American engineering units

$$C_S = 0.240 + 0.45(\mathscr{H}) \qquad \text{Btu/(°F)(lb dry air)} \tag{4.59}$$

or in SI units,

$$C_S' = 1.00 + 1.88(\mathscr{H}) \qquad \text{kJ/(K)(kg dry air)} \tag{4.59a}$$

(b) The ***humid volume*** is the volume of 1 lb or kg of dry air plus the water vapor in the air. In the American engineering system.

$$\hat{V} = \frac{359\ \text{ft}^3}{1\ \text{lb mol}}\left|\frac{1\ \text{lb mol air}}{29\ \text{lb air}}\right|\frac{T_{°F} + 460}{32 + 460}$$

$$+ \frac{359\ \text{ft}^3}{1\ \text{lb mol}}\left|\frac{1\ \text{lb mol H}_2\text{O}}{18\ \text{lb H}_2\text{O}}\right|\frac{T_{°F} + 460}{32 + 460}\left|\frac{\mathscr{H}\ \text{lb H}_2\text{O}}{\text{lb air}}\right. \tag{4.60}$$

$$= (0.730T_{°F} + 336)\left(\frac{1}{29} + \frac{\mathscr{H}}{18}\right)$$

where $\hat{V}$ is in ft³/lb dry air. In the SI system,

$$\hat{V} = \frac{22.4\ \text{m}^3}{1\ \text{kg mol}}\left|\frac{1\ \text{kg mol air}}{29\ \text{kg air}}\right|\frac{T_K}{273}$$

$$+ \frac{22.4\ \text{m}^3}{1\ \text{kg mol}}\left|\frac{1\ \text{kg mol H}_2\text{O}}{18\ \text{kg H}_2\text{O}}\right|\frac{T_K}{273}\left|\frac{\mathscr{H}\ \text{kg H}_2\text{O}}{\text{kg air}}\right. \tag{4.60a}$$

$$= 2.83 \times 10^{-3}T_K + 4.56 \times 10^{-3}\mathscr{H}$$

where $\hat{V}$ is in m³/kg dry air.

(c) The ***dry-bulb temperature*** (T_{DB}) is the ordinary temperature you always have been using for a gas in °F or °C (or °R or K).

(d) The ***wet-bulb temperature*** (T_{WB}) you may guess, even though you may never have heard of this term before, has something to do with water (or other liquid, if we are concerned not with humidity but with saturation) evaporating from around an ordinary mercury thermometer bulb. Suppose that you put a wick, or porous cotton cloth, on the mercury bulb of a thermometer and wet the wick. Next you either (1) whirl the thermometer in the air as in Fig. 4.23 (this apparatus is called a sling psychrometer when the wet-bulb and dry-bulb thermometers are mounted together), or (2) set up a fan to blow rapidly on the bulb at 1000 ft^3/min or more. What happens to the temperature recorded by the wet-bulb thermometer?

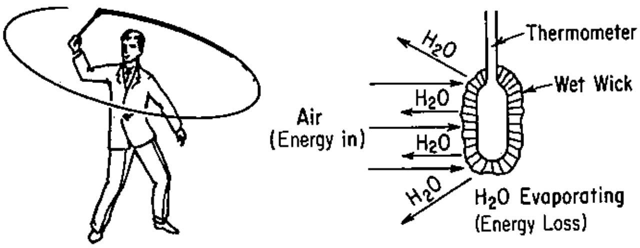


Figure 4.23 Wet-bulb temperature obtained with a sling psychrometer.

As the water from the wick evaporates, the wick cools down and continues to cool until the rate of energy transferred to the wick by the air blowing on it equals the rate of loss of energy caused by the water evaporating from the wick. We say that the temperature of the bulb with the wet wick at equilibrium is the wet-bulb temperature. (Of course, if water continues to evaporate, it eventually will all disappear, and the wick temperature will rise.) The equilibrium temperature for the process described above will lie on the 100% relative humidity curve (saturated-air curve).

Suppose that we prepare a graph on which the vertical axis is the humidity and the horizontal axis is the dry-bulb temperature. We want to plot the temperature of the thermometer as it changes to reach T_{WB}. This line is the so called **wet-bulb line.**

The equation for the wet-bulb lines is based on a number of assumptions, a detailed discussion of which is beyond the scope of this book. Nevertheless, the idea of the wet-bulb temperature is based on the equilibrium between the *rates* of energy transfer to the bulb and evaporation of water. Rates of processes are a topic that we have not discussed. The fundamental idea is that a large amount of air is brought into contact with a little bit of water and that presumably the evaporation of the water leaves the temperature and humidity of the air unchanged. Only the temperature of the water changes. The equation of the wet-bulb line is an energy balance

$$\underbrace{h_c(T - T_{WB})}_{\text{heat transfer to water}} = \underbrace{k'_g \Delta\hat{H}_{vap}(\mathcal{H}_{WB} - \mathcal{H})}_{\text{heat transfer from water}} \tag{4.61}$$

where h_c = heat transfer coefficient for convection to the bulb
T = temperature of moist air
k'_g = mass transfer coefficient
$\Delta\hat{H}_{vap}$ = latent heat of vaporization
$\mathcal{H}$ = humidity of moist air

Next, we can form the ratio

$$\frac{\mathcal{H}_{\mathrm{WB}} - \mathcal{H}}{T_{\mathrm{WB}} - T} = -\frac{h_c}{(k_g')\Delta\hat{H}_{\mathrm{vap}}} \tag{4.62}$$

to get the slope of the wet bulb line. For water only, it so happens that $h_c/k_g' \simeq C_S$ (i.e., the numerical value is about 0.25), which gives the wet-bulb lines the slope of

$$\frac{\mathcal{H}_{\mathrm{WB}} - \mathcal{H}}{T_{\mathrm{WB}} - T} = -\frac{C_S}{\Delta\hat{H}_{\mathrm{vap}}} \tag{4.63}$$

For other substances, the value of h_c/k_g' can be as much as twice the value of C_S for water.

Figure 4.24 shows the plot of the wet-bulb line as $T_{\mathrm{DB}} \rightarrow T_{\mathrm{WB}}$. The line is approximately straight and has a negative slope. Does this result agree with Eq. (4.63)?

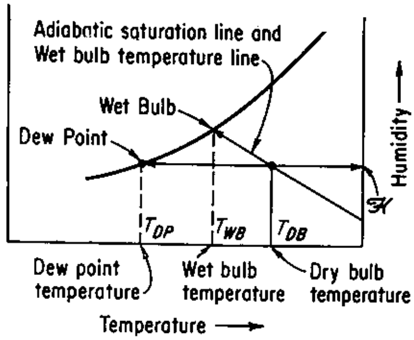


Figure 4.24 General layout of the humidity chart showing the location of the wet-bulb and dry-bulb temperatures, the dew point and dew-point temperature, and the adiabatic saturation line and wet-bulb line.

Another type of process of some importance occurs when adiabatic cooling or humidification takes place between air and water that is recycled as illustrated in Fig. 4.25. In this process the air is both cooled and humidified (its water content rises) while a little bit of the recirculated water is evaporated. At *equilibrium,* in the steady state, the temperature of the air is the same as the temperature of the water, and the exit air is saturated at this temperature. By making an overall energy balance around the process ($Q = 0$), we can obtain the equation for the adiabatic cooling of the air.

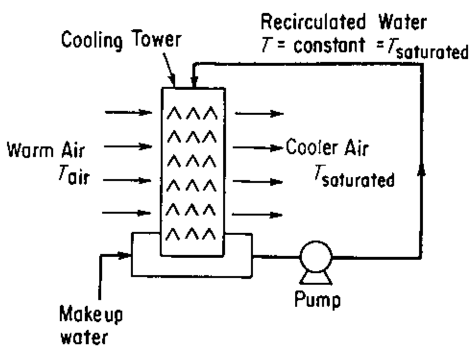


Figure 4.25 Adiabatic humidification with recycle of water.

The equation, when plotted on the humidity chart, yields what is known as an adiabatic cooling line. We take the equilibrium temperature of the water, T_S, as a reference temperature rather than 0°C or 32°F. Do you see why? We ignore the small amount of makeup water or assume that it enters at T_S. The energy balance is

$$\overbrace{C_{p\text{air}}(T_{\text{air}} - T_S)}^{\text{enthalpy of air entering}} + \overbrace{\mathcal{H}_{\text{air}}[\Delta\hat{H}_{\text{vap}\,H_2O\,\text{at}\,T_S} + C_{pH_2O\,\text{vapor}}(T_{\text{air}} - T_S)]}^{\text{enthalpy of water vapor in air entering}} \tag{4.64}$$

$$= \overbrace{C_{p\text{air}}(T_S - T_S)}^{\text{enthalpy of air leaving}} + \overbrace{\mathcal{H}_S[\Delta\hat{H}_{\text{vap}\,H_2O\,\text{at}\,T_S} + C_{pH_2O\,\text{vapor}}(T_S - T_S)]}^{\text{enthalpy of water vapor in air leaving}}$$

Equation (4.64) can be reduced to

$$T_{\text{air}} = \frac{\Delta\hat{H}_{\text{vap}\,H_2O\,\text{at}\,T_S}(\mathcal{H}_S - \mathcal{H}_{\text{air}})}{C_{p\text{air}} + C_{pH_2O\,\text{vapor}}\mathcal{H}_{\text{air}}} + T_S \tag{4.65}$$

which is the equation for adiabatic cooling.

Notice that this equation can be written as

$$\frac{\mathcal{H}_S - \mathcal{H}}{T_S - T_{\text{air}}} = -\frac{C_S}{\Delta\hat{H}_{\text{vap at}\,T_S}} \tag{4.66}$$

Compare Eq. (4.66) with Eq. (4.63). Can you conclude that the wet-bulb process equation, for water only, is essentially the same as the adiabatic cooling equation?

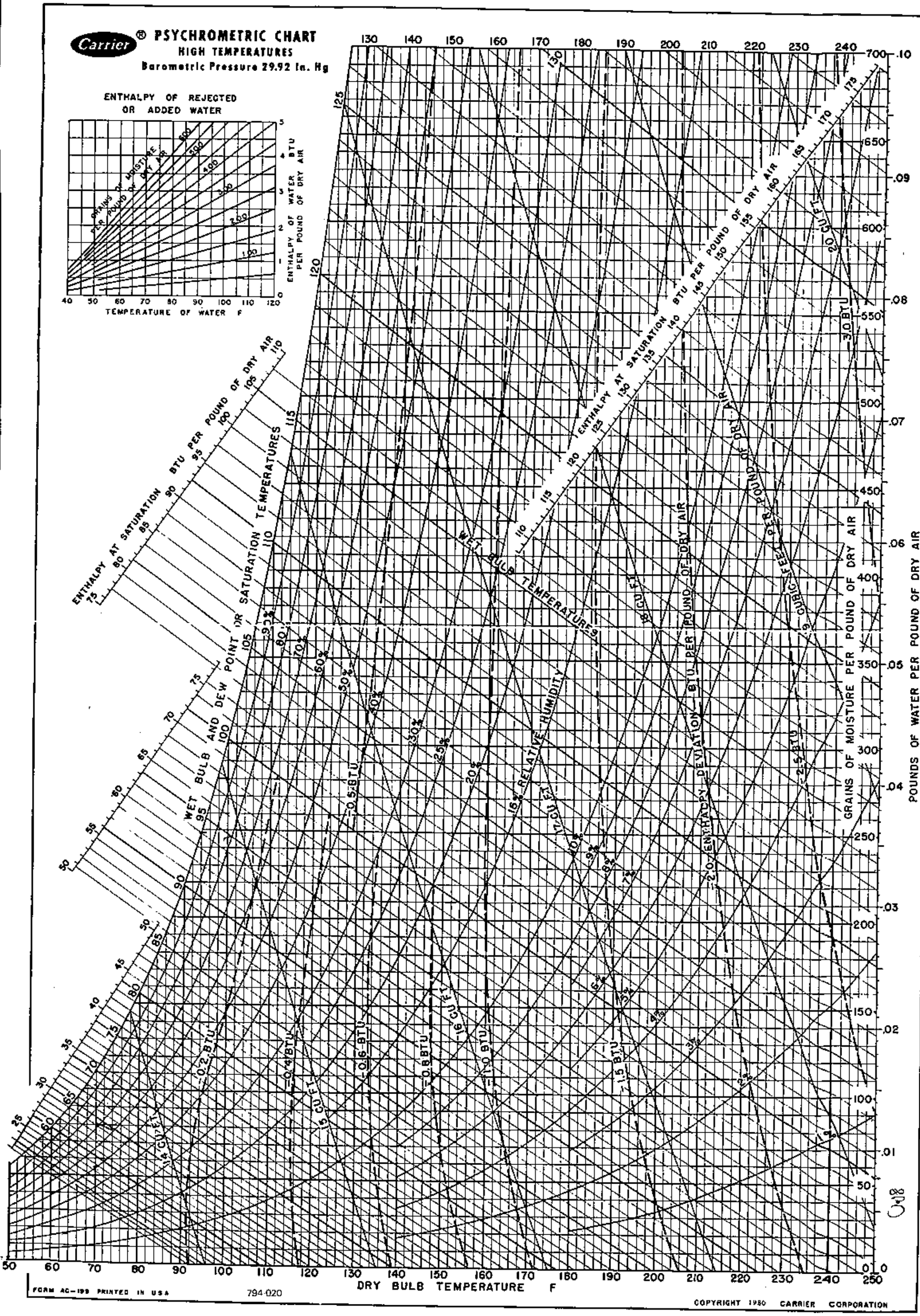
Of course! We have the nice feature that two processes can be represented by the same set of lines. For a detailed discussion of the uniqueness of this coincidence, consult any of the references at the end of the chapter. For most other substances besides water, the two equations will have different slopes.

Now that you have an idea of what the various features portrayed on the ***humidity chart*** (psychrometric chart) are, let us look at the chart itself (Fig. 4.26). It is nothing more than a graphical means to assist in the presentation of material and energy balances in water vapor–air mixtures and various associated parameters. Its skeleton consists of a humidity ($\mathcal{H}$)-temperature (T_{DB}) set of coordinates together with the additional parameters (lines) of

- **(a)** Constant relative humidity indicated in percent
- **(b)** Constant moist volume (humid volume)
- **(c)** Adiabatic cooling lines which are the same (for water vapor only) as the wet-bulb or psychrometeric lines
- **(d)** The 100% relative humidity (identical to the 100% absolute humidity) curve (i.e., saturated-air curve)

With any two values known, you can pinpoint the air-moisture condition on the chart and determine all the other associated values.

Off to the left of the 100% relative humidity line you will observe scales showing the enthalpy per mass of dry air of a saturated air–water vapor mixture. Enthalpy



0.357

0.08

Figure 4.26a (Reprinted by permission of Carrier Corporation.)

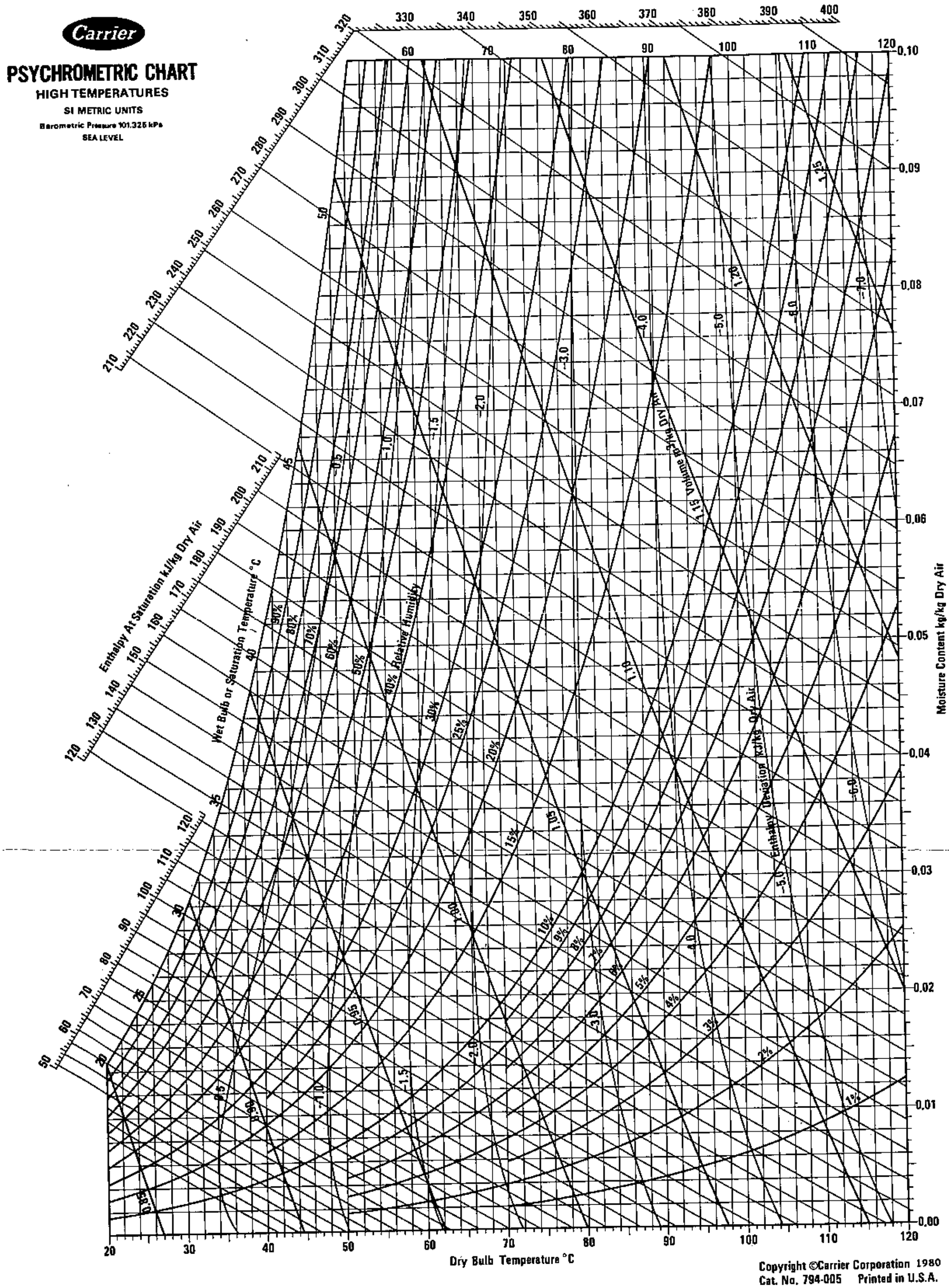


Figure 4.26b (Reprinted by permission of Carrier Corporation.)

adjustments for air less than saturated (identified by minus signs) are shown on the chart itself by a series of curves. The enthalpy of the wet air, in energy/mass of dry air, is

$$\Delta\hat{H} = \Delta\hat{H}_{\text{air}} + \Delta\hat{H}_{\text{H}_2\text{O vapor}}(\mathcal{H})$$

We should mention at this point that the reference conditions for the humidity chart are liquid water at 32°F (0°C) and 1 atm (not the vapor pressure of H_2O) for water, and 0°F and 1 atm for air. The chart is suitable for use only at normal atmospheric conditions and must be modified[27] if the pressure is significantly different than 1 atm. If you wanted to, you could calculate the enthalpy values shown on the chart directly from tables listing the enthalpies of air and water vapor or you could compute the enthalpies with reasonable accuracy from the following equation

$$\Delta\hat{H} = \underbrace{C_{p\text{air}}(T - T_{\text{ref}})_{\text{air}}}_{\text{enthalpy for air}} + \underbrace{\mathcal{H}[\overbrace{\Delta\hat{H}_{\text{vap}}}^{\substack{\text{heat of vaporation}\\ \text{of water at } T_{\text{ref}}}} + C_{p\text{air}}(T - T_{\text{ref}})_{\text{water}}]}_{\text{enthalpy for water vapor}} \tag{4.67}$$

In American engineering units, Eq. (4.67) is

$$\Delta\hat{H} = 0.240T_{°\text{F}} + \mathcal{H}(1061 + 0.45T_{°\text{F}}) \tag{4.68}$$

because $C_S = 0.240 + 0.45\mathcal{H}$ and $T_{\text{WB}} = T_S$. Thus the wet-bulb process equation, for water only, is essentially the same as the adiabatic cooling equation. For other materials these two equations have different slopes.

Only two of the quantities in Eq. (4.67) are variables, if T_S is known, because $\mathcal{H}_S$ is the humidity of saturated air at T_S and $\Delta\hat{H}_{\text{vap H}_2\text{O at } T_S}$ is fixed by T_S. Thus, for any value of T_S, you can make a plot of Eqs. (4.65) and/or (4.68) on the humidity chart in the form of $\mathcal{H}$ vs. T_{air}. These curves, which are essentially linear, will intersect the 100% relative humidity curve at $\mathcal{H}_S$ and T_S, as described earlier.

The adiabatic cooling lines are lines of almost constant enthalpy for the entering air–water mixture, and you can use them as such without much error (1 or 2%). However, if you want to correct a saturated enthalpy value for the deviation which exists for a less-than-saturated air–water vapor mixture, you can employ the enthalpy deviation lines which appear on the chart and which can be used as illustrated in the examples below. Any process that is not a wet-bulb process or an adiabatic process with recirculated water can be treated by the usual material and energy balances, taking the basic data for the calculation from the humidity charts. If there is any increase or decrease in the moisture content of the air in a psychrometric process, the small enthalpy effect of the moisture added to the air or lost by the air may be included in the energy balance for the process to make it more exact as illustrated in Examples 4.47 and 4.49.

You can find further details regarding the construction of humidity charts in the references at the end of the chapter. Tables are also available listing all the ther-

[27] See G. E. McElroy, *U.S. Bur. Mines Rep. Invest. No. 4615*, December 1947, or other Carrier charts. See *Educational Materials*, Carrier Corp., Syracuse, N.Y., 1980.

modynamic properties (p, $\hat{V}$, $\mathcal{H}$, and $\Delta\hat{H}$) in great detail.[28] Although we shall be discussing humidity charts exclusively, charts can be prepared for mixtures of any two substances in the vapor phase, such as CCl_4 and air or acetone and nitrogen, by use of Eqs. (4.57)–(4.67) if all the values of the physical constants for water and air are replaced by those of the desired gas and vapor. The equations themselves can be used for humidity problems if charts are too inaccurate or are not available. In the pocket in the back of this book is a Fortran computer program from which psychrometic data can be retrieved.

EXAMPLE 4.46 Properties of Moist Air from the Humidity Chart

List all the properties you can find on the humidity chart in American engineering units for moist air at a dry-bulb temperature of 90°F and a wet-bulb temperature of 70°F.

Solution

A diagram will help explain the various properties obtained from the humidity chart. See Fig. E4.46. You can find the location of point A for 90°F DB (dry bulb) and 70°F WB (wet bulb) by following a vertical line at T_{DB} = 90°F until it crosses the wet-bulb line for 70°F. This wetbulb line can be located by searching along the 100% humidity line until the saturation temperature of 70°F is reached, or, alternatively, by proceeding up a vertical line at 70°F until it intersects the 100% humidity line. From the wet-bulb temperature of 70°F, follow the adiabatic cooling line (which is the same as the wet-bulb temperature line on the humidity chart) to the right until it intersects the 90°F DB line. Now that point A has been fixed, you can read the other properties of the moist air from the chart.

(a) *Dew point*. When the air at A is cooled at constant pressure (and in effect at *constant humidity*), as described in Chap. 3, it eventually reaches a temperature at which the moisture begins to condense. This is represented by a horizontal line, a constant-humidity line, on the humidity chart, and the dew point is located at B, or about 60°F.

(b) *Relative humidity*. By interpolating between the 40% $\mathcal{RH}$ and 30% $\mathcal{RH}$ lines you can find that point A is at about 37% $\mathcal{RH}$.

(c) *Humidity* ($\mathcal{H}$). You can read the humidity from the right-hand ordinate as 0.0112 lb H_2O/lb dry air.

[28] Byron Engelbach, *Microfilms of Psychrometric Tables*, University Microfilms, Ann Arbor, Mich., 1953.

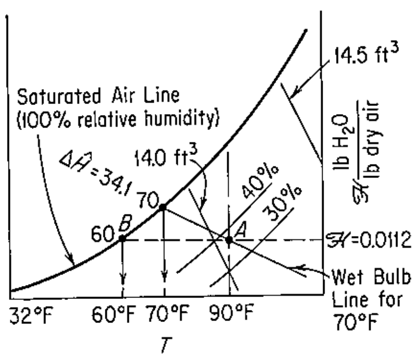


Figure E4.46

(d) *Humid volume*. By interpolation again between the 14.0- and the 14.5-ft³ lines, you can find the humid volume to be 14.097 ft³/lb dry air.

(e) *Enthalpy*. The enthalpy value for saturated air with a wet-bulb temperature of 70°F is $\Delta\hat{H} = 34.1$ Btu/lb dry air (a more accurate value can be obtained from psychrometric tables if needed). The enthalpy deviation (not shown in Fig. E4.46—see Fig. 4.26) for less-than-saturated air is about −0.2 Btu/lb of dry air; consequently, the actual enthalpy of air at 37% $\mathcal{RH}$ is 34.1 − 0.2 = 33.9 Btu/lb of dry air.

EXAMPLE 4.47 Heating at Constant Humidity

Moist air at 38°C and 48% $\mathcal{RH}$ is heated in your furnace to 86°C. How much heat has to be added per cubic meter of initial moist air, and what is the final dew point of the air?

Solution

As shown in Fig. E4.47, the process goes from point *A* to point *B* on a horizontal line of constant humidity. The initial conditions are fixed at $T_{DB} = 38$°C and 48% $\mathcal{RH}$. Point *B* is fixed by the intersection of the horizontal line from *A* and the vertical line at 86°C. The dew point is unchanged in this process and is located at *C* at 24.8°C.

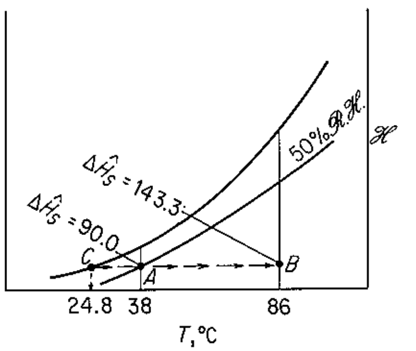


Figure E4.47

The enthalpy values are as follows (all in kJ/kg of dry air):

Point	$\Delta\hat{H}_{satd}$	δH	$\Delta\hat{H}_{actual}$
A	90.0	−0.5	89.5
B	143.3	−3.3	140.0

Also, at *A* the volume of the moist air is 0.91 m³/kg of dry air. Consequently, the heat added is $(Q = \Delta\hat{H})140.0 - 89.5 = 50.5$ kJ/kg of dry air.

$$\frac{50.5\ \text{kJ}}{\text{kg dry air}}\left|\frac{1\ \text{kg dry air}}{0.91\ \text{m}^3}\right. = 55.5\ \text{kJ/m}^3 \text{ initial moist air}$$

EXAMPLE 4.48 Cooling and Humidification

One way of adding moisture to air is by passing it through water sprays or air washers. See Fig. 4.48a. Normally, the water used is recirculated rather than wasted. Then, in the steady state, the water is at the adiabatic saturation temperature, which is the same as the wet-bulb

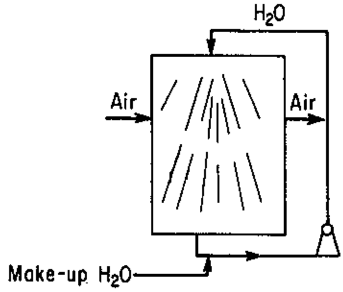


Figure E4.48a

temperature. The air passing through the washer is cooled, and if the contact time between the air and the water is long enough, the air will be at the wet-bulb temperature also. However, we shall assume that the washer is small enough so that the air does not reach the wet-bulb temperature; instead, the following conditions prevail:

	T_{DB}(°C)	T_{WB}(°C)
Entering air:	40	22
Exit air:	27	

Find the moisture added per kilogram of dry air.

Solution

The whole process is assumed to be *adiabatic*, and, as shown in Fig. E4.48b, takes place between points A and B along the adiabatic cooling line. The wet-bulb temperature remains constant at 22°C. Humidity values are

	$\mathcal{H}\left(\dfrac{\text{kg } H_2O}{\text{kg dry air}}\right)$
B	0.0145
A	0.0093
Difference:	$0.0052 \dfrac{\text{kg } H_2O}{\text{kg dry air}}$ added

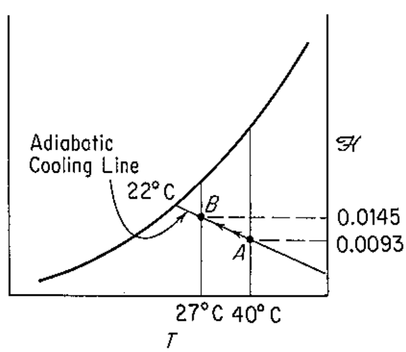


Figure E4.48b

EXAMPLE 4.49 Cooling and Dehumidification

A process that takes moisture out of the air by passing the air through water sprays sounds peculiar but is perfectly practical as long as the water temperature is below the dew point of the air. Equipment such as shown in Fig. E4.49a would do the trick. If the entering air has a dew point of 70°F and is at 40% $\mathcal{RH}$, how much heat has to be removed by the cooler, and how much water vapor is removed, if the exit air is at 56°F with a dew point of 54°F?

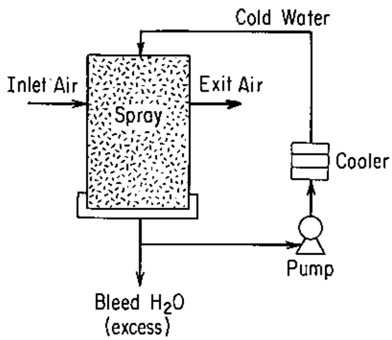


Figure E4.49a

Solution

From Fig. 4.26 in American engineering units the initial and final values of the enthalpies and humidities are

	A	B
$\mathcal{H}\left(\frac{\text{grains } H_2O}{\text{lb dry air}}\right)$	111	62
$\Delta\hat{H}\left(\frac{\text{Btu}}{\text{lb dry air}}\right)$	$41.3 - 0.2 = 41.1$	$23.0 - 0 = 23.0$

Look at Fig. E4.49b for the relative positions of A and B. The grains of H_2O removed are

$$111 - 62 = 49 \text{ grains/lb dry air}$$

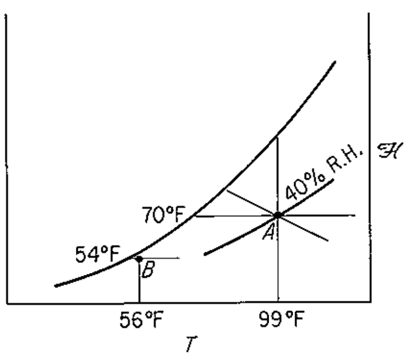


Figure E4.49b

The cooling duty is approximately

$$41.1 - 23.0 = 18.1 \text{ Btu/lb dry air}$$

In the upper left of the humidity chart is a little insert that gives the value of the small correction factor for the water condensed from the air which leaves the system. Assuming that the water leaves at the dew point of 54°F, read for 49 grains a correction of −0.15 Btu/lb of dry air. You could calculate the same value by taking the enthalpy of liquid water from the steam tables and saying,

$$\frac{22\text{ Btu}}{\text{lb } H_2O}\left|\frac{1\text{ lb } H_2O}{7000\text{ grains}}\right|\frac{49\text{ grains rejected}}{1\text{ lb dry air}} = 0.154\text{ Btu/lb dry air}$$

The energy (enthalpy) balance will then give us the cooling load:

$$\underbrace{41.1}_{\substack{\Delta H\\\text{air in}}} - \underbrace{23.0}_{\substack{\Delta H\\\text{air out}}} - \underbrace{0.15}_{\substack{\Delta H\\H_2O\text{ out}}} = 17.9\text{ Btu/lb dry air}$$

EXAMPLE 4.50 Combined Material and Energy Balances for a Cooling Tower

You have been requested to redesign a water-cooling tower that has a blower with a capacity of 8.30×10^6 ft³/hr of moist air (at 80°F and a wet-bulb temperature of 65°F). The exit air leaves at 95°F and 90°F wet bulb. How much water can be cooled in pounds per hour if the water to be cooled is not recycled, enters the tower at 120°F, and leaves the tower at 90°F?

Solution

Enthalpy, humidity, and humid volume data for the air taken from the humidity chart are as follows (see Fig. E4.50):

	A	B
$\mathcal{H}\left(\frac{\text{lb } H_2O}{\text{lb dry air}}\right)$	0.0098	0.0297
$\mathcal{H}\left(\frac{\text{grains } H_2O}{\text{lb dry air}}\right)$	69	208
$\Delta\hat{H}\left(\frac{\text{Btu}}{\text{lb dry air}}\right)$	30.05 − 0.12 = 29.93	55.93 − 0.10 = 55.83
$\hat{V}\left(\frac{\text{ft}^3}{\text{lb dry air}}\right)$	13.82	14.65

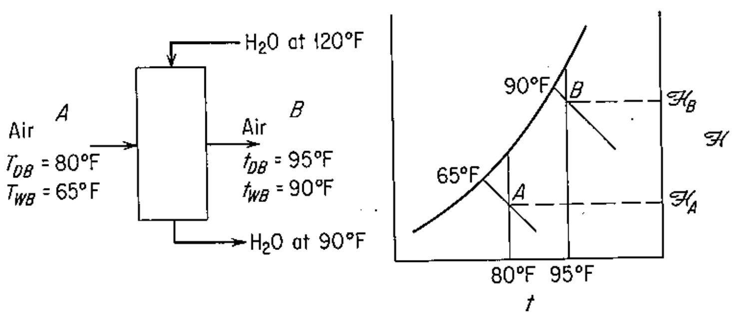


Figure E4.50

The cooling-water exit rate can be obtained from an energy balance around the process.

Basis: 8.30×10^6 ft^3/hr of moist air

$$\frac{8.30 \times 10^6 \text{ ft}^3}{\text{hr}} \left| \frac{\text{lb dry air}}{13.82 \text{ ft}^3} \right. = 6.00 \times 10^5 \text{ lb dry air/hr}$$

The relative enthalpy of the entering water stream is (reference temperature is 32°F and 1 atm)

$$\Delta\hat{H} = C_{p_{H_2O}}\,\Delta T = 1(120 - 32) = 88 \text{ Btu/lb } H_2O$$

and that of the exit stream is 58 Btu/lb H_2O. [The value from the steam tables at 120°F for liquid water of 87.92 Btu/lb H_2O is slightly different since it represents water at its vapor pressure (1.69 psia) based on reference conditions of 32°F and liquid water at its vapor pressure.] Any other datum could be used instead of 32°F for the liquid water. For example, if you chose 90°F, one water stream would not have to be taken into account because its relative enthalpy would be zero.

The loss of water to the air is

$$0.0297 - 0.0098 = 0.0199 \text{ lb } H_2O\text{/lb dry air}$$

(a) *Material balance for water stream:* Let W = lb H_2O entering the tower in the water stream per lb dry air. Then

$$W - 0.0199 = \text{lb } H_2O \text{ leaving tower in the water stream per lb dry air}$$

(b) *Energy balance (enthalpy balance) around the entire process:*

air and water in air entering **water stream entering**

$$\frac{29.93 \text{ Btu}}{\text{lb dry air}} \left| \frac{6.00 \times 10^5 \text{ lb dry air}}{} \right. + \frac{88 \text{ Btu}}{\text{lb } H_2O} \left| \frac{W \text{ lb } H_2O}{\text{lb dry air}} \right| \frac{6.00 \times 10^5 \text{ lb dry air}}{}$$

air and water in air leaving

$$= \frac{55.83 \text{ Btu}}{\text{lb dry air}} \left| \frac{6.00 \times 10^5 \text{ lb dry air}}{} \right.$$

water stream leaving

$$+ \frac{58 \text{ Btu}}{\text{lb } H_2O} \left| \frac{(W - 0.0199)\text{lb } H_2O}{\text{lb dry air}} \right| \frac{6.00 \times 10^5 \text{ lb dry air}}{}$$

$$29.93 + 88W = 55.83 + 58(W - 0.0199)$$

$$W = 0.825 \text{ lb } H_2O\text{/lb dry air}$$

$$W - 0.0199 = 0.805 \text{ lb } H_2O\text{/lb dry air}$$

The total water leaving the tower is

$$\frac{0.805 \text{ lb } H_2O}{\text{lb dry air}} \left| \frac{6.00 \times 10^5 \text{ lb dry air}}{\text{hr}} \right. = 4.83 \times 10^5 \text{ lb/hr}$$

Self-Assessment Test

1. What is the difference between the wet- and dry-bulb temperatures?
2. Can the wet-bulb temperature ever be higher than the dry-bulb temperature?
3. Explain why the slope of the wet-bulb lines are essentially the same as the slope of the adiabatic cooling lines for gaseous air and water mixtures.
4. Estimate for air at 70°C dry-bulb temperature, 1 atm, and 15% relative humidity the:
   (a) kg H_2O/kg of dry air
   (b) m^3/kg of dry air
   (c) Wet-bulb temperature (in °C)
   (d) Specific enthalpy
   (e) Dew point (in °C)
5. Calculate the following properties of moist air at 1 atm and compare with values read from the humidity chart.
   (a) The humidity of saturated air at 120°F
   (b) The enthalpy of air in part (a) per pound of dry air
   (c) The volume per pound of dry air of part (a)
   (d) The humidity of air at 160°F with a wet-bulb temperature of 120°F
6. Humid air at 1 atm and 200°F, and containing 0.0645 lb of H_2O/lb of dry air, enters a cooler at the rate of 1000 lb of dry air per hour (plus accompanying water vapor). The air leaves the cooler at 100°F, saturated with water vapor (0.0434 lb of H_2O/lb of dry air). Thus 0.0211 lb H_2O is condensed per pound of dry air. How much heat is transferred to the cooler?
7. A cooling tower that uses a cold-water spray provides a method of cooling and dehumidifying a school. During the day, the average number of students in the school is 100 and the average heat-generation rate per person is 800 Btu/hr. Suppose that the ambient conditions outside the school in the summer are expected to be 100°F and 95% $\mathcal{RH}$. You run this air through the cooler-dehumidifier and then mix the saturated exit air with recirculated air from the exhaust of the school building. You need to supply the mixed air to the building at 70°F and 60% $\mathcal{RH}$ and keep the recirculated air leaving the building at not more than 72°F. Leakage occurs from the building of the 72°F air also. Calculate:
   (a) The volumetric rate of air recirculation per hour in cubic feet at 70°F and 60% $\mathcal{RH}$
   (b) The volume of fresh air required at entering conditions
   (c) The heat transferred in the cooler-dehumidifier from the inlet air per hour

Thought Problems

1. The use of home humidifiers has recently been promoted in advertisements as a means of providing more comfort in houses with the thermostat turned down. "Humidification makes life more comfortable and prolongs the life of furniture." Many advertisers describe a humidifier as an energy-saving device because it allows lower temperatures (4 or 5°F lower) with comfort.

   Is this true?
2. In cold weather, water vapor exhausted from cooling towers condenses and fog is formed as a plume. What are one or two *economically practical* methods of preventing such cooling tower fog?

SUPPLEMENTARY REFERENCES

Concepts and Units

BRADSHAW, J. A., "Historical Background and Foundations of Thermodynamics," in *Thermodynamics,* Vol. 5D, p. 1, AIChE Modular Instruction Series, American Institute of Chemical Engineers, New York, 1984.

EDMISTER, W. C., *Applied Hydrocarbon Thermodynamics,* 2nd ed., Gulf Publishing Company, Houston, 1984.

LINDSAY, R. B., *Energy-Historical Development of the Concept,* Academic Press, New York, 1975.

MOTT-SMITH, M., *The Concept of Energy Simply Explained,* Dover, New York, 1964.

National Academy of Sciences, *Energy in Transition,* W.H. Freeman, San Francisco, 1980.

National Research Council, *Energy in Transition, 1985–2010,* W.H. Freeman, San Francisco, 1980.

STOBAUGH, R., and D. YERGIN, *Energy Future,* Random House, New York, 1979.

TAUBE, M., *Evolution of Matter and Energy,* Springer-Verlag, New York, 1985.

Heat Capacity

Department of Energy, *Coal Conversion Systems Technical Data Book,* U.S.D.O.E. Report HCP/T2286-01, 1978.

DOMALSKI, E. S., W. H. EVANS, and E. D. HEARING, *Heat Capacities and Entropies of Organic Compounds in the Condensed Phase,* American Chemical Society, Washington, D.C., 1984.

TOULOUKIAN, Y. S., and T. MAKITA, *Specific Heat,* Plenum, New York, 1977.

WERT, C. A., and R. M. THOMSON, *Physics of Solids,* McGraw-Hill, New York, 1964.

YAWS, C.L., H.M., NI, and P.Y. CHANG, "Heat Capacities for 700 Compounds," *Chem. Eng.*, p. 91 (May 9, 1988).

Enthalpy Calculations and Properties

BRULÉ, M. R., L. L. LEE, and K. E. STARLING, "Predicting Thermodynamic Properties for Fossil-Fuel Chemicals," *Chem. Eng.*, p. 155 (November 19, 1979).

FRATZSCHER, W., H.-P. PICHT, and J. H. BITTRICH, "The Acquisition, Collection, and Tabulation of Substance Data on Fluid Systems for Calculations in Chemical Engineering," *Int. Chem. Eng.*, v. 20, p. 19 (1980).

MCMULLAN, J. T., D. W. HUGHES, and R. MORGAN, "A Suite of Computer Programs for Calculating Refrigerant Properties," *J. Heat Recovery Syst.*, v. 5, p. 143 (1985).

RIEMER, D. H., H. R. JACOBS, and R. F. BOEHM, *Computer Programs for Determining the Thermodynamic Properties,* IDO/1549-1, -2, -3, -4, National Technical Information Service, Springfield, Va., 1978.

SCHMIDT, E., *Properties of Water and Steam in SI Units,* Springer-Verlag, New York, 1979.

Enthalpy for Phase Change

EDMONDS, B., "Enthalpies of Vaporization," *Chem. Eng.*, p. 109 (February 1979); p. 179 (March 1979); p. 357 (May 1979).

TORQUATO, S., and P. SMITH, "The Latent Heat of Vaporization of a Widely Diverse Class of Fluids," *Trans. ASME,* v. 106, p. 252 (1984).

Energy Balance

BRUSH, S. G., *A History of Heat and Thermodynamics*, North-Holland, Amsterdam, 1976.

ERLICHSON, H., "Internal Energy in the First Law of Thermodynamics," *Am. J. Phys.*, v. 52, p. 623 (1984).

RAMAN, V. V., "Where Credit Is Due—The Energy Conservation Principle," *Phys. Teacher*, p. 80 (February 1975).

SMITH, J. M., and H. C. VAN NESS, *Introduction to Chemical Engineering Thermodynamics*, 4th ed., McGraw-Hill, New York, 1987.

Energy Balances with Reaction

JANAF Thermochemical Tables and supplements 1–33 available from U.S. Government Printing Office, Cat. No. C13.48:37; supplements 34–52 available from American Chemical Society, Washington, D.C. (various dates). Also magnetic tape available.

KARLSEN, L. G., and J. VILLADSEN, "Isothermal Reaction Calorimeters—I. A Literature Review," *Chem. Eng. Sci.*, v. 42, 1153 (1987).

NELSON, R., "Material Properties in SI Units, Part 3," *Chem. Eng. Prog.*, p. 86 (April 1980).

STULL, D. E., E. F. WESTRUM, and G. C. SINKE, *The Chemical Thermodynamics of Organic Compounds*, Wiley, New York, 1969.

YAWS, C.L., and P.Y. CHANG, "Enthalpy of Formation for 700 Major Organic Compounds," *Chem. Eng.*, p. 81 (September 26, 1988).

Heat of Mixing and Enthalpy–Concentration Charts

CHRISTENSEN, J. J., R. W. HANKS, and R. M. IZATT, *Handbook of Heats of Mixing*, Wiley, New York, 1982.

CHRISTENSEN, C., J. GMEHLING, and P. RASMUSSEN, *Heats of Mixing Data Collection*, Dechema, Frankfurt, West Germany, 1984.

JOHNSON, J. E., and D. J. MORGAN, "Graphical Techniques for Process Engineering," *Chem. Eng.*, p. 72 (July 8, 1985).

KEHIAIAN, H. V., ed., *International Data, Series A*, Texas A & M University, College Station, Tex. (annually).

SMITH, B. D., et al., "Critical Evaluation of Vapor–Liquid Equilibrium, Heat of Mixing, and Volume Change of Mixing Data," *J. Phys. Chem. Ref. Data*, v. 11, p. 941 (1982).

STEPHAN, K., and H. HILDWEIN, *Recommended Data of Selected Compounds and Binary Mixtures*, Dechema, Frankfurt, West Germany, 1987.

Psychrometric Charts

BULLOCK, C. E., "Psychrometric Tables," in *ASHRAE Handbook Product Directory*, paper No. 6, American Society of Heating, Refrigeration, and Air Conditioning Engineers, Atlanta, 1977.

MCMILLAN, H. K., and J. KIM, "A Computer Program to Determine Thermodynamic Properties of Moist Air," *Access*, p. 36 (January 1986).

NELSON, R., "Material Properties in SI Units, Part 4," *Chem. Eng. Prog.*, p. 83 (May 1980).

TREYBAL, R. E., *Mass Transfer Operations*, McGraw-Hill, New York, 1980.

WILHELM, L. R., "Numerical Calculation of Psychrometric Properties in SI Units," *Trans. ASAE*, v. 19, p. 318 (1976).

PROBLEMS

An asterisk designates problems appropriate for solution using a computer. Refer also to the problems that require writing computer programs at the end of the chapter.

Section 4.1

4.1. Convert 45.0 Btu/lb_m to the following:
(a) cal/kg (b) J/kg
(c) kWh/kg (d) $(ft)(lb_f)/lb_m$

4.2. Convert the following physical properties of liquid water at 0°C and 1 atm from the given SI units to the equivalent values in the listed American engineering units.

(a) Heat capacity of 4.184 J/(g)(K)	Btu/(lb)(°F)
(b) Enthalpy of −41.6 J/kg	Btu/lb
(c) Thermal conductivity of 0.59 $(kg)(m)/(s^3)(K)$	Btu/(ft)(hr)(°F)

4.3. Convert the following quantities as specified.
(a) A rate of heat flow of 6000 $Btu/(hr)(ft^2)$ to $cal/(s)(cm^2)$.
(b) A heat capacity of 2.3 Btu/(lb)(°F) to cal/(g)(°C)
(c) A thermal conductivity of 200 Btu/(hr)(ft)(°F) to cal/(s)(cm)(°C).
(d) The gas constant, 10.73 $(psia)(ft^3)/(lb\ mol)(°R)$ to cal/(g mol)(K).

4.4. A simplified equation for the heat transfer coefficient from a pipe to air is

$$h = \frac{0.026G^{0.6}}{D^{0.4}}$$

where h = heat transfer coefficient, $Btu/(hr)(ft^2)(°F)$
G = mass rate of flow, $lb_m/(hr)(ft^2)$
D = outside diameter of the pipe, (ft)

If h is to be expressed in $J/(min)(cm^2)(°C)$, what should the new constant in the equation be in place of 0.026?

4.5. A problem for many people in the United States is excess body weight stored as fat. Many persons have tried to capitalize on this problem with fruitless weight-loss schemes. However, since energy is conserved, an energy balance reveals only two real ways to lose weight (other than water loss); (1) reduce the caloric intake, and/or (2) increase the caloric expenditure. In answering the following questions, assume that fat contains approximately 7700 kcal/kg (1 kcal is called a "dietic calorie" in nutrition, or commonly just "calorie").
(a) If a normal diet containing 2400 kcal/day is reduced by 500 kcal/day, how many days does it take to lose 1 lb of fat?
(b) How many miles would you have to run to lose 1 lb of fat if running at a moderate pace of 12 km/hr expends 400 kJ/km?
(c) Suppose that two joggers each run 10 km/day. One runs at a pace of 5 km/hr and the other at 10 km/hr. Which will lose more weight (ignoring water loss)?

4.6. The energy from the sun incident on the surface of the earth averages 2.0 cal/(min) (cm^2). It has been proposed to use space stations in synchronous orbits 36,000 km from earth to collect solar energy. How large a collection surface is needed (in m^2) to obtain 10^{11} watts of electricity? Assume that 10% of the collected energy is converted to electricity. Is this a reasonable size?

4.7. Explain specifically what the system is for each of the following processes; indicate if the energy transfer that takes place is by heat or work (use the symbols Q and W, respectively) or is zero.

(a) A liquid inside a metal can, well insulated on the outside of the can, is shaken very rapidly in a vibrating shaker.

(b) Hydrogen is exploded in a calometric bomb and the water layer outside the bomb rises in temperature by 1°C.

(c) A motor boat is driven by an outboard-motor propeller.

(d) Water flows through a pipe at 1.0 m/min, and the temperature of the water and the air surrounding the pipe are the same.

4.8. Draw a simple sketch of each of the following processes, and, in each, label the system boundary, the system, the surroundings, and the streams of material and energy that cross the system boundary.

(a) Water enters a boiler, is vaporized, and leaves as steam. The energy for vaporization is obtained by combustion of a fuel gas with air outside the boiler surface.

(b) Steam enters a rotary steam turbine and turns a shaft connected to an electric generator. The steam is exhausted at a low pressure from the turbine.

(c) A battery is charged by connecting it to a source of current.

4.9. Draw a simple sketch of the following processes; indicate the system boundary; and classify the system as open or closed.

(a) Automobile engine
(b) Water wheel
(c) Pressure cooker
(d) Human being
(e) River
(f) The earth and its atmosphere
(g) Air compressor
(h) Coffee pot

4.10. Are the following variables intensive or extensive variables? Explain for each.

(a) Pressure
(b) Volume
(c) Specific volume
(d) Refractive index
(e) Surface tension

4.11. Classify the following measurable physical characteristics of a gaseous mixture of two components as (1) an intensive property, (2) an extensive property, (3) both, or (4) neither:

(a) Temperature
(b) Composition
(c) Pressure
(d) Mass

4.12. Find the kinetic energy in (ft)(lb_f)/(lb_m) of water moving at the rate of 10 ft/s through a pipe 2 in. ID.

4.13. A windmill converts the kinetic energy of the moving air into electrical energy at an efficiency of about 30%, depending on the windmill design and speed of the wind. Estimate the power in kW for a wind flowing perpendicular to a windmill with blades 15 m in diameter when the wind is blowing at 20 mi/hr at 27°C and 1 atm.

4.14. Before it lands, a vehicle returning from space must convert its enormous kinetic energy to heat. To get some idea of what is involved, a vehicle returning from the moon at 25,000 mi/hr can, in converting its kinetic energy, increase the internal energy of the vehicle sufficiently to vaporize it. Obviously, a large part of the total kinetic energy must be transferred from the vehicle. How much kinetic energy does the vehicle have (in Btu)? How much energy must be transferred by heat if the vehicle is to heat up only 20°F/lb?

4.15. If benzene (sp gr = 0.879) flows at 4630 L/hr with a pressure differential of 3450 kPa, what is the power requirement in watts of the pump (if 100% efficient)?

4.16. If benzene (sp gr = 0.879) flows at 1220 gal/hr and a pressure differential of 500 psia, what is the power requirement of the pump in hp (if 100% efficient)?

4.17. You have a positive-displacement pump with a piston area of 1 m^2. The piston moves 4.63 m/hr to pump 4.63 m^3/hr of fluid. The required force will be 3450 kN. What is the power needed in kW?

4.18. Benzene (sp gr = 0.879) flows through an orifice that is 0.8 cm in diameter at the rate of 4.63 m^3/hr. What is the kinetic energy of the fluid in J/kg?

4.19. You pump benzene at the rate of 1220 gal/hr through an orifice that is 0.315 in.^{2} in diameter. What is the kinetic energy of the fluid in Btu/lb_m?

4.20. Constant advances in laser technology are aiding ongoing research in nuclear fusion at the Lawrence Livermore National Laboratory, California. Lasers are being used to heat deuterium–tritium fuel pellets, causing the surface to explode. The explosion heats and compresses the interior of the pellet, which causes fusion to take place. The laboratory has acquired a new laser system which combines the power of several lasers and has a maximum capability of delivering an energy pulse of 200 to 300 kJ (depending on the frequency of the light) in just one billionth of a second. What is the equivalent range of power expressed in watts?

4.21. The world's largest plant that obtains energy from tidal changes is at Saint Malo, France. The plant uses both the rising and falling cycle (one period in and out is 6 hr 10 min in duration). The tidal range from low to high is 14 m, and the tidal estuary (the LaRance River) is 21 km long with an area of 23 km^2. Assume that the efficiency of the plant in converting potential to electrical energy is 85%, and estimate the average power produced by the plant. (*Note:* Also assume that after high tide, the plant does not release water until the sea level drops 7 m, and after a low tide does not permit water to enter the basin until the level outside the basin rises 7 m, and the level differential is maintained during discharge and charge.)

4.22. Steam is used to cool a polymer reaction. The steam in the steam chest of the apparatus is found to be at 250.5°C and 4000 kPa absolute during a routine measurement at the beginning of the day. At the end of the day the measurement showed that the temperature was 650°C and the pressure 10,000 kPa absolute. What was the internal energy change of 1 kg of steam in the chest during the day? Obtain your data from the steam tables.

4.23. **(a)** Ten pound moles of an ideal gas are originally in a tank at 100 atm and 40°F. The gas is heated to 440°F. The specific molal enthalpy, $\Delta\hat{H}$, of the ideal gas is given by the equation

$$\Delta\hat{H} = 300 + 8.00T$$

where $\Delta\hat{H}$ is in Btu/lb mol and T is the temperature in °F.

(1) Compute the volume of the container (ft^3).

(2) Compute the final pressure of the gas (atm).

(3) Compute the enthalpy change of the gas.

(b) Use the equation above to develop an equation giving the molal internal energy, ΔU, in cal/g mol as a function of temperature, T, in °C.

4.24. You have calculated that specific enthalpy of 1 kg mol of an ideal gas at 300 kN/m^2 and 100°C is 6.05×10^5 J/kg mol (with reference to 0°C and 100 kN/m^2). What is the specific internal energy of the gas?

4.25. One hundred and fifty pounds of CO_2 has a specific enthalpy of 45,000 (ft)(lb_f)/lb_m with reference to 0°F and 1 psia. What is the enthalpy of this CO_2 in Btu/lb mol?

What is the total internal energy if the pressure is 6 atm and the temperature is 50°F?

4.26. One kilogram mole of steam at 150°C and 150 kPa is expanded to 210°C and 100 kPa. Thereafter it is condensed to water at 25°C and 100 kPa, and finally heated back to 150°C and 150 kPa. What is the overall **(a)** enthalpy change and **(b)** internal energy change per kilogram mole of steam for the process?

4.27. State which of the following variables are point, or state, variables, and which are not; explain your decision in one sentence for each one:

(a) Pressure **(b)** Density
(c) Molecular weight **(d)** Heat capacity
(e) Internal energy **(f)** Ionization constant

Section 4.2

4.28.* Experimental values calculated for the heat capacity of ammonia from −40 to 1200°C are:

T (°C)	C_p° [cal/(g mol)(°C)]	T (°C)	C_p° [cal/(g mol)(°C)]
−40	8.180	500	12.045
−20	8.268	600	12.700
0	8.371	700	13.310
18	8.472	800	13.876
25	8.514	900	14.397
100	9.035	1000	14.874
200	9.824	1100	15.306
300	10.606	1200	15.694
400	11.347		

Fit the data by least squares for the following two functions:

$$\left.\begin{aligned} C_p^\circ &= a + bT + cT^2 \\ C_p^\circ &= a + bT + cT^2 + dT^3 \end{aligned}\right\} \quad \text{where } T \text{ is in °C}$$

4.29. Experimental values of the heat capacity C_p have been determined in the laboratory as follows; fit a second-order polynomial in temperature to the data ($C_p = a + bT + cT^2$):

T (°C)	C_p [J/(g mol)(°C)]
100	40.54
200	43.81
300	46.99
400	49.33
500	51.25
600	52.84
700	54.14

(The data are for carbon dioxide.)

4.30. A laboratory analysis from No. 46 well yields the following composition by volume for a product gas.

Component	Vol. %	Component	Vol. %
CH_4	90.05	N_2	5.518
C_2H_6	1.984	O_2	1.467
C_3H_8	0.202	H_2	0.31
C_4H_{10}	0.069	CO_2	0.662
C_5H_{12}	0.018		

What is the heat capacity (in Btu/(lb mol)(°F) of this gas at room temperature?

4.31. In a computer program you find that the relation for the specific enthalpy (J/g mol) of a gas is given by

$$\hat{H} = 32.97T + 6.694 \times 10^{-3}T^2 - 6.745 \times 10^{-7}T^3 + 5.481 \times 10^{-5}p$$

where T is in °C and p is in kPa, and the reference state is unspecified.

(a) Determine the value of the heat capacity C_p of the gas in J/(kg mol)(°C) at 100°C and 100 kPa.

(b) What is a function that expresses the specific internal energy (for the same reference state) assuming that the gas acts as an ideal gas.

(c) Determine the value of the heat capacity C_v of the gas in J/(kg mol)(°C) at 100°C and 100 kPa.

4.32. An equation for the heat capacity of acetone vapor is

$$C_p = 71.96 + 20.10 \times 10^{-2}T - 12.78 \times 10^{-5}T^2 + 34.76 \times 10^{-9}T^3$$

where C_p is in J/(g mol)(°C) and T is in °C. Convert the equation so that C_p is in Btu/(lb mol)(°F) and T is in °F.

4.33. Your assistant has developed the following equation to represent the heat capacity of air (with C_p in cal/(g mol) (K) and T in K):

$$C_p = 6.39 + 1.76 \times 10^{-3}T - 0.26 \times 10^{-6}T^2$$

(a) Derive an equation giving C_p but with the temperature expressed in °C.

(b) Derive an equation giving C_p in terms of Btu per pound per degree Fahrenheit with the temperature being expressed in degrees Fahrenheit.

4.34. To evaluate the suitability of Kopp's rule for the heat capacities of solids, compute the heat capacities of sodium sulfate (Na_2SO_4), dextrose ($C_6H_{12}O_6$), and copper ammonium sulfate [$CuSO_4(NH_4)_2\ SO_4 \cdot 6H_2O$] and compare your values with the experimental ones at 25°C.

4.35. One hundred pounds of a 35°API distillate with an average boiling point of 250°F is cooled from 400°F to 300°F. Estimate the heat capacity of the distillate at each temperature using the equation in the text and the charts in Appendix K.

4.36. Estimate the heat capacity of gaseous isobutane at 1000 K and 200 mm Hg by using the Kothari–Doraiswamy relation

$$C_p = A + B \log_{10} T_r$$

from the following experimental data at 200 mm Hg:

C_p[cal/(K)(g mol)]	23.25	35.62
Temp. (K)	300	500

The experimental value is 54.40; what is the percentage error in the estimate?

Section 4.3

4.37. The heat capacity of carbon monoxide is given by the following equation:

$$C_p = 6.935 + 6.77 \times 10^{-4}T + 1.3 \times 10^{-7}T^2$$

where C_p = cal/(g mol)(°C)
T = °C

What is the enthalpy change associated with heating carbon monoxide from 500°C to 1000°C?

4.38. Two gram moles of nitrogen are heated from 50°C to 250°C in a cylinder. What is ΔH for the process? The heat capacity equation is

$$C_p = 27.32 + 0.6226 \times 10^{-2}T - 0.0950 \times 10^{-5}T^2$$

where T is in kelvin and C_p is in J/(g mol)(°C).

4.39. Calculate the enthalpy change (in J/kg mol) that takes place in raising the temperature of 1 kg mol of the following gas mixture from 50°C to 550°C.

Component	Mol %
CH_4	80
C_2H_6	20

4.40. Can you find the enthalpy change at constant pressure of substance such as CO_2 from the solid to the gaseous state by integrating $\int_{T_1}^{T_2} C_p\,dT$ from T_1 (the solid temperature) to T_2 (the gas temperature) for a constant-pressure path?

4.41.* Calculate the change in enthalpy for 5 kg mol of CO which is cooled from 927°C to 327°C using the program ENTH on the disk included with the book.

4.42.* Hydrogen sulfide is heated from 77°C to 227°C. What is the enthalpy change due to the heating? Use the program ENTH on the disk at the back of the book.

4.43.* What is the enthalpy change for acetylene when heated from 37.8°C to 93.3°C? Use the program ENTH on the disk in the back of the book.

4.44.* Use the steam tables to calculate the enthalpy change (in joules) of 2 kg mol of steam when heated from 400 K and 100 kPa to 900 K and 100 kPa. Repeat using the table in the text for the enthalpies of combustion gases. Repeat using the heat capacity for steam. Compare your answers. Which is most accurate?

4.45. A closed vessel contains steam at 1000.0 psia in a 4-to-1 vapor volume-to-liquid volume ratio. What is the steam quality?

4.46. Use the tables for the heat capacities of the combustion gases to compute the enthalpy change (in Btu) that takes place when a mixture of 6.00 lb mol of gaseous H_2O and 4.00 lb mol of CO_2 is heated from 60°F to 600°F.

4.47. Chemical vapor deposition (CVD) is an important technique in producing solid-state materials because of its potential application to a variety of materials with controlled properties. Although this method has advantages, the film formation rates are so low, usually 0.1 to 1.0 nm/s, that the application of this method has been limited to date to very expensive materials.

In one trial to test film growth the temperature was controlled by use of steam on the underside of the film surface in the reactor assembly. Titanium tetraisopropoxide [$Ti(OC_3H_7)_4$] was carried by a helium stream over the condensing surface, where it decomposed to TiO_2 plus gases.

The steam entered the reactor at 523 K and 130 kPa and exited at 540 K and 100 kPa. What was the internal energy change per kg of steam going through the reactor? Use the steam tables.

4.48. In a proposed molten-iron coal gasification process (*Chemical Engineering*, p. 17, July 1985), pulverized coal of up to 3 mm size is blown into a molten iron bath, and oxygen and steam are blown in from the bottom of the vessel. Materials such as lime for settling the slag, or steam for bath cooling and hydrogen generation, can be injected at the same time. The sulfur in the coal reacts with lime to form calcium sulfide, which dissolves into the slag. The process operates at atmospheric pressure and 1400 to 1500°C. Under these conditions, coal volatiles escape immediately and are cracked. The carbon conversion rate is said to be above 98%, and the gas is typically 65 to 70% CO, 25 to 35% H_2, and less than 2% CO_2. Sulfur content of the gas is less than 20 ppm.

Assume that the product gas is 68% CO, 30% H_2, and 2% CO_2, and calculate the enthalpy change that occurs on the cooling of 1000 m^3 of gaseous product from 1400°C to 25°C at 101 kPa. Use the table for the enthalpies of the combustion gases.

4.49. Use the CO_2 chart in the following calculations.

(a) Four pounds of CO_2 are heated from saturated liquid at 20°F to 600 psia and 180°F.

(1) What is the specific volume of the CO_2 at the final state?

(2) Is the CO_2 in the final state gas, liquid, solid, or a mixture of two or three phases?

(b) The 4 lb of CO_2 is then cooled at 600 psia until the specific volume is 0.07 ft^3/lb.

(1) What is the temperature of the final state?

(2) Is the CO_2 in the final state gas, liquid, solid, or a mixture of two or three phases?

4.50. The heat capacity of chlorine has been determined experimentally as follows:

T (°C)	C_p [cal/(g mol)(K)]
0	8.00
18	8.08
25	8.11
100	8.36
200	8.58
300	8.71
400	8.81
500	8.88
600	8.92
700	8.95

T (°C)	C_p [cal/(g mol)(K)]
800	8.98
900	9.01
1000	9.03
1100	9.05
1200	9.07

Calculate the enthalpy change required to raise 1 g mol of chlorine from 0°C to 1200°C.

Section 4.4

4.51. Water can exist in various states, depending on its temperature, pressure, and so on. From what you have learned in Sec. 3.7, what is the state (gaseous, liquid, solid, or combinations thereof) for water at the following conditions:
(a) 250°F and 1 atm
(b) −10°C and 720 mm Hg
(c) 32°F and 100 psia

4.52. Calculate the enthalpy change (in joules) that occurs when 1 kg of benzene vapor at 150°C and 100 kPa condenses to a solid at −20.0°C and 100 kPa.

4.53. The vapor pressure of zinc in the range 600 to 985°C is given by the equation

$$\log_{10} p = -\frac{6160}{T} + 8.10$$

where p = vapor pressure, mm Hg
T = temperature, K

Estimate the latent heat of vaporization of zinc at its normal boiling point of 907°C.

4.54. After graduation you accept a job as a chemical engineer at the Muleshoe Natural Gasoline Co. The laboratory technician brings you the following data for n-butane (C_4H_{10}) made at 30°C. Are these data internally consistent or not? Explain your answer in complete detail.

Data: Liquid specific gravity (water = 1.00 at 4°C) is 0.564
Vapor specific gravity (air = 1.00 at 0°C and 1 atm) is 6.50
Latent heat of vaporization is 85.2 cal/g
Vapor pressure relation (p is in atm and T is in K)

$$\log_{10} p = 1.767 - \frac{1337}{T} + 1.75 \log_{10} T - 0.004\,T$$

4.55. The vapor pressure of phenylhydrazine has been found to be (from 25° to 240°C)

$$\log_{10} p^* = 7.9046 - \frac{2366.4}{T + 230}$$

where p^* is in atm and T is in °C. What is the heat of vaporization of phenylhydrazine in Btu/lb at 200°F? Use the Clausius–Clapeyron equation.

4.56. Check the precision of the empirical relation proposed by Watson to calculate heats of vaporization. Take water as a test example. With boiling water at 100°C as the reference temperature, estimate the heat of vaporization of water at 300°C from Watson's relation. Compare your result with the actual value from the steam tables

4.57.* Make an Othmer plot for refrigerant 12, and calculate the latent heat of vaporization at 100°F; compare this value with the one calculated from the following experimental data:

T (°F)	p^* (psia)	$V_{(l)}$ (ft³/lb)	$V_{(g)}$ (ft³/lb)
86	108.04	0.01240	0.37657
90	114.49	0.01248	0.35529
100	131.86	0.01269	0.30794
120	172.35	0.01317	0.23326
140	221.32	0.01375	0.17799

The experimental value is 55.93 Btu/lb.

4.58. Make an Othmer plot to determine the heat of vaporization of ____________ at ____________ °C. Compare with an experimental value if one can be found in a handbook.

4.59. (a) Using the following data on the vapor pressure of Cl_2, plot $\log p^*$ against $1/T$, and calculate the latent heat of vaporization of Cl_2 as a function of temperature:

T (°F)	p^* (psia)
−22	17.8
32	53.5
86	126.4
140	258.5
194	463.5
248	771
284	1050

How satisfactory are these data?

(b) The vapor pressure of Cl_2 is given by Lange by the equation

$$\log p^* = A - \frac{B}{C + T}$$

where p^* is the vapor pressure in mm Hg, T is in °C, and for chlorine, $A = 6.86773$, $B = 821.107$, and $C = 240$. Calculate the latent heat of vaporization of chlorine at the normal boiling point (in Btu/lb). Compare with the experimental value.

4.60. Use of the steam tables:

(a) What is the enthalpy change needed to change 3 lb of liquid water at 32°F to steam at 1 atm and 300°F?

(b) What is the enthalpy change needed to heat 3 lb of water from 60 psia and 32°F to steam at 1 atm and 300°F?

(c) What is the enthalpy change needed to heat 1 lb of water at 60 psia and 40°F to steam at 300°F and 60 psia?

(d) What is the enthalpy change needed to change 1 lb of a water-steam mixture of 60% quality to one of 80% quality if the mixture is at 300°F?
(e) Calculate the ΔH value for an isobaric (constant pressure) change of steam from 120 psia and 500°F to saturated liquid.
(f) Repeat part (e) for an isothermal change to saturated liquid.
(g) Does an enthalpy change from saturated vapor at 450°F to 210°F and 7 psia represent an enthalpy increase or decrease? a volume increase or decrease?
(h) In what state is water at 40 psia and 267.24°F? At 70 psia and 302°F? At 70 psia and 304°F?
(i) A 2.5-ft^3 tank of water at 160 psia and 363.5°F has how many cubic feet of liquid water in it? Assume that you start with 1 lb of H_2O. Could it contain 5 lb of H_2O under these conditions?
(j) What is the volume change when 2 lb of H_2O at 1000 psia and 200°F expands to 245 psia and 460°F?
(k) Ten pounds of wet steam at 100 psia has an enthalpy of 9000 Btu. Find the quality of the wet steam.

Section 4.5

4.61. A cylinder contains 1 lb of steam at 600 psia and a temperature of 500°F. It is connected to another equal-sized cylinder which is evacuated. A valve between the cylinders is opened. If the steam expands into the empty cylinder, and the final temperature of the steam in both cylinders is 500°F, calculate Q, W, ΔU, and ΔH for the system comprised of both cylinders.

4.62. One pound of steam at 130 psia and 600°F is expanded isothermally to 75 psia in a closed system. Thereafter it is cooled at constant volume to 60 psia. Finally, it is compressed adiabatically to back to its original state. For each of the three steps of the process, compute ΔU and ΔH. For each of the three steps, where possible, also calculate Q and W.

4.63. Two 1-m^3 tanks submerged in a constant-temperature water bath of 77°C are connected by a globe valve. One tank contains steam at 40 kPa, while the second is completely evacuated. The valve is opened and the pressure in the two tanks equilibrates. Calculate:
(a) The work done in the expansion of the steam
(b) The heat transferred to the tanks from the water bath
(c) The change in internal energy of the combined steam system

4.64. You have 0.37 lb of CO_2 at 1 atm abs and 100°F contained in a cylinder closed by a piston. The CO_2 is compressed to 70°F and 1000 psia. During the process, 40.0 Btu of heat is removed. Compute the work required for the compression.

4.65. A system consists of 25 lb of water vapor at the dew point. The system is compressed isothermally at 400°F, and 988 Btu of work are done on the system by the surroundings. What volume of liquid was present in the system before and after compression?

4.66. Steam fills tank 1 at 1000 psia and 700°F. The volume of tank 1 equals the volume of tank 2. A vacuum exists in tank 2 initially. See Fig. P4.66. The valve connecting

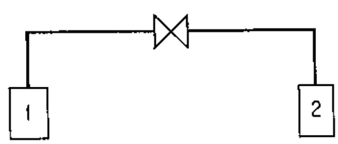


Figure P4.66

the two tanks is opened, and isothermal expansion of the steam occurs from tank 1 to tank 2. Find the enthalpy change per pound of steam for the entire process.

4.67. (a) A large piston does 12,500 (ft)(lb_f) of work in compressing 3 ft^3 of air to 25 psia. Five pounds of water circulates into and out of the water jacket around the piston and increases in temperature 2.3°F by the end of the process. What is the change of internal energy of the air?

(b) Suppose that the water does not circulate but is stationary and increases in temperature the same amount. What is the change now in the internal energy of the air?

(c) Suppose that the piston after compressing the gas returns to its original position, and all the 12,500 (ft)(lb_f) of work is recovered. What will the water temperature be in part (b)?

4.68. A household freezer is placed inside an insulated sealed room. If the freezer door is left open with the freezer operating, will the temperature of the room increase or decrease? Explain your answer.

4.69. One pound of water at 70°F is put into a closed tank of constant volume. If the temperature is brought to 80°F by heating, find Q, W, ΔU, and ΔH for the process (in Btu). If the same temperature change is accomplished by a stirrer that agitates the water, find Q, W, ΔE, and ΔH. What assumptions are needed?

4.70. On a winter day (38°F or 3°C outside average temperature) a "standard" house loses heat at the following rates:

(1) 2.1 kW is lost through partially insulated walls and roof by conductance,
(2) 0.3 kW is lost through the floor by conductance, and
(3) 1.9 kW is lost through the windows (no storm windows) by conductance.

Energy is also needed at the following rates:

(1) 2.3 kW to heat the air entering the house through cracks, flues, etc. (infiltration losses) and
(2) 1.1 kW to humidify the incoming air (because warm air must contain more water vapor than cold air for people to be comfortable).

On the same winter day, some energy is supplied to the house in the following amounts (on a day with average cloudiness):

(1) Sunlight through windows	0.5 kW
(2) People's heat transfer	0.2 kW
(3) Appliances' heat transfer	1.2 kW

If the house temperature is to be maintained at a constant value, what is the rate of energy that must be provided by the heating system in the house?

Suppose that the following design changes are made in the house:

(1) Added insultation of walls, roof, and floors cuts the losses incurred there by 60%.
(2) Tightly fitting double-glazed windows with selective coatings to reduce the passage of infrared light, or with special shutters, cut conductance losses by 70%.

(3) Elimination of cracks, closing of flues, and so on, cuts infiltration losses by 70%.

Now what is the total rate at which energy is lost from this house, all other factors remaining the same?

4.71. In the vapor-recompression evaporator shown in Fig. P4.71, the vapor produced on evaporation is compressed to a higher pressure and passed through the heating coil to provide the energy for evaporation. The steam entering the compressor is 98% vapor and 2% liquid, at 10 psia; the steam leaving the compressor is at 50 psia and 400°F; and 6 Btu of heat are lost from the compressor per pound of steam throughout. The condensate leaving the heating coil is at 50 psia, 200°F.

(a) Compute: The Btu of heat supplied for evaporation in the heating coil per Btu of work needed for compression by the compressor.

(b) If 1,000,000 Btu per hour of heat is to be transferred in the evaporator, what must be the intake capacity of the compressor in ft^3 of wet vapor per minute?

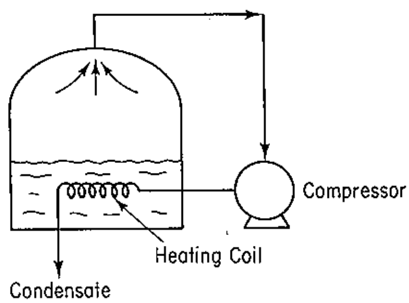


Figure P4.71

4.72. A 55-gal drum of fuel oil (15°API) is to be heated from 70°F to 180°F by means of an immersed steam coil. Steam is available at 220°F. How much steam is required if it leaves the coil as liquid water at 1 atm pressure?

4.73. Oil of average C_p = 0.8 Btu/(lb)(°F) flows at 2000 lb/min from an open reservoir standing on a hill 1000 ft high into another open reservoir at the bottom. To ensure rapid flow, heat is put into the pipe at the rate of 100,000 Btu/hr. What is the enthalpy change in the oil per pound? Suppose that a 1-hp pump (50% efficient) is added to the pipeline to assist in moving the oil. What is the enthalpy change per pound of the oil now?

4.74. Write the simplified energy balances for the following changes:

(a) A fluid flows steadily through a poorly designed coil in which it is heated from 70°F to 250°F. The pressure at the coil inlet is 120 psia, and at the coil outlet is 70 psia. The coil is of uniform cross section, and the fluid enters with a velocity of 2 ft/sec.

(b) A fluid is expanded through a well-designed adiabatic nozzle from a pressure of 200 psia and a temperature of 650°F to a pressure of 40 psia and a temperature of 350°F. The fluid enters the nozzle with a velocity of 25 ft/sec.

(c) A turbine directly connected to an electric generator operates adiabatically. The working fluid enters the turbine at 1400 kPa absolute and 340°C. It leaves the turbine at 275 kPa absolute and at a temperature of 180°C. Entrance and exit velocities are negligible.

(d) The fluid leaving the nozzle of part (b) is brought to rest by passing through the blades of an adiabatic turbine rotor and leaves the blades at 40 psia and at 400°F.

4.75. Your company produces small power plants that generate electricity by expanding waste process steam in a turbine. One way to ensure good efficiency in turbine operation is to operate adiabatically. For one turbine, measurements showed that for 1000 lb/hr steam at the inlet conditions of 500°F and 250 psia, the work output from the turbine was 86.5 hp and the exit steam leaving the turbine was at 14.7 psia with 15% wetness (i.e., with a quality of 85%).

Check whether the turbine is operating adiabatically by calculating the value of Q.

4.76. Feedwater heaters are used to increase the efficiency of steam power plants. A particular heater is used to preheat 10 kg/s of boiler feed water from 20°C to 188°C at a pressure of 1200 kPa by mixing it with saturated steam bled from a turbine at 1200 kPa, as shown in Fig. P4.76. Although insulated, the heater loses heat at the rate of 50 J per gram of exiting mixture. What fraction of the exit stream is steam?

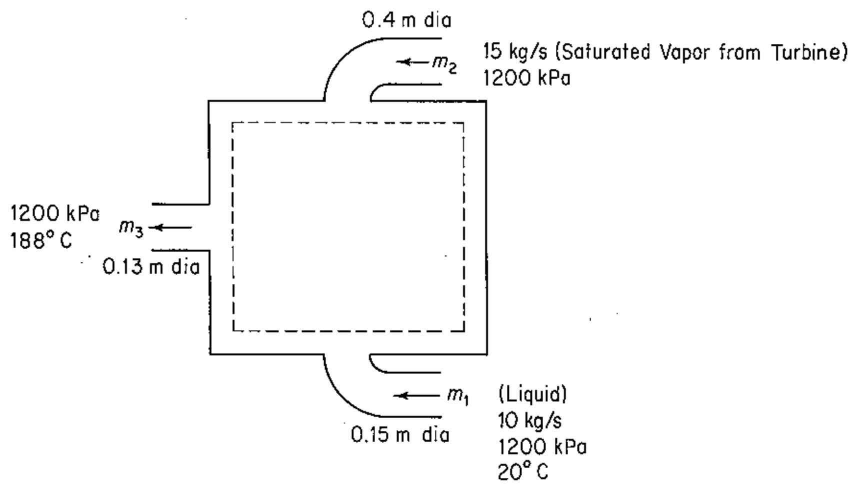


Figure P4.76

4.77. Hot reaction products (assume that the products have same properties as air) at 1000°C leave a reactor (see Fig. P4.77). To prevent further reaction, the process is designed to reduce the temperature of the products to 400°C by immediately spraying liquid water at 25°C into the gas stream.

(a) How many moles of water at 25°C are required per mole of products, and how many lb of water per 1000 ft^3 of products (measured at 400°C and 0.41 in. Hg pressure gauge—the barometer is 29.75 in. Hg)?

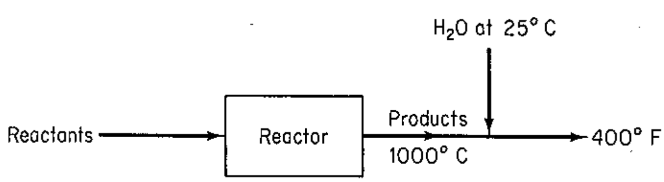


Figure P4.77

(b) Based on the data in part (a), explain in words and if possible with specific numbers how you would make the following adjustments or calculations.
   (1) What was the work done by or on the system if the system is the reactor?
   (2) What would be the increase or decrease in the lb of water required if the velocity of the 1000°C gas was assumed to be 20 ft/sec?
   (3) What would be the change in lb of water required if the reactor were not insulated and as a result lost 2000 Btu/lb mol of the 1000°C exit products?
   (4) What would be the change in lb of water required if the water temperature were 40°C instead of 25°C?
   (5) What would happen in your calculations if the hot gases contained a product that condensed at 500°C and 1 atm?

4.78. A plant is proposing to utilize waste gas, now being discarded at 750°F, to supply the heat for a waste heat boiler which is to produce saturated steam at 400°F from feed water at 70°F. The average molal heat capacity of the gas is 8.2. Under ideal conditions (i.e., no heat losses to the surroundings and very large heat exchange surfaces), what is the maximum possible production of steam, expressed as pounds of steam per pound mole of gas?

4.79. In a continuous reactor, the oxygen feed is compressed adiabatically from 100 to 200 kPa at a rate of 2 kg/min. The entering temperature of the oxygen is 20°C and the exit temperature is 100°C. If 7.675 kJ/hr of work are done on a system and the change in kinetic energy (ΔK) is a negative 10.0 kJ/kg, calculate the change in internal energy (ΔU) per 1.0 kg O_2 flowing through the compressor for the process. The change in potential energy (ΔP) is negligible. Oxygen behaves ideally at these conditions.

4.80. Cell growth liberates energy which must be removed; otherwise, the temperature will rise so much that the cells may be killed. In a continuous fermenter for the production of *Penicillium chrysogenum,* the cells generate 27.6 kJ/L per hour, and the volume of the well-insulated fermenter is 2 L. The feed temperature is 25°C and the exit temperature is equal to the temperature in the fermenter. *Penicillium chrysogenum* cannot grow above 42°C. Will the cells survive? Assume for simplicity that the inlet and outlet streams have a heat capacity of 4 J/(g)(°C) and the mass flow rates of 1025 g/hr are constant.

4.81. Three hundred kilograms per hour of air flow through a countercurrent heat exchanger as shown in Fig. P4.81. Two hundred thirty kilograms per hour of potassium carbonate solution are heated by the air. Assume that the heat exchanger has negligible heat losses. The terminal temperatures are given in Fig. P4.81. Calculate the temperature, in kelvin, of the exit potassium carbonate stream.

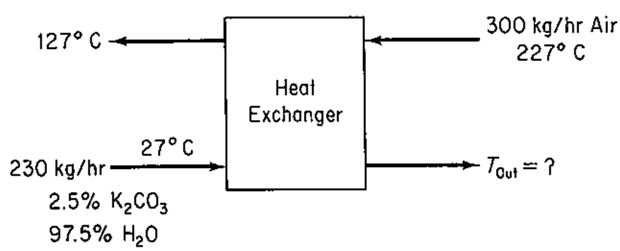


Figure P4.81

4.82. Boilers such as those used for generating steam must be purged periodically to prevent solids buildup. Solids in the boiler will lower both the efficiency and quality of the product steam. This process and the liquid removed are refered to as the blowdown. So that this energy is not wasted, the blowdown from a boiler operating at 1000 kPa is run into a flash chamber from which steam at 200 kPa is withdrawn. What fraction of the blowdown becomes available as steam at 200 kPa?

4.83. In a recycle fermenter, a cell separater is used to concentrate the cells and return them to the fermenter. The steady-state rate of cell production in the fermentation vessel is $Q = \mu XV$, where μ is a growth rate constant, X the (dry) cell concentration, and V the volume of the vessel. The volumetric flow rate into the fermenter and out of the separater is F. For the recycle fermenter shown in Fig. P4.83:

(a) Calculate the constant μ as a function of R, F, V, X_r, X_1.

(b) If $X_r/X_1 = 1.5$, calculate μ/D for $R/F = 0.5$, where D is the dilution rate (F_{in}/V).

(c) For $X_r/X_1 = 1.5$ and $R/F = 0.5$, calculate X_2/X_1.

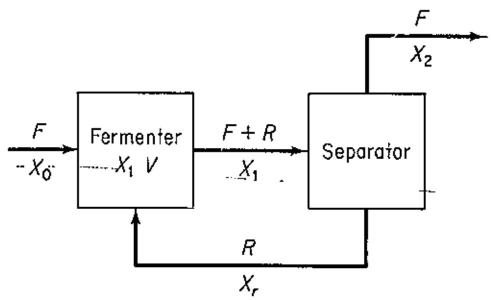


Figure P4.83

4.84. A chemical plant has just perfected a process for the manufacture of a new revolutionary drug. A large plant must be designed even before complete data are available. The research laboratory has obtained the following data for the drug.

boiling point at 101.3 kPa	250°C
vapor pressure at 230°C	55 kPa
specific gravity, 28/4	1.316
melting point	122.4°C
solubility in water at 76°C	2.2/100 parts water by weight
molecular weight	122
heat capacity of liquid	at 150°C is 2.09 kJ/(kg)(°C)
heat capacity of liquid	at 240°C is 2.13 kJ/(kg)(°C)

Calculate the heat duty for a vaporizer that will be required to vaporize 10,000 kg/hr of this chemical at atmospheric pressure (assume that no superheating of the vapor occurs). The entering temperature of the drug will be 130°C.

4.85. Water at 250 psia and 70°F enters a boiler, where it is converted to superheated steam at 250 psi absolute and 700°F. The superheated steam passes through an adiabatic turbine and emerges at a pressure of 15 psi gauge, 98% quality (i.e., 98% vapor). The exit wet steam is then condensed and subcooled to 100°F at a constant pressure of 15 psi gauge by heating air from 60°F to 200°F.

Per pound of steam, calculate Q, W, ΔH, and ΔU for (a) the boiler and (b) the turbine.

4.86. Saturated steam at 227°C is used in a heat exchanger to preheat a reactor feed stream.

(a) If the condensate is removed from the heat exchanger at 152°C and the flow rate is 1000 kg/hr, how many kJ are transferred to the feed stream in 1 hr?

(b) How much additional steam would be required if the condensate is removed at 227°C with the exit reactor feed stream temperature the same?

4.87. A process involving catalytic dehydrogenation in the presence of hydrogen is known as *hydroforming*. Toluene, benzene, and other aromatic materials can be economically produced from naphtha feed stocks in this way. After the toluene is separated from the other components, it is condensed and cooled in a process such as the one shown in Fig. P4.87. For every 100 kg of stock charged into the system, 27.5 kg of a toluene and water mixture (9.1% water) are produced as overhead vapor and condensed by the charge stream. Calculate:

(a) The temperature of the charge stream after it leaves the condenser

(b) The kilograms of cooling water required per hour

Additional data:

Stream	C_p [kJ/(kg)(°C)]	B.P. (°C)	ΔH_{vap} (kJ/kg)
$H_2O(l)$	4.2	100	2260
$H_2O(g)$	2.1	—	—
$C_7H_8(l)$	1.7	111	230
$C_7H_8(g)$	1.3	—	—
Stock (l)	2.1	—	—

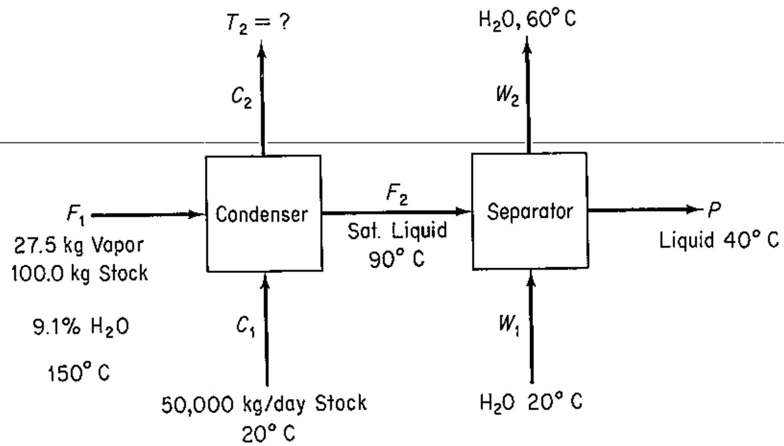


Figure P4.87

4.88. Simplify the general energy balance so as to represent the process in each of the following cases. Number each term in the general balance, Eq. (4.24a), and state why you retained or deleted it.

(a) A bomb calorimeter is used to measure the heating value of natural gas. A measured volume of gas is pumped into the bomb. Oxygen is then added to give a total pressure of 10 atm, and the gas mixture is exploded using a hot wire. The resulting heat transfer from the bomb to the surrounding water bath is measured. The final products in the bomb are CO_2 and water.

(b) Cogeneration (generation of steam both for power and heating) involves the use of gas turbines or engines as prime movers, with the exhausted steam going to the process to be used as a heat source. A typical installation is shown in Fig. P4.88.

(c) In a mechanical refrigerator the Freon liquid is expanded through a small insulated orifice so that part of it flashes into vapor. Both the liquid and vapor exit at a lower temperature than the temperature of the liquid entering.

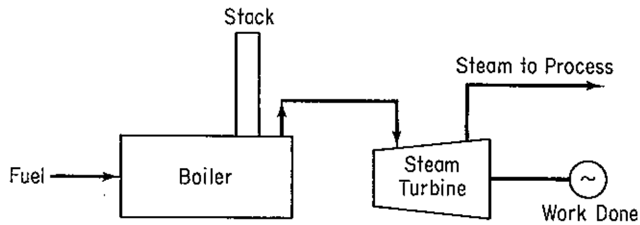


Figure P4.88

4.89. A desalted crude oil (40° API) is being heated in a parallel-flow heat exchanger from 70°F to 200°F with a straight-run gasoline vapor available at 280°F. If 400 gal/hr of gasoline is available (measured at 60°F, 63°API), how much crude can be heated per hour? Would countercurrent flow of the gasoline to the crude change your answer to this problem? Assume that $K = 11$ for the oil and 12.2 for the gasoline. C_p of the vapor is 0.95 Btu/(lb)(°F).

4.90. The process of throttling (i.e., expansion of a gas through a small orifice) is an important part of refrigeration. If wet steam (fraction vapor is 0.975) at 700 kPa absolute is throttled to 1 atm, find the following for the process:

(a) Heat transferred to or from steam
(b) Work done
(c) Enthalpy change
(d) Internal energy change
(e) Exit temperature

4.91. Solid CO_2 is being produced by adiabatically expanding high-pressure CO_2 at 70°F through a valve to a pressure of 20.0 psia.

(a) If 30% conversion of the initial CO_2 to the solid is desired, at what pressure must the CO_2 be before expansion?
(b) Is the high-pressure CO_2 gas, liquid, or a mixture of each?
(c) What is the approximate volume change per pound of CO_2 during the expansion?

4.92. Examine the simplified refinery power plant flow sheet in Fig. P4.92 and calculate the energy loss (in Btu/hr) for the plant. Also calculate the energy loss as a percentage of the energy input.

4.93. Start with the general energy balance, and simplify it for each of the processes listed below to obtain an energy balance that represents the process. Label each term in the general energy balance by number, and list by their numbers the terms retained or deleted followed by your explanation. (You do not have to calculate any quantities in this problem.)

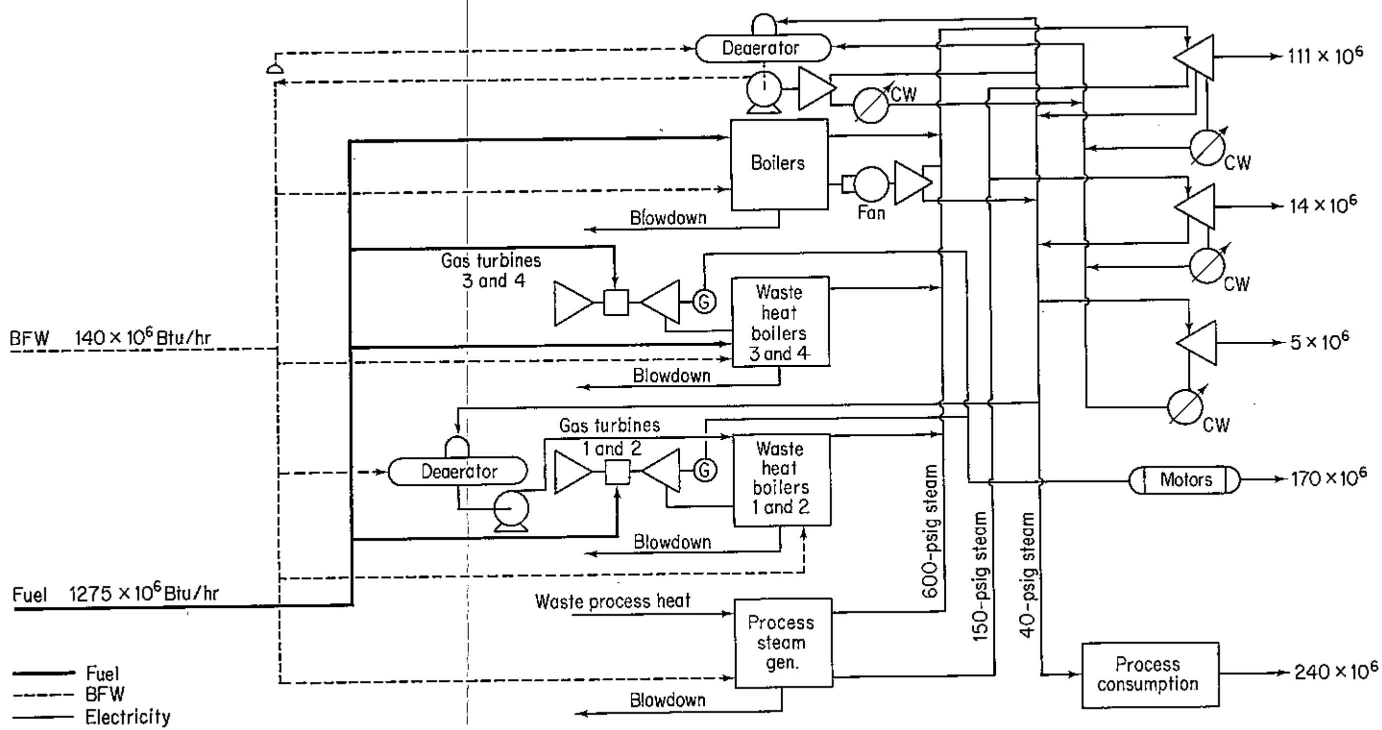

Deaerator
CW
Boilers
Fan
Blowdown
Gas turbines
3 and 4
G
Waste
heat
boilers
3 and 4
BFW 140 × 10^6 Btu/hr
Blowdown
Gas turbines
1 and 2
Deaerator
Waste
heat
boilers
1 and 2
Blowdown
Fuel 1275 × 10^6 Btu/hr
Waste process heat
Process
steam
gen.
Blowdown
600-psig steam
150-psig steam
40-psig steam
111 × 10^6
CW
14 × 10^6
CW
5 × 10^6
CW
Motors
170 × 10^6
Process
consumption
240 × 10^6
Fuel
BFW
Electricity

Figure P4.92

(a) One hundred kilograms per second of water is cooled from 100°C to 50°C in a heat exchanger. The heat is used to heat up 250 kg/s of benzene entering at 20°C. Calculate the exit temperature of benzene.

(b) A feedwater pump in a power generation cycle pumps water at the rate of 500 kg/min from the turbine condensers to the steam generation plant, raising the pressure of the water from 6.5 kPa to 2800 kPa. If the pump operates adiabatically with an overall mechanical efficiency of 50% (including both pump and its drive motor), calculate the electric power requirement of the pump motor (in kW). The inlet and outlet lines to the pump are of the same diameter. Neglect any rise in temperature across the pump due to friction (i.e., the pump may be considered to operate isothermally).

(c) A caustic soda solution is placed in a mixer together with water to dilute it from 20% NaOH in water to 10%. What is the final temperature of the mixture if the materials initially are at 25°C and the process is adiabatic?

4.94. A distillation process has been set up to separate an ethylene-ethane mixture as shown in Fig. P4.94. The product stream will consist of 98% ethylene and it is desired to recover 97% of the ethylene in the feed. The feed, 1000 lb per hour of 35% ethylene, enters the preheater as a subcooled liquid (temperature = −100°F, pressure = 250 psia). The feed experiences a 20°F temperature rise before it enters the still. The heat capacity of liquid ethane may be considered to be constant and equal to 0.65 Btu/(lb)(°F) and the heat capacity of ethylene may be considered to be constant and equal to 0.55 Btu/(lb)(°F). Heat capacities and saturation temperatures of mixtures may be determined on a weight fraction basis. An optimum reflux ratio of 6.1 lb reflux/lb product has been previously determined and will be used. Operating pressure in the still will be 250 psia. Additional data are as follows:

Pressure = 250 psia

Component	Temp. sat. (°F)	Heat of vaporizatioin (Btu/lb)
C_2H_6	10°	140
C_2H_4	−30°	135

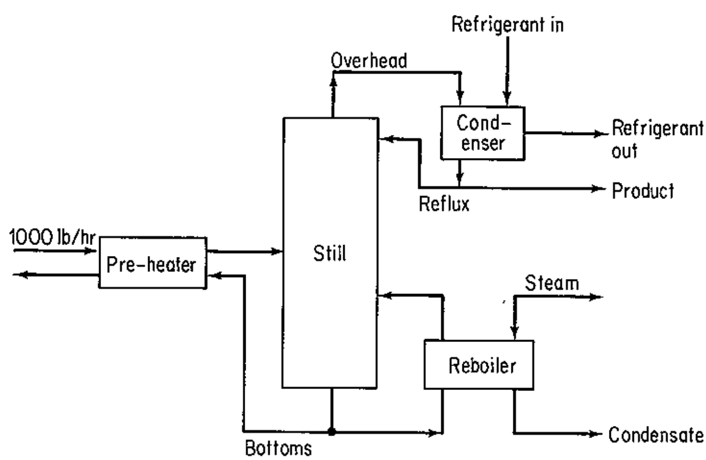


Figure P4.94

Determine:

(a) The pounds of 30 psig steam required in the reboiler per pound of feed.

(b) The gallons of refrigerant required in the condenser per hour assuming a 25°F rise in the temperature of the refrigerant. Heat capacity is approximately 1.0 Btu/(lb)(°F) and density = 50 lb/ft^3.

(c) The temperature of the bottoms as it leaves the preheater.

4.95. A dryer with a rating of 5.0×10^5 J/s is available to dry the products leaving the separator of a plant, as shown in Fig. P4.95. Is the heat input to the dryer large enough to do the job?

	C_p [J/(kg)(°C)]	H_{vap} (J/kg)
Urea	2500	
Solvent	2800	4.50×10^5

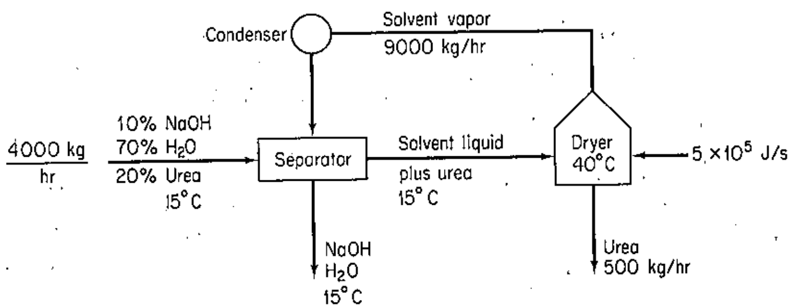


Figure P4.95

Section 4.6

4.96. A phase change (condensation, melting, etc.) of a pure component is an example of a reversible process because the temperature and pressure remain constant during the change. Use the definition of work to calculate the work done by butane when 1 kg of saturated liquid butane at 70 kPa vaporizes completely. Can you calculate the work done by the butane from the energy balance alone?

4.97. Calculate the work done when 1 lb mol of water in an open vessel evaporates completely at 212°F. Express your result in Btu.

4.98. One kilogram of steam goes through the following reversible process. In its initial state (state 1) it is at 2700 kPa and 540°C. It is then expanded isothermally to state 2, which is at 700 kPa. Then it is cooled at constant volume to 400 kPa (state 3). Next it is cooled at constant pressure to a volume of 0.4625 m^3/kg (state 4). Then it is compressed adiabatically to 2700 kPa, and 425°C (state 5), and finally it is heated at constant pressure back to the original state.

(a) Sketch the path of each step in a p–V diagram.

(b) Compute ΔU and ΔH for each step and for the entire process.

(c) Compute Q and W whenever possible for each step of the process.

4.99. Calculate the work done when 1 lb mol of water evaporates completely at 212°F in the following cases. Express your results in Btu/lb mol.

(a) A steady-state flow process: water flowing in a pipeline at 1 lb mol/min neglecting the potential and kinetic energy changes.

(b) A nonflow process: water contained in a constant-pressure, variable-volume tank.

4.100. A system consists of 25 lb of water vapor at the dew point. The system is compressed isothermally at 400°F, and 988 Btu of work is done on the system by the surroundings. What volume of liquid was present in the system before and after compression?

4.101. An inventor claims that his new device will compress steam isothermally from 14.7 psia and 500°F to 200 psia and 500°F with 100% efficiency. If so, what is the work in Btu per pound of steam calculated
(a) For a batch (closed) process?
(b) For a flow (open) process?

4.102. In the transport of pulverized coal by pipeline, water is the usual fluid to carry the coal. However, problems exist in obtaining water in arid western states so that liquid CO_2 (obtained from burning some of the coal) has been proposed as the transport fluid. To liquefy the pure CO_2 gas obtained after separation from N_2 and H_2O, the CO_2 is compressed isothermally and essentially reversibly from 15 psia and 60°F to saturated liquid. State assumptions about the states of the compounds.
(a) What is the specific volume of the saturated liquid?
(b) What is the final pressure on the saturated liquid?
(c) What is the work of compression per lb of CO_2?
(d) If the efficiency of the compressor is actually 90% (111% of the reversible work is the actual work required), and the cost of compression is \$0.034/kWh, how much does it cost to liquefy 1 lb of CO_2?

4.103. In a processing plant, milk flows from a storage tank maintained at 5°C through a valve to a pasteurizer via an insulated 10-cm-diameter pipe at the rate of 1000 L/min. The upstream pressure is 290 kPa and downstream pressure is 140 kPa. Determine the lost work (E_v) in J/min and the temperature change which occurs in the milk as a result of this throttling process. (Milk and water are sufficiently equivalent in properties for you to use those of water.)

4.104. A power plant is as shown in Fig. P4.104. If the pump moves 100 gal/min into the boiler with an overall efficiency of 40%, find the horsepower required for the pump. List all additional assumptions required.

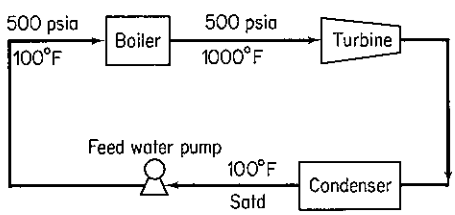


Figure P4.104

4.105. An office building requires water at two different floors. A large pipe brings the city water supply into the building in the basememt level, where a booster pump is located. The water leaving the pump is transported by smaller insulated pipes to the second and fourth floors, where the water is needed. Calculate the *minimum* amount of work per unit time (in horsepower) that the pump must do in order to deliver the necessary water, as indicated in Fig. P4.105. (*Minimum* refers to the fact that you should neglect the friction and pump energy losses in your calculations.) The water does not change temperature.

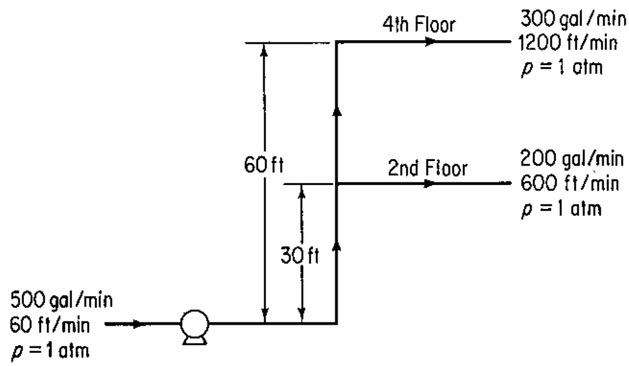


Figure P4.105

4.106. Water at 20°C is being pumped from a constant-head tank open to the atmosphere to an elevated tank kept at a constant pressure of 1150 kPa in an experiment as shown in Fig. P4.106. If water is flowing in the 5.0-cm line at a rate of 0.40 m^3/min, find
(a) The rating of the pump in joules per kilogram
(b) The rating of the pump in joules per minute
The pump and motor have an overall efficiency of 70% and the energy loss in the line can be determined to be 60.0 J/kg.

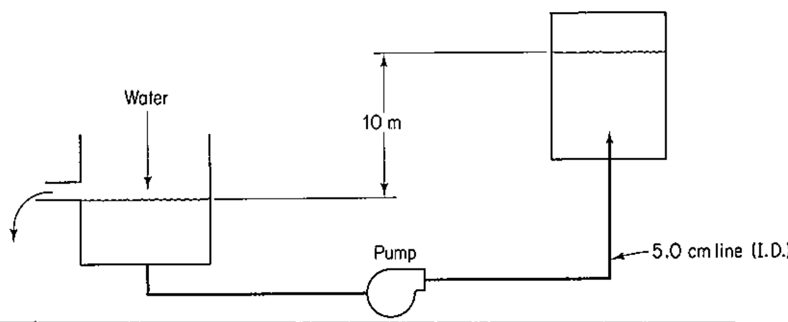


Figure P4.106

Section 4.7

4.107. Calculate the heat of reaction at 77°F and 1 atmosphere for the following reactions:
(a) $Fe_2O_3(s) + 2Al(s) \longrightarrow Al_2O_3(s) + 2Fe(s)$
Data: ΔH_f° for $Al_2O_3 = -399.09$ kcal/g mol
(b) $4FeS_2(s) + 15O_2(g) \longrightarrow 2Fe_2O_3(s) + 8SO_3(g)$
(c) $C_6H_5CHO(l) + 8O_2(g) \longrightarrow 7CO_2(g) + 3H_2O(l)$

4.108. Calculate the heat of reaction in the standard state for 1 mole of $C_3H_8(g)$ for the following reaction from the given data:

$$C_3H_8(l) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$$

Compound	$-\Delta\hat{H}_f^\circ$ (kcal/g mol)	Vaporization at 25°C kcal/g mol
$C_3H_8(g)$	24.820	3.823
$CO_2(g)$	94.052	1.263
$H_2O(g)$	57.798	10.519

4.109. At Texas City, Amoco operates a hydroformer in which cycloparaffin hydrocarbon vapors plus recycled hydrogen are passed at about 500°C and 20 atm pressure over a catalyst that produces an aromatic hydrocarbon. Cyclohexane produces benzene and hydrogen.

$$C_6H_{12}(g) \longrightarrow C_6H_6(g) + 3H_2(g)$$

The cyclohexane vapors are condensed, separated by distillation, and recycled in the process. Three volumes of hydrogen are recycled for every volume of feed stock, which consists of cyclohexane containing 5 mol % benzene. To initiate a design for the reactor, calculate standard heat of reaction per g mol of C_6H_{12}.

4.110. Hydrogen is used in many industrial processes, such as the production of ammonia for fertilizer. Hydrogen also has been considered to have a potential as an energy source because its combustion yields a clean product and it is easily stored as metal hydrides. Thermochemical cycles (a series of reactions resulting in a recycle of some of the reactants) can be used in the production of hydrogen from an abundant natural compound—water. One process involving a series of five steps is outlined below. State assumptions about the states of the compounds.

$$3Fe_2O_3 + 18HCl \xrightarrow{120°C} 6FeCl_3 + 9H_2O \qquad (1)$$
$$6FeCl_3 \xrightarrow{420°C} 6FeCl_2 + 3Cl_2 \qquad (2)$$
$$6FeCl_2 + 8H_2O \xrightarrow{650°C} 2Fe_3O_4 + 12HCl + 2H_2 \qquad (3)$$
$$2Fe_3O_4 + \tfrac{1}{2}O_2 \xrightarrow{350°C} 3Fe_2O_3 \qquad (4)$$
$$3H_2O + 3Cl_2 \xrightarrow{800°C} 6HCl + \tfrac{3}{2}O_2 \qquad (5)$$

(a) Calculate the standard heat of reaction for each step. State whether the reaction is endothermic or exothermic.
(b) What is the overall reaction? Is it exothermic or endothermic?

4.111. J. D. Park *et al.* [JACS *72*, 331–3 (1950)] determined the heat of hydrobromination of propene and cyctopropane. For hydrobromination (addition of HBr) of propene to 2-bromopropane (C_3H_7Br), they found that $\Delta H = -84{,}441$ J/g mol. The heat of hydrogenation of propene to propane is $\Delta H = -126{,}000$ J/g mol. N.B.S. Circ. 500 gives the heat of formation of HBr(g) as $-36{,}233$ J/g mol when the bromine is liquid and the heat of vaporization of bromine as 30,710 J/g mol. Calculate the heat of bromination of propane using gaseous bromine to form 2-bromopropane using the above data?

4.112. Compare prices of five fuels:

(1) Natural gas (all CH_4) at \$2.20 per 10^3 ft^3
(2) No. 2 fuel oil (33°API) at \$0.85 per gallon
(3) Dry yellow pine at \$95 per cord
(4) High-volatile A bituminous coal at \$55.00 per ton
(5) Electricity at \$0.032 per kilowatt

List the cost of each in descending order per 10^6 Btu. As far as energy consumption is concerned, would you be better off with a gas dryer or an electric dryer? Should you heat your house with wood, No. 2 fuel oil, or coal? (*Note:* A cord of wood is a pile 8 ft long, 4 ft high, and 4 ft wide.)

4.113. How would you determine the heat of formation of gaseous fluorine at 25°C and 1 atm?

4.114. Calculate the heat of reaction at the standard reference state for the following reactions:

(a) $CO_2(g) + H_2(g) \longrightarrow CO(g) + H_2O(l)$
(b) $2CaO(s) + 2MgO(s) + 4H_2O(l) \longrightarrow 2Ca(OH)_2(s) + 2Mg(OH)_2(s)$
(c) $Na_2SO_4(s) + C(s) \longrightarrow Na_2SO_3(s) + CO(g)$
(d) $NaCl(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HCl(g)$
(e) $NaCl(s) + 2SO_2(g) + 2H_2O(l) + O_2(g) \longrightarrow 2Na_2SO_4(s) + 4HCl(g)$
(f) $SO_2(g) + \frac{1}{2}O_2(g) + H_2O(l) \longrightarrow H_2SO_4(l)$
(g) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$
(h) $Na_2CO_3(s) + 2Na_2S(s) + 4SO_2(g) \longrightarrow 3Na_2S_2O_3(s) + CO_2(g)$
(i) $NaNO_3(s) + Pb(s) \longrightarrow NaNO_2(s) + PbO(s)$
(j) $Na_2CO_3(s) + 2NH_4Cl(s) \longrightarrow 2NaCl(s) + 2NH_3(g) + H_2O(l) + CO_2(g)$
(k) $Ca_3(PO_4)_2(s) + 3SiO_2(s) + 5C \longrightarrow 3CaSiO_3(s) + 2P(s) + 5CO(g)$
(l) $P_2O_5(s) + 3H_2O(l) \longrightarrow 2H_3PO_4(l)$
(m) $CaCN_2(s) + 2NaCl(s) + C(s) \longrightarrow CaCl_2(s) + 2NaCN(s)$
(n) $NH_3(g) + HNO_3(aq) \longrightarrow NH_4NO_3(aq)$
(o) $2NH_3(g) + CO_2(g) + H_2O(l) \longrightarrow (NH_4)_3CO_2(aq)$
(p) $(NH_4)_2CO_3(aq) + CaSO_4 \cdot 2H_2O(s) \longrightarrow CaCO_3(s) + 2H_2O(l) + (NH_4)_2SO_4(aq)$
(q) $CuSO_4(aq) + Zn(s) \longrightarrow ZnSO_4(aq) + Cu(s)$
(r) $C_6H_6(l) + Cl_2(g) \longrightarrow C_6H_5Cl(l) + HCl(g)$
benzene — chlorobenzene
(s) $CS_2(l) + Cl_2(g) \longrightarrow S_2Cl_2(l) + CCl_4(l)$
(t) $C_2H_4(g) + HCl(g) \longrightarrow CH_3CH_2Cl(g)$
ethylene — ethylchloride
(u) $CH_3OH(g) + \frac{1}{2}O_2(g) \longrightarrow H_2CO(g) + H_2O(g)$
methyl alcohol — formaldehyde
(v) $C_2H_2(g) + H_2O(l) \longrightarrow CH_3CHO(l)$
acetylene — acetaldehyde
(w) $n\text{-}C_4H_{10}(g) \longrightarrow C_2H_4(g) + C_2H_6(g)$
butane — ethylene, ethane
(x) $C_2H_4(g) + C_6H_6(l) \longrightarrow$ ethyl benzene(l)
ethylene, benzene
(y) $2C_2H_4(g) \longrightarrow C_4H_8(l)$
ethylene — 1-butane
(z) $C_3H_6(g) + C_6H_6(l) \longrightarrow$ cumene(l)
propene, benzene — isopropyl benzene

4.115. The process of converting SO_2 to sulfuric acid with the aid of nitrogen oxides can be expressed on an overall basis as follows:

$$SO_2(g) + NO_2(g) + H_2O(l) \longrightarrow H_2SO_4(l) + NO(g)$$

This process has been used by the chemical industry for over a century in what is known as the chamber process. With the high dilution of N_2 that accompanies the SO_2, the raw gases have to pass at low velocities through large lead chambers in order to react sufficiently. If 10,000 ft^3/hr of NO (measured at S.C.) and 2750 lb/hr of H_2SO_4 are produced, what is the cooling required if the reaction is to proceed isothermally at 77°F?

4.116. Determine the heat of reaction at 25°C and 1 atm for the following reactions from heat of combustion data:

(a) $H_2S(g) + \frac{3}{2}O_2(g) \longrightarrow SO_2(g) + H_2O(g)$
(b) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$
(c) $NO(g) + \frac{5}{2}H_2(g) \longrightarrow NH_3(g) + H_2O(g)$

4.117. A mixture of gaseous methane (CH_4) and hydrogen (H_2) was found to transfer heat in a reaction at 25°C of −150.0 kcal/g mol mixture; however, this value was determined under conditions such that the water produced left as a vapor rather than as a liquid as required for the standard values. Calculate the relative compositions of the mixture of CH_4 and H_2.

4.118. The following enthalpy changes are known for reactions at 25°C in the standard thermochemical state:

No.			$\Delta H°$(kJ)
1	$C_3H_6(g) + H_2(g)$	$\longrightarrow C_3H_8(g)$	−123.8
2	$C_3H_8(g) + 5O_2(g)$	$\longrightarrow 3CO_2(g) + 4H_2O(l)$	−2220.0
3	$H_2(g) + \frac{1}{2}O_2(g)$	$\longrightarrow H_2O(l)$	−285.8
4	$H_2O(1)$	$\longrightarrow H_2O(g)$	43.9
5	C(diamond) + $O_2(g)$	$\longrightarrow CO_2(g)$	−395.4
6	C(graphite) + $O_2(g)$	$\longrightarrow CO_2(g)$	−393.5

Calculate:
(a) The heat of formation of propylene
(b) The heat of combustion of propylene

4.119. When 2.00 g of MgO (mol. wt. = 40.32) are formed by burning Mg in air, the laboratory reports that 31,780 J were absorbed by the calorimeter. What is the heat of combustion of Mg per g mol?

4.120. About 30% of crude oil is processed eventually into automobile gasoline. As petroleum prices rise and resources dwindle, alternatives must be found. However, automobile engines can be tuned so that they will run on simple alcohols. Methanol and ethanol can be derived from coal or plant biomass, respectively. While the alcohols produce fewer pollutants than gasoline, they also reduce the travel radius of a tank of fuel. What percent increase in the size of fuel tank is needed to give an equivalent travel radius if gasoline is replaced by alcohol? Base your calculations on 40 kJ/g of gasoline having a sp gr of 0.84, and with the product water as a gas. Make the calculations for **(a)** methanol; **(b)** ethanol.

4.121. Calculate the heat of formation at 25°C of ethyl mercaptan (C_2H_5SH). The following thermal values at 25°C are available.

$$C_2H_5SH(g) + 4.5O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) + SO_2(g) \qquad \Delta H = -1{,}895{,}600 \text{ J/g mol}$$

$$H_2(g) + 0.5O_2(g) \longrightarrow H_2O(g) \qquad \Delta H = -241{,}826 \text{ J/g mol}$$

Latent heat of vaporization of water at 25°C is 44,012 J/g mol.

$$C(\beta) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -393{,}510 \text{ J/g mol}$$

$$S(s) + O_2(g) \longrightarrow SO_2(g) \qquad \Delta H = -296{,}900 \text{ J/g/ mol}$$

If the first reaction were carried out in a constant-volume calorimeter, what would be the amount of heat liberated?

4.122. A consulting laboratory is called upon to determine the heating value of a natural gas in which the combustible is entirely methane. They do not have a Sargent flow calorimeter, but do have a Parr bomb calorimeter. They pump a measured volume of the natural gas into the Parr bomb, add oxygen to give a total pressure of 1000 kPa, and explode the gas–O_2 mixture with a hot wire. From the data they calculate that the gas has a heating value of 3.44×10^4 kJ/m^3. Should they report this value? Explain.

4.123. Calculate the heat of reaction at 25°C for the following reaction if it takes place at constant volume.

$$C_2H_4(g) + 2H_2(g) \longrightarrow 2CH_4(g)$$

4.124. If 1 lb mol of Cu and 1 lb mol of H_2SO_4(100%) react together completely in a bomb calorimeter, how many Btu is absorbed (or evolved)? Assume that the products are H_2(g) and solid $CuSO_4$. The initial and final temperatures in the bomb are 25°C.

4.125. In the reaction

$$4FeS_2(s) + 11O_2(g) \longrightarrow 2Fe_2O_3(s) + 8SO_2(g)$$

the conversion of FeS_2(s) to Fe_2O_3(s) is only 80% complete. If the standard heat of reaction for the reaction is calculated to be -197.7 kcal/g mol FeS_2(s), what value of ΔH^0_{rxn} will you use in the energy balance per kg of FeS_2 fed?

4.126. Find the higher (gross) heating value of H_2(g) at 0°C.

4.127. The chemist for a gas company finds a gas analyzes 9.2% CO_2, 0.4% C_2H_4, 20.9% CO, 15.6% H_2, 1.9% CH_4, and 52.0% N_2. What should the chemist report as the gross heating value of the gas?

4.128. What is the higher heating value of 1 m^3 of n-propylbenzene measured at 25°C and 1 atm and with a relative humidity of 40%?

4.129. An off-gas from a crude oil topping plant has the following composition:

Component	Vol. %
Methane	88
Ethane	6
Propane	4
Butane	2

(a) Calculate the higher heating value on the following bases: (1) Btu per pound, (2) Btu per mole, and (3) Btu per cubic foot of off-gas measured at 60°F and 760 mm Hg.

(b) Calculate the lower heating value on the same three bases indicated in part (a).

4.130. Calculate the lower heating value of methane at 100°C.

4.131. Compute the heat of reaction at 600 K for the following reaction:

$$S(\ell) + O_2(g) \longrightarrow SO_2(g)$$

4.132. Calculate the heat of reaction at 500°C for the decomposition of propane.

$$C_3H_8 \longrightarrow C_2H_2 + CH_4 + H_2$$

4.133. In a new process for the recovery of tin from low-grade ores, it is desired to oxidize stannous oxide, SnO, to stannic oxide, SnO_2, which is then soluble in caustic solution. What is the heat of reaction at 90°C and 1 atm for the reaction $SnO + \frac{1}{2}O_2 \longrightarrow SnO_2$? Data are:

	ΔH_f^0 (kJ/g mol)	C_p [J/(g mol)(K)], T in K
SnO	−283.3	$39.33 + 15.15 \times 10^{-3}\,T$
SnO_2	−577.8	$73.89 + 10.04 \times 10^{-3}T - \dfrac{2.16 \times 10^6}{T^2}$

4.134. Develop an equation to be used in a computer code from which the heat of the following reaction in kJ/g mol FeO can be calculated at any temperature T (in K).

$$FeO(s) + CO(g) \longrightarrow Fe(s) + CO_2(g)$$

	C_p [J/(g mol)(K)]
Fe	$14.1 + 29.71 \times 10^{-3}T + 1.80 \times 10^5T^{-2}$
FeO	$38.79 + 20.08 \times 10^{-3}T$
CO	$27.61 + 5.02 \times 10^{-3}T$
CO_2	$32.21 + 22.18 \times 10^{-3}T - 3.47 \times 10^{-6}T^2$

4.135. To take care of peak loads in the winter, the local gas company wishes to supplement its limited supply of natural gas (CH_4) by catalytically hydrogenating a heavy liquid petroleum oil that it can store in tanks. The supplier says that the oil has a U.O.P. Characterization Factor of 12.0 and an API gravity of 30°API. To minimize carbon deposition on the catalyst, twice the required amount of hydrogen is introduced with the oil. The reactants enter at 70°F and after heat exchange the products leave at 70°F. Calculate the following, per 1000 ft^3 of CH_4 at the standard conditions in the gas industry:
(a) The pounds of oil required
(b) The heat input or output to the catalytic reactor

4.136. Calculate the heat of reaction of the following reactions at the stated temperature:

(a) $MgO(s) + C(s) + Cl_2(g) \xrightarrow{850°C} MgCl_2(s) + CO(g)$

(b) $CH_3OH(g) + \frac{1}{2}O_2(g) \xrightarrow{200°C} H_2CO(g) + H_2O(g)$
methyl alcohol — formaldehyde

(c) $SO_2(g) + \frac{1}{2}O_2(g) \xrightarrow{300°C} SO_3(g)$

4.137. Sulfur can be recovered from the H_2S in natural gas by the following reaction:

$$3H_2S(g) + 1\tfrac{1}{2}O_2(g) \longrightarrow 3S(s) + 3H_2O(g)$$

For the stoichiometric ratio of materials, with H_2S and O_2 entering the process at 300°F and the products leaving the reactor at 100°F, calculate the heat of reaction in British thermal units per pound of sulfur formed. Assume that the reaction is complete. If it is only 75% complete, will this change your answer, and if so, how much?

4.138. Propane, butane, or liquefied petroleum gas (LPG) has seen practical service in passenger automobiles for 30 years or more. Because it is used in the vapor phase, it pollutes less than gasoline but more than natural gas. A number of cars in the Clean Air Car Race ran on LPG. The table below lists their results and those for natural gas. It must be kept in mind that these vehicles were generally equipped with platinum catalyst reactors and with exhaust-gas recycle. Therefore, the gains in emission control did not come entirely from the fuels.

	Natural gas, avg. 6 cars	LPG, avg. 13 cars	Fed std.
HC (g/mi)	1.3	0.49	0.22
CO (g/mi)	3.7	4.55	2.3
NO_x (g/mi)	0.55	1.26	0.6

Suppose that in a test butane gas at 100°F is burned completely with the stoichiometric amount of heated air which is at 400°F and a dew point of 77°F in an engine. To cool the engine, 12.5 lb of steam at 100 psia and 95% quality was generated from water at 77°F per pound of butane burned. It may be assumed that 7% of the gross heating value of the butane is lost as radiation from the engine. Will the exhaust gases leaving the engine exceed the temperature limit of the catalyst of 1500°F?

4.139. Liquid hydrazine (N_2H_4) is injected into a jet combustion chamber at 100°C, where it is ignited and burns in an airstream with 100% excess air preheated to 500°C. The hot gases issue directly from the end of the reactor. What is their temperature?

$$N_2H_4(l) + O_2(g) \rightarrow N_2(g) + 2H_2O(g) \qquad \begin{aligned} C_p\ N_2H_4(l) &= 33.2 \text{ cal/(g mol)(°C)} \\ \Delta H_v\ N_2H_4(l) &= 10.7 \text{ kcal/g mol at 25°C} \end{aligned}$$

4.140. The data for this problem have been taken from D. Dyer, G. Maples, and J. C. Maxwell, *Measuring and Improving the Efficiency of Boilers* (Cambridge, Mass.: MIT Press, 1982), Chapter 2. A boiler (see Fig. P4.140) burns the following fuel:

CH_4	90.05%	N_2	5.518%
C_2H_6	1.98%	O_2	1.467%
C_3H_8	0.202%	H_2	0.31%
C_4H_{10}	0.069%	CO_2	0.662%
C_5H_{12}	0.018%		

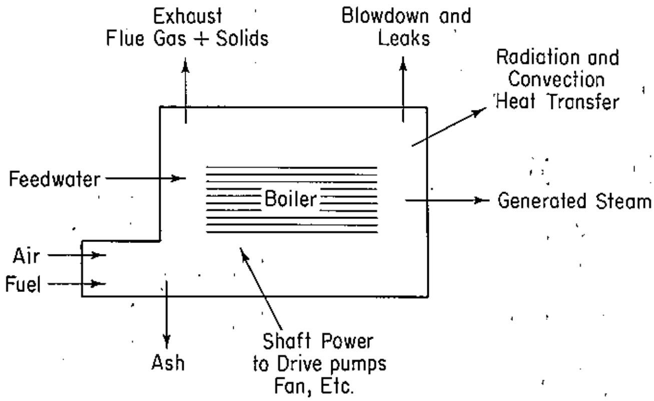


Figure P4.140

In a test for boiler efficiency, the following data were collected:

Variable	Value
Feedwater mass flow rate	50,603 lb/hr
Feedwater inlet temperature	228°F
Steam exit pressure	347.5 psig
Mass loss due to blowdown	0
Enthalpy of steam	1196.4 Btu/lb
Mass flow rate of fuel	2790.2 lb/hr
Inlet temperature of the fuel	55°F
Inlet pressure of the fuel	8.5 psig
Inlet temperature of the air	70°F
Inlet pressure of the air	14.7 psia
Relative humidity of the air	20%
Flue-gas temperature	345°F
Orsat analysis:	
CO_2 volume in flue gas	10.1%
CO volume in flue gas	0.1%
O_2 volume in flue gas	1.9%

Carry out material and energy balances, and calculate the boiler efficiency defined by

$$\text{efficiency} = \frac{(\text{mass of water evaporated/hr})(\Delta \hat{H}_{\text{steam}} - \Delta \hat{H}_{\text{feed water}})}{(\text{mass of fuel flow/hr})(\text{HHV}_{\text{fuel}})}$$

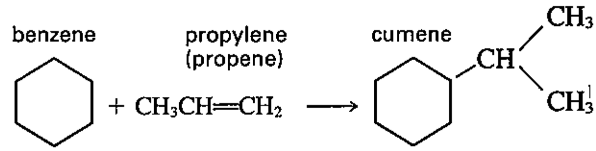
Also calculate the dew point of the exit flue gas and the air-to-fuel ratio in lb/lb.

4.141. Formaldehyde can be made by the partial oxidation of natural gas using pure oxygen made industrially from liquid air. The natural gas must be in large excess.

$$CH_4 + O_2 \longrightarrow CH_2O + H_2O$$

The CH_4 is heated to 400°C and the O_2 to 300°C and introduced into a reaction chamber. The products leave the cooled chamber at 600°C and show an Orsat analysis of CO_2 1.9, CH_2O 11.7, O_2 3.8, and CH_4 82.6%. How much heat is removed from the reaction chamber per 1000 kg of formaldehyde produced?

4.142. In a new process for making phenol, cumene (isopropylbenzene) is oxidized with air to form cumene hydroperoxide, which is then changed to phenol and acetone. The cumene is made by the direct alkylation of gaseous benzene, using a phosphoric acid–kieselguhr catalyst and operating conditions of 500 K.



Calculate the heat of reaction at 500 K.

Enthalpy data for benzene (C_6H_6) amd cumene (C_9H_{12}) are as follows:

	ΔH (J/g mol)	
T (K)	C_6H_6	C_9H_{12}
273	0	0
298	1,946	3,628
400	2,109	21,585
500	24,372	43,786

4.143. A catalytic converter for the production of SO_3 from SO_2 is operating as illustrated in Fig. P4.143. The material balance for 1 hr of operation is also shown in Fig. P4.143.

$$SO_2(g) + 0.5O_2(g) \longrightarrow SO_3 \qquad \Delta H^\circ_{25} = -98{,}280 \text{ J/g mol}$$

The unit is insulated and heat losses are negligible. It has been found that corrosion can be reduced considerably by keeping the discharge temperature at approximately 400°C.

Determine the heat duty of a cooler for the converter that will accomplish this purpose.

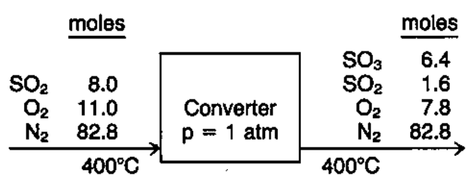


Figure P4.143

4.144. It is desired to volatilize the iron from a reaction mixture by treating it with chlorine gas and driving off the ferric chloride formed. The reaction is

$$Fe_2O_3(s) + 3C(s) + 3Cl_2(g) \longrightarrow 3CO(g) + 2FeCl_3(g)$$

The reaction mixture containing 10% Fe_2O_3 plus C is introduced into the reactor at 50°C. The Cl_2 gas is used in 10% excess and is preheated to 200°C by the products leaving. The reactor is kept at the vaporization point of $FeCl_3$ (300°C) at which the liquid $FeCl_3$ is vaporized and swept out by the gases formed.

Calculate the heat that must be added or removed from the reactor per gram of charge in order that the temperature will be maintained at 300°C. Use the following data if applicable.

$$FeCl_3: \Delta H_{\substack{\text{fusion}^\circ \\ 280^\circ C}} \quad c \longrightarrow l \qquad 43{,}095 \text{ J/g mol}$$

$$\Delta H^\circ_{\substack{\text{vaporization} \\ 300^\circ C}} \quad l \longrightarrow g \qquad 26{,}108 \text{ J/g mol}$$

$$\text{Approximate } C_p(s) = 121 \text{ J/(g mol)(K)}$$

$$\text{Approximate } C_p(l) = 102.5 \text{ J/(g mol)(K)}$$

$$\text{Approximate } C_p \text{ of the charge} = 1.26 \text{ J/(g)(K)}$$ mixt. of Fe_2O_3 + C.

4.145. Three fuels are being considered as a heat source for a metallurgical process:

	Components	Percent
Gas	CH_4	96.0
	CO_2	3.0
	N_2	1.0
Oil (liquid)	$C_{16}H_{34}$	99.0
	S	1.0
Coke	C	95.0
	ash	5.0

Calculate the maximum flame temperature (i.e., combustion with the theoretically required amount of air) for each of the three fuels assuming that the fuels and air enter at 18°C, and the ash (C_p = 1.15 J/(g)(°C)) leaves the combustion chamber containing no carbon and at 527°C. Which fuel gives the highest temperature?

4.146.* Calculate the adiabatic flame temperature of $C_3H_6(g)$ at 1 atm when burned with 20% excess air, and the reactants enter at 25°C.

4.147.* A fuel oil (liquid) composed of C (86.0%), H (10.0%), and S (4.0%) is burned with 20% excess air. The oil enters the process at 200°F and the air at 300°F. The water vapor in the air is at a dew point of 101°F when the barometer reads 750 mm Hg. Compute the adiabatic flame temperature in the process. The mean heat capacity of the oil can be assumed to be 0.50 Btu/(lb)(°F) and the heat of reaction [to form $CO_2(g)$, $H_2O(g)$, and $SO_2(g)$] is 17,860 Btu/lb at 25°C and 1 atm.

4.148.* If CO at constant pressure is burned with excess air and the theoretical flame temperature is 1255 K, what was the percentage of excess air used? The reactants enter at 93°C.

4.149.* Which substance will give the higher theoretical flame temperature if the inlet percentage of excess air and temperature conditions are identical: (1) CH_4, (2) C_2H_6, or (3) C_4H_8?

Section 4.8

4.150. A mixture containing 65% H_2SO_4 (and 35% water) is at 320°F and 1 atm. Answer the following questions.

(a) How many phases are present in the mixture?

(b) What is the percent liquid and/or vapor in the mixture, and what is the mass fraction of each component?
(c) What is the enthalpy of the mixture?
(d) If 100 lb of this mixture is added to 50 lb of a 20% solution of H_2SO_4 at 220°F, what is the (1) mass, (2) composition, and (3) temperature and enthalpy of the final solution?

4.151. (a) In a petrochemical plant 1 ton/hr of 50% by weight H_2SO_4 is being produced at 250°F by concentrating 30% H_2SO_4 supplied at 100°F. From the data provided, find the heat that has to be supplied per hour.

Btu evolved at 100°F per lb mol H_2SO_4	Moles of H_2O per mol of H_2SO_4
18,000	2
21,000	3
23,000	4
24,300	5
27,800	10
31,200	22

C_p [Btu/(lb)(°F)]	30% H_2SO_4	50% H_2SO_4
100	0.80	0.70
250	0.75	0.60

The C_p of water vapor is about 0.50, and for water use, 1.0.
(b) Using the same data as in part (a), what is the temperature of a solution of 30% by weight H_2SO_4 if it is made from 50% H_2SO_4 at 100°F and pure water at 200°F if the process is adiabatic?

4.152.* (a) From the data plot the enthalpy of 1 *mole of solution* at 27°C as a function of the weight percent HNO_3. Use as reference states liquid water at 0°C and liquid HNO_3 at 0°C. You can assume that C_p for H_2O is 75 J/(g mol)(°C) and for HNO_3, 125 J/(g mol)(°C).

$-\Delta H_{soln}$ at 27°C (J/g mol HNO_3)	Moles H_2O added to 1 mole HNO
0	0.0
3,350	0.1
5,440	0.2
6,900	0.3
8,370	0.5
10,880	0.67
14,230	1.0
17,150	1.5
20,290	2.0
24,060	3.0

$-\Delta H_{soln}$ at 27°C (J/g mol HNO_3)	Moles H_2O added to 1 mole HNO
25,940	4.0
27,820	5.0
30,540	10.0
31,170	20.0

(b) Compute the energy absorbed or evolved at 27°C on making a solution of 4 moles of HNO_3, and 4 moles of water by mixing a solution of $33\frac{1}{3}$ mol % acid with one of 60 mol % acid.

4.153.* *National Bureau of Standards Circular 500* gives the following data for calcium chloride (mol. wt. 111) and water:

Formula	State	$-\Delta H_f$ at 25°C (kcal/g mol)
H_2O	Liquid	68.317
	Gas	57.798
$CaCl_2$	Crystal	190.0
	in 25 moles of H_2O	208.51
	50	208.86
	100	209.06
	200	209.20
	500	209.30
	1000	209.41
	5000	209.60
	∞	209.82
$CaCl_2 \cdot H_2O$	Crystal	265.1
$CaCl_2 \cdot 2H_2O$	Crystal	335.5
$CaCl_2 \cdot 4H_2O$	Crystal	480.2
$CaCl_2 \cdot 6H_2O$	Crystal	623.15

Calculate the following:

(a) The energy evolved when 1 lb mol of $CaCl_2$ is made into a 20% solution at 77°F

(b) The heat of hydration of the dihydrate to the hexahydrate

(c) The energy evolved when a 20% solution containing 1 lb mol of $CaCl_2$ is diluted with water to 5% at 77°F

4.154. A vessel contains 100 g of an NH_4OH–H_2O liquid mixture at 25°C and 1 atm that is 15.0% by weight NH_4OH. Just enough aqueous H_2SO_4 is added to the vessel from an H_2SO_4 liquid mixture at 25°C and 1 atm (25.0 mole % H_2SO_4) so that the reaction to $(NH_4)_2SO_4$ is complete. After the reaction, the products are again at 25°C and 1 atm. How much heat (in J) is absorbed or evolved by this process? It may be assumed that the final volume of the products is equal to the sum of the volumes of the two initial mixtures.

4.155. An ammonium hydroxide solution is to be prepared at 77°F by dissolving gaseous NH_3 in water. Prepare charts showing:

(a) The amount of cooling needed (in Btu) to prepare a solution containing 1 lb mol of NH_3 at any concentration desired

(b) The amount of cooling needed (in Btu) to prepare 100 gal of a solution of any concentration up to 35% NH_3

(c) If a 10.5% NH_3 solution is made up without cooling, at what temperature will the solution be after mixing?

4.156. Prepare an enthalpy–concentration chart for the sulfuric acid–water system, and compare your chart with the one in Appendix I. You will have to look up the necessary data in reference books or the literature.

4.157. Draw the saturated-liquid line (1 atm isobar) on an enthalpy concentration chart for caustic soda solutions. Show the 200°F, 250°F, 300°F, 350°F, and 400°F isotherms in the liquid region, and draw tie lines for these temperatures to the vapor region. Using the data from *National Bureau of Standards Circular 500,* show where molten NaOH would be on this chart. Take the remaining data from the following table and the steam tables.

Temp. (°F)	x, Satd. liquid conc. wt fraction	$\Delta\hat{H}$, Enthalpy, satd. liquid (Btu/lb)
200	0	168
250	0.34	202
300	0.53	314
350	0.68	435
400	0.78	535

C_p for NaOH solutions [Btu/(lb)(°F)]

Wt % NaOH	Temperature (°F)				
	32	60	100	140	180
10	0.882	0.897	0.911	0.918	0.922
20	0.842	0.859	0.875	0.884	0.886
30		0.837	0.855	0.866	0.869
40		0.815	0.826	0.831	0.832
50			0.769	0.767	0.765

NOTE: All H_2O tie lines extend to pure water vapor.

SOURCE: J. W. Bertetti and W. L. McCabe, *Ind. Eng. Chem.*, v. 28, p. 375 (1936).

4.158. An evaporator at atmospheric pressure is designed to concentrate 10,000 lb/hr of a 10% NaOH solution at 70°F into a 40% solution. The steam pressure inside the steam chest is 40 psig. Determine the pounds of steam needed per hour if the exit strong caustic preheats the entering weak caustic in a heat exchanger, leaving the heat exchanger at 100°F.

4.159. A 50% by weight sulfuric acid solution is to be made by mixing the following:

(1) Ice at 32°F
(2) 80% H_2SO_4 at 100°F
(3) 20% H_2SO_4 at 100°F

How much of each must be added to make 1000 lb of the 50% solution with a final temperature of 100°F if the mixing is adiabatic?

4.160. Saturated steam at 300°F is blown continuously into a tank of 30% H_2SO_4 at 70°F. What is the highest concentration of liquid H_2SO_4 that can result from this process?

4.161. One thousand pounds of 10% NaOH solution at 100°F is to be fortified to 30% NaOH by adding 73% NaOH at 200°F. How much 73% solution must be used? How much cooling must be provided so that the final temperature will be 70°F?

4.162. A mixture of ammonia and water in the vapor phase, saturated at 250 psia and containing 80% by weight ammonia, is passed through a condenser at a rate of 10,000 lb/hr. Heat is removed from the mixture at the rate of 5,800,000 Btu/hr while the mixture passes through a cooler. The mixture is then expanded to a pressure of 100 psia and passes into a separator. A flow sheet of the process is given Fig. P4.162. If the heat loss from the equipment to the surroundings is neglected, determine the composition of the liquid leaving the separator by a material and energy balance set of equations.

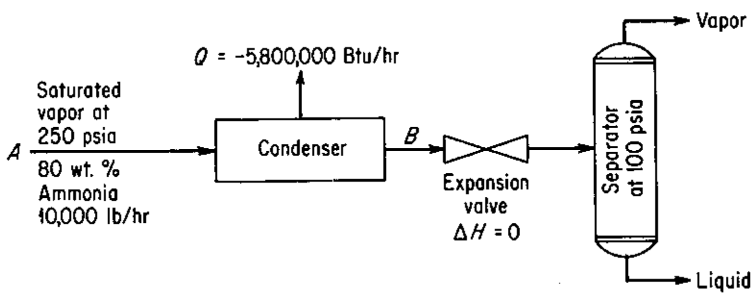


Figure P4.162

Section 4.9

4.163. Autumn air in the deserts of the southwestern United States during the day will typically be moderately hot and dry. If a dry-bulb temperature of 27°C and a wet-bulb temperature of 17°C is measured for the air at noon:
(a) What is the dew point?
(b) What is the percent relative humidity?
(c) What is the percent humidity?

4.164. The air supply for a dryer has a dry-bulb temperature of 32°C and a wet-bulb temperature of 25.5°C. It is heated to 90°C by coils and blown into the dryer. In the dryer, it cools along an adiabatic cooling line as it picks up moisture from the dehydrating material and leaves the dryer fully saturated.
(a) What is the dew point of the initial air?
(b) What is its humidity?
(c) What is its percent relative humidity?

(d) How much heat is needed to heat 100 m^3 to 90°C?
(e) How much water will be evaporated per 100 m^3 of air entering the dryer?
(f) At what temperature does the air leave the dryer?

4.165. Moist air at 100 kPa, a dry-bulb temperature of 90°C, and a wet-bulb temperature of 46°C is enclosed in a rigid container. The container and its contents are cooled to 43°C.
(a) What is the molar humidity of the cooled moist air?
(b) What is the final total pressure in atm in the container?
(c) What is the dew point in °C of the cooled moist air?

4.166. Calculate:
(a) The humidity of air saturated at 120°F
(b) The saturated volume at 120°F
(c) The adiabatic saturation temperature and wet-bulb temperature of air having a dry-bulb T = 120°F and a dew point = 60°F
(d) The percent saturation when the air in (3) is cooled to 82°F
(e) The pounds of water condensed/100 lb of moist air in (3) when the air is cooled to 40°F

4.167. A rotary dryer operating at atmospheric pressure dries 10 tons/day of wet grain at 70°F, from a moisture content of 10% to 1% moisture. The air flow is countercurrent to the flow of grain, enters at 225°F dry-bulb and 110°F wet-bulb temperature, and leaves at 125°F dry-bulb. See Fig. P4.167. Determine:
(a) The humidity of the entering and leaving air
(b) The water removal in pounds per hour
(c) The daily product output in pounds per day
(d) The heat input to the dryer
Assume that there is no heat loss from the dryer, that the grain is discharged at 110°F, and that its specific heat is 0.18.

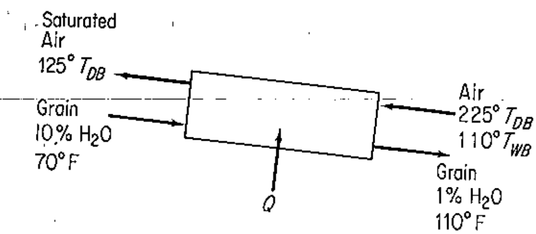


Figure P4.167

4.168. Temperatures (in °F) taken around a forced-draft cooling tower are as follows:

	In	Out
Air	85	90
Water	102	89

The wet-bulb temperature of the entering air is 77°F. Assuming the air leaving the tower to be saturated, calculate with the aid of the data given below:
(a) The humidity of the entering air
(b) The pounds of dry air through the tower per pound of water into the tower
(c) The percentage of water vaporized in passing through the tower

T (°F)	77	85	89	90	100	102
Saturated $\mathcal{H}$	0.0202	0.0264	0.030	0.031	0.0425	0.0455

C_p air = 0.24; C_p water vapor = 0.45; ΔH_v = 1040 Btu/lb at any of the temperatures encountered in this problem.

4.169. A dryer produces 180 kg/hr of a product containing 8% water from a feed stream that contains 1.25 g of water per gram of dry material. The air enters the dryer at 100°C dry-bulb and a wet bulb temperature of 38°C; the exit air leaves at 53°C dry-bulb and 60% relative humidity. Part of the exit air is mixed with the fresh air supplied at 21°C, 52% relative humidity, as shown in Fig. P4.169. Calculate the air and heat supplied to the heater, neglecting any heat lost by radiation, used in heating the conveyor trays, and so forth. The specific heat of the product is 0.18.

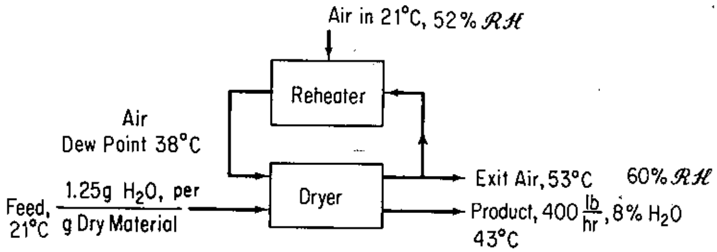


Figure P4.169

4.170. Air, dry bulb 38°C, wet bulb 27°C, is scrubbed with water to remove dust. The water is maintained at 24°C. Assume that the time of contact is sufficient to reach complete equilibrium between air and water. The air is then heated to 93°C by passing it over steam coils. It is then used in an adiabatic rotary drier from which it issues at 49°C. It may be assumed that the material to be dried enters and leaves at 46°C. The material loses 0.05 kg H_2O per kilogram of product. The total product is 1000 kg/hr.

(a) What is the humidity:
  (1) Of the initial air?
  (2) After the water sprays?
  (3) After reheating?
  (4) Leaving the drier?

(b) What is the percent humidity at each of the points in part (a)?

(c) What is the total weight of dry air used per hour?

(d) What is the total volume of air leaving the drier?

(e) What is the total amount of heat supplied to the cycle in joules per hour?

PROBLEMS THAT REQUIRE WRITING COMPUTER PROGRAMS

4.1. The enthalpy of a mixture of constant composition at any constant temperature T is

$$\Delta H_{\text{mixture}}(T, p) = \sum_{i=1}^{n} x_i \sum H_i^0(T, p) - \Delta H^D_{\text{mixture}}(T, p)$$

where the deviation, ΔH^D, is a correction because of high pressure. Use of the Benedict-Webb-Rubin (BWR) equation (see Table 3.2) gives the following for ΔH^D:

$$\Delta H^D = \left(B_0RT - 2A_0 - 4\frac{C_0}{T^2}\right)\rho + \frac{1}{2}(2bRT - 3a)\rho^2 + \frac{6}{5}(a\alpha\rho^5)$$
$$+ \frac{c\rho}{T^2}\left[\frac{3[1 - \exp(-\gamma\rho^2)]}{\gamma\rho^2} - \left(\frac{1}{2} - \gamma\rho^2\right)\exp(-\gamma\rho^2)\right]$$

Use the density values calculated in Computer Problem 3.3 for ρ, and prepare an additional subroutine to give $\Delta H^D_{\text{mixture}}(T, p)$ for selected T and p pairs. Refer to Computer Problem 3.3 for other pertinent information.

4.2. Prepare a computer program to find the adiabatic flame temperature of a gas such as in Problem 4.129. Assume that the principal reactant is completely burned.

4.3. Prepare a more general program than the one in Computer Problem 4.2 to compute the adiabatic flame temperature. Assume that several hydrocarbons of the form CH_n can be burned with various amounts of excess air. The percentage of excess air is to be read into the program as well as the type of fuel gas or solid. The only combustion products that need to be considered are carbon dioxide, carbon monoxide, water, hydrogen, oxygen, and nitrogen. Make provision for the following additional parameters:

(a) Pressure in the burner
(b) Humidity of the incoming air
(c) Number of moles of each hydrocarbon and of nitrogen in the fuel stream
(d) Standard heats of formation for all the hydrocarbon gases burned and the products (Assume that gaseous water is formed in the reaction.)
(e) Heat capacities of all components
(f) Inlet reactant temperature, which may be different for each reactant.

The gases can be treated as an ideal mixture.

4.4. Gonzales-Pozo listed the coefficients that can be used to predict the properties of saturated liquid water and steam based on the equation

$$Y = Ap + B/p + Cp^{1/2} + D\ln p + Ep^2 + Fp^3 + G$$

where p is the pressure in psia. Prepare a computer program to calculate the liquid and vapor specific volume, temperature, liquid enthalpy, heat of vaporization, vapor enthalpy, liquid internal energy, and vapor internal energy based on the set of coefficients in Table P4.4. Check your program calculations with the steam tables.

4.5. Prepare a graph in which the vertical axis shows the fraction of the gross energy input from the fuel in a combustion process that is available for transfer to the process as a fraction of the stack gas temperature in °F on the horizontal axis. Use as a third parameter lines of percent excess air that are used in the combustion. The graph you prepare can be used to answer such questions as:

(a) How much energy could be saved if the stack gas temperature were reduced 10% or the excess air reduced 10%?
(b) What conditions limit the extent to which the efficiency can be improved?

4.6. Prepare a Fortran program that will use the input number of atoms of C, H, and O in the fuel, the reactant moles and temperatures, and product composition and temperature, to make both material and energy balances for a combustion process. Assume that the process is adiabatic. Take the heat capacity equations and ΔH_f° data from Appendix K and Table D.1.

4.7. Prepare a Fortran program that uses either (a) the dry-bulb temperature plus the wet-bulb temperature, or (b) the dry-bulb temperature and the dew point, or (c) the dry-bulb temperature and the relative humidity of (1) the inlet air and (2) the outlet air, respectively, to calculate the amount of water condensed, the heat added or removed, and the dew point temperature of outlet air for a cooling unit given the input of wet air in m^3/min.

TABLE P4.4

Property	*A*	*B*	*C*
Temperature (°F)	−0.17724	3.83986	11.48345
Liquid specific volume (ft^3/lb)	-5.280126×10^{-7}	2.99461×10^{-5}	1.521874×10^{-4}
Vapor specific volume (ft^3/lb)			
1–200 psia	−0.48799	304.717614	9.8299035
200–1,500 psia	2.662×10^{-3}	457.5802	−0.176959
Liquid enthalpy (Btu/lb)	−0.15115567	3.671404	11.622558
Vaporization enthalpy (Btu/lb)	0.008676153	−1.3049844	−8.2137368
Vapor enthalpy (Btu/lb)	−0.14129	2.258225	3.4014802
Liquid entropy [Btu/(lb)(°R)]	-1.67772×10^{-4}	4.272688×10^{-3}	0.01048048
Vaporization entropy [Btu/(lb)(°R)]	3.454439×10^{-5}	-2.75287×10^{-3}	-7.33044×10^{-3}
Vapor entropy [Btu/(lb)(°R)]	-1.476933×10^{-4}	1.2617946×10^{-3}	3.44201×10^{-3}
Liquid internal energy (Btu/lb)	−0.1549439	3.662121	11.632628
Vapor internal energy (Btu/lb)	−0.0993951	1.93961	2.428354

Property	*D*	*E*	*F*	*G*
Temperature (°F)	31.1311	8.762969×10^{-5}	-2.78794×10^{-8}	86.594
Liquid specific volume (ft^3/lb)	6.62512×10^{-5}	8.408856×10^{-10}	1.86401×10^{-14}	0.01596
Vapor specific volume (ft^3/lb)				
1–200 psia	−16.455274	9.474745×10^{-4}	-1.363366×10^{-6}	19.53953
200–1,500 p	0.82682	-4.601876×10^{-7}	6.3181×10^{-11}	−2.3928
Liquid enthalpy (Btu/lb)	30.832667	8.74117×10^{-5}	-2.62306×10^{-8}	54.55
Vaporization enthalpy (Btu/lb	−16.37649	-4.3043×10^{-5}	9.763×10^{-9}	1,045.81
Vapor enthalpy (Btu/lb)	14.438078	4.222624×10^{-5}	-1.569916×10^{-8}	1,100.5
Liquid entropy [Btu/(lb)(°R)]	0.05801509	9.101291×10^{-8}	-2.7592×10^{-11}	0.11801
Vaporization entropy [Btu/(lb)(°R)]	−0.14263733	-3.49366×10^{-8}	7.433711×10^{-12}	1.85565
Vapor entropy [Btu/(lb)(°R)]	−0.08494128	6.89138×10^{-8}	-2.4941×10^{-11}	1.97364
Liquid internal energy (Btu/lb	30.82137	8.76248×10^{-5}	-2.646533×10^{-8}	54.56
Vapor internal energy (Btu/lb)	10.9818864	2.737201×10^{-5}	-1.057475×10^{-8}	1,040.03

SOURCE: V. Gonzalez-Pozo, *Chem. Eng.*, p. 123 (May 12, 1986).

SOLVING MATERIAL AND ENERGY BALANCES VIA COMPUTER CODES

5

Now that you have accumulated some experience in making energy balances and in the principles of thermochemistry, it is time to apply this knowledge to more complex problems involving both material and energy balances. You have already encountered some simple examples of combined material and energy balances, as, for

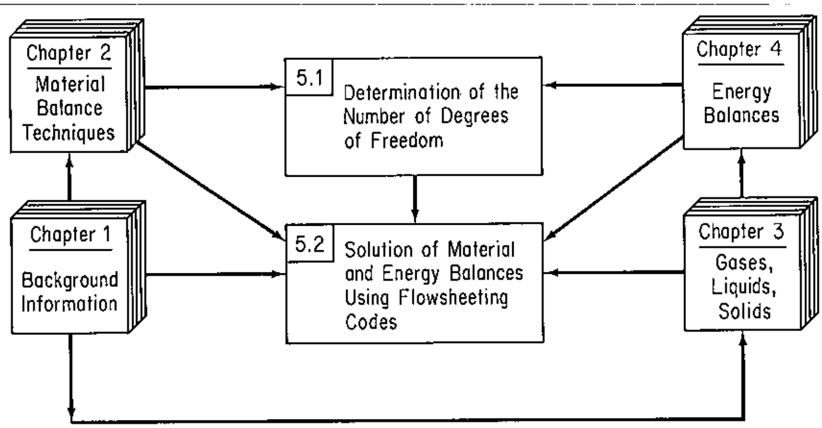


Figure 5.0 Hierarch of topics to be studied in this chapter (section numbers are in the upper left-hand corner of the boxes).

example, in the calculation of the adiabatic reaction temperature, where a material balance provides the groundwork for the writing of an energy balance. In fact, in all energy balance problems, however trifling they may seem, you must know the amount of material entering and leaving the process if you are to apply successfully the appropriate energy balance equation(s).

In this chapter we first examine how to make sure that a problem is properly and completely specified by extending our earlier treatment in Chap. 2 of the analysis of the degrees of freedom. Then we review how computer codes and, in particular, flowsheeting codes can be employed to solve complex problems. Figure 5.0 shows the relationships among two topics to be discussed in this chapter, and the relationships with topics in previous chapters.

5.1 ANALYSIS OF THE DEGREES OF FREEDOM

Your objectives in studying this section are to be able to:

1. Identify the names and numbers of variables in the streams entering and leaving a processing unit, and the variables associated with the unit itself.
2. Identify the types and numbers of all the equations (= 0) (and any inequality constraints (≤0)) that relate to the variables.
3. Determine the number of independent equations for each processing unit.
4. Calculate the number of degrees of freedom (decision variables) for single units and combination of units both without and with a reaction taking place.
5. Specify the values of variables equal to the number of degrees of freedom for a unit.
6. Accomplish numbers 1 through 5 for combinations of units, including bypass and recycle streams.

An important aspect of combined material and energy balance problems is how to ensure that the process equations or sets of modules are determinate, that is, have at least one solution, and hopefully no more than one solution. The question is: How many variables are unknown, and how many must have their values specified in any problem? The **number of degrees of freedom** is the number of variables in a set of *independent* equations to which values must be assigned to make the number of unknown variables plus the variables assigned values equal to the number of equations. We will first discuss the number of degrees of freedom associated with a process stream, next evaluate how many degrees of freedom exist for a unit module, and finally examine the degrees of freedom in a complete flowsheet.

Let N_d = number of degrees of freedom, N_v = number of variables, and N_r = number of equations (restrictions, constraints). Then for N_r independent equations in general

$$N_d = N_v - N_r \tag{5.1}$$

and we conclude that $N_v - N_r$ variables must be specified **as long as the N_r equations are still independent.** The latter is an important point because, as we showed in Chap. 2, it is not always easy to write the set of equations that exactly model a unit operation (ignoring problems with the theory underlying the equations). You can write too many equations, but not notice that one or more of them are not independent (i.e., can be derived by combining others). If you make a mistake in the count for N_d, you may formulate a set of equations that has an infinite number of solutions or a least-squares solution (for a set of indeterminate equations), and not understand why, or perhaps not even notice, that the results returned by executing a computer code are wrong.

How many variables are required to specify a process stream completely? To characterize a process stream we can select among temperature, pressure, total flow, overall mole fractions, phase fractions, mole fractions, total enthalpy, phase enthalpies, and so on. How many variables are needed to fix completely the state of the stream if we assume that phase and chemical equilibrium exist in the stream? A review of the phase rule shows that for a single phase, the degrees of freedom associated with a stream are equal to $C - 1 + 2 = C + 1$ intensive variables. A little thought will indicate that in the absence of reaction, you must in addition add the total flow rate of a stream to the count to specify the stream completely. The variables often associated with a stream are p, T, the total molar (mass) flow, and $C - 1$ overall mole (mass) fractions for the components. An equivalent set would be p, T, and the individual component molar flow rates. You cannot specify C compositions, and the molar flow rate as one composition is redundant.

Thus we can conclude that the number of variables needed to specify the condition of a stream completely is given by

$$N_v = N_{sp} + 2 \tag{5.2}$$

where N_{sp} is the number of chemical species in the stream. You should keep in mind that in a binary system, for example, in which one stream component is zero, for consistency you would count $N_{sp} = 2$ with the one component having a zero value treated as a restriction.

Next, let us look at how to ascertain the number of degrees of freedom for a complete process unit. We will see that each process unit has a different number of degrees of freedom associated with it. In particular we need to pay close attention as to how to count N_v and N_r in Eq. (5.1). The total number of variables for all streams entering or leaving the boundaries of a process unit would be

$$N_v = N_s(N_{sp} + 2) + N_Q + N_W + N_p \tag{5.3}$$

where N_s = total number of streams
N_Q = number of heat transfer variables
N_W = number of work variables
N_p = number of equipment parameters, such as kinetic rate constants, equilibrium constants, recycle ratio, and so on

What types of constraints do you have to include in the count for N_r for a process unit? Here is a list of the common restrictions:

(a) Material balances
(b) One energy balance
(c) Equipment constraints involving the equipment parameters and equilibrium relations
(d) Variables with prespecified values—one constraint for each specification (as long as the material and energy balances are still independent), or fixed ratios of components and so on. Note that if the energy balance is not involved in the process, the energy balance as a constraint will not be independent. It will be the same as an overall material balance.

We now take up five simple processes in Example 5.1, and evaluate the number of degrees of freedom for each.

EXAMPLE 5.1 Determining the Degrees of Freedom in a Process

We shall consider five typical processes, as depicted by the respective figures below, and for each ask the question: How many variables have to be specified [i.e., what are the degrees of freedom (N_d)] to make the problem of solving the combined material and energy balances determinate? All the processes will be steady-state ones, and the entering and exit streams single-phase streams.

(a) Stream splitter (Fig. E5.1a): We assume that $Q = W = 0$, and that the energy balance is not involved in the process. By implication of a splitter, the temperatures, pressures, and compositions of the inlet and outlet streams are identical, and $N_p = 0$. The count of the total number of variables, total number of constraints, and degrees of freedom is as follows.

Total number of variables		
$N_v = N_s(N_{sp} + 2) =$		$3(N_{sp} + 2)$
Number of independent equality constraints		
Material balances	1	
Compositions of Z, P_1, and P_2 are the same	$2(N_{sp} - 1)$	
$T_{P_1} = T_{P_2} = T_Z$	2	
$p_{P_1} = p_{P_2} = p_Z$	2	$2N_{sp} + 3$
Total no. d.f.		$N_{sp} + 3$

Did you note that we did not count N_{sp} material balances but only one? Why? Let us look at

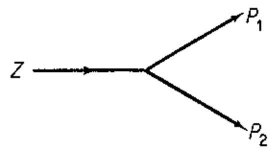


Figure E5.1a

some of the balances:

$$\text{Component 1: } Zx_{1Z} = P_1x_{1P_1} + P_2x_{1P_2}$$

$$\text{Component 2: } Zx_{2Z} = P_1x_{2P_1} + P_2x_{2P_2}$$

etc.

If $x_{1Z} = x_{1P_1} = x_{1P_2}$ and the same is true for x_2, and so on, only one independent material exists.

Do you understand the counts resulting from making the compositions equal in Z, P_1, and P_2? Write down the expressions for each component: $x_{iZ} = x_{iP_1} = x_{iP_2}$. Each set represents $2N_{sp}$ constraints. But you cannot specify every x_i in a stream, only $N_{sp} - 1$ of them. Do you remember why?

To make the problem determinate we might specify the values of the following decision variables:

Flow rate Z	1
Composition of Z	$N_{sp} - 1$
T_Z	1
p_Z	1
Ratio of split $\alpha = P_1/P_2$	1
Total no. d.f.	$N_{sp} + 3$

(b) Mixer (Fig. E5.1b): For this process we assume that $W = 0$, but Q is not.

Total number of variables (3 streams + Q)	$3(N_{sp} + 2) + 1$
Number of independent equality constraints	
Material balances	N_{sp}
Energy balance	1

$$\text{Total no. d.f.} = 3(N_{sp} + 2) + 1 - (N_{sp} + 1) = 2N_{sp} + 6$$

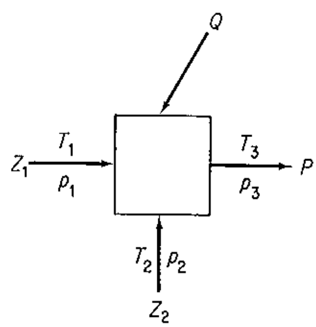


Figure E5.1b

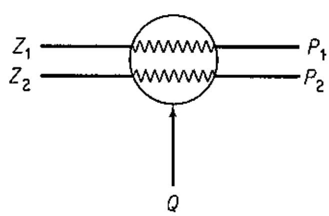


Figure E5.1c

(c) Heat exchanger (Fig. E5.1c): For this process we assume that $W = 0$ (but not Q) and $N_p = 0$.

Total number of variables		
$N_v = N_s(N_{sp} + 2) + 1 =$		$4N_{sp} + 9$
Number of independent equality constraints		
Material balances (streams 1 and 2)	2	
Energy balance	1	
Composition of inlet and outlet streams the same	$2(N_{sp} - 1)$	$2N_{sp} + 1$
Total no. d.f.		$2N_{sp} + 8$

One might specify four temperatures, four pressures, $2(N_{sp} - 1)$ compositions in Z_1 and Z_2, and Z_1 and Z_2 themselves to use up $2N_{sp} + 8$ degrees of freedom.

(d) Pump (Fig. E5.1d): Here $Q = 0$ but W is not; $N_p = 0$.

Total number of variables		
$N_v = N_s(N_{sp} + 2) + 1$		$2N_{sp} + 5$
Number of independent equality constraints		
Material balances	1	
Composition of inlet and outlet streams the same	$N_{sp} - 1$	
Energy balance	1	$N_{sp} + 1$
Total no. d.f.		$N_{sp} + 4$

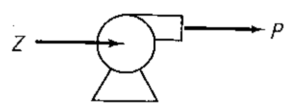


Figure E5.1d

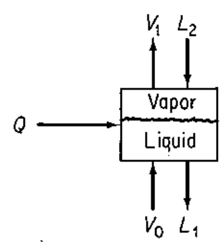


Figure E5.1e

(e) Two-phase well-mixed tank (stage) at equilibrium (Fig. E5.1e) (L, liquid phase; V, vapor phase): Here $W = 0$. Although two phases exist (V_1 and L_1) at equilibrium inside the system, the streams entering and leaving are single-phase streams. By equilibrium we mean that both phases are at the same temperature and pressure and that an equation is known that relates the composition in one phase to that in the other for each component.

Total number of variables (4 streams + Q)	$4(N_{sp} + 2) + 1$	
Number of independent equality constraints		
Material balances		N_{sp}
Energy balance		1
Composition relations at equilibrium		N_{sp}
Temperatures of the streams V_1 and L_1 in the two phases equal		1
Pressures of the streams V_1 and L_1 in the two phases equal		1

Total no. d.f. $= 4(N_{sp} + 2) + 1 - 2N_{sp} - 3 = 2N_{sp} + 6$

In general you might specify the following variables to make the problem determinate:

Input stream L_2	$N_{sp} + 2$
Input stream V_0	$N_{sp} + 2$
Pressure	1
Q	1
Total	$2N_{sp} + 6$

Other choices are of course possible, but such choices must leave the equality constraints independent.

EXAMPLE 5.2 Proper Process Specification

Figure E5.2 shows an isothermal separations column. At present the column specifications call for the feed mass fractions to be $\omega_{C_4} = 0.15$, $\omega_{C_5} = 0.20$, $\omega_{iC_5} = 0.30$, and $\omega_{C_6} = 0.35$; the overhead mass fractions to be $\omega_{C_5} = 0.40$ and $\omega_{C_6} = 0$; and the residual product mass fraction to be $\omega_{C_4} = 0$. Unless specified as 0, the component exists in a stream.

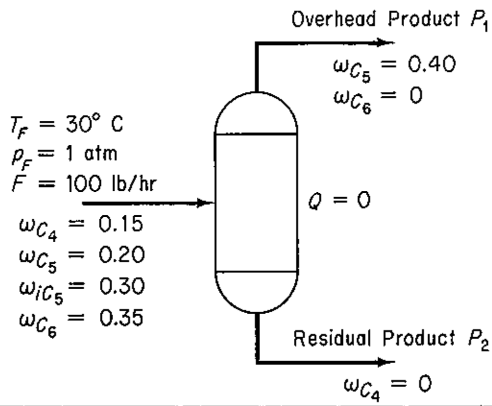


Figure E5.2

Given that $F = 100$ lb/hr, is the separator completely specified, that is, are the degrees of freedom $N_d = 0$? The streams P_1 and P_2 are not in equilibrium.

Solution

First we calculate N_v and then N_r. We will assume that all the stream temperatures and pressures are identical. Number of variables N_v:

$$N_v = (N_{sp} + 2)(3) = (4 + 2)(3) = \qquad 18$$

Number of restrictions and equations for N_r:

Component mass balances (no reaction) $= N_{sp} =$	4
$T_F = T_{P_1} = T_{P_2}$	2
$p_F = p_{P_1} = p_{P_2}$	2

Initial column specifications:

(4 in F, 2 in P_1, and 1 in P_2 plus $T_F = 30°$C and $p_F = 1$ atm)	9	
Total for N_r		17

Number of degress of freedom N_d: $18 - 17 =$ 1

Note that only four factors can be specified in stream F, namely the rate of F itself and three ω's; one of the ω's is redundant. One more variable must be specified for the process, but one that will not reduce the number of independent equations and restrictions already enumerated.

So far we have examined single units without a reaction occurring in the unit. How is the count for N_d affected by the presence of a reaction in the unit? The way N_v is calculated does not change. As to N_r, all restrictions and constraints are deducted from N_v that represent *independent* restrictions on the unit. Thus the number of material balances is not necessarily equal to the number of species (H_2O, O_2, CO_2, etc.) but instead is the number of independent material balances that exist determined in the same way as we did in Secs. 2.2 to 2.4, usually (but not always) equal to the number of elemental balances (H, O, C, etc.). Fixed ratios of materials such as the O_2/N_2 ratio in air or the CO/CO_2 ratio in a product gas would be a restriction, as would be a specified conversion fraction or a known molar flow rate of a material. If some degrees of freedom exist still to be specified, improper specification of a variable may disrupt the independence of equations and/or specifications previously enumerated in the unit of N_r, so be careful.

EXAMPLE 5.3 Degrees of Freedom with a Reaction Taking Place in the System

A classic reaction for producing H_2 is the so-called "water gas shift" reaction:

$$CO + H_2O \rightleftharpoons CO_2 + H_2$$

Figure E5.3 shows the process data and the given information. How many degrees of freedom remain to be satisfied? For simplicity assume that the temperature and pressures of all entering and exit streams are the same and that all streams are gases. The amount of water which is in excess of that needed to convert all the CO completely to CO_2 is prespecified.

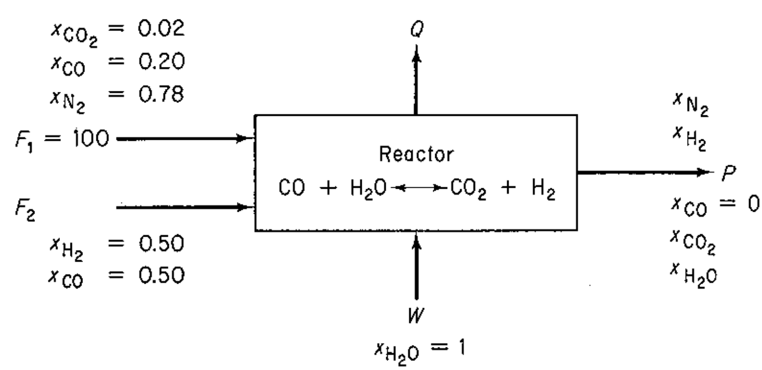


Figure E5.3

Solution

$N_v = N_s(N_{sp} + 2) + 1 = 4(5 + 2) + 1 =$ (+1 is for Q)			29
N_r: Independent material balances			
(C, O, N, H)	4		
Energy balance	1		
$T_{F_1} = T_{F_2} = T_W = T_P$	3		
$p_{F_1} = p_{F_2} = p_W = p_P$	3	11	
Compositions and flows specified:			
In F_1 ($x_{H_2O} = x_{H_2} = 0$)	5		
In F_2 ($x_{N_2} = x_{CO_2} = x_{H_2O} = 0$)	4		
In W (all but x_{H_2O} are 0)	4		
In P ($x_{CO} = 0$)	1	14	
Excess W given		1	
			26
$N_d = 29 - 26 =$			3

In the streams F_2 and W, only four compositions can be specified; a fifth specification is redundant. The total flows are not known. The given value of the excess water provides the information about the reaction products. Certainly, the temperature and pressure need to be specified, absorbing two degrees of freedom. The remaining degree of freedom might be the N_2/H_2 ratio in P, or the value of F_2, or the ratio of F_1/F_2, and so on.

EXAMPLE 5.4 Degrees of Freedom for the Case of Multiple Reactions

Methane burns in a furnace with 10% excess air, but not completely, so some CO exits the furnace, but no CH_4 exits. The reactions are:

$$CH_4 + 1.5O_2 \longrightarrow CO + 2H_2O$$
$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$
$$CO + 0.5O_2 \longrightarrow CO_2$$

Carry out a degree-of-freedom analysis for this combustion problem.

Solution

Figure E5.4 shows the process; all streams are assumed to be gases. Only two of the reactions are independent. Q is a variable here. For simplicity assume that the entering and exit streams are at the same temperatures and pressures.

$N_v = 3(6 + 2) + 1 =$ (+1 is for Q)		25
N_r:		
Material balances (C, H, O, N)	4	
Energy balance	1	
$T_A = T_F = T_P$	2	
$p_A = p_F = p_P$	2	
Compositions specified:		
In A ($N_{sp} - 1$)	5	
In F ($N_{sp} - 1$)	5	
In P	1	11
Percent excess air:		1
		21

$N_d = 25 - 21 = 4$

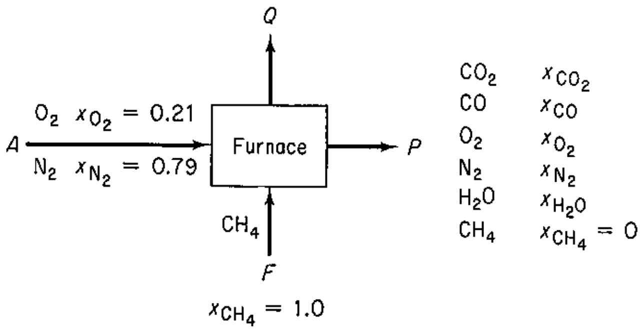


Figure E5.4

To have a well-defined problem you should specify (a) the temperature, (b) the pressure, (c) either the feed rate, or the air rate, or the product rate, and (d) either the CO/CO_2 ratio or the fraction of CH_4 converted to CO or alternatively to CO_2.

You can compute the degrees of freedom for combinations of like or different simple processes by proper combination of their individual degrees of freedom. In adding the degrees of freedom for units, you must eliminate any double counting either for variables or constraints and take proper account of interconnecting streams whose characteristics are often fixed only by implication.

Examine the mixer-separator in Fig. 5.1. For the mixer considered as a separate unit, from Example 5.1b, $N_d = 2N_{sp} + 6$. For the separator, an equilibirum unit:

$$
\begin{array}{lll}
N_v = 3(N_{sp} + 2) + 1 = & & 3N_{sp} + 7 \\
N_r: & & \\
\quad \text{Material balances} & N_{sp} & \\
\quad \text{Equilibrium relations} & N_{sp} & \\
\quad \text{Energy balance} & 1 & \\
\quad T_Z = T_{P_1} = T_{P_2} & 2 & \\
\quad p_Z = p_{P_1} = p_{P_2} & \underline{2} & \underline{2N_{sp} + 5} \\
N_d = (3N_{sp} + 7) - (2N_{sp} + 5) = & & N_{sp} + 2
\end{array}
$$

The sum of the mixer and separator is $3N_{sp} + 8$.

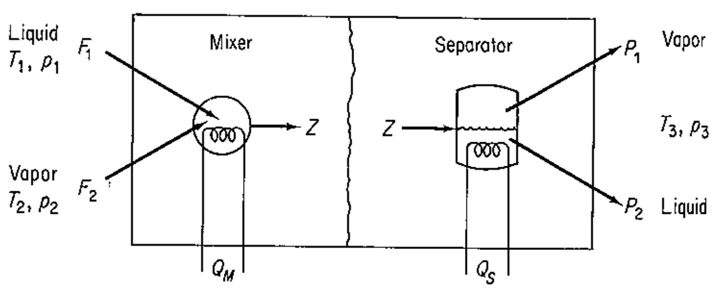


Figure 5.1 Degrees of freedom in combined units.

We must deduct redundant variables and add redundant restrictions as follows:

Redundant variables:	
Remove 1 Q	1
Remove Z	$N_{sp} + 2$
Redundent constraints:	
1 energy balance	1

Then $N_d = (3N_{sp} + 8) - (N_{sp} + 3) + 1 = 2N_{sp} + 6$, the same as in Example 5.1e.

EXAMPLE 5.5 Degrees of Freedom in a System Composed of Several Units

Ammonia is produced by reacting N_2 and H_2:

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

Figure E5.5a shows a simplified flowsheet. All the units except the separator and lines are adiabatic. The liquid ammonia product is essentially free of N_2, H_2, and A, and assume that the purge gas is free of NH_3. Treat the process as four separate units for a degree-of-freedom analysis, and then remove redundant variables and add redundant constraints to obtain the degrees of freedom for the overall process. The fraction conversion in the reactor is 25%.

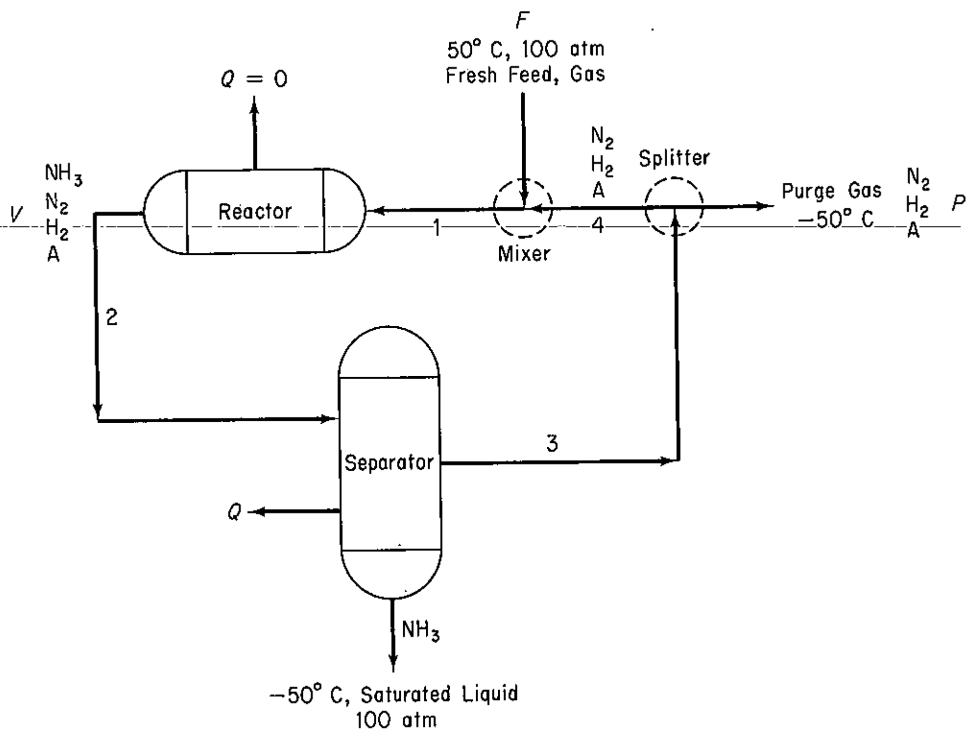


Figure E5.5a

Solution

Mixer:

$N_v = 3(N_{sp} + 2) + 1 = 3(6) + 1 =$		19
N_r:		
Material balances (H_2, N_2, A only)	3	
Energy balance	1	
Specifications:		
NH_3 concentration is zero	3	
$T_P = -50°C$	1	
$T_F = 50°C$	1	
Assume that $p_F = p_{\text{mix out}} = p_{\text{split}} = 100$	3	
$Q = 0$	1	13
N_d: $19 - 13 =$		6

Reactor:

$N_v = 2(N_{sp} + 2) + 1 = 2(6) + 1 =$		13
N_r:		
Material balances (H, N, A)	3	
Energy balances	1	
Specifications:		
NH_3 entering $= 0$	1	
$Q = 0$	1	
Fraction conversion	1	
$p_{\text{in}} = p_{\text{out}} = 100$ atm	2	
Energy balance	1	10
$N_d = 13 - 10$		3

Separator:

$N_v = 3(N_{sp} + 2) + 1 = 3(6) + 1 =$		19
N_r:		
Material balances	4	
Energy balance	1	
Specifications:		
$T_{\text{out}} = -50°C$	1	
$p_r = p_{\text{in}} = p_{NH_3} = 100$	3	
NH_3 concentration is 0 in recycle gas	1	
N_2, H_2, A are 0 in liquid NH_3	3	13
$N_d = 19 - 13$		6

Splitter:

$N_v = 3(N_{sp} + 2) = 3(6) =$		18
N_r:		
Material balances	1	
Specifications:		
NH_3 concentration $= 0$	1	
Compositions same $2(N_{sp} - 1)$	6	
Stream temperatures same $= -50°C$	3	
Stream pressures same $= 100$ atm	3	14
$N_d = 18 - 14$		4